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REFERENCE

# Pediatric & Neonatal Common & Emergency Care Guide

A searchable, algorithmic bedside companion for pediatricians and primary care physicians — every medication listed separately with weight-based dosing, first- to third-line therapy, and the can't-miss diagnoses. Tap any topic to open it.

Prepared and proposed to the Kuwait Pediatric Council by

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**Medical responsibility (State of Kuwait).** In Kuwait, the practice of medicine, the standard of care, and professional accountability are governed by **Law No. 70 of 2020** on the practice of the medical and allied health professions, patients' rights, and health facilities,<sup>51</sup> together with the decisions of the **Authority for Medical Responsibility** established under it. Nothing in this guide alters or lessens the treating physician's legal duties under that law — to hold a valid license and practice within their scope, to follow established medical rules and sound diagnostic and therapeutic procedures, to obtain informed consent, and to keep accurate, confidential medical records. Responsibility for every clinical decision, and for verifying it against the law and current Ministry of Health regulations, rests with the treating physician and the treating institution.

**Age & scope.** This guide covers the newborn through the adolescent. A **neonate** is a baby aged **≤ 28 days**; a **pediatric patient** is any child from birth **up to 18 years** of age. This mirrors Kuwait's **Child Rights Law No. 21 of 2015**, which defines a child as every person who has not completed 18 years.<sup>144</sup> In Kuwait the age of full legal (civil) majority is **21 years**; for any patient under 18, **consent to assessment and treatment is given by a parent or legal guardian**, while the older child/adolescent is involved according to their capacity. Individual

services may apply their own operational age cut-offs, so confirm local policy.



**Preparation.** This guide was drafted, edited, and formatted with the assistance of an AI-based large language model used under direct human supervision. The tool was not an author and made no clinical judgments. All clinical recommendations, medication names and doses, diagnostic thresholds, and references were written, verified against the primary sources and current guidelines, and approved by the author and the review panel named below, who take full responsibility for the accuracy and integrity of the content. Prepared in line with ICMJE guidance on the use of AI-assisted technologies and with the Ministry of Health **Practical Guide for the Use of Language-based Artificial Intelligence Tools in Healthcare Practice**, adopted and published as a guiding framework for staff of all MOH health facilities by Administrative Decision No. 6189 of 2026 of the Undersecretary of the Ministry of Health (issued under Law No. 70/2020 and Ministerial Decision No. 373/2025 on the code of ethics for health-sector workers).

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## What the colors mean

- 01 · Resuscitation** — cardiopulmonary arrest & the universal approach
- 02 · Behavioral & Psychiatric** — agitation, self-harm, drug-induced syndromes
- 03 · Cardiac & Shock** — arrhythmias, all types of shock, hypercyanotic spell
- 04 · Common Acute Conditions** — everyday clinic & ED infections and presentations
- 05 · Complex & Technology-Dependent Care** — tracheostomy, feeding tubes, ECMO, post-ICU, transport
- 06 · Endocrine & Metabolic** — diabetes, adrenal, inborn errors, electrolytes
- 07 · Focused Specialty Emergencies** — eye, dental, dermatologic, ENT & neurosurgical
- 08 · Gastroenterology & Hepatology** — liver failure, GI bleeding, pancreatitis, hepatitis, cholestasis
- 09 · Hematologic & Oncologic** — neutropenia, tumor lysis, sickle cell, bleeding
- 10 · Neonatal** — newborn-specific emergencies
- 11 · Neurologic** — seizures & status epilepticus
- 12 · Orthopedic & Musculoskeletal** — the limping child, SCFE, compartment syndrome, fractures

-  **13 · Pediatric Trauma & TBI** — primary/secondary survey, traumatic brain injury
-  **14 · Renal & Urology** — AKI, HUS, hypertensive emergency, acute scrotum & testicular torsion
-  **15 · Respiratory, Allergy & Immunology** — airway, breathing, anaphylaxis, allergy & immunology
-  **16 · Rheumatologic** — cytokine storm (HLH/MAS) & Kawasaki
-  **17 · Surgical Emergencies** — recognize, resuscitate, refer
-  **18 · Toxicology & Environmental** — poisoning, burns, heat, drowning, smoke inhalation

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Each topic also carries an **urgency tag** showing how fast to act:

**MINUTES**

immediately life-threatening — act now

**HOURS**

urgent — stabilize and treat early

**ASSESS**

evaluate and manage on its own pathway

**Call vs refer** — in a **hospital**, **call** the in-house specialist team (on site); in a **polyclinic** — or any facility **where the specialty is not available** — **refer** the patient to a facility that has it.

## Abbreviations

**CPS** — Canadian Pediatric Society

**GINA** — Global Initiative for Asthma

**KDIGO** — Kidney Disease: Improving Global Outcomes

**KNCPP** — Kuwait National Child Protection Program

**NASPGHAN** — North American Society for Pediatric Gastroenterology, Hepatology and Nutrition

**PES** — Pediatric Endocrine Society

**PIDS** — Pediatric Infectious Diseases Society

**NTSP** — National Tracheostomy Safety Project

**AAP** — American Academy of Pediatrics

**NRP** — Neonatal Resuscitation Program

**AARC** — American Association for Respiratory Care

**AUA** — American Urological Association

**BTF** — Brain Trauma Foundation

**CDC** — Centers for Disease Control and Prevention

**ELSO** — Extracorporeal Life Support Organization

**EMA** — European Medicines Agency

**ESPGHAN** — European Society for Pediatric Gastroenterology, Hepatology and Nutrition

**EULAR** — European Alliance of Associations for Rheumatology

**FDA** — US Food and Drug Administration

**AHA** — American Heart Association

**IDSA** — Infectious Diseases Society of America

**ASHP** — American Society of Health-System Pharmacists

**ANMF** — Australasian Neonatal Medicines Formulary

**ESC** — European Society of Cardiology

**ISMP** — Institute for Safe Medication Practices

**SIDP** — Society of Infectious Diseases Pharmacists

**AES** — American Epilepsy Society

**EAN** — European Academy of Neurology

**PNS** — Peripheral Nerve Society

**RCPCH** — Royal College of Paediatrics and Child Health (UK)

**ILAE** — International League Against Epilepsy

**ISPAD** — International Society for Pediatric and Adolescent Diabetes

**MOH** — Ministry of Health

**BTS** — British Thoracic Society (UK)

**NAEMSP** — National Association of EMS Physicians (US)

**NICE** — National Institute for Health and Care Excellence (UK)

**WHO** — World Health Organization

**DRACMA** — Diagnosis and Rationale for Action against Cow's Milk Allergy (WAO)

**WAO** — World Allergy Organization

**AAF** — amino acid formula

**ABC** — airway, breathing, circulation

**ABCDE** — airway, breathing, circulation, disability, exposure

**ABO** — ABO blood-group system

**ACS** — acute chest syndrome

**ADAMTS13** — von Willebrand factor-cleaving protease (ADAMTS13)

**ADHD** — attention-deficit / hyperactivity disorder

**AED** — automated external defibrillator; also antiepileptic drug

**aEEG** — amplitude-integrated electroencephalography

**AEIOU** — alcohol, epilepsy, insulin, opioids/overdose, uremia (coma mnemonic)

**AGEP** — acute generalized exanthematous pustulosis

**AHI** — apnea-hypopnea index

**AHO** — acute hematogenous osteomyelitis

**AHT** — abusive head trauma

**aHUS** — atypical hemolytic uremic syndrome

**AKI** — acute kidney injury

**ALT** — alanine aminotransferase

**ALTE** — apparent life-threatening event

**AMPLE** — allergies, medications, past history, last meal, events (history mnemonic)

**AMR** — antimicrobial resistance

**ANC** — absolute neutrophil count

**AOM** — acute otitis media

**APGAR** — Appearance, Pulse, Grimace, Activity, Respiration (score)

**APML** — acute promyelocytic leukemia

**aPTT** — activated partial thromboplastin time

**ARDS** — acute respiratory distress syndrome

**ASD** — atrial septal defect

**AST** — aspartate aminotransferase

**ATOMAC** —

Arizona-Texas-Oklahoma-Memphis-Arkansas Consortium (pediatric trauma guidelines)

**ASH** — American Society of Hematology

**ASO** — antistreptolysin O

**ATA** — American Thyroid Association

**ATP** — adenosine triphosphate

**AUC** — area under the (concentration–time) curve

**AV** — atrioventricular

**AVM** — arteriovenous malformation

**AVPU** — alert / voice / pain / unresponsive (conscious-level scale)

**AVSD** — atrioventricular septal defect

**B symptoms** — lymphoma constitutional symptoms (Ann Arbor staging): unexplained fever > 38 °C, drenching night sweats, weight loss > 10%

**BCG** — bacille Calmette–Guérin (vaccine)

**BID** — twice daily

**BMI** — body mass index

**BNP** — B-type natriuretic peptide

**BP** — blood pressure

**BRUE** — brief resolved unexplained event

**BSPED** — British Society for Paediatric Endocrinology and Diabetes

**BWPS** — Burch–Wartofsky Point Scale (for thyrotoxicosis)

**BXO** — balanitis xerotica obliterans

**C1-INH** — C1-esterase inhibitor

**CAH** — congenital adrenal hyperplasia

**CAP** — community-acquired pneumonia

**CBC** — complete blood count

**CCHD** — critical congenital heart disease

**CDH** — congenital diaphragmatic hernia

**CFU** — colony-forming units

**CHD** — congenital heart disease

**CICO** — can't intubate, can't oxygenate

**CKD** — chronic kidney disease

**CMC** — children with medical complexity

**CMPA** — cow's milk protein allergy

**CMV** — cytomegalovirus

**CNO** — chronic nonbacterial osteomyelitis

**CNS** — central nervous system

**CO** — carbon monoxide

**COG** — Children's Oncology Group

**COHb** — carboxyhemoglobin

**CPAP** — continuous positive airway pressure

**CPEG** — Canadian Pediatric Endocrine Group

**CPR** — cardiopulmonary resuscitation

**CRMO** — chronic recurrent multifocal osteomyelitis

**CRP** — C-reactive protein

**CRASH-2 / CRASH-3** — Clinical Randomisation of an Antifibrinolytic in Significant Haemorrhage trials (tranexamic acid in trauma / head injury)

**CRRT** — continuous renal replacement therapy

**CSF** — cerebrospinal fluid

**CT** — computed tomography

**CVC** — central venous catheter

**CVID** — common variable immunodeficiency

**CVS** — cardiovascular system

**CXR** — chest radiograph (chest X-ray)



**DAT** — direct antiglobulin (Coombs) test

**DDAVP** — desmopressin

**DEFG** — don't ever forget glucose (coma mnemonic)

**DIC** — disseminated intravascular coagulation

**DIVA** — difficult intravenous access

**DKA** — diabetic ketoacidosis

**DL** — direct laryngoscopy

**DMD** — Duchenne muscular dystrophy

**DNase** — deoxyribonuclease (dornase alfa)

**DOPE** — displacement, obstruction, pneumothorax, equipment (deterioration mnemonic)

**DOPES** — displacement, obstruction, pneumothorax, equipment, stacked breaths (ventilator mnemonic)

**DRESS** — drug reaction with eosinophilia and systemic symptoms

**DSD** — difference / disorder of sex development

**DTaP** — diphtheria, tetanus, acellular pertussis (pediatric)

**EBV** — Epstein–Barr virus

**ECG** — electrocardiogram

**ECMO** — extracorporeal membrane oxygenation

**ECPR** — extracorporeal cardiopulmonary resuscitation

**ED** — emergency department

**EEG** — electroencephalogram

**eGFR** — estimated glomerular filtration rate

**EHF** — extensively hydrolyzed formula

**EMLA** — eutectic mixture of local anesthetics

**ENT** — ear, nose and throat (otolaryngology)

**ERCP** — endoscopic retrograde cholangiopancreatography

**ESBL** — extended-spectrum  $\beta$ -lactamase

**ESETT** — Established Status Epilepticus Treatment Trial

**ESC-NOW** — Eat, Sleep, Console approach for Neonatal Opioid Withdrawal (trial)

**ESR** — erythrocyte sedimentation rate

**ET** — endotracheal (route)

**ETT** — endotracheal tube

**EVD** — external ventricular drain

**eFAST** — extended Focused Assessment with Sonography for Trauma (adds lung views for pneumothorax)

**FAST** — Focused Assessment with Sonography for Trauma (bedside ultrasound for free fluid)

**FBC** — full blood count

**FEIBA** — factor eight inhibitor bypassing activity

**FHH** — familial hypocalciuric hypercalcemia

**FiO<sub>2</sub>** — fraction of inspired oxygen

**FN** — febrile neutropenia

**FPE** — food protein-induced enteropathy

**FPIAP** — food protein-induced allergic proctocolitis

**FPIES** — food protein-induced enterocolitis syndrome

**FTT** — failure to thrive

**FVC** — forced vital capacity

**G6PD** — glucose-6-phosphate dehydrogenase

**GA<sup>2</sup>LEN** — Global Allergy and Asthma European Network (anaphylaxis criteria)

**GAS** — group A *Streptococcus*

**GBS** — group B *Streptococcus*; also Guillain-Barré syndrome

**GCS** — Glasgow Coma Scale

**GER** — gastroesophageal reflux

**GERD** — gastroesophageal reflux disease

**GFR** — glomerular filtration rate

**GGT** — gamma-glutamyl transferase

**GH** — growth hormone

**GI** — gastrointestinal

**GIR** — glucose infusion rate

**GJ** — gastrojejunal (tube)

**HA** — hypoallergenic (partially hydrolyzed formula label)

**HAE** — hereditary angioedema

**HAST** — hypoxia-altitude simulation test

**HbA1c** — glycated hemoglobin

**HBO** — hyperbaric oxygen

**HCN** — hydrogen cyanide

**HCT** — hematocrit

**HDU** — high-dependency unit

**HEEADSSS** — Home, Education, Eating, Activities, Drugs, Sexuality, Suicide, Safety (adolescent interview)

**HFNC** — high-flow nasal cannula

**HHS** — hyperosmolar hyperglycemic state

**HHV-6** — human herpesvirus 6

**HLA** — human leukocyte antigen

**HLH** — hemophagocytic lymphohistiocytosis

**HME** — heat and moisture exchanger

**HR** — heart rate

**HSP** — Henoch–Schönlein purpura (IgA vasculitis)

**HSV** — herpes simplex virus

**HUS** — hemolytic uremic syndrome

**IABP** — intra-aortic balloon pump

**IBI** — invasive bacterial infection (bacteremia or bacterial meningitis)

**ICP** — intracranial pressure

**ICU** — intensive care unit

**ID** — infectious diseases

**IDM** — infant of a diabetic mother

**IEI** — inborn errors of immunity

**IgA** — immunoglobulin A

**IgE** — immunoglobulin E

**IgG** — immunoglobulin G

**IgM** — immunoglobulin M

**IGRA** — interferon- $\gamma$  release assay

**IM** — intramuscular

**iNO** — inhaled nitric oxide

**INR** — international normalized ratio

**IO** — intraosseous

**IPNA** — International Pediatric Nephrology Association

**ITP** — immune thrombocytopenia

**IU** — international units

**IUGR** — intrauterine growth restriction

**IV** — intravenous

**IVA** — isovaleric acidemia

**IVIG** — intravenous immunoglobulin

**KCl** — potassium chloride

**KPCC** — Kuwait Poison Control Center

**LAD** — leukocyte adhesion deficiency

**LCHADD** — long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency

**LDH** — lactate dehydrogenase

**LFTs** — liver function tests

**LGMD** — limb-girdle muscular dystrophy

**LGA** — large for gestational age

**LMA** — laryngeal mask airway

**LP** — lumbar puncture

**LOC** — loss of consciousness

**LRINEC** — Laboratory Risk Indicator for Necrotizing Fasciitis (score)

**MAS** — macrophage activation syndrome

**MCAD** — medium-chain acyl-CoA dehydrogenase deficiency

**MCP** — membrane cofactor protein (CD46)

**MDRO** — multidrug-resistant organism

**MEDIF** — airline Medical Information Form

**MEN** — multiple endocrine neoplasia

**MenACWY** — quadrivalent meningococcal conjugate vaccine (serogroups A, C, W, Y)

**MenB** — serogroup B meningococcal vaccine

**MEP** — maximal expiratory pressure

**MIC** — minimum inhibitory concentration

**MIP** — maximal inspiratory pressure

**min** — minutes

**MMA** — methylmalonic acidemia

**MMR** — measles, mumps and rubella (vaccine)

**MRI** — magnetic resonance imaging

**MRP** — most responsible physician

**MRSA** — methicillin-resistant *Staphylococcus aureus*

**MSSA** — methicillin-susceptible *Staphylococcus aureus*

**MTP** — massive transfusion protocol

**MSUD** — maple syrup urine disease

**NAAT** — nucleic acid amplification test

**NAC** — N-acetylcysteine

**NaCl** — sodium chloride

**NAGS** — N-acetylglutamate synthase

**NAI** — non-accidental injury

**NEC** — necrotizing enterocolitis

**NEJM** — The New England Journal of Medicine

**NEWS2** — National Early Warning Score 2

**NG** — nasogastric

**NGT** — nasogastric tube

**NICU** — neonatal intensive care unit

**NIF** — negative inspiratory force

**NIV** — non-invasive ventilation

**NJT** — nasojejunal tube

**NK** — natural killer (cell)

**NMS** — neuroleptic malignant syndrome

**NPA** — nasopharyngeal airway

**NPO** — nil per os (nothing by mouth)

**NSAID** — non-steroidal anti-inflammatory drug

**NT-proBNP** — N-terminal pro-B-type natriuretic peptide

**NTM** — nontuberculous mycobacteria

**OGT** — orogastric tube

**OHP** — 17-hydroxyprogesterone (17-OHP)

**OI** — oxygenation index

**OME** — otitis media with effusion

**OPA** — oropharyngeal airway

**OR** — odds ratio

**ORS** — oral rehydration solution

**ORT** — oral rehydration therapy

**OSA** — obstructive sleep apnea

**PA** — propionic acidemia; also posteroanterior (radiograph view)

**PALF** — pediatric acute liver failure

**PALS** — Pediatric Advanced Life Support

**PaO<sub>2</sub>** — arterial partial pressure of oxygen

**PAT** — Pediatric Assessment Triangle

**PCR** — polymerase chain reaction

**PDA** — patent ductus arteriosus

**PE** — phenytoin equivalents

**PEA** — pulseless electrical activity

**PECARN** — Pediatric Emergency Care Applied Research Network

**PED-TRAX** — Pediatric Trauma and Tranexamic Acid study (observational)

**PEG** — percutaneous endoscopic gastrostomy

**PEN-FAST** — penicillin-allergy risk-assessment score

**PEP** — post-exposure prophylaxis

**PEWS** — Pediatric Early Warning Score

**PICC** — peripherally inserted central catheter

**PICU** — pediatric intensive care unit

**PIGN** — post-infectious glomerulonephritis

**PJP** — *Pneumocystis jirovecii* pneumonia

**PLP** — pyridoxal-5-phosphate

**PMA** — postmenstrual age

**PO** — by mouth (oral)

**POC** — portable oxygen concentrator

**POCUS** — point-of-care ultrasound

**PPHN** — persistent pulmonary hypertension of the newborn

**PPI** — proton pump inhibitor

**PPV** — positive-pressure ventilation

**PR** — per rectum (rectal route); also PR interval on ECG

**PRAM** — Pediatric Respiratory Assessment Measure

**PRBC** — packed red blood cells

**PRN** — as required (pro re nata)

**PROPPR** — Pragmatic, Randomized Optimal Platelet and Plasma Ratios trial

**PSG** — polysomnography

**PSGN** — post-streptococcal glomerulonephritis

**PT** — prothrombin time

**PTH** — parathyroid hormone

**PTHrP** — parathyroid hormone-related peptide

**PTU** — propylthiouracil

**PUCAI** — Pediatric Ulcerative Colitis Activity Index

**PVC** — premature ventricular contraction

**pVT** — pulseless ventricular tachycardia

**QID** — four times daily

**QRS** — QRS complex (ECG)

**QT** — QT interval (ECG)

**QTc** — corrected QT interval

**RADT** — rapid antigen detection test

**RAPD** — relative afferent pupillary defect

**RBC** — red blood cell

**RBS** — random blood sugar

**RBUS** — renal and bladder ultrasound



**RCT** — randomized controlled trial

**RDS** — respiratory distress syndrome

**rFVIIa** — recombinant activated factor VII

**RIG** — rabies immunoglobulin

**RLQ** — right lower quadrant

**ROSC** — return of spontaneous circulation

**ROTEM** — rotational thromboelastometry

**RR** — respiratory rate

**RSI** — rapid sequence intubation

**RSV** — respiratory syncytial virus

**RT** — respiratory therapist

**RV** — right ventricle

**s** — seconds

**SA** — septic arthritis

**SBP** — systolic blood pressure

**SC** — subcutaneous

**SCFE** — slipped capital femoral epiphysis

**sCD25** — soluble CD25 (soluble interleukin-2 receptor- $\alpha$ )

**SCID** — severe combined immunodeficiency

**SCIWORA** — spinal cord injury without radiographic abnormality

**SGA** — small for gestational age

**SIDS** — sudden infant death syndrome

**SIP** — spontaneous intestinal perforation

**SIPA** — shock index, pediatric age-adjusted

**sJIA** — systemic juvenile idiopathic arthritis

**SJS** — Stevens–Johnson syndrome

**SOPA** — suction, open mouth, pressure increase, airway alternative (NRP corrective steps)

**SpO<sub>2</sub>** — peripheral oxygen saturation

**SPUR** — severe, persistent, unusual, recurrent (immunodeficiency pattern)

**SSSS** — staphylococcal scalded skin syndrome

**SSTI** — skin and soft-tissue infection

**STAT** — immediately (statim)

**STEC** — Shiga toxin-producing E. coli

**STI** — sexually transmitted infection

**SVC** — superior vena cava

**SVR** — systemic vascular resistance

**SVT** — supraventricular tachycardia

**T1DM** — type 1 diabetes mellitus

**T2DM** — type 2 diabetes mellitus

**TB** — tuberculosis

**TBI** — traumatic brain injury

**TBSA** — total body surface area

**TBW** — total body water

**TCA** — tricyclic antidepressant

**TcB** — transcutaneous bilirubin

**Tdap** — tetanus, diphtheria, acellular pertussis (adolescent/adult)

**TEN** — toxic epidermal necrolysis

**TEN-4-FACESp** — bruising clinical decision rule (Torso, Ears, Neck in a child  $\leq 4$  y or any bruise  $< 4$  months; Frenulum, Angle of jaw, Cheek, Eyelid, Sclera; patterned)

**TEG** — thromboelastography

**TFI** — total fluid intake

**TID** — three times daily

**TIF** — tracheo-innominate fistula

**TLS** — tumor lysis syndrome

**TM** — tympanic membrane

**TMA** — thrombotic microangiopathy

**TMP-SMX** — trimethoprim-sulfamethoxazole

**TOF** — tetralogy of Fallot

**tPA** — tissue plasminogen activator

**TRAb** — TSH-receptor antibody

**TREC** — T-cell receptor excision circle

**TREKK** — Translating Emergency Knowledge for Kids

**TSB** — total serum bilirubin

**TSH** — thyroid-stimulating hormone

**TSI** — thyroid-stimulating immunoglobulin

**TTP** — thrombotic thrombocytopenic purpura

**TTS** — tight-to-shaft (tracheostomy cuff)

**TWIST** — Testicular Workup for Ischemia and Suspected Torsion (score)

**TXA** — tranexamic acid

**UCD** — urea cycle disorder

**ULN** — upper limit of normal

**UTI** — urinary tract infection

**V<sub>d</sub>** — volume of distribution

**VA** — venoarterial (ECMO)

**VAD** — ventricular assist device

**VATS** — video-assisted thoracoscopic surgery

**VCUG** — voiding cystourethrogram

**VF** — ventricular fibrillation

**VL** — videolaryngoscopy

**VLCADD** — very-long-chain acyl-CoA dehydrogenase deficiency

**VP** — ventriculoperitoneal (shunt)

**VRE** — vancomycin-resistant enterococci

**VSD** — ventricular septal defect

**VT** — ventricular tachycardia

**VTE** — venous thromboembolism

**VUR** — vesicoureteral reflux

**VV** — venovenous (ECMO)

**WBC** — white blood cell (count)

**WOB** — work of breathing

**WPW** — Wolff–Parkinson–White (syndrome)

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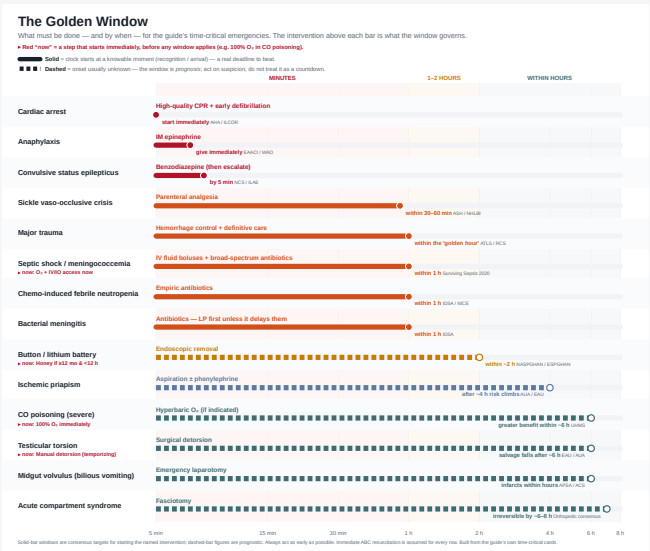
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# Time-critical emergencies at a glance

Solid bar = clock starts at a knowable moment (a real deadline).

Dashed bar = onset usually unknown – prognosis, act on suspicion.





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# Find It by Symptom

The front door — start here: a presentation routes you to the card

## Presentation → card index

NAVIGATOR

Presentation → card

You see a presentation, not a diagnosis. Start here and jump to the card.

| Presentation                           | Go to                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Bilious vomiting (green)               | <u>Malrotation / Midgut Volvulus</u>                                                                                                             |
| The limping / non-weight-bearing child | <u>The Limping Child</u> · <u>Transient Synovitis vs Septic Arthritis</u> · <u>Slipped Capital Femoral Epiphysis (SCFE)</u>                      |
| Cyanosis / the blue baby               | <u>Hypercyanotic (Tet) Spell</u> · <u>Neonatal &amp; Infant CHD</u> · <u>Critical CHD – Pulse Oximetry Screening</u>                             |
| Floppy infant / acute weakness         | <u>Guillain-Barré Syndrome / Acute Flaccid Paralysis</u> · <u>The Child with Neuromuscular Disease</u> · <u>Neonatal Sepsis (rapid approach)</u> |
| Petechiae / purpura                    | <u>Immune Thrombocytopenia (ITP)</u> · <u>IgA Vasculitis (Henoch-Schönlein Purpura)</u> · <u>Septic Shock</u>                                    |

| Presentation                    | Go to                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stridor                         | <u>Croup</u> • <u>Choking</u><br>( <u>Foreign-Body Obstruction</u> )<br>• <u>Anaphylaxis</u> •<br><u>Retropharyngeal Abscess</u>                     |
| Wheeze                          | <u>Acute Asthma</u> •<br><u>Bronchiolitis</u> •<br><u>Anaphylaxis</u>                                                                                |
| Seizure                         | <u>Convulsive Status</u><br><u>Epilepticus</u> • <u>Febrile</u><br><u>Seizure</u> • <u>Neonatal</u><br><u>Seizures</u>                               |
| Fever +<br>non-blanching rash   | <u>Septic Shock</u> • <u>Bacterial</u><br><u>Meningitis &amp; Encephalitis</u>                                                                       |
| Acute scrotal pain              | <u>Acute Scrotum</u> • <u>Testicular</u><br><u>Torsion</u>                                                                                           |
| Jaundiced neonate               | <u>Neonatal</u><br><u>Hyperbilirubinemia</u> •<br><u>Neonatal Cholestasis</u>                                                                        |
| Bloody diarrhea                 | <u>Hemolytic Uremic Syndrome</u><br>• <u>Acute IBD Flare &amp; Toxic</u><br><u>Megacolon</u> •<br><u>Gastroenteritis &amp;</u><br><u>Dehydration</u> |
| Apnea / brief<br>resolved event | <u>BRUE (Brief Resolved</u><br><u>Unexplained Event)</u> •<br><u>Pertussis (Whooping Cough)</u>                                                      |
| Severe abdominal<br>pain        | <u>Acute Appendicitis</u> •<br><u>Intussusception</u> •<br><u>Malrotation / Midgut</u><br><u>Volvulus</u> • <u>Diabetic</u><br><u>Ketoacidosis</u>   |

| Presentation                  | Go to                                                                                                                                                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bruising / abnormal bleeding  | <u><a href="#">Immune Thrombocytopenia (ITP)</a></u> · <u><a href="#">Acute Bleeding in Hemophilia</a></u> · <u><a href="#">Vitamin K Deficiency Bleeding (VKDB)</a></u> · <u><a href="#">Child Protection &amp; Safeguarding</a></u> |
| Fever in the young infant     | <u><a href="#">Fever in the Young Infant</a></u>                                                                                                                                                                                      |
| Shock / poor perfusion        | <u><a href="#">Shock – Recognition &amp; Type</a></u> · <u><a href="#">Septic Shock</a></u>                                                                                                                                           |
| Poisoning / ingestion         | <u><a href="#">Poisoning &amp; Toxic Ingestion</a></u> · <u><a href="#">Button (Lithium) Battery Ingestion</a></u> · <u><a href="#">Caustic (Corrosive) Ingestion</a></u>                                                             |
| High glucose / ketones        | <u><a href="#">Diabetic Ketoacidosis</a></u> · <u><a href="#">Hyperglycemic Hyperosmolar State (HHS)</a></u>                                                                                                                          |
| Hypoglycemia                  | <u><a href="#">Hypoglycemia (Beyond the Newborn)</a></u> · <u><a href="#">Metabolic Crisis (Inborn Error)</a></u> · <u><a href="#">Neonatal Hypoglycemia</a></u>                                                                      |
| Painful swollen joint + fever | <u><a href="#">Transient Synovitis vs Septic Arthritis</a></u> · <u><a href="#">Osteoarticular Infections</a></u>                                                                                                                     |
| Dark / tea-colored urine      | <u><a href="#">Rhabdomyolysis</a></u> · <u><a href="#">Acute Kidney Injury</a></u> · <u><a href="#">Gross Hematuria</a></u>                                                                                                           |
| Agitation / behavioral crisis | <u><a href="#">Acute Severe Agitation</a></u> · <u><a href="#">Self-Harm / Intentional Ingestion</a></u>                                                                                                                              |
| Major hemorrhage / trauma     | <u><a href="#">Massive Transfusion &amp; Blood Products</a></u> · <u><a href="#">Major Trauma – Primary Survey &amp; Resuscitation</a></u>                                                                                            |



| Presentation                          | Go to                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anaphylaxis /<br>urticaria + collapse | <u><a href="#">Anaphylaxis</a></u>                                                                                                                                      |
| The unwell child<br>with a device     | <u><a href="#">Tracheostomy Emergencies</a></u> · <u><a href="#">Gastrostomy (Feeding) Tube Problems</a></u> · <u><a href="#">The Child with Medical Complexity</a></u> |

#### ◆ HOW TO USE

- This routes the commonest bedside **PRESENTATIONS** to the relevant cards — it is a starting point, not a differential.
- Tap a card name to jump straight to it.
- The full alphabetical index and the section colors remain the other two ways in.

## Can't-miss patterns – recognition map

See it → name the danger → act. Each node is a sentinel pattern  
– tap it to jump to the full card (workup + doses). Not exhaustive;  
the section cards remain the source of truth.

### HEART / NEONATE

---

#### Duct-dependent CHD / IEM

Neonate collapse / shock

---

#### Cyanotic CHD

Cyanosis unresponsive to O<sub>2</sub>

---

#### Myocarditis

Tachycardia + gallop, hepatomegaly

### FEVER, RASH & SEPSIS

---

#### Meningococcal sepsis

Fever + non-blanching rash

---

#### Serious bacterial infection

Fever, infant < 90 days

---

#### ITP vs meningococemia

Petechiae — well or unwell?

---

#### Epiglottitis / deep-neck

Droling + tripod + stridor

## **SURGICAL / ABDOMEN**

---

### **Malrotation + volvulus**

Bilious (green) vomiting, infant

---

### **Testicular torsion**

Acute scrotal pain / swelling

---

### **Caustic injury**

Button / lithium battery

## **BRAIN, CORD & BP**

---

### **Raised ICP / brain tumor**

Morning headache + vomiting

---

### **Cord compression**

Night back pain + leg weakness

---

### **Neuromuscular / botulism**

Floppy infant, weak cry

---

### **Hypertensive emergency**

Very high BP + seizure

## **METABOLIC, BONE & BLOOD**

---

### **DKA ± cerebral edema**

Vomiting + polyuria + high glucose

---

### **Septic arthritis**

Limp + fever, won't bear weight

---

### **Acute hemolysis**

Sudden pallor + jaundice

---

### **Inflicted injury**

Bruise, non-mobile infant

---

### **Reconsider diagnosis**

"Gastroenteritis", no diarrhea

## Cardiopulmonary Arrest MINUTES

Unresponsive, apneic, pulseless within 10 seconds → CPR now.

+ Calculators: [Resuscitation dose calculator](#) · [ETT size & depth](#)

### ▴ HIGH-QUALITY CPR IS THE PRIORITY

- Rate **100–120 per minute**; depth  $\frac{1}{3}$  of chest depth (~4 cm infant, ~5 cm child); full recoil; minimal interruptions<sup>1</sup>.
- Ratio **30:2** (single rescuer) · **15:2** (two rescuers). **Adolescent (post-pubertal): 30:2 for both single and two rescuers** (adult BLS)<sup>1,157</sup>.
- **With an advanced airway (all ages):** continuous chest compressions with **1 breath every 2–3 seconds (20–30 breaths/min)**<sup>1</sup>, without pausing compressions.

#### DURING CPR

- Attach a monitor/defibrillator; establish access — **immediate IO in cardiac arrest if an IV is not already in place** (for other emergencies, go IO if IV fails after 2 attempts or 90 s); consider an **advanced airway** (then continuous compressions with one breath every 2–3 s); give oxygen.<sup>1</sup>
- Do not delay epinephrine or defibrillation for access or intubation<sup>1</sup>.



#### DECISION

**Rhythm on the monitor?**

**Defibrillate as early as possible** — 2 J/kg → 4 J/kg →  $\geq 4$  J/kg (subsequent shocks; up to 10 J/kg or the adult dose<sup>1</sup>,  $\approx$  200 J biphasic / 360 J monophasic).

- **Resume high-quality CPR** immediately after every shock<sup>1</sup>.
- **Reassess** rhythm and a central pulse (brachial in infants; femoral/carotid in children)<sup>1</sup> **every 2 min**, pausing compressions  $\leq 10$  s. If VF/pVT persists, shock again and resume CPR.
- **Epinephrine** during CPR **after the 2nd shock**<sup>1</sup> (don't pause compressions), then every 3–5 min.
- **Amiodarone** (or lidocaine) **after the 3rd shock**<sup>1</sup> for refractory VF/pVT.

**Sequence (a continuous cycle):** shock → shock → epinephrine → shock → amiodarone → **↺ back to shock** — keep repeating the 2-min CPR–shock cycle, epinephrine every 3–5 min and a further amiodarone dose as needed<sup>1</sup>.

Give epinephrine during CPR as soon as possible, then every 3–5 minutes. Continue high-quality CPR without unnecessary interruptions; check rhythm every 2 minutes. If a shockable rhythm appears, **cross over to the shockable<sup>1</sup> (VF/pVT) pathway** and defibrillate.

**EVERY CYCLE – REVERSIBLE CAUSES**

- **6 H's:** hypoxia, hypovolemia, hydrogen ion (acidosis), hypo-/hyperkalemia, hypoglycemia, hypothermia<sup>1</sup>.
- **5 T's:** tension pneumothorax, tamponade (cardiac), toxins, thrombosis (pulmonary), thrombosis (coronary)<sup>1</sup>.

  
ROSC



Post-arrest: start post-arrest care · secure the airway · wean oxygen to target SpO<sub>2</sub> 94–99% (avoid hyperoxic reperfusion injury<sup>1</sup>) and normal carbon dioxide · systolic and/or mean arterial BP > 10th centile for age · targeted normothermia (36–37.5 °C) · correct metabolic derangements · treat seizures (EEG if persistent encephalopathy).

## MEDICATIONS & ENERGY

| Drug / action            | Dose                  | Route & notes                                                                                                                                     |
|--------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Epinephrine (adrenaline) | 0.01 mg/kg (max 1 mg) | IV/IO, 0.1 mg/mL (1:10,000) <sup>1</sup> concentration; give during CPR without pausing compressions, every 3–5 min                               |
| Amiodarone               | 5 mg/kg (max 300 mg)  | IV/IO bolus for refractory VF/pVT; may repeat up to twice (3 doses total; max 15 mg/kg). Avoid in prolonged QT (further prolongs QT) <sup>1</sup> |
| Lidocaine                | 1 mg/kg               | IV/IO alternative to amiodarone <sup>1</sup>                                                                                                      |

| Drug / action  | Dose            | Route & notes                                                                                  |
|----------------|-----------------|------------------------------------------------------------------------------------------------|
| Defibrillation | 2 → 4 → ≥4 J/kg | Unsynchronized; <sup>1</sup> up to 10 J/kg or adult dose (≈ 200 J biphasic / 360 J monophasic) |

### ⚠ REMEMBER

- Respiratory failure is the leading cause of cardiac arrest in pediatrics — oxygenate and ventilate early<sup>1</sup>.
- Intraosseous access is fast — in **cardiac arrest go straight to IO if an IV is not already in place** (in other emergencies, IO if IV fails after 2 attempts or 90 s); do not delay epinephrine waiting for a vein<sup>1</sup>.

### ◆ ALGORITHMS

- Full AHA PALS flowcharts (cardiac arrest
- bradycardia
- tachycardia
- post-arrest) — [cpr.heart.org](https://cpr.heart.org)



## ▲ ADMIT · DISCHARGE · REFER

- **Admit** — every child with return of circulation → **start post-arrest care** (oxygenation/ventilation targets, hemodynamic support, treat the cause, neuroprotection, avoid fever and hypo/hyperglycemia)<sup>1</sup>.
- **Discharge** — not applicable — care after unsuccessful resuscitation follows the death-confirmation and bereavement pathway.
- **Refer** — the specialty for the reversible cause (6 Hs / 5 Ts): cardiology, surgery, toxicology, or metabolic; involve the resuscitation team early.

◆ **SEE ALSO** Situation-specific resuscitation:

**Neonatal Resuscitation** · **Shock — Recognition & Type** · **Anaphylaxis** · **Difficult Airway** · **Major Trauma — Primary Survey**.

## Code Blue – team roles & positions

Assign each role out loud and close the loop: Team Leader (hands-off), Airway Manager, Compressor, Defib Operator / CPR Coach, Medications & Vascular Access (×2), and Recorder / Documenter.



*Human-prompted, AI-generated image, reviewed and approved by the panel.*

02

## Behavioral & Psychiatric

Agitation · self-harm · drug-induced syndromes

### Acute Severe Agitation MINUTES

Keep child and staff safe; look for a treatable cause.

**+ Calculator: QTc**

**① EXCLUDE AN ORGANIC CAUSE FIRST**

Check glucose, oxygenation, head injury, intoxication/withdrawal, pain. Agitation is a symptom — treat the cause where present<sup>221</sup>.



**② NON-DRUG MEASURES FIRST**

Calm low-stimulation environment, verbal de-escalation, familiar caregiver; offer oral medication, including the child's own regular medication, when available.<sup>221</sup>



severe / unsafe



**③ PHARMACOLOGIC (PER LOCAL PATHWAY)**

Second-generation antipsychotics first-line; benzodiazepines where intoxication/anxiety predominates. Physical restraint = brief last resort, continuous monitoring<sup>221</sup>.

| Drug            | Dose (approximate)                    | Route & notes                                                                                                                                                                                |
|-----------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Olanzapine      | 2.5–5 mg (< 40 kg); 5–10 mg (≥ 40 kg) | PO/IM/ODT; the <b>ODT</b> (orally disintegrating — Zyprexa Zydis) is easiest in an agitated child. Do <b>not</b> give within 1 h of a benzodiazepine (respiratory depression) <sup>221</sup> |
| Risperidone     | 0.25–1 mg                             | PO; well tolerated <sup>221</sup>                                                                                                                                                            |
| Lorazepam       | 0.05–0.1 mg/kg (max 2 mg)             | PO/IM/IV; for substance- or anxiety-driven agitation <sup>221</sup>                                                                                                                          |
| Haloperidol     | 0.025–0.05 mg/kg (max 5 mg)           | IM; watch for dystonia/QT — second-generation agents preferred <sup>221</sup>                                                                                                                |
| Diphenhydramine | 1–2 mg/kg/dose (max 50 mg)            | IV/IM/PO; <b>antidote for an acute dystonic reaction</b> from an antipsychotic (haloperidol/risperidone) — not a treatment for agitation itself <sup>221</sup>                               |

### ⚠ CAUTION WITH BENZODIAZEPINES

- Avoid benzodiazepines and diphenhydramine in **delirium** — can worsen confusion (paradoxical disinhibition)<sup>221</sup>.

- Also avoid or use with caution in **developmental delay and autism**, where benzodiazepines commonly cause **paradoxical reactions (more agitation)**<sup>221</sup>.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — ongoing risk, a medical cause, or need for continued sedation/monitoring.
- **Discharge** — settled, medically cleared, with a safety plan and follow-up.
- **Refer** — psychiatry; medical/toxicology for organic causes.

## Self-Harm / Intentional Ingestion

### HOURS

Stabilize the patient medically first, then address psychological safety.

+ Calculators: [Paracetamol & iron](#) · [QTc](#)

### ① MEDICAL STABILIZATION

ABCDE, glucose, ECG, and a pregnancy test ( $\beta$ -hCG) in all post-menarchal females<sup>225</sup>. For any ingestion: identify agent and timing, check acetaminophen and salicylate levels, contact poisons information<sup>225</sup>. Treat per the specific antidote pathway.



### ② SAFETY

Continuous one-to-one observation, safe environment, involve a trusted adult/caregiver. Do not discharge before a mental-health assessment<sup>225</sup>.



Refer for psychiatric/psychosocial assessment; arrange follow-up and support before discharge<sup>225</sup>.

◆ **CROSS-REFERENCE** General approach, antidotes, and Kuwait Poison Control hotline: see [Poisoning & Toxic Ingestion](#) (Toxicology, Trauma & Environmental).

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — medical risk from the ingestion/injury, or ongoing suicidality → admit and monitor.



- **Discharge** — medically cleared AND psychiatrically assessed as safe, with follow-up and means restriction<sup>225</sup>.
- **Refer** — psychiatry (risk assessment), social work/safeguarding, and toxicology.

## Drug-Induced Hyperthermia Syndromes

### MINUTES

Three look-alikes with different triggers — and different antidotes.

| Syndrome                       | Clue                                                                                         | Key treatment                                                                     |
|--------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Neuroleptic malignant syndrome | Antipsychotic; <b>"lead-pipe" rigidity</b> , slow onset, high creatine kinase <sup>223</sup> | Stop drug; cool; fluids; benzodiazepine; dantrolene; bromocriptine <sup>223</sup> |
| Serotonin syndrome             | Serotonergic drug; <b>clonus &amp; hyperreflexia</b> , rapid onset, agitation <sup>222</sup> | Stop drug; cool; benzodiazepine; cyproheptadine <sup>222</sup>                    |

| Syndrome                      | Clue                                                                                  | Key treatment                                                                                                                                                            |
|-------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Malignant hyperthermia</b> | Volatile anesthetic / succinylcholine <sup>224</sup> rigidity, rising CO <sub>2</sub> | Stop trigger; <b>dantrolene</b> ; aggressive cooling; treat hyperkalemia/rhabdomyolysis; urgent PICU/toxicology/anesthesia — antidotes per local protocol <sup>224</sup> |

### ⚠ ALL THREE

- Active cooling, IV fluids; treat hyperkalemia and rhabdomyolysis<sup>224</sup>.
- Secure airway; transfer to PICU.

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU (cooling, sedation, dantrolene as indicated).
- **Discharge** — not acutely; after resolution and review of the causative agent.
- **Refer** — toxicology/PICU; psychiatry/pharmacy for a medication review.

# Cardiac & Shock

03

Shock recognition & types · septic shock · arrhythmias · myocarditis / heart failure · congenital heart disease · hypercyanotic spell

## Shock – Recognition & Type MINUTES

Circulatory failure: tissues not getting enough oxygen. Hypotension is a late sign.

+ Calculators: Maintenance fluids & bolus · Infusion dose ↔ rate · SIPA shock index

### RECOGNIZE EARLY (BEFORE HYPOTENSION)

- Tachycardia, prolonged or flash capillary refill, cool or mottled skin, weak pulses, altered mental status, reduced urine output, rising lactate.<sup>6</sup>

### UNIVERSAL FIRST STEPS

**Start with ABCDEs** · Oxygen · IV/IO access · monitor · check glucose & lactate · identify the type below<sup>6</sup>.



### DECISION

**Which type of shock?**

| Type                | Cause                                                                      | Key treatment                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hypovolemic</b>  | Vomiting/diarrhea, hemorrhage, burns                                       | Crystalloid or balanced solution 10–20 mL/kg boluses; <b>packed red cells</b> if hemorrhage; stop ongoing losses <sup>6</sup>                                                                                                  |
| <b>Distributive</b> | Sepsis, anaphylaxis, neurogenic                                            | Fluids + early vasoactive; treat the cause (see <b>Septic shock</b> , <b>Anaphylaxis</b> )                                                                                                                                     |
| <b>Cardiogenic</b>  | Myocarditis, arrhythmia, heart disease                                     | <b>Cautious</b> fluid 5 mL/kg slowly; inotrope/inodilator; treat arrhythmia; early cardiology consult and PICU escalation <sup>227</sup>                                                                                       |
| <b>Obstructive</b>  | Tension pneumothorax, tamponade, duct-dependent lesion, pulmonary embolism | Relieve the obstruction (decompress a tension pneumothorax, drain a tamponade, and — for example — prostaglandin for a duct-dependent lesion; massive/high-risk PE → thrombolysis + heparin, specialist-directed) <sup>1</sup> |

### ⚠️ MIMICS SHOCK – NOT CIRCULATORY FAILURE

- **Carbon monoxide, cyanide, hydrogen sulfide, and methemoglobinemia** impair oxygen **delivery or use at the**

**hemoglobin / cellular level**, not the circulation — so this is **not true shock**, and the usual endpoints (blood pressure, capillary refill) can look normal. **Suspect it** when perfusion and BP appear adequate but there is **unexplained lactic acidosis / tissue hypoxia** with a suggestive history (fire / smoke, enclosed space, oxidizing drugs). **SpO<sub>2</sub> is unreliable** — confirm with **co-oximetry / blood gas<sup>1</sup>**.

- It will **not respond to fluids or vasopressors**: give **100% oxygen** and the specific antidote<sup>1</sup> (see **Smoke Inhalation & CO** and **Poisoning**).

## RESUSCITATION DRUGS

| Drug                                    | Dose                                 | Notes                                                                                                                                                                                                          |
|-----------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Balanced crystalloid                    | 10–20 mL/kg (5 mL/kg if cardiogenic) | Ringer's lactate preferred; normal saline if a balanced fluid is unavailable. Bolus; reassess for fluid overload after each                                                                                    |
| Epinephrine (adrenaline) infusion       | 0.05–0.3 mcg/kg/min                  | Titrate to response; first-line vasoactive — start epinephrine or norepinephrine early (PALS) <sup>6</sup> . May start peripherally with close observation for extravasation until central access is obtained  |
| Norepinephrine (noradrenaline) infusion | 0.05–0.3 mcg/kg/min                  | Titrate to response; first-line alternative — warm/cold classification no longer directs the choice <sup>6</sup> . May start peripherally with close extravasation monitoring until central access is obtained |

### ⚠ FLUID CAUTION

- Reassess after every bolus.
- Vasoactives may start alongside fluids if evidence of

life-threatening end-organ hypoperfusion.

- Slow or stop fluids at first **signs of fluid overload** — new crackles, enlarging liver, gallop, worsening work of breathing, or overload on POCUS<sup>6</sup> — especially suspected cardiogenic shock.

#### ◆ THERAPEUTIC ENDPOINTS (AIM FOR)

- Heart rate and blood pressure normal for age, capillary refill  $\leq 2$  s, warm extremities with normal pulses<sup>6</sup>, normal mental status, urine output  $\geq 1$  mL/kg/h, falling / normalizing lactate (central venous O<sub>2</sub> saturation  $\geq 70\%$  where available).
- Reassess after every fluid bolus and every change in vasoactive support.

#### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — any child in shock → **call the PICU team to evaluate** (high-dependency/PICU); begin resuscitation before the type is confirmed<sup>6</sup>.

- **Discharge** — only once the cause is treated and perfusion is normal off vasoactive support.
- **Refer** — by type — septic → PICU/infectious diseases; cardiogenic → cardiology; obstructive → treat the cause (tension pneumothorax, tamponade); hypovolemic/hemorrhagic → surgery/trauma. **Involve PICU in parallel** for any child in shock.

## Septic Shock

MINUTES

SSC

Infection + circulatory dysfunction. Defined by the [Phoenix Sepsis Score](#)  ( $\geq 2$  points = sepsis;  $\geq 1$  cardiovascular point = septic shock).

**+ Calculators: Maintenance fluids & bolus · Infusion dose ↔ rate**

### ⚠ THE PHOENIX SCORE IS NOT A SCREENING TOOL

- The Phoenix Sepsis Score identifies **organ dysfunction** — by the time it is positive, sepsis is already established<sup>226</sup>.



- It is **not** an early-screening tool.
- Keep using your standard pediatric **early-warning / sepsis-screening** triggers to catch the child **before** organ dysfunction develops<sup>226</sup>.

**FIRST HOUR — every minute counts**



## RESUSCITATE

- **Impaired perfusion** = altered mental status; tachycardia with weak or bounding pulses and wide pulse pressure ( $\pm$  hypotension); capillary refill  $> 2$  s, mottled cool skin or pallor; oliguria  $< 1$  mL/kg/h (adolescents  $< 0.5$  mL/kg/h).
- 100% oxygen via non-rebreather mask
- two IV/IO
- **antibiotics within 1 hour**<sup>6</sup>
- IV fluid boluses.
- **Labs:** glucose, CBC + differential, PCT or CRP, electrolytes, ionized calcium, renal and liver function, coagulation, blood gas, lactate, cultures (blood  $\pm$  urine).
- **Hypotension** (late sign) = systolic BP below, by age:
  - $< 60$  mmHg — neonate
  - $< 70$  mmHg — 1–12 mo
  - $< 70 + 2 \times \text{age}(\text{yr})$  — 1–10 yr
  - $< 90$  mmHg —  $\geq 10$  yr.<sup>6</sup>



## FLUIDS

- **Balanced crystalloid** (Ringer's lactate preferred over 0.9% saline, except with hyponatremia or raised-ICP concern) 10–20 mL/kg boluses; reassess after each, repeat up to ~40 mL/kg<sup>6</sup>.
- **Watch for fluid overload** — new hepatomegaly, crackles, gallop, or B-lines on lung POCUS. Third bolus only if still dehydrated without overload<sup>161</sup>.
- Start vasoactives if shock persists after ~40 mL/kg, or earlier if overload develops<sup>6</sup>.

fluid-refractory



## VASOACTIVE

- **Call PICU immediately.**
- Start **epinephrine or norepinephrine** (either is acceptable first-line; not dopamine)<sup>6</sup> — the **warm/cold picture** (norepinephrine if warm/vasodilated with bounding pulses, epinephrine if cold/low-output) can **guide but no longer strictly directs**<sup>6</sup> the choice; a well-sited peripheral IV is acceptable, with close extravasation monitoring while central access is arranged.
- **Catecholamine-resistant shock** → consider **early** hydrocortisone, especially with adrenal-insufficiency risk (purpura fulminans, recent/chronic steroids, hypothalamic–pituitary–adrenal disease): stress dose  $\approx 50 \text{ mg/m}^2$  then  $50\text{--}100 \text{ mg/m}^2/\text{day}$ <sup>6</sup>
- OR  $1\text{--}2 \text{ mg/kg}$  then  $\approx 2\text{--}4 \text{ mg/kg/day}$ .
- Give stress-dose hydrocortisone for suspected/documented adrenal insufficiency. In catecholamine-resistant shock, hydrocortisone may restore vascular responsiveness to

catecholamines — adult data suggest a mortality benefit, but pediatric evidence remains limited<sup>6</sup>. Call PICU (SSC).

- **Refractory shock** — persisting despite fluids,  $\geq 2$  vasoactives, and hydrocortisone, or **low cardiac output on high inotropic support** → **consider VA-ECMO early** and activate an ECMO center<sup>6</sup>; survival falls the longer cannulation is delayed. SSC 2020 suggests VA-ECMO in children with septic shock refractory to all other treatments<sup>6</sup>.

## MEDICATIONS

| Drug                         | Dose                       | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ceftriaxone (empiric)</b> | <b>100 mg/kg (max 4 g)</b> | <p>IV; empiric regimen by focus</p> <p>(immunocompetent, community-acquired):</p> <p>no obvious focus →</p> <p>ceftriaxone ±</p> <p>vancomycin ±</p> <p>amikacin;</p> <p>meningitis/encephalitis → ceftriaxone +</p> <p>vancomycin ±</p> <p>acyclovir; pneumonia → ceftriaxone ±</p> <p>vancomycin;</p> <p>skin/soft-tissue →</p> <p>cefotaxime +</p> <p>vancomycin ±</p> <p>clindamycin; toxic shock or necrotizing fasciitis →</p> <p>piperacillin-tazobactam + vancomycin +</p> <p>clindamycin;</p> <p>intra-abdominal →</p> <p>piperacillin-tazobactam;</p> <p>neonate → cefotaxime + ampicillin (±</p> <p>acyclovir if HSV suspected, ±</p> <p>vancomycin if MRSA suspected).</p> <p>Immunocompromised or hospital-acquired: add antipseudomonal cover and consider antifungal. Send MRSA screening</p> |

| Drug                           | Dose                      | Notes                                                                                                                                                                                                                           |
|--------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                           | swabs. Meningitic ceftriaxone dose 50 mg/kg/dose every 12 h (max 2 g/dose) <sup>6</sup>                                                                                                                                         |
| Balanced crystalloid           | 10–20 mL/kg               | Ringer's lactate preferred; normal saline if unavailable. Bolus, reassess after each                                                                                                                                            |
| Epinephrine (adrenaline)       | Start 0.05–0.1 mcg/kg/min | IV/IO infusion; escalate every 5–10 min to target — <b>escalation and titration must be done by the PICU team.</b> Central line preferred; peripheral is a bridge only with close extravasation monitoring until central access |
| Norepinephrine (noradrenaline) | Start 0.05–0.1 mcg/kg/min | IV/IO infusion; escalate every 5–10 min to target — <b>escalation and titration must be done by the PICU team.</b> Central line preferred; peripheral is a bridge only with close extravasation monitoring until central access |

| Drug            | Dose                      | Notes                                                                                                                        |
|-----------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Hydro-cortisone | 1–2 mg/kg<br>(max 100 mg) | IV; give for suspected/documentated adrenal insufficiency. No clear evidence for or against in catecholamine-resistant shock |

◆ Vasoactive infusions — how to give them safely

- **ADRENALINE AND NORADRENALINE INFUSIONS SHOULD BE GIVEN VIA A CENTRAL VENOUS CATHETER WHEREVER POSSIBLE. DO NOT DELAY** initiation of vasoactive support to obtain central access (SSC Pediatric 2020)<sup>6</sup>.
- Peripheral use is a **BRIDGE ONLY** — via a large vein, a dedicated dilute infusion, with site review **EVERY 15 MINUTES FOR THE FIRST HOUR THEN HOURLY**, and conversion to central access as soon as feasible.
- **DOSES EXCEEDING THE STATED CENTRAL OR PERIPHERAL MAXIMUM REQUIRE CONSULTANT APPROVAL**, documented in the notes. Escalation beyond **1 MCG/KG/MIN** should prompt reassessment of the shock phenotype and consideration of **VASOPRESSIN AND HYDROCORTISONE**<sup>171</sup>,



rather than relying solely on dose escalation.

◆ Preparing the infusion — where you have no local formulary

- **AMPOULES (GULF/ME):** epinephrine (adrenaline) **1 MG/ML (1:1,000)** is the stock ampoule (0.1 mg/mL / 1:10,000 is the pre-filled code syringe); norepinephrine (noradrenaline) as the **BITARTRATE 1 MG/ML** (2 mg/2 mL, 4 mg/4 mL). **DOSE NOREPINEPHRINE AS BASE** — a 4 mg/4 mL bitartrate ampoule = **2 MG BASE**.
- **RULE OF 6 (WEIGHT-BASED):** put **0.3 MG/KG IN 50 ML** (or 0.6 mg/kg in 100 mL) of glucose 5% → **1 ML/H = 0.1 MCG/KG/MIN** (0.05 = 0.5 mL/h; 0.3 = 3 mL/h).
- **STANDARD CONCENTRATION (SAFETY-PREFERRED):** epinephrine **10 / 20 / 40 MCG/ML**; norepinephrine (base) **16 / 32 / 64 MCG/ML**. **RATE (ML/H) = DOSE (MCG/KG/MIN) × WEIGHT (KG) × 60 ÷ CONCENTRATION (MCG/ML)**. Use the most dilute for a peripheral bridge.
- **DILUENT & SAFETY:** dilute in **GLUCOSE 5%** (or 0.9% saline) — **NOT** with bicarbonate/alkaline fluids; **PROTECT FROM LIGHT** and discard if discolored; dedicated line; **NEVER BOLUS OR FLUSH** a vasoactive line; independent

double-check of drug, concentration, weight, and rate.

- **SOURCES:** BNF for Children; ASHP/ISMP Standardize 4 Safety pediatric infusion standards; ANMF adrenaline & noradrenaline standard-concentration monographs (2025–26).

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU/high-dependency; fluid-refractory or vasoactive-requiring shock is always PICU.
- **Discharge** — not from the acute episode — step down once off vasoactives with a controlled source.
- **Refer** — infectious diseases for source control and antimicrobials; surgery for a drainable focus; an ECMO center if refractory.

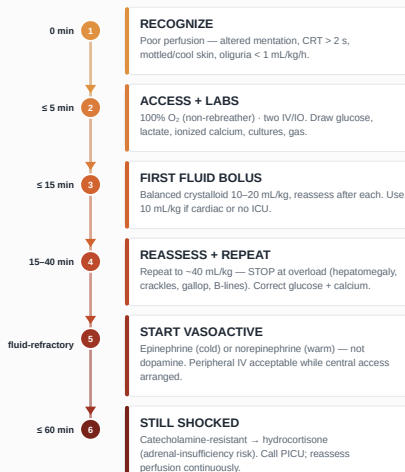
## Septic shock – the first hour

The first hour, step by step: recognize → access + labs → fluids → reassess → vasoactive → hydrocortisone/PICU, with antibiotics inside the hour. Details in the card above.



### ANTIBIOTICS within 60 minutes

Two agents IV/IO, after cultures — do not delay for the lab.



## Myocarditis / Acute Heart Failure

### HOURS

The "gastroenteritis" or "asthma" that is really the heart — look for it.

### + Calculator: Infusion dose ↔ rate

#### RECOGNIZE

- Viral prodrome, then **tachycardia out of proportion** to the fever, poor feeding, lethargy, grunting or

respiratory distress.

- **Hepatomegaly, gallop rhythm, weak pulses, cool peripheries.** Older child: chest pain, syncope, palpitations.
- Commonly mislabeled as gastroenteritis, sepsis, asthma, pneumonia, or bronchiolitis.<sup>227</sup>

#### CONFIRM

ECG (sinus tachycardia, low voltages, ST/T changes, arrhythmia, or heart block), chest X-ray (cardiomegaly, pulmonary edema), troponin and BNP raised, echocardiogram for reduced function.<sup>227</sup>

#### ⚠ RESUSCITATE WITH CARE

- Oxygen and monitor with caution; may need positive-pressure ventilation — **call PICU.**
- Do **not** give large fluid boluses.<sup>227</sup>
- Use 5 mL/kg slowly and reassess for **signs of fluid overload** (worsening hepatomegaly, new crackles, a gallop, or respiratory distress).
- Start inotropes early (cardiology/PICU-directed) —

milrinone (preferred if blood pressure allows — an inodilator that lowers afterload and can drop BP) or dobutamine.

- Tailor to hemodynamics: normotensive with cold peripheries and rising lactate → dobutamine or milrinone; hypotensive with poor perfusion → epinephrine.
- Check and correct electrolytes (calcium, potassium, magnesium, phosphate).
- Treat arrhythmia, give oxygen and ventilatory support.
- Early PICU and pediatric cardiology; refractory shock needs urgent PICU/cardiology transfer — advanced mechanical circulatory support is a specialist/PICU decision.<sup>227</sup>

## INOTROPES / VASOACTIVES

| Drug        | Dose                             | Notes                                                                                                                                                       |
|-------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Milrinone   | 0.25–0.75 mcg/kg/min IV infusion | Inodilator — preferred if the blood pressure allows. <b>Consider omitting the 50 mcg/kg loading dose in acute shock<sup>227</sup></b> to avoid hypotension. |
| Dobutamine  | 2–20 mcg/kg/min IV infusion      | Normotensive with cold peripheries and rising lactate.                                                                                                      |
| Epinephrine | 0.05–0.3 mcg/kg/min IV infusion  | Hypotensive with poor perfusion.                                                                                                                            |

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected cases → PICU/cardiology; arrhythmia, low output, or a troponin/BNP rise mandates continuous monitoring.
- **Discharge** — only after cardiology review, recovered function, and a heart-failure plan.
- **Refer** — cardiology urgently (echo, rhythm); a transplant/ECMO/mechanical-support center if low output or refractory arrhythmia.

# Bradycardia + Poor Perfusion

## MINUTES

Slow heart rate with shock — usually hypoxic.

### + Calculator: Resuscitation dose calculator

#### FIRST

100% oxygen, support airway & breathing, monitor, IV/IO. **Ventilation corrects most pediatric bradycardia.**<sup>1</sup>



#### DECISION

**HR < 60 with poor perfusion despite oxygenation & ventilation? Poor perfusion = acutely altered mental status, signs of shock, or low BP.**

#### YES

Start CPR. Give epinephrine. Atropine for a vagal cause or heart block; consider pacing.<sup>1</sup> Treat the reversible cause; call PICU/cardiology early.

#### NO

Support, observe, treat the cause, seek cardiology input.

## LOOK FOR THE CAUSE

- **Hypoxia is the commonest — oxygenate and ventilate first.**<sup>1</sup> Also: hypothermia; drugs/toxins; raised intracranial pressure (Cushing's triad — hypertension, bradycardia, irregular breathing) (see **Raised ICP**); increased vagal tone (e.g., suctioning, intubation); AV / heart block.
- A well, athletic, or sleeping child may be **physiologically** bradycardic.<sup>1</sup>

| Drug                     | Dose                             | Notes                                                                                  |
|--------------------------|----------------------------------|----------------------------------------------------------------------------------------|
| Epinephrine (adrenaline) | 0.01 mg/kg (max 1 mg)            | IV/IO, 0.1 mg/mL (1:10,000) concentration, every 3–5 min                               |
| Atropine                 | 0.02 mg/kg (min 0.1, max 0.5 mg) | IV/IO; for vagal cause, AV block, or cholinergic/organophosphate toxicity <sup>1</sup> |

## ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU with continuous monitoring, including any child needing atropine, epinephrine, or pacing.
- **Discharge** — only after a reversible cause is corrected, the rhythm is stable, and cardiology has reviewed.



- **Refer** — cardiology for pacing or structural/conduction disease; treat reversible causes (hypoxia, vagal, raised ICP, toxins, electrolytes).

#### ◆ READING THE TRACING

- New to pediatric rhythms?
- See a [stepwise approach to the](#)

[pediatric ECG](#)



and

[common pediatric arrhythmias](#)



(LITFL ECG Library).

## Supraventricular Tachycardia

MINUTES

Narrow QRS, no rate variability. HR > 220 infant / > 180 child.

+ Calculator: [Resuscitation dose calculator](#)

#### ▶ THROUGHOUT

- Run a continuous ECG rhythm strip during every intervention — documents the moment of

conversion; can unmask underlying WPW.<sup>1</sup>

#### DECISION

**Hemodynamically unstable (shock, poor perfusion, reduced consciousness)?**

#### UNSTABLE

Synchronized cardioversion; sedate only if it will not delay — use an agent that gives **amnesia**, not fentanyl alone (see sedation note). If IV/IO access is already in place, one rapid adenosine dose may be given while the defibrillator charges<sup>1</sup> — never delay the shock.

#### STABLE

Vagal maneuvers first — ice to the face in infants; modified Valsalva in older children<sup>1</sup> — then adenosine.

#### ✦ MODIFIED VALSALVA

- Sit child semi-recumbent; blow into a 10 mL syringe hard enough to just move the plunger (~15 s); then immediately lay flat and raise legs to ~45° for 15 s.<sup>3</sup>

- More effective than standard strain in the adult REVERT trial and a 2025 pediatric randomized trial.<sup>2,3</sup>

| Drug / action         | Dose                  | Notes                                                                                                                                                                                                     |
|-----------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adenosine (1st dose)  | 0.1 mg/kg (max 6 mg)  | ⚠ Technique is critical: use the most proximal IV; two-syringe or 3-way-stopcock method — rapid push, then an instant 5–10 mL saline flush while raising the arm <sup>1</sup> . Often needs two people.   |
| Adenosine (2nd dose)  | 0.2 mg/kg (max 12 mg) | If no response to the first dose.                                                                                                                                                                         |
| If the 2nd dose fails | Consult cardiology    | Next steps: <b>synchronized cardioversion</b> , or an antiarrhythmic infusion — <b>amiodarone 5 mg/kg over 20–60 min OR procainamide 15 mg/kg over 30–60 min. Never mix antiarrhythmics.</b> <sup>1</sup> |

| Drug / action               | Dose           | Notes                                                                                                                                                                    |
|-----------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synchro nized cardioversion | 0.5–1 → 2 J/kg | For the unstable child. If it will not delay the shock, <b>sedate with an agent that provides amnesia</b> — do not rely on fentanyl alone (see the sedation note below). |

### ⚠ SEDATION FOR CARDIOVERSION – AMNESIA, NOT JUST ANALGESIA

**Fentanyl alone is a pitfall:** it gives analgesia but **no amnesia**<sup>1</sup>, so the child may be awake and remember the shock; pushed rapidly it can cause **chest-wall rigidity**. **Ketamine**'s myocardial depression matters mainly in **prolonged, catecholamine-depleted** shock; in most children it is a net **sympathomimetic (positive inotrope)**, and ketamine or **etomidate** is standard for brief pediatric procedural sedation — there are no RCTs for this ~60-second window. **Involve a pediatric anesthesiologist** / follow your procedural-sedation protocol.

### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — first presentation, hemodynamic instability, or need for adenosine/cardioversion → monitored bed/PICU.

- **Discharge** — a stable child who reverts and stays in sinus rhythm, feeding and well, with cardiology follow-up and a clear return plan.
- **Refer** — cardiology for all confirmed SVT (WPW screen, ablation planning) and for recurrent episodes.

#### ◆ READING THE TRACING

- Align terminology with PALS/cardiology: SVT = **narrow-complex tachycardia**, VT = **wide-complex tachycardia** (each with or without a pulse); cross-check the label with cardiology.

- New to pediatric rhythms?
- See a [stepwise approach to the pediatric ECG](#)



and

[common pediatric arrhythmias](#)



(LITFL ECG Library).

## Ventricular Tachycardia (with a pulse)

MINUTES

Wide-QRS tachycardia. Pulseless VT → use the arrest algorithm.

+ Calculators: Resuscitation dose calculator · QTc

DECISION

Stable or unstable?

UNSTABLE

Synchronized cardioversion 0.5–1 → 2 J/kg. Sedate if no delay (short-acting agent, e.g., IV fentanyl). Expert input.

If **regular and monomorphic**, consider adenosine (diagnostic/therapeutic — may unmask SVT with aberrancy; not for irregular or polymorphic rhythms).

**Polymorphic VT / torsades →**

**magnesium sulfate 25–50 mg/kg IV (max 2 g)** and avoid QT-prolonging antiarrhythmics<sup>1</sup>. **Contact cardiology before starting antiarrhythmics** — a single infusion of amiodarone **or** procainamide (not both: additive QT prolongation)<sup>1</sup>.

| Drug         | Dose                 | Notes                                                          |
|--------------|----------------------|----------------------------------------------------------------|
| Amiodarone   | 5 mg/kg (max 300 mg) | IV over 20–60 min if perfusing; monitor BP <sup>1</sup> & QT   |
| Procainamide | 15 mg/kg             | IV over 30–60 min; do not combine with amiodarone <sup>1</sup> |

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU with continuous monitoring and a defibrillator at the bedside.
- **Discharge** — not from the acute episode — only after cardiology work-up and a management plan.

- **Refer** — cardiology urgently (structural disease, channelopathy, QT/electrolytes); consider genetics and family screening.

#### ◆ READING THE TRACING

- New to pediatric rhythms?
- See a [stepwise approach to the](#)

[pediatric ECG](#)



and

[common pediatric arrhythmias](#)



(LITFL ECG Library).

## Neonatal & Infant CHD

MINUTES

### RECOGNIZE BY PRESENTATION

Three ways congenital heart disease declares itself in the newborn — cyanosis, shock, or heart failure.

#### ① CYANOSIS – DUCT-DEPENDENT PULMONARY FLOW / MIXING ("5 T'S + 1")

- Transposition of the great arteries (duct-dependent for mixing)
- Truncus arteriosus



- Total anomalous pulmonary venous drainage (obstructed type = emergency)
- Tricuspid atresia
- Tetralogy of Fallot — **only the critical / pulmonary-atresia type is duct-dependent** (ordinary TOF presents with tet spells; see below).
- **+ 1:** critical pulmonary stenosis / atresia.
- **Assess the response to oxygen — a formal hyperoxia test is no longer routinely recommended.**<sup>235</sup> High  $\text{FiO}_2$  can accelerate ductal closure, dangerous in a duct-dependent lesion. Use pre-/post-ductal  $\text{SpO}_2$ , chest X-ray + blood gas, and echocardiography instead. In suspected duct-dependent CHD, **do not delay alprostadil** while waiting for a hyperoxia test<sup>235</sup>.

## ② CARDIOGENIC SHOCK / COLLAPSE — DUCT-DEPENDENT SYSTEMIC FLOW (LEFT-HEART OBSTRUCTION)

- Hypoplastic left heart
- critical coarctation
- interrupted aortic arch
- critical aortic stenosis.

- Gray, mottled shock with **weak/absent femoral pulses**, a pre-/post-ductal saturation<sup>235</sup> & blood-pressure gap, and metabolic acidosis. **Avoid high inspired oxygen** — it lowers pulmonary vascular resistance and steals from systemic flow<sup>235</sup>. Start alprostadil.

### ③ HEART FAILURE – LEFT-TO-RIGHT SHUNTS

- VSD
- AVSD
- PDA
- (large ASD — rarely).
- **Present after the first weeks as pulmonary vascular resistance falls — not as immediate newborn emergencies;** ASD rarely causes neonatal heart failure.<sup>235</sup>
- Tachypnea, poor feeding, sweating with feeds, hepatomegaly, faltering growth.

**Suspect a duct-dependent lesion →  
start Prostaglandin E1 (alprostadil)  
before the echocardiogram<sup>235</sup>**



## ANTICIPATE

**Apnea** (be ready to intubate), hypotension, fever. Urgent cardiology + transfer; carry the infusion on transfer.

| Drug                           | Dose                                              | Route & notes                                                                                                                                                                                                                                                                     |
|--------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alprostadil (prostaglandin E1) | start 0.05–0.1 → maintenance 0.01–0.02 mcg/kg/min | IV infusion; two IV lines; never interrupt; once responding, titrate to the lowest effective dose (low dose maintains patency in ~80% with less apnea and fever <sup>235</sup> ). Watch for apnea (~12%), hypotension, and fever <sup>235</sup> — be ready to support ventilation |

## ⚠ DON'T BE CAUGHT OUT

- The collapsed neonate labeled “sepsis,” “bronchiolitis,” or a “cyanotic spell” may be a closing duct.
- If shock or cyanosis does not respond to oxygen and fluids, **start alprostadil empirically** while arranging an urgent echocardiogram<sup>235</sup> — do not wait for imaging.

- In duct-dependent **systemic** lesions keep SpO<sub>2</sub> targets modest.

#### ◆ AVAILABILITY

- Prescribe and label as **ALPROSTADIL** (the single unambiguous name<sup>235</sup> — avoid the ambiguous "prostaglandin E1" and any trade name).
- Stocked in neonatal and cardiac units.

#### ◆ ARRHYTHMIA

- Tachycardia — **SVT** (narrow QRS) and **VENTRICULAR TACHYCARDIA** (wide QRS): see the Supraventricular Tachycardia and Ventricular Tachycardia cards; adenosine dosing is given on the SVT card.
- Screen every well newborn before discharge — see **Critical CHD — Pulse Oximetry Screening** (Neonatal).

#### ▶ ADMIT · DISCHARGE · REFER

- **Admit** — any duct-dependent lesion or cyanosis/shock → NICU/PICU; keep the duct open with alprostadil.

- **Discharge** — only under cardiology, with a feeding/oxygen/weight plan and clear red-flag advice.
- **Refer** — pediatric cardiology urgently (echo) and a cardiac-surgical center; arrange safe transfer.

◆ **SEE ALSO** Newborn screening & pulmonary hypertension: [Critical CHD — Pulse Oximetry Screening](#) · [Persistent Pulmonary Hypertension \(PPHN\)](#) (Neonatal).

## Hypercyanotic (Tet) Spell MINUTES

### TETRALOGY OF FALLOT

Sudden deep cyanosis, distress, and a quieter murmur in Tetralogy of Fallot.

#### ① POSITION & CALM

Knee-to-chest position (or let the child squat). Keep calm — minimal handling. Give 100% oxygen.<sup>389</sup>



#### ② INCREASE RV PRELOAD

Morphine to settle the cycle; fluid bolus to increase filling.<sup>389</sup>



### ③ INCREASE SYSTEMIC RESISTANCE / SLOW THE HEART

- Phenylephrine; beta-blocker (propranolol or esmolol). **Avoid agents that increase contractility** — they worsen the outflow obstruction.<sup>389</sup>
- Phenylephrine, IV beta-blocker, and bicarbonate (for persistent acidosis) are specialist-supported — **call PICU/cardiology early.**

| Drug               | Dose                                                           | Notes                                                     |
|--------------------|----------------------------------------------------------------|-----------------------------------------------------------|
| Morphine           | 0.1–0.2 mg/kg                                                  | IV/IM/subcut; settles the spell <sup>389</sup>            |
| Crystalline bolus  | 10–20 mL/kg                                                    | IV, increases preload                                     |
| Phenylephrine      | 5–20 mcg/kg                                                    | IV; raises systemic vascular resistance                   |
| Propranolol        | 0.05–0.1 mg/kg                                                 | slow IV; alternative to esmolol                           |
| Esmolol            | 500 mcg/kg IV load over 1 min, then 50–300 mcg/kg/min infusion | IV beta-blocker alternative to propranolol <sup>389</sup> |
| Sodium bicarbonate | 1 mmol/kg                                                      | IV, if significant metabolic acidosis                     |

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — any spell → monitored bed/PICU; recurrent or severe spells need urgent intervention.
- **Discharge** — only after cardiology review, a spell-management plan (knee-chest, hydration, propranolol as advised), and caregiver education.
- **Refer** — pediatric cardiology urgently — a spell often brings surgical repair forward.

◆ **SEE ALSO** Neonatal & Infant CHD for duct-dependent lesions and alprostadil.

# Blocked Systemic-to-Pulmonary Shunt (BT / Sano)

MINUTES

Shunt-dependent CHD

Sudden cyanosis with a **lost shunt murmur** — the shunt is occluding; a peripheral-hospital emergency.

## RECOGNIZE IT – THE SHUNT IS BLOCKING

- Known **shunt-dependent** cyanotic CHD (modified **Blalock-Taussig** or **Sano** shunt) with **acute, severe cyanosis**.
- **Loss of the continuous shunt murmur** — the single most useful sign (a patent shunt murmurs continuously)<sup>236</sup>.
- **Sudden desaturation**, then **hypotension and metabolic acidosis**; falling PaO<sub>2</sub>, rising lactate, falling cerebral NIRS from baseline<sup>236</sup>.
- **Echo: reduced or absent flow** across the shunt confirms it — but do **not** delay treatment for imaging.



**⚠ CALL FOR HELP NOW**

- **Call cardiac ICU/PICU + cardiothoracic surgery + cardiology immediately**, and activate your retrieval service.
- The physiology to fix it: **drive more blood across the shunt — raise systemic vascular resistance (SVR) and lower pulmonary vascular resistance (PVR) — while you anticoagulate.**

**① RESUSCITATE (airway & oxygen) —**

Secure the airway and ventilate; give **100% oxygen** ( $\text{FiO}_2 \approx 10\%$  above the child's baseline) and optimize oxygenation. Urgent echo — but do not delay treatment.



**② RAISE SVR (push flow across the shunt) —** Fluid bolus **5 mL/kg** aliquots (4% albumin), reassessing; add a vasoconstrictor — **phenylephrine 3–10 mcg/kg slow IV**, or **norepinephrine 0.1 mcg/kg/min**, or **epinephrine up to 0.05 mcg/kg/min**<sup>236</sup>.



③ **LOWER PVR (unload the lungs)** —

Deepen **sedation + paralysis**; **hyperventilate** to lower  $\text{PaCO}_2$  (mild respiratory alkalosis); keep well oxygenated; **magnesium sulfate 0.4 mmol/kg (50%)**<sup>236</sup>.



④ **ANTICOAGULATE (stop the clot)** —

**Heparin 100 units/kg IV bolus**, then **20 units/kg/h infusion**<sup>236</sup>; an optional **second 50 units/kg bolus** at 15 min. Restart **prostaglandin (alprostadil)** if the arterial duct has not been surgically ligated<sup>235</sup>.



⑤ **DEFINITIVE RESCUE** — **cardiac catheterization** (balloon angioplasty, stent, or thrombolysis) or **emergency surgical shunt revision**; **ECMO** to bridge a deteriorating child. Confirm with echo and transfer to the cardiac center.

## MEDICATIONS – CONFIRM AGAINST YOUR CARDIAC CENTER'S PROTOCOL

| Drug / product                                      | Dose                                                              | Notes                                                                             |
|-----------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>SVR</b><br><b>Phenylephrine</b>                  | 3–10 mcg/kg<br>slow IV                                            | Raises SVR; dilute before use (e.g., 10 mg in 100 mL, then 1 mL made up to 10 mL) |
| <b>SVR</b><br><b>Norepinephrine (noradrenaline)</b> | 0.1 mcg/kg/min<br>infusion                                        | Raises SVR; central line preferred, peripheral as a bridge                        |
| <b>SVR</b><br><b>Epinephrine (adrenaline)</b>       | up to 0.05 mcg/kg/min<br>infusion                                 | Inotrope/vasopressor support                                                      |
| <b>SVR</b><br><b>Albumin 4%</b>                     | 5 mL/kg<br>aliquots                                               | Volume to support preload/SVR; reassess after each                                |
| <b>PVR</b><br><b>Magnesium sulfate 50%</b>          | 0.4 mmol/kg<br>IV                                                 | Pulmonary vasodilator — helps lower PVR                                           |
| <b>CLOT</b><br><b>Heparin</b>                       | 100 units/kg<br>IV bolus → 20 units/kg/h; ± 50 units/kg at 15 min | Prevents clot propagation; monitor APTT / anti-Xa                                 |

| Drug / product                                                                         | Dose                 | Notes                                                               |
|----------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------|
| <b>I DUCT</b><br><b>Prostagl</b><br><b>andin E1</b><br><b>(alprosta</b><br><b>dil)</b> | per unit<br>protocol | If the duct is not<br>ligated — may reopen<br>ductal pulmonary flow |

### ✦ THE TRAP

The instinct to give a vasodilator, or deep vasodilating sedation without vasopressor support, **drops SVR and worsens shunt flow**. Do the opposite — **raise SVR, lower PVR, heparinize**, and get the child to the cath lab or theatre. **A lost shunt murmur in a cyanotic child is the shunt occluding until proven otherwise.**<sup>236</sup>

### ◆ SOURCES

- NHS Greater Glasgow & Clyde paediatric guideline — immediate management of the modified Blalock–Taussig / central shunt (Paediatrics 386); PIER Network — acute management of the cyanosed infant with a systemic-to-pulmonary (BT/Sano) shunt in the ED; shunt-thrombolysis case & review data (heparin/urokinase; catheter-directed thrombolysis).

- Confirm all doses against your formulary and cardiac center's protocol.

## Acute Rheumatic Fever

### DIAGNOSE

Jones 2015 • AHA

Prevent the second attack — every recurrence adds valve damage.

#### ▲ DIAGNOSE – JONES 2015 (REVISED)

- Need **evidence of preceding group A strep** (rising/elevated ASO or anti-DNase B, or positive throat swab/RADT, or recent scarlet fever) **plus** either **2 major** or **1 major + 2 minor**<sup>228</sup>.
- Recurrence also allows **3 minor**<sup>228</sup>. **Chorea** and **indolent carditis** can be diagnostic on their own.
- Get an **echo in every suspected case** (repeat even if first is normal) — subclinical valve regurgitation counts as carditis<sup>228</sup>.

### ⚠️ USE THE MODERATE/HIGH-RISK COLUMN

- Gulf / Middle-East children are a **moderate-to-high-risk** population — applying low-risk thresholds under-diagnoses ARF<sup>228</sup>.
- In this group a **single joint (mono-arthritis or even mono-arthralgia)** and a **lower fever/ESR threshold** already count<sup>228</sup>.

| Jones 2015   | Low-risk population                                                                                                                                                | Moderate / high-risk (Gulf)                                                                                                                                       |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Major</b> | Carditis (clinical <b>or</b> subclinical on echo) · <b>migratory polyarthritis</b> (flitting, large joints) · chorea · erythema marginatum · subcutaneous nodules  | Same, except joints = <b>mono-arthritis, polyarthritis, or polyarthralgia</b>                                                                                     |
| <b>Minor</b> | Polyarthralgia · fever $\geq 38.5\text{ }^{\circ}\text{C}$ · ESR $\geq 60\text{ mm/h}$ and/or CRP $\geq 3.0\text{ mg/dL}$ · prolonged PR (if carditis not a major) | Mono-arthralgia · fever $\geq 38\text{ }^{\circ}\text{C}$ · ESR $\geq 30\text{ mm/h}$ and/or CRP $\geq 3.0\text{ mg/dL}$ · prolonged PR (if carditis not a major) |

A manifestation counts once — polyarthritis as a major cannot also be counted as a minor<sup>228</sup>.

## TREAT

| Target                                                    | What to give                                                                                                                                                                                     |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eradicate GAS</b><br>(everyone, even if swab negative) | <b>Benzathine penicillin G IM once:</b> < 27 kg → 600,000 units; ≥ 27 kg → 1,200,000 units. Alt: penicillin V PO ×10 d; if allergic, azithromycin 12 mg/kg/day (max 500 mg) ×5 d. <sup>229</sup> |
| <b>Arthritis / mild carditis</b>                          | <b>Aspirin 80–100 mg/kg/day PO</b> divided every 6 h for 2–6 weeks, then taper; watch for tinnitus/hyperpnea (salicylate toxicity) <sup>228</sup> .                                              |
| <b>Moderate–severe carditis / heart failure</b>           | Add <b>prednisone/prednisolone ~2 mg/kg/day</b> (max ~60 mg) 2–3 weeks then taper, overlapping aspirin. Diuretic + ACE inhibitor for heart failure; bed rest. <sup>228</sup>                     |
| <b>Sydenham chorea</b>                                    | Usually self-limited — quiet environment, protect from injury. If disabling: valproate or carbamazepine (off-label). Salicylates/steroids do <b>not</b> help the chorea itself. <sup>228</sup>   |

## SECONDARY PROPHYLAXIS – BENZATHINE

### PENICILLIN G IM EVERY 3–4 WEEKS (< 27 KG 600,000 U; ≥ 27 KG 1,200,000 U)

| Category (from last attack)          | Duration                                                 |
|--------------------------------------|----------------------------------------------------------|
| ARF without carditis                 | 5 years or until age 21 (whichever longer)               |
| Carditis, no residual valve disease  | 10 years or until age 21 (whichever longer)              |
| Carditis with residual valve disease | 10 years or until age 40 — often lifelong <sup>229</sup> |

◆ **IF THE IM INJECTION IS DECLINED** Oral **PENICILLIN V 250 MG TWICE DAILY** is an accepted alternative for secondary prophylaxis (sulfadiazine if penicillin-allergic) — but adherence is worse and it is **LESS EFFECTIVE THAN IM BENZATHINE**, which remains preferred<sup>229</sup>.

#### ✦ THE POINT OF THE CARD

- Treatment settles the attack; **prophylaxis prevents rheumatic heart disease.**
- Recurrences are cumulative valve damage — document the injection schedule and hand it over so it is never missed.



### ◆ SOURCES

- AHA 2015 Revised Jones Criteria (Gewitz, Circulation 2015); AHA/AAP prevention statement (Gerber, Circulation 2009); WHO RHD guidance; SPIDS Gulf prophylaxis guidance.
- Confirm doses against your formulary.

## Pericarditis & Cardiac Tamponade

MINUTES

ESC 2015

Tamponade is a clinical diagnosis — drain, don't wait for the scan.

### ▶ PERICARDITIS – 2 OF 4

- Pleuritic chest pain **relieved by sitting forward**
- pericardial **friction rub**
- **widespread ST elevation / PR depression** on ECG (not one coronary territory)<sup>230</sup>
- new/worsening **pericardial effusion** on echo.
- Supportive: raised CRP/ESR.

- Causes — viral/idiopathic (commonest), purulent, post-pericardiotomy, autoimmune (SLE, JIA), uremic, TB, malignancy<sup>230</sup>.

### ⚠ **TAMPONADE – RECOGNIZE AND ACT**

- **Beck's triad** (hypotension, muffled heart sounds, and raised JVP) + **tachycardia** and a **narrow pulse pressure; pulsus paradoxus > 10 mmHg** (inspiratory drop in systolic BP)<sup>230</sup>.
- Children compensate then crash — tachycardia is the early sign. **A small effusion can tamponade if it accumulates fast** — volume on echo does not equal severity; the clinical picture does<sup>230</sup>.
- Beck's triad and pulsus are often incomplete — their absence does not exclude it<sup>230</sup>.

#### DECISION

**Hemodynamic compromise (tamponade)?**

## YES

**Emergency echo-guided pericardiocentesis** — removing even a small volume rescues output<sup>230</sup>. **IV fluid 10–20 mL/kg is a bridge only**, never a substitute. **Maintain systemic vascular resistance** with a vasopressor — **norepinephrine** ( $\pm$  **vasopressin**) — as a temporizing bridge to support coronary and systemic perfusion; keep the circulation **full, fast, and tight** (preload, avoid bradycardia, support afterload). **Avoid intubation / positive-pressure ventilation** — it drops preload and can cause arrest<sup>230</sup>; if unavoidable, drain first, low tidal volume/PEEP, keep volume-loaded. Avoid diuretics/vasodilators. Call **cardiology + cardiac surgery + PICU** in parallel.

## NO

Uncomplicated pericarditis: **NSAID (ibuprofen) + colchicine** — colchicine halves recurrence<sup>230</sup>. Treat the cause; restrict strenuous exercise until symptom-free with normal CRP/ECG/echo. Steroids are second-line (increase recurrence)<sup>230</sup>.

| Drug                     | Pediatric dose                                                                   | Notes                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Ibuprofen                | 30–40 mg/kg/day<br>divided every 6–8<br>h (max ~2.4 g/day)                       | First-line<br>anti-inflammatory;<br>give with<br>gastroprotection,<br>taper on<br>CRP <sup>230</sup> . |
| Colchicine               | < 5 yr: 0.5 mg/day ·<br>> 5 yr: 1–1.5<br>mg/day divided                          | First-line<br>adjunct —<br>halves<br>recurrence;<br>reduce in renal<br>impairment.                     |
| Purulent<br>pericarditis | Urgent drainage +<br>IV vancomycin +<br>3rd-gen<br>cephalosporin, ~3–<br>4 weeks | Surgical<br>emergency —<br>NSAIDs alone<br>are inadequate;<br>narrow to<br>culture <sup>230</sup> .    |

### ✦ THE TRAP

Don't reach for the ventilator in a crashing tamponade — positive-pressure ventilation can be the last straw. **Drain first.** And treat **purulent pericarditis** as the surgical emergency it is.

### ◆ SOURCES

- ESC 2015 pericardial diseases guidelines (Adler, Eur Heart J); pediatric echo/tamponade reviews (Frontiers Pediatr 2017); pediatric

colchicine data (Pediatr Cardiol 2024).

- Confirm doses against your formulary.

## Infective Endocarditis (incl. Subacute)

HOURS

Modified Duke / AHA

Fever + a new murmur + positive cultures — think endocarditis; diagnose by the modified Duke criteria.

### ▲ SUSPECT IT – SUBACUTE CLUES

- Persistent/unexplained fever with a **new or changed regurgitant murmur** — the classic pairing.
- Subacute course over weeks: malaise, weight loss, night sweats, anorexia. **Viridans-group streptococci** are the commonest cause of subacute native-valve disease; **S. aureus** dominates acute disease<sup>233</sup>.
- Peripheral stigmata (late, immune-mediated): **splinter hemorrhages**, tender **Osler nodes**,

painless **Janeway lesions**, **Roth spots**<sup>233</sup>, splenomegaly, embolic phenomena.

- At-risk: **congenital heart disease**, prosthetic valve/material, previous IE, indwelling central lines, IV drug use.

#### ▶ MODIFIED DUKE CRITERIA – HOW TO CLASSIFY

- **DEFINITE** (clinical) = **2 major**, or **1 major + 3 minor**, or **5 minor**.
- **POSSIBLE** = **1 major + 1 minor**, or **3 minor**.
- **REJECTED** = firm alternative diagnosis, or resolution with  $\leq 4$  days of antibiotics<sup>233</sup>.

#### DECISION

##### The two MAJOR criteria

#### MICROBIOLOGY

**Typical organism** from  $\geq 2$  separate **blood cultures** (viridans streptococci, *S. gallolyticus*/bovis, HACEK group, *S. aureus*, or community-acquired enterococci with no primary focus), **or persistently positive** cultures of a compatible organism<sup>233</sup>.

## ENDOCARDIAL

Echo evidence of **endocardial involvement: vegetation, abscess, new partial dehiscence of a prosthetic valve, or new valvular regurgitation** (worsening of an existing murmur is not enough)<sup>233</sup>.

### ▲ THE FIVE MINOR CRITERIA

- **Predisposition** — predisposing heart condition or IV drug use.
- **Fever**  $\geq 38^{\circ}\text{C}$ .
- **Vascular phenomena** — arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial or conjunctival hemorrhage, **Janeway lesions**.
- **Immunologic phenomena** — glomerulonephritis, **Osler nodes**, **Roth spots**, rheumatoid factor.
- **Microbiologic** evidence not meeting a major criterion<sup>234</sup>.

### ⚠ WORKUP – GET IT RIGHT THE FIRST TIME

- **Draw  $\geq 3$  sets of blood cultures** from separate venipunctures **before** starting antibiotics — the

single most important step for diagnosis and tailoring therapy<sup>234</sup>.

- Echo: **TTE first**; proceed to **TEE** if TTE is non-diagnostic<sup>231</sup>, a prosthetic valve is present, or suspicion stays high.
- Add ESR/CRP, FBC, and urinalysis (microscopic hematuria supports it). Do **not** delay antibiotics in a septic/unstable child once cultures are drawn.

◆ Endocarditis prophylaxis (AHA / ESC 2023) — who

Prophylaxis needs **BOTH** a highest-risk cardiac condition **AND** a high-risk procedure. The **HIGHEST-RISK CARDIAC CONDITIONS**:

1. **PROSTHETIC HEART VALVE** (surgical or transcatheter) or prosthetic material used for valve repair.
2. **PREVIOUS INFECTIVE ENDOCARDITIS**.
3. **UNREPAIRED CYANOTIC CHD** (including palliative shunts/conduits).
4. Completely **REPAIRED CHD WITH PROSTHETIC MATERIAL/DEVICE** — during the **FIRST 6 MONTHS** after the procedure.
5. **REPAIRED CHD WITH A RESIDUAL DEFECT** at or adjacent to the prosthetic patch/device.
6. **CARDIAC TRANSPLANT** recipients who develop **VALVULOPATHY**.



7. **VENTRICULAR ASSIST DEVICE (VAD)** — added by **ESC 2023**<sup>231</sup>.

◆ **ESC 2023 EMPHASIS:** for all **INTERMEDIATE- AND HIGH-RISK** patients, **GOOD ORAL HYGIENE AND REGULAR DENTAL REVIEW** prevent more endocarditis than any single antibiotic dose<sup>231</sup>.

◆ Endocarditis prophylaxis (AHA) — which procedures

- **DENTAL** — any procedure that manipulates **GINGIVAL TISSUE** or the **PERIAPICAL REGION** of teeth, or **PERFORATES THE ORAL MUCOSA** (extractions, scaling, periodontal work, implants). Routine fillings, radiographs, and shedding of deciduous teeth do **NOT** count<sup>232</sup>.
- **RESPIRATORY TRACT** — procedures with **INCISION OR BIOPSY OF THE RESPIRATORY MUCOSA** (e.g., **TONSILLECTOMY, ADENOIDECTOMY**, bronchoscopy **WITH BIOPSY**). Cover staphylococci if the airway procedure treats an established infection (add anti-staphylococcal agent).
- **INFECTED SKIN, SKIN STRUCTURES, OR MUSCULOSKELETAL TISSUE** — regimen should cover **STAPHYLOCOCCI AND B-HEMOLYTIC STREPTOCOCCI**.

- **NOT RECOMMENDED** for **GI OR GU PROCEDURES**, or for bronchoscopy/endoscopy **WITHOUT** incision/biopsy, solely to prevent endocarditis.
- **◆ ESC 2023** is more restrictive for non-dental procedures — it advises prophylaxis for respiratory, GI, GU, skin, or musculoskeletal procedures **ONLY WHEN PERFORMED AT AN INFECTED OR COLONIZED SITE**<sup>231</sup>. Where the two guidelines differ, follow your local protocol.

## PROPHYLAXIS REGIMEN – SINGLE DOSE 30-60 MIN BEFORE THE PROCEDURE

| Situation                        | Pediatric dose (single)                                                                                            | Max                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Able to take oral</b>         | <b>Amoxicillin</b> 50 mg/kg PO <sup>232</sup>                                                                      | 2 g                                                                      |
| <b>Unable to take oral</b>       | Ampicillin 50 mg/kg IM/IV, or cefazolin / ceftriaxone 50 mg/kg IM/IV                                               | Ampicillin 2 g · cefazolin/ceftriaxone 1 g                               |
| <b>Penicillin allergy — oral</b> | Cephalexin 50 mg/kg, or azithromycin / clarithromycin 15 mg/kg, or doxycycline (< 45 kg 2.2 mg/kg; ≥ 45 kg 100 mg) | Cephalexin 2 g · azithromycin/clarithromycin 500 mg · doxycycline 100 mg |

| Situation                    | Pediatric dose (single)                 | Max |
|------------------------------|-----------------------------------------|-----|
| Penicillin allergy — no oral | Cefazolin or ceftriaxone 50 mg/kg IM/IV | 1 g |

### ✦ THE TRAP

**Clindamycin is no longer recommended** (dropped in the 2021 AHA update<sup>232</sup> — *C. difficile* risk and adverse effects). Prophylaxis is **not** indicated for most CHD, nor for routine GI/GU procedures. Avoid cephalosporins if there is a history of anaphylaxis, angioedema, or urticaria to penicillin<sup>232</sup>. **Good oral hygiene and regular dental care prevent more endocarditis than any single antibiotic dose.**

### ◆ SOURCES

- AHA prevention of IE statement (Wilson, *Circulation* 2007) and 2021 update (Wilson, *Circulation* 2021), and AHA IE wallet card; **2023 ESC ENDOCARDITIS GUIDELINES** (Delgado & Ajmone Marsan, *Eur Heart J* 2023;44:3948); Modified Duke criteria (Li, *Clin Infect Dis* 2000); 2023 Duke-ISCVID criteria (Fowler, *Clin Infect Dis* 2023).

- Confirm doses against your formulary.

## Common Acute

04

## Conditions

Everyday clinic & ED infections and presentations

### Fever in a Well-Appearing Infant < 2 Months

HOURS

≤ 60 DAYS • CPS 2026 / AAP 2021

Most are viral — but ~8–13% have a serious bacterial infection<sup>16</sup>. Define fever as rectal  $\geq 38.0^{\circ}\text{C}$ ; stratify risk and spare the low-risk from over-testing.

#### ► SCOPE & FIRST MOVE

- This pathway is for the **well-appearing, full-term ( $\geq 37$  wk), previously healthy** infant with a documented **rectal temperature  $\geq 38.0^{\circ}\text{C}$ . Hypothermia  $< 36.0^{\circ}\text{C}$**  carries the same risk — manage identically<sup>16</sup>.

- **This pathway does not apply to the ill-appearing at any age.**
- **Risk factors that are not applicable to this pathway:** prematurity < 37 wk, prior or prolonged hospitalization, a chronic / genetic / technology-dependent condition, immunodeficiency, recent antibiotics, or any focal bacterial infection (e.g., omphalitis, cellulitis).

#### **⚠ KEEP A HIGH INDEX OF SUSPICION**

- Even a well-appearing infant can deteriorate quickly.
- The youngest (**≤ 21–28 days**) have the highest risk of **invasive bacterial infection (IBI)** — bacteremia or bacterial meningitis) and the **lowest threshold** for full work-up, LP, empiric antibiotics, and admission<sup>16</sup>.

#### **▶ EVERYONE GETS**

- **All ages:** history + exam, **urinalysis + urine culture** (catheter or suprapubic; clean-catch acceptable), **blood culture, CBC + differential**, and an **inflammatory marker** — procalcitonin (PCT) or CRP.

- **Risk-stratify with a validated rule:** **PECARN** or **Step-by-Step** (both use PCT), or **Aronson** when PCT is unavailable — keep a **lower threshold for LP** with Aronson<sup>16</sup>.
- **CSF pleocytosis** =  $> 15 \text{ cells/mm}^3$  ( $\leq 28 \text{ d}$ ) or  $> 9 \text{ cells/mm}^3$  ( $> 28 \text{ d}$ ) → treat as bacterial meningitis. Don't delay the LP if there are neurological signs.<sup>16</sup>

## RISK STRATIFICATION – VALIDATED LOW-RISK RULES

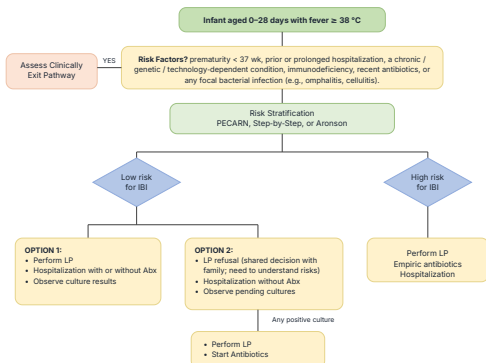
| Rule                                         | Low risk if ALL criteria are met                                                                                                                                                                                       |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PECARN (uses PCT)                            | 1) Urinalysis negative (leukocyte esterase, nitrite, $\leq 5 \text{ WBC/hpf}$ ); 2) $\text{ANC} \leq 4000/\mu\text{L}$ ; 3) $\text{PCT} \leq 0.5 \text{ ng/mL}$ <sup>16</sup>                                          |
| Step-by-Step (uses PCT)                      | 1) Well-appearing; 2) 22–90 days; 3) urinalysis negative for leukocytes; 4) $\text{PCT} < 0.5 \text{ ng/mL}$ ; 5) $\text{CRP} \leq 20 \text{ mg/L}$ and $\text{ANC} \leq 10,000/\mu\text{L}$ <sup>16</sup>             |
| Aronson (no PCT — keep a lower LP threshold) | Low risk if $\leq 1$ point: age $< 21 \text{ d}$ (1) · ED temp $38.0\text{--}38.4^\circ\text{C}$ (2) or $\geq 38.5^\circ\text{C}$ (4) · $\text{ANC} \geq 5185/\mu\text{L}$ (2) · urinalysis positive (3) <sup>16</sup> |

### ◆ APPLYING THE RULES – PCT TIMING & ALTERNATIVES

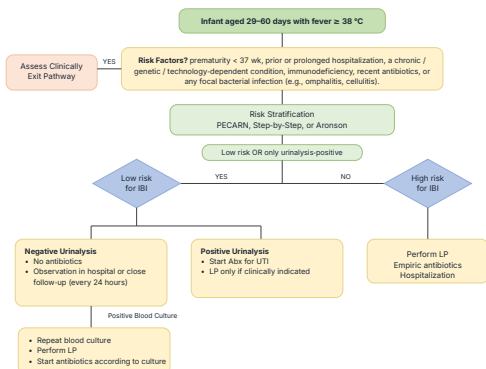
- Where **PROCALCITONIN IS NOT READILY AVAILABLE**, PECARN and Step-by-Step (both require PCT) **CANNOT BE FULLY APPLIED**. Use an alternative: the **ARONSON** rule (no PCT needed) and keep a **LOWER THRESHOLD FOR LP; CRP  $\leq$  20 MG/L** with a normal ANC and urinalysis still support lower risk.
- When PCT **IS** obtained, draw it  **$\geq$  2 H AFTER FEVER ONSET** — it rises ~3–4 h after a bacterial trigger, so very early samples can read falsely low<sup>16</sup>. It typically **STAYS LOW ( $< 0.5$  NG/ML)** in isolated viral illness.
- Missed invasive infection with Step-by-Step clusters in the **22–28-DAY** group and in infants tested very early — keep a low threshold in these.<sup>16</sup>

## FEVER PATHWAY ALGORITHMS (WELL-APPEARING, ≥ 37 WEEKS)

### Algorithm 1 – infant 0–28 days with fever ≥ 38°C



### Algorithm 2 – infant 29–60 days with fever ≥ 38°C





## EMPIRIC ANTIBIOTICS – WELL-APPEARING, ≥ 37 WEEKS

| Age        | First-line IV                                                                                                           | Modifiers                                                                                                                                                                  |
|------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–7 days   | Ampicillin 100 mg/kg/dose q8h (reduced neonatal renal clearance; Lexicomp) <sup>16</sup> + gentamicin 4 mg/kg/dose q24h | Meningitis: add — or replace the aminoglycoside with — cefotaxime 50 mg/kg/dose q8h. HSV risk: add acyclovir 20 mg/kg/dose q8h.                                            |
| 8–28 days  | Ampicillin 75 mg/kg/dose q6h + gentamicin 5 mg/kg/dose q24h                                                             | Meningitis: add — or replace the aminoglycoside with — cefotaxime 50 mg/kg/dose q6h (200 mg/kg/day for meningitis at this age). HSV risk: add acyclovir 20 mg/kg/dose q8h. |
| 29–60 days | Ceftriaxone 100 mg/kg/dose q24h                                                                                         | Meningitis: add vancomycin 15 mg/kg/dose q6h. HSV risk: add acyclovir. Low-risk UTI: cefixime 8 mg/kg/dose q24h PO. <sup>16</sup>                                          |

### SOURCE

Canadian Pediatric Society — Management of well-appearing febrile young infants aged ≤ 90 days (updated May 2026); AAP — Pantell et al.,

## Evaluation and Management of Well-Appearing Febrile Infants 8–60 Days Old, *Pediatrics* 2021.

◆ **BEYOND 60 DAYS** The well-appearing infant **61–90 DAYS** falls outside this pathway: risk of invasive infection is lower and routine blood culture / full septic work-up is **NOT REQUIRED**<sup>16</sup>. Test **SELECTIVELY** (urinalysis ± inflammatory markers) and use clinical judgment; escalate only the ill-appearing or those with a positive screen.

◆ **SEE ALSO** For the neonate ( $\leq 28$  days), see **Neonatal Sepsis — rapid approach** for neonatal-specific empiric therapy (ampicillin + gentamicin/cefotaxime ± acyclovir).

## Gastroenteritis & Dehydration

### HOURS

Assess hydration: alertness, eyes, mucosa, skin turgor, capillary refill, urine output.

✚ **Calculators:** **Maintenance fluids & bolus** · **Corrected Na** · **anion gap** · **Ca** · **osmolality**

### ▴ **CLINICAL SIGNS OF DEHYDRATION**

- Altered mental status

- sunken eyes
- dry oral mucosa
- poor skin turgor
- delayed capillary refill
- decreased urine output
- tachycardia
- abnormal peripheral perfusion.
- The more signs present, the greater the fluid deficit; no single sign is reliable alone.<sup>88</sup>

#### DECISION

### Degree of dehydration?

#### NONE–MODERATE

Oral rehydration with ORS; a single ondansetron dose can reduce vomiting and the need for IV.<sup>88</sup>

#### SEVERE / SHOCK

IV/IO crystalloid bolus, repeat to reverse shock; then deficit + maintenance. Check glucose & electrolytes.

| Drug / fluid                    | Dose                    | Route & notes                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oral rehydration solution       | ~50–100 mL/kg over 4 h  | Oral/NG + replace ongoing losses.<br>First-line for minimal <sup>88</sup> –moderate dehydration.                                                                                                                                                                                                                                          |
| 0.9% sodium chloride (bolus)    | 20 mL/kg                | IV/IO rapid bolus for hypovolemia/shock; reassess during & after each bolus and repeat until perfusion is restored. If shock seems out of proportion to the diarrhea/vomiting, consider sepsis, DKA, a surgical abdomen, or myocarditis. <b>Balanced crystalloid (e.g., Ringer's lactate) is an acceptable alternative.</b> <sup>88</sup> |
| 0.9% sodium chloride (moderate) | 10 mL/kg over 30–60 min | Moderate dehydration not tolerating oral — then reassess for a further bolus or transition back to ORT if tolerated.                                                                                                                                                                                                                      |

| Drug / fluid | Dose                                                                                                                                     | Route & notes                                                                                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ondansetron  | <p>0.15 mg/kg/dose – single dose, max 8 mg (either route).</p> <p>Weight-based: 2 mg (8–15 kg) • 4 mg (15–30 kg) • 8 mg (&gt; 30 kg)</p> | <p>PO preferred; <b>age ≥ 6 months only</b>. Reduces vomiting and the need for IV. <b>Restart ORT 15–30 min after the dose.</b></p> <p>Ondansetron prolongs the QT interval dose-dependently — do not exceed <b>8 mg</b> in a single pediatric dose (the 16 mg cap is an adult chemotherapy figure, not for gastroenteritis)<sup>88</sup>.</p> |

| Drug / fluid             | Dose                                     | Route & notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zinc<br>(adjunct to ORS) | 10 mg/day (< 6 mo)<br>20 mg/day (≥ 6 mo) | Oral, <b>10–14 days</b> . A WHO/UNICEF measure for low- and middle-income settings where zinc deficiency is common — shortens diarrhea by ~½ day (~1 day if malnourished) concentrates at <b>≥ 6 months</b> in zinc-deficient or malnourished children. <b>Not routine in high-income, well-nourished populations including Kuwait</b> ; reserve for malnutrition, documented zinc deficiency, or persistent / chronic diarrhea. Little benefit under 6 months. (ESPGHAN/ESPID acute-gastroenteritis guideline: zinc not recommended for well-nourished European children.) <sup>14</sup> |

| Drug / fluid                    | Dose                                                                                                                                                                 | Route & notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Racecadotril — optional adjunct | 1.5 mg/kg/dose PO TID with ORS; sachets 10 mg ( $\leq$ ~12 kg) / 30 mg ( $>$ ~13 kg). Age $>$ 3 months. Continue until diarrhea stops (2 normal stools), max 7 days. | Antisecretory enkephalinase inhibitor — reduces stool water loss <b>without slowing gut motility</b> (unlike loperamide, which is contraindicated). <b>Adjunct to ORS, not a substitute.</b> Licensed and used in Kuwait/Europe/Asia; <b>not FDA/NICE/AAP/CPS/WHO-ESPGHAN/ESPID 2014: “may be considered”</b> (II-B, weak recommendation, moderate-quality evidence; responder NNT $\approx$ 4). Cochrane 2019 found it safe with limited benefit (low-certainty). <sup>88</sup> |

## PEARLS

- Resume normal feeding (including breastfeeding) early. Antibiotics only for specific pathogens.<sup>88</sup>
- Red flags: bilious or bloody vomit, severe pain/distension (think surgical), inability to tolerate ORS.

- Zinc dosing & evidence: [WHO eLENA](#)



#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — shock or severe dehydration, failed oral/NG rehydration, intractable vomiting, dangerous electrolyte derangement, or diagnostic uncertainty.
- **Discharge** — minimal or mild dehydration tolerating oral rehydration, with carers confident on fluids and red-flag advice.
- **Refer** — surgical review if a surgical abdomen is suspected; PICU for refractory shock or severe electrolyte disturbance.

## Community-Acquired Pneumonia

HOURS

CPS 2024

The healthy, immunized child with fever, cough, and tachypnea — treat *S. pneumoniae*, mostly orally, and keep it short.

+ Calculator: [PARDS oxygenation index](#)



## RECOGNIZE & ASSESS SEVERITY

- Acute fever + cough + **tachypnea** (the most useful sign) ± increased work of breathing, chest or abdominal pain, poor feeding. Wheeze + hypoxia points to bronchiolitis or asthma, not bacterial CAP.<sup>58</sup>
- **Tachypnea cut-offs (breaths/min):**
  - < 2 mo > 60
  - 2–12 mo > 50
  - 1–5 yr > 40
  - > 5 yr > 30.<sup>58</sup>
- Check SpO<sub>2</sub> in everyone — hypoxemia may not be clinically obvious; **give oxygen for SpO<sub>2</sub> < 92%.**
- **Likely cause by age:**  
infants/preschool — mostly **viral** (RSV, influenza, human metapneumovirus); *Streptococcus pneumoniae* is the leading bacterial cause at all ages; school-age — add atypicals (<sup>58</sup>*Mycoplasma*, *Chlamydophila*).

## KEEP INVESTIGATIONS TARGETED

- A chest radiograph is not essential for a non-severe outpatient with a typical picture; obtain posteroanterior + lateral when the diagnosis is uncertain or the child is admitted (extent, effusion, abscess).<sup>173</sup>
- Do not image typical bronchiolitis or asthma.
- In hospitalized children: complete blood count (CBC)/C-reactive protein (CRP) + blood culture before antibiotics; consider respiratory viral testing in influenza season.

#### DECISION

### Outpatient or admit?

#### OUTPATIENT • NON-SEVERE

Well-hydrated, tolerating oral intake, no respiratory compromise. **Oral amoxicillin** is the mainstay — high-dose covers *S. pneumoniae*. Most do not need a chest radiograph.

Inadequate oral intake or intolerance, **hypoxemia / grunting / nasal flaring / apnea**, severe or complicated pneumonia, or **age < 6 months**. Start **IV ampicillin**.

**Amoxicillin (oral) or ampicillin (IV) is first-line — cover *S. pneumoniae*.**

## EMPIRIC THERAPY

| Setting / drug                             | Dose                                    | Notes                                                                                                                                      |
|--------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Amoxicillin — outpatient, first-line       | 25–30 mg/kg/dose PO q8h (max 4 g/day)   | Drug of choice for non-severe CAP. <b>TID</b> dosing (not BID). 5 days is enough if responding. <sup>173</sup>                             |
| Ampicillin — admitted, first-line          | 50 mg/kg/dose IV q6h (max 12 g/day)     | IV first-line for hospitalized, non-life-threatening CAP. Step down to oral amoxicillin on improvement. <sup>58</sup>                      |
| Ceftriaxone — severe / respiratory failure | 50 mg/kg/dose IV q12–24h (max 2 g/dose) | For respiratory failure or septic shock (or cefotaxime 50 mg/kg/dose q6–8h). Broader cover incl. $\beta$ -lactamase <i>H. influenzae</i> . |

| Setting / drug                                                    | Dose                                                     | Notes                                                                                                                                       |
|-------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| +<br>Vancomycin — if<br>MRSA<br>suspected                         | 15 mg/kg/dose<br>IV q6h                                  | Add for rapidly progressing multilobar disease or pneumatocoles until cultures return; send nasal MRSA screen first; stop if MRSA excluded. |
| Azithromycin —<br>atypical<br>( <i>Mycoplasma</i><br><i>sma</i> ) | 10 mg/kg day<br>1, then 5<br>mg/kg × 4 d<br>(max 500 mg) | School-age with atypical features. Macrolide resistance rising; doxycycline (any age, short courses). <sup>173</sup>                        |

### ⚠ DURATION, COURSE & FOLLOW-UP

- **5 days** for non-severe CAP (ambulatory, or admitted and responding within 24–36 h); **7–10 days** for more significant disease.<sup>173</sup>
- Expect improvement (appetite, fever, tachypnea, O<sub>2</sub> need) within **48 h**.
- If **not improving by 48–72 h** → re-image (radiograph/ultrasound) for empyema or abscess and reconsider the diagnosis (foreign body, tuberculosis, atypical).<sup>58</sup>
- Routine end-of-therapy radiograph is not needed — radiographic

clearance lags<sup>58</sup> 4–6 weeks.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — hypoxia ( $\text{SpO}_2 < 90\%$ ;  $< 92\%$  in high-risk infants), respiratory distress, dehydration or feeding failure, an effusion/empyema, or failure of outpatient therapy.
- **Discharge** — well child maintaining saturations in air, feeding, on oral antibiotics with follow-up arranged.
- **Refer** — PICU for respiratory failure or sepsis; surgical or respiratory review for empyema or non-resolving disease.

◆ **DIFFICULT IV ACCESS?** Reassess for an early IV→oral switch at every line failure rather than re-sticking — see [IV → Oral Switch & the DIVA Patient](#) (Quick Reference).

# Complicated Pneumonia – Empyema

HOURS

IDSA/PIDS 2026 • CPS 2024

Pneumonia + pus or a moderate-to-large pleural effusion — antibiotics plus, when needed, drainage with a fibrinolytic.

+ Calculator: PARDS oxygenation index

## ◆ DEFINITION & STAGES

- Empyema = intrapleural pus, or a moderate-to-large exudative parapneumonic effusion.
- It evolves: **STAGE 1** exudative (free-flowing) → **STAGE 2** fibrinopurulent (loculated) → **STAGE 3** organized (thick fibrinous peel)<sup>57</sup>.
- Small parapneumonic effusions are common in uncomplicated pneumonia and do **NOT** need drainage.<sup>57</sup>
- Lung abscess and necrotizing pneumonia are separate complications.

## ▶ RECOGNIZE & DIAGNOSE

- Suspect it when an initially uncomplicated pneumonia is **poorly responsive** — persistent fever after 48–72 h, worsening distress or

hypoxia, or new effusion signs (decreased breath sounds, dullness, reduced expansion).

- **Imaging:** chest radiograph first, then **chest ultrasound** (confirms effusion, estimates size, distinguishes free-flowing from loculated). Computed tomography (CT) is not routine — reserve for a suspected alternative diagnosis (e.g., malignancy).<sup>57</sup>
- **Pathogens:** *S. pneumoniae* (most common), *S. aureus* (incl. MRSA), *S. pyogenes* (group A streptococcus). Send drained fluid for **culture and Gram stain** ( $\pm$  pneumococcal PCR where available); blood cultures positive in only  $\sim 10\%$ <sup>57</sup>. Labs often show high white-cell count/ESR/CRP, thrombocytosis, hypoalbuminemia — these normalize with recovery.

#### DECISION

**How big is the effusion, and how sick is the child?**

### SMALL / MILD – ANTIBIOTICS ALONE

Small effusion, mild distress, no mediastinal shift → a trial of **antibiotics alone** is reasonable with close monitoring.

### MODERATE-LARGE / COMPROMISE – DRAIN

Moderate-to-large effusion, significant respiratory distress, or mediastinal shift → **drainage plus antibiotics**, early. Consult thoracic surgery / interventional radiology promptly.

### ▴ DRAIN WHEN

Moderate-to-large effusion or empyema causing respiratory compromise, or occupying much of the hemithorax ± **mediastinal shift; loculated (stage 2) or organized** collection on ultrasound, or frank **pus** on aspiration; or **failure of antibiotics alone** (not improving, or progressing, after an adequate trial).

### ◆ FIBRINOLYTIC (ALTEPLASE / TPA)

- First-line, most cost-effective drainage = **SMALL-BORE ( $\leq 12$  FR) PERCUTANEOUS CHEST TUBE + INTRAPLEURAL FIBRINOLYTIC**, preferred over surgical debridement (VATS) in



most cases — reserve VATS for refractory disease or extensive loculation. **ALTEPLASE 0.1 MG/KG (MAX 4 MG)** in normal saline instilled via the chest tube, **CLAMP/DWELL 1 HOUR, THEN DRAIN – ONCE DAILY FOR UP TO 3 DAYS** (a fixed 4 mg dose daily × 3 is an accepted alternative). **TPA ALONE IS NOW PREFERRED OVER TPA + DNASE** (no added benefit, higher cost).<sup>57</sup>

- Prescribe and label as **ALTEPLASE**.

## ANTIBIOTICS FOR EMPYEMA

| Agent                                 | Dose                                                                 | Notes                                                                                  |
|---------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Cefotaxime / ceftriaxone — first-line | Cefotaxime 50 mg/kg/dose q6–8h;<br>ceftriaxone 50 mg/kg/dose q12–24h | Empiric IV cover for the usual pathogens; choose per local antibiogram. <sup>173</sup> |
| + Vancomycin — MRSA                   | 15 mg/kg/dose IV q6h                                                 | Reserve for culture-proven or strongly suspected MRSA / severe disease.                |
| Oral step-down — culture-negative     | Amoxicillin (high-dose) or amoxicillin-clavulanate                   | Switch once drainage complete, improving, and off oxygen.                              |

### ⚠ DURATION, COURSE & FOLLOW-UP

- Total **3–4 weeks** is reasonable with adequate drainage and no further complication; transition to **oral** at or just before discharge once drained, improving, and off oxygen. **Persistent fever > 72 h is common** and — if the child is otherwise improving — usually **not** treatment failure; serial CRP can confirm improvement.<sup>57</sup>
- Outcome is excellent: lung function and radiograph normalize over weeks–months (a repeat film at 2–3 months is reasonable).
- Watch necrotizing pneumonia for **bronchopleural fistula** → involve surgery / respirology / infectious diseases.<sup>57</sup>

### ▶ ADMIT · DISCHARGE · REFER

- **Admit** — clinically significant empyema or a moderate-to-large effusion, for IV antibiotics ± drainage. A **small parapneumonic effusion in a well child** can be managed as community-acquired pneumonia with close follow-up rather than admitted.

- **Discharge** — not acutely; complete therapy with follow-up imaging.
- **Refer** — respiratory/surgery for drainage (chest drain ± fibrinolytics or VATS).

## Acute Otitis Media ASSESS

INFANT-CHILD • CPS 2016 / AAP 2013

Diagnose by the drum, not the symptoms — then choose between watchful waiting and antibiotics.

### ▲ CLINICAL DIAGNOSIS – EFFUSION + ACUTE INFLAMMATION

- Acute onset (otalgia, ear-tugging, or non-specific signs in the non-verbal child) **plus** a middle-ear effusion **plus** signs of acute inflammation.
- **A bulging tympanic membrane is the key sign** — highest predictive value for bacterial AOM, especially if yellow, cloudy, or hemorrhagic.<sup>43</sup>  
Acute perforation with purulent otorrhea also confirms a bacterial cause — distinguish from otitis externa, chronic drainage, or tube discharge.

- **Not AOM:** a non-bulging drum, or erythema / air-fluid level alone without bulging. Effusion without inflammation is OME<sup>43</sup> — do not give antibiotics.
- **Scope:** the **watchful-waiting** option applies only to healthy children  $\geq 6$  **months** (CPS 2016 / AAP 2013). **Infants < 6 months are excluded from observation**<sup>43</sup> — treat all with antibiotics. Excludes tympanostomy tubes, craniofacial anomalies, immunocompromise, or recurrent AOM.

## DECISION BY AGE & SEVERITY

### DECISION

How old is the child, and how severe?

#### < 6 MONTHS

**Antibiotics for all — no watchful waiting** (below the CPS/AAP observation range; the exam is less reliable and complications are more likely). In a **febrile infant  $\leq 90$  days**, do **not** attribute the fever to AOM alone — evaluate for invasive bacterial infection (see [Fever in the Well-Appearing Infant](#)).

**≥ 6 MO – TREAT NOW**

Bulging, erythematous TM **and** any of:  
highly febrile ( $\geq 39^{\circ}\text{C}$ ),  
moderately-to-severely ill, severe  
otalgia, symptoms  $\geq 48$  h, perforation  
with otorrhea, or complicated disease  
(mastoiditis, facial palsy).<sup>43</sup>

**≥ 6 MO – WATCHFUL WAITING (24–48 H)**

Otherwise-healthy child, mildly ill, alert,  
responding to antipyretics, low-grade  
fever ( $< 39^{\circ}\text{C}$ ), mild otalgia — or when  
the diagnosis is uncertain. Needs  
assured follow-up.

**◆ HOW TO WATCHFUL-WAIT (CHOOSE ONE)**

- Reassess within 24–48 h to document the course; **OR** have the caregiver return if no improvement or worsening by 48 h; **OR** give a “safety-net” prescription to fill only if not improved by 48 h.
- Always give analgesia advice; ensure ready access to care.

| Drug                       | Dose                                          | Route & notes                                                                                                                            |
|----------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Amoxicillin (first choice) | 40–45 mg/kg/dose BID                          | PO. Drug of choice — covers <i>S. pneumoniae</i> ; excellent middle-ear penetration. <sup>43</sup>                                       |
| Amoxicillin-clavulanate    | 40–45 mg/kg/dose BID (amoxicillin component)  | PO. If AOM + purulent conjunctivitis, amoxicillin in the last 30 days, or failure at 48–72 h. Specify the 14:1 suspension. <sup>43</sup> |
| Cefuroxime-axetil          | 15 mg/kg/dose BID (30 mg/kg/day; max 1 g/day) | PO, twice daily. Non-anaphylactic penicillin allergy (cefprozil is an alternative).                                                      |
| Ceftriaxone                | 50 mg/kg IM/IV once daily × 3 days            | If oral not tolerated or amoxicillin-clavulanate failed. Reserve.                                                                        |

#### ◆ DURATION – BY AGE

- < 2 years: 10 days. ≥ 2 years, uncomplicated: 5 days.<sup>43</sup>
- Also 10 days for a perforated TM or recurrent AOM.
- Symptoms should improve within 24 h and resolve in 2–3 days — reassess if they persist or worsen.

### ⚠ **ANALGESIA IS PART OF EVERY PLAN**

- Pain control (acetaminophen or ibuprofen) is essential whether or not antibiotics are given — it is often the child's main need.
- Persistent effusion after treatment does **not** warrant a change of antibiotic; a middle-ear effusion can linger for months.<sup>43</sup>

### ▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — a complication (mastoiditis, intracranial spread, facial palsy), a toxic child, or failure of second-line therapy.
- **Discharge** — uncomplicated disease — analgesia, with observation or antibiotics per age and severity, and safety-netting.
- **Refer** — ENT for suspected mastoiditis, recurrent or complicated disease, or a non-resolving effusion.

# GAS Pharyngitis

ASSESS

≥ 3 YR • IDSA 2025

Most sore throats are viral. Score first to decide who needs a test — then treat the confirmed case.

## ▶ WHEN TO TEST – SCORE FIRST

- Sore throat is usually viral, and clinician judgment alone is unreliable. GAS causes up to ~30% of childhood (and ~15% of adult) pharyngitis.<sup>44</sup>
- IDSA 2025 suggests a **clinical scoring system to decide who to test**. A score is most useful for spotting **low-probability** patients in whom testing is unlikely to help — prefer a score that needs no lab (Centor / McIsaac).<sup>44</sup>
- **Does not apply under 3 years** — GAS in this age may not show the typical features; test only for an outbreak, scarlet fever, or a household case.<sup>45</sup>



- Score criteria point to GAS: fever, tonsillar swelling or exudate, tender anterior cervical nodes, and **absence of cough**. Cough, coryza, or hoarseness point to a virus.

| Centor / Mclsaac –<br>one point each     | Point |
|------------------------------------------|-------|
| Tonsillar exudate or swelling            | +1    |
| Tender / swollen anterior cervical nodes | +1    |
| History of fever                         | +1    |
| No cough                                 | +1    |
| Age 3–14 years (Mclsaac)                 | +1    |

#### ◆ HOW TO TEST

- Read the score as a probability of GAS: **≤1 LOW** (~1–10%) → testing can be deferred; **2–3 INTERMEDIATE** (~11–35%) → test; **≥4 HIGH** (~50%) → test, and still confirm before treating.<sup>44</sup>
- Test with a rapid antigen test (RADT), a molecular test (NAAT), or throat culture.
- In children, confirm a **NEGATIVE** rapid antigen test with culture, or use a molecular test.<sup>45</sup>

- The score selects *who to test* — never a treatment threshold.
- Do **NOT** use ASO titers to diagnose acute pharyngitis.<sup>44</sup>
- Start treatment within 9 days of onset.

### ⚠ TEST EVEN IF THE SCORE IS LOW

Test regardless of a low score when the patient is high-risk: a **household GAS case** (shared indoor living or sleeping space), prior **rheumatic fever**, or features of complicated or systemic GAS — peritonsillar or retropharyngeal abscess, scarlet fever, or toxic shock.<sup>44</sup>

| Lab-confirmed GAS       | Drug & dose                                                              | Duration |
|-------------------------|--------------------------------------------------------------------------|----------|
| First choice            | Amoxicillin 50 mg/kg once daily (max 1 g), or penicillin V <sup>45</sup> | 10 days  |
| Poor adherence expected | Benzathine penicillin G IM – < 27 kg: 600,000 U; ≥ 27 kg: 1,200,000 U    | 1 dose   |

| Lab-confirmed GAS                                                     | Drug & dose                                                                                                                                                                                                                                                                                                                                                                        | Duration  |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Non-immediate (delayed / non-IgE) penicillin allergy                  | <p>Cephalexin 20 mg/kg/dose BID (max 500 mg/dose) ♦ if unavailable: cefuroxime-axetil 20 mg/kg/day ÷ BID (max 500 mg/day) PO</p>                                                                                                                                                                                                                                                   | 10 days   |
| Documented penicillin anaphylaxis / IgE-mediated (immediate) reaction | <p>Azithromycin 12 mg/kg on day 1 (max 500 mg) then 6 mg/kg (max 250 mg) once daily on days 2–5 (IDSA, as corrected 2014)<sup>45</sup> – the FDA-labeled 12 mg/kg once daily × 5 days (max 500 mg/day) is an accepted alternative<sup>229</sup>; or clarithromycin 7.5 mg/kg/dose BID (max 250 mg/dose) × 10 days; or clindamycin 7 mg/kg/dose TID (max 300 mg/dose) × 10 days</p> | 5–10 days |

### ⚠ WHY WE TREAT

- Antibiotics prevent suppurative complications and acute rheumatic fever (but not post-streptococcal glomerulonephritis). **All GAS is penicillin-susceptible**, but macrolide/clindamycin resistance reaches ~20% and varies locally — reserve those for true penicillin anaphylaxis and confirm local susceptibility.<sup>45</sup>
- Acetaminophen or ibuprofen for pain and fever; corticosteroids are not recommended as analgesic adjuncts.<sup>44</sup>

### ▶ ADMIT · DISCHARGE · REFER

- **Admit** — rarely — complications (peritonsillar abscess, dehydration, systemic illness).
- **Discharge** — most, with antibiotics and safety-net advice.
- **Refer** — ENT for abscess or recurrent tonsillitis.

# Urinary Tract Infection

HOURS

> 2 MONTHS

A common occult source of fever in infants — get a reliable sample before antibiotics.

## ◀ WHEN TO SUSPECT & TEST

- **Preverbal child** with unexplained fever ( $> 39^{\circ}\text{C}$ , no source) — especially **fever > 48 h** without another focus. A clear viral source (coryza, cough, diarrhea, rash) makes UTI unlikely.<sup>46</sup>
- **Verbal child ( $\geq 3$  yr):** dysuria, frequency, hematuria, abdominal/back pain, or new daytime incontinence. Beware bubble-bath / hygiene vulvovaginitis — sterile cultures, often mis-treated as UTI.
- **Usual pathogens:** *E. coli* (most), then *Klebsiella*, *Enterobacter*, *Citrobacter*, *Serratia*; *S. saprophyticus* in adolescent girls.<sup>46</sup>
- Do not over-diagnose: low colony counts, mixed growth, or **no pyuria** usually mean contamination or asymptomatic bacteriuria.

## SAMPLING – BEFORE ANTIBIOTICS

- **Toilet-trained:** clean-catch midstream. **Not toilet-trained:** catheter or suprapubic aspirate for culture.
- A bag specimen may screen the urinalysis but its culture is unreliable (contamination up to 63%) — never use it to diagnose.<sup>46</sup>
- A single antibiotic dose sterilizes urine; always collect first.<sup>46</sup>

◆ **INTERPRETING THE URINALYSIS** UTI is unlikely if the urinalysis is completely normal.

**LEUKOCYTE ESTERASE OR NITRITE POSITIVE** ≈ 93% sensitive; **NITRITE ALONE** is highly specific (~98%) but often falsely negative (frequent voiding, Gram-positive / non-nitrate organisms)<sup>46</sup>. **PYURIA** (> 5 WBC/HPF, or ≥ 10/μL) supports infection.

## DIAGNOSTIC COLONY-COUNT THRESHOLDS

| Collection method       | Minimum count (CFU/mL) | Note                                         |
|-------------------------|------------------------|----------------------------------------------|
| Clean catch / midstream | ≥ 10 <sup>5</sup>      | Mixed growth usually = contamination.        |
| In-and-out catheter     | ≥ 5 × 10 <sup>4</sup>  | Indwelling-catheter specimens less reliable. |

| Collection method   | Minimum count (CFU/mL) | Note                                |
|---------------------|------------------------|-------------------------------------|
| Suprapubic aspirate | Any growth             | Most reliable method. <sup>46</sup> |

#### ADMIT · DISCHARGE · REFER

- **Admit** — young infants, ill/septic, unable to tolerate oral, or complicated/obstructive UTI.
- **Discharge** — well child on oral antibiotics with follow-up.
- **Refer** — nephrology/urology for recurrent, atypical, or obstructive UTI, or abnormal imaging.

## UTI – Treatment & Follow-up HOURS

### LOCAL SUSCEPTIBILITY

Oral is fine for most non-toxic children — pick the empiric agent from your antibiogram; plan for ESBL.

#### DECISION

**Cystitis or febrile UTI / pyelonephritis?**

## CYSTITIS • AFEBRILE LOWER TRACT

Dysuria/frequency, no fever (mainly post-pubertal girls). **2–4 days** of an oral agent chosen by local *E. coli* susceptibility. No imaging needed.<sup>48</sup>

## FEBRILE UTI / PYELONEPHRITIS

Treat **7–10 days**. **Oral is as effective as IV** in non-toxic children likely to tolerate every dose.<sup>46</sup> Use IV initially if toxic, < 2–3 months, vomiting, or unable to tolerate oral — then step down.

### ◆ ESBL RISK – PLAN FOR IT

- ESBL-producing Enterobacterales are increasingly isolated even without prior antibiotics — often resistant to cephalosporins and many oral agents.
- Where local ESBL rates are high or the child is high-risk: **AMIKACIN (IV)** is the most reliable aminoglycoside — active against the large majority of pediatric ESBL isolates, whereas gentamicin is variable (roughly half)<sup>93,385,386</sup>. **TMP-SMX (PO)** is a useful oral step-down once susceptibility is confirmed, though ESBL isolates are often co-resistant.



- Fluoroquinolones are not routine in prepubertal children but may be used if the isolate is resistant to other oral options.

## ANTIMICROBIAL OPTIONS – IF THE ISOLATE IS SUSCEPTIBLE

| Drug                       | Dose                             | Notes                                                                                                        |
|----------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Amikacin (IV)</b>       | 15 mg/kg once daily              | <b>ESBL-active</b> empiric choice; monitor levels <sup>93,385,386</sup> & renal function if > 48 h.          |
| <b>Gentamicin (IV/IM)</b>  | 5–7.5 mg/kg once daily           | ESBL activity variable (~half of isolates) — confirm susceptibility. Monitor levels / renal function > 48 h. |
| <b>Ceftriaxone (IV/IM)</b> | 50–75 mg/kg once daily (max 2 g) | Less nephrotoxic but broader; <b>not</b> reliable if ESBL.                                                   |
| <b>Ampicillin (IV)</b>     | 50 mg/kg/dose q6h (max 2 g/dose) | ± aminoglycoside; many <i>E. coli</i> now resistant. (Lexicomp)                                              |
| <b>TMP-SMX (PO)</b>        | 4–5 mg/kg/dose (TMP) BID         | <b>Preferred oral step-down once susceptible</b> — covers many ESBL isolates if tested S. <sup>46</sup>      |
| <b>Cefixime (PO)</b>       | 8 mg/kg/dose once daily          | Common first oral choice pending susceptibilities (if low ESBL).                                             |

| Drug                         | Dose                                                     | Notes                                                                                                                |
|------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Cefdinir (PO)                | 7 mg/kg/dose BID or 14 mg/kg once daily (max 600 mg/day) | Palatable oral step-down; <b>not always on the MOH formulary</b> . Like cefixime, <b>not reliable against ESBL</b> . |
| Cephalexin (PO)              | 12.5 mg/kg/dose QID                                      | Oral cephalosporin option.                                                                                           |
| Amoxicillin-clavulanate (PO) | 13 mg/kg/dose (7:1) TID                                  | If susceptible; amoxicillin alone often resisted.                                                                    |
| Ciprofloxacin (PO) *         | 15 mg/kg/dose BID                                        | * Not routine in prepubertal children — reserve for resistant isolates.                                              |

### ◀ **COMPLICATED UTI – ESCALATE & IMAGE EARLY**

- Hemodynamic instability, **rising creatinine**, abdominal/bladder mass, poor urine flow, or **no improvement by 24 h / fever not settling by 48 h** on appropriate antibiotics.
- Use **IV until clearly improving**; start with a **renal & bladder ultrasound (RBUS)** to look for obstruction or abscess.<sup>46</sup>

### ⚠️ IMAGING & FOLLOW-UP

- **RBUS** after a first febrile UTI in a child **< 2 yr** (during, or within ~2 weeks of, the illness)<sup>46</sup>.
- **VCUG is not routine** — obtain it only if RBUS shows **marked hydronephrosis, a dilated ureter, a duplex system / ureterocele, or obstruction**, or after a **second febrile UTI**. A normal RBUS does not exclude VUR (**VCUG** grades reflux; **DMSA** for scarring)<sup>48</sup>.
- Routine antibiotic **prophylaxis is not recommended** while awaiting RBUS.<sup>46</sup>
- Repeat urinalysis / culture to confirm cure is **not recommended** unless symptoms persist.
- Counsel families to test urine early in any future unexplained fever.
- Recurrent afebrile UTI is often driven by **bladder–bowel dysfunction** — assess and treat constipation (a major, modifiable risk factor).<sup>46</sup>

## Skin & Soft Tissue Infections

ASSESS

Match the antibiotic to the layer and the pathogen — most are oral, short-course, and cephalexin-first.

### ▀ TYPES – BY DEPTH & MORPHOLOGY

- **Impetigo** (epidermis): non-bullous (honey-crusted) or bullous; highly contagious. Mostly group A streptococcus (GAS) / *S. aureus*.
- **Erysipelas / cellulitis** (dermis → subcutis): spreading erythema, warmth, tenderness. Non-purulent cellulitis is more often **GAS**; purulent points to *S. aureus*.<sup>187</sup>
- **Folliculitis / furuncle / carbuncle / abscess**: discrete pus collections — *S. aureus* (methicillin-susceptible, MSSA, or methicillin-resistant, MRSA). Source control is the priority.
- **Necrotizing infection** (fascia/muscle): pain out of proportion, rapid spread, necrosis, crepitus<sup>187</sup> — surgical emergency. GAS ± polymicrobial.

### DECISION

**How severe, and is there pus?**

#### MILD / NON-PURULENT

Impetigo or small localized lesion, systemically well — often resolves with skin/wound care alone. Limited: topical

**mupirocin 2% TID × 5 days.**

Widespread/multiple lesions or systemic features: oral **cephalexin**.

#### PURULENT – ABSCESS / FURUNCLE

**Incision & drainage is the treatment.**

For a small, drained abscess in a healthy host without surrounding cellulitis, antibiotics are usually **not** needed; stop them once source control is achieved.<sup>187</sup>

#### SEVERE / DEEP / NECROTIZING

Extensive or rapidly progressive, systemic toxicity, necrosis or crepitus

→ **admit, IV antibiotics, urgent surgery.** Add an anti-toxin agent (clindamycin) and consult infectious diseases.

**Oral cephalexin is the antimicrobial of choice when antibiotics are warranted.**

## BY PATHOGEN / SETTING

| Target                               | Drug & dose                                                                      | Notes                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>MSSA / GAS — oral, first-line</b> | Cephalexin 25 mg/kg/dose (max 1 g) PO QID; high-dose TID for moderate cellulitis | Drug of choice. Covers MSSA <b>and</b> GAS. High-dose enables outpatient moderate cellulitis. <sup>187</sup> |
| <b>MSSA / GAS — IV (admitted)</b>    | Cefazolin 33 mg/kg/dose IV q8h (max 2 g/dose)                                    | IV first-line for non-toxic inpatients. Step down to oral cephalexin on improvement.                         |
| <b>GAS — streptococcal</b>           | Amoxicillin 25 mg/kg/dose PO BID; penicillin G IV if severe                      | For confirmed/likely pure GAS (erysipelas, non-purulent cellulitis).                                         |
| <b>MRSA — oral, first-line</b>       | Trimethoprim-sulfamethoxazole 4–6 mg/kg/dose (trimethoprim component) PO BID     | Preferred oral MRSA agent <sup>187</sup> — also covers GAS. Use when colonized or MRSA strongly suspected.   |
| <b>MRSA — alternative</b>            | Clindamycin 10 mg/kg/dose PO/IV q8h                                              | Only where local clindamycin susceptibility is high. Anti-toxin role in severe disease.                      |

| Target                   | Drug & dose                           | Notes                                                                                                                                                                                                                                                                                                  |
|--------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MRSA —<br>IV<br>(severe) | Vancomycin 15<br>mg/kg/dose IV<br>q6h | Add when MRSA suspected in toxic/deep infection; stop once MRSA excluded. <i>K. kingae</i> is resistant to vancomycin and clindamycin and shows reduced susceptibility (high MICs) to cloxacillin — so cefazolin/cephalosporins, which cover it, are preferred empirically in under-4s. <sup>187</sup> |

#### ◆ DURATION & STEP-DOWN

- **5-7 DAYS IS USUALLY ENOUGH**, even for many severe SSTIs (rarely > 7 days).<sup>187</sup>
- Failure to improve → re-evaluate diagnosis and source control, not simply broaden.
- Avoid routine combination therapy (e.g., cephalexin + TMP-SMX) and broad outpatient IV (ceftriaxone).
- If a child improves on a non-MRSA agent but is later found MRSA-colonized/positive, **DO NOT SWITCH** — clinical response governs.

### ⚠ PREVENTING RECURRENCE

- Protect the skin barrier: optimize **eczema** control, diligent wound care, cover open lesions, treat scabies/lice, and encourage varicella vaccination.
- Reserve **decolonization** (5-day nasal mupirocin + chlorhexidine washes, or dilute bleach baths) for recurrent moderate–severe SSTIs or household clusters — success is variable and re-colonization common.<sup>187</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — systemic illness, rapid spread, an abscess needing drainage, immunocompromise, or failed oral therapy; suspected necrotizing infection → emergency.
- **Discharge** — localized, well child on oral antibiotics with review.
- **Refer** — surgery for drainage or necrotizing infection.

## Cervical Lymphadenopathy

ASSESS



## COMMON

Very common and usually reactive — your job is to separate the benign node from the few that need antibiotics, imaging, or a malignancy work-up.

### ▀ WHAT COUNTS AS ABNORMAL

- A small, mobile, palpable node is **normal** in most healthy children. Treat a **cervical node > 1 cm** as enlarged (inguinal > 1.5 cm; epitrochlear / post-auricular > 0.5 cm).<sup>200</sup>
- A **supraclavicular** node is **never normal** — it is the single highest-risk site for malignancy (up to ~75%). Image and investigate every one.<sup>199</sup>
- Sort by **tempo & laterality** — it points to the cause.

## PATTERN → LIKELY CAUSE

| Pattern                                                  | Think                                                                                                              | Action                                                                                                                                                                   |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute, bilateral, tender                                 | Reactive viral URTI (commonest); EBV / CMV if prolonged with splenomegaly                                          | Supportive, no antibiotics, safety-net. EBV serology if a glandular-fever picture.                                                                                       |
| Acute, unilateral, tender, warm ± fluctuant              | Bacterial lymphadenitis – <i>S. aureus</i> (incl. MRSA) & group A strep                                            | Empiric oral antibiotics 10 d; ultrasound if fluctuant → drain an abscess. Review at 48 h.                                                                               |
| Subacute (2–6 wks) / chronic (> 6 wks), often non-tender | Nontuberculous (atypical) mycobacteria (violaceous, "cold", young child), cat-scratch (Bartonella), TB, toxoplasma | Don't just re-course antibiotics — investigate. NTM → complete surgical excision; cat-scratch → usually self-limited (azithromycin if bulky/symptomatic). <sup>200</sup> |

### ⚠ CAN'T-MISS – RED FLAGS FOR MALIGNANCY (LYMPHOMA / LEUKEMIA / METASTASIS)

- **Supraclavicular** node; **hard**, **matted**, **fixed**, or **painless / rubbery**; **> 2 cm and enlarging**; persisting or progressive **beyond 4–6 weeks**; **generalized** nodes (> 2 non-contiguous regions) or **hepatosplenomegaly**; **B symptoms** — unexplained fever >

38°C, drenching night sweats, weight loss > 10%; a **mediastinal mass** on chest X-ray.

- Any one → urgent bloods + imaging and refer. **Strongest independent predictors (2024 cohort):** a larger node — each extra centimeter roughly **doubles** the odds, a node > **3 cm** ≈ **12% malignant** — and **older age** (> 12 yr ≈ 20% vs ≤ 2.5 yr < 1%); firm consistency, weight loss, and an abnormal chest X-ray also raise the odds.<sup>199</sup>

#### ▶ **WORK-UP (WHEN IT ISN'T SIMPLE REACTIVE)**

- **Bloods:** CBC + differential + **blood film** (blasts, cytopenias), **ESR / CRP, LDH**; EBV / CMV serology; a **TB test (IGRA or tuberculin)**; Bartonella / toxoplasma serology if exposed.
- **Imaging: ultrasound first-line** (size, abscess, architecture); chest X-ray if any red flag (mediastinal mass).
- **Biopsy** (excisional preferred) for a red-flag or supraclavicular node, **failure to regress by 4–6 weeks**, or progression despite treatment.<sup>197</sup>

## EMPIRIC ANTIBIOTICS – ACUTE BACTERIAL LYMPHADENITIS (10 DAYS)

| Setting                        | Oral                                                                               | Notes                                                                                           |
|--------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Cover <i>S. aureus</i> + GAS   | Cephalexin<br>25–50<br>mg/kg/day ÷<br>3–4 <b>OR</b><br>amoxicillin-<br>clavulanate | Review at 48 h; most settle within days. Add source control if an abscess forms. <sup>200</sup> |
| MRSA risk / penicillin allergy | Clindamycin<br>20–40<br>mg/kg/day ÷<br>3–4                                         | Also covers GAS; use per local MRSA rates.                                                      |
| Fluctuant / abscess            | Incision & drainage + culture                                                      | Ultrasound-guided if deep; send pus for routine <b>and</b> mycobacterial culture.               |

### ✦ DON'T FORGET

- A cervical node **> 1.5 cm** is a clinical criterion of Kawasaki disease — reassess if fever has lasted  $\geq 5$  days.<sup>35</sup>
- Tender anterior nodes with exudative pharyngitis suggest GAS pharyngitis.
- A firm supraclavicular or matted node in a child with pallor or bruising → treat as possible malignancy and refer **same day**.

## ✦ YOUR ROLE – THE GENERAL PEDIATRICIAN

- **Most nodes are yours to manage.** Do the first-line assessment and reassurance; classify by duration (**acute  $\leq 2$  wks** · **subacute 2–6 wks** · **chronic  $> 6$  wks**) and actively look for a red flag.
- Treat acute bacterial lymphadenitis empirically and **review at 48 h. Safety-net every child and set a definite review point** — a node that has not begun to shrink by **~4 weeks**, or is growing, needs work-up.
- You can start the first-line bloods and request the ultrasound yourself. **Refer early rather than re-course antibiotics** for a persistent or red-flag node: **same-day to hematology-oncology** for a hard/matted/supraclavicular node or systemic features; **surgery/ENT** for excision biopsy (persisting  $> 6$  weeks or progressive); **infectious diseases** for suspected TB or atypical mycobacterial disease.

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — systemically unwell, a large / fluctuant node needing drainage, airway or deep-space concern, or a strong malignancy suspicion.
- **Discharge** — the well child with a small, mobile, reactive node → safety-net and review in 2–4 weeks.
- **Refer** — surgery / ENT for excision (NTM, abscess, biopsy); hematology-oncology for a red-flag node; infectious diseases for TB or atypical infection.

### SOURCE

Pediatric cervical lymphadenopathy: an evidence-based approach for GPs (*Br J Gen Pract* 2026;76(764):137); Refining indications for pediatric cervical lymph-node biopsy (*Eur J Pediatr* 2026); Risk factors for malignancy in subacute/chronic focal craniocervical lymphadenopathy (*Front Pediatr* 2024); AAFP — Management of Cervical Lymphadenitis in Children (*Am Fam Physician* 2008); AAP Red Book 2024 (nontuberculous mycobacteria, cat-scratch disease).

# Fever – Assessment & Antipyretics

ASSESS

NICE NG143

Fever  $\geq 38^{\circ}\text{C}$ . Treat the child's discomfort and risk — not the number.

## ▲ RISK-ASSESS BY TRAFFIC LIGHT (AND BY AGE)

- Any infant **under 3 months** with fever is high-risk → use the **Fever in the Young Infant** pathway.
- **Red (high-risk):**  
pale/mottled/ashen/blue, no response to social cues, unrousable, weak/high-pitched cry, grunting, respiratory rate  $> 60$ , reduced skin turgor, bulging fontanelle, neck stiffness, non-blanching rash, focal seizures/neurology.<sup>237</sup>
- **Amber (intermediate):** pallor, decreased activity, poor feeding, dry mucous membranes, reduced urine, nasal flaring, tachycardia, fever  $\geq 5$  days, rigors, non-weight-bearing/limb swelling.<sup>237</sup>
- **Green (low-risk):** normal color, responsive, content, moist membranes — manage at home with safety-netting.

## ANTIPYRETIC DOSING

| Drug · route       | Dose                             | Max · notes                                                                                             |
|--------------------|----------------------------------|---------------------------------------------------------------------------------------------------------|
| Acetaminophen — PO | 10–15 mg/kg/dose every 4–6 h PRN | ≤ 1 g/dose, ≤ 5 doses/24 h; max 75 mg/kg/day (not > 4 g/day). Onset 30 <sup>238</sup> –60 min.          |
| Acetaminophen — IV | 15 mg/kg/dose every 6 h          | < 50 kg: max 15 mg/kg (750 mg/dose); ≥ 50 kg: 1 g/dose. ≤ 75 mg/kg/day.                                 |
| Ibuprofen — PO     | 5–10 mg/kg/dose every 6–8 h      | <b>Age ≥ 6 months.</b> Max 400 mg/dose, 40 mg/kg/day. Give with food + ensure hydration. <sup>238</sup> |
| Ibuprofen — IV     | 10 mg/kg/dose every 6–8 h        | <b>Age ≥ 6 months.</b> Max 400 mg/dose, 40 mg/kg/day.                                                   |

### ◆ AVOID

- Acetaminophen — severe liver failure.
- Ibuprofen — dehydration, renal impairment, active GI bleeding / ulcer disease, or any poorly hydrated child.

**⚠ USE ANTIPYRETICS FOR COMFORT, NOT THE**



## THERMOMETER

- Give only while the child is uncomfortable.<sup>238</sup>
- Combining or alternating the two clears fever faster and gives more fever-free time (PITCH trial), but does **not** improve comfort — so for comfort one agent is usually enough.<sup>162</sup>
- Switch agents if discomfort is not relieved; alternate only if discomfort recurs before the next dose; counsel carefully to avoid dosing errors.
- Tepid sponging is not recommended; don't over- or under-wrap. **Antipyretics do not prevent febrile seizures.**<sup>238</sup>  
Encourage fluids; safety-net for red/amber signs.

## The Rash – Approach & Red Flags

### MINUTES

Two questions first: does the child look unwell, and does the rash blanch?

### ▲ STEP 1 – DOES THE CHILD LOOK UNWELL?

### (BEFORE FOCUSING ON THE RASH)

- Rapid ABCDE-style screen: work of breathing; color and perfusion (capillary refill > 2–3 s, mottled, cold peripheries); heart rate and respiratory rate for age; mental status (irritable, lethargic); fever; neck stiffness, photophobia, bulging fontanelle in infants.
- Any red flag → urgent assessment + empiric IV antibiotics (see **Septic Shock**), **in parallel** with the rash exam. Looks well → Step 2.
- **A blanching rash does not rule out sepsis** — meningococcal disease often begins as a blanching maculopapular rash before turning petechial or purpuric, and the classic non-blanching rash can be a late sign.<sup>119</sup> Escalate an unwell child whatever the rash looks like.
- **Darkly pigmented skin:** erythema, petechiae and purpura may appear dark brown or violaceous and be masked — check palms, soles, conjunctivae and mucosa, and lower the threshold to escalate.<sup>119</sup>

## STEP 2 – DOES THE RASH BLANCH? (PRESS A CLEAR GLASS AGAINST IT)

### DECISION

Does the rash blanch under glass?

#### NON-BLANCHING

Petechiae (< 2 mm) or purpura (> 2 mm) — **purpura carries higher risk.**<sup>119</sup>

Unwell or febrile → treat as invasive meningococcal disease / sepsis until excluded. A **well** child still needs a diagnosis and follow-up (see below) — do not dismiss just because they look well.

#### BLANCHING

Usually benign — viral exanthem, urticaria, or eczema. Treat symptomatically; safety-net with clear advice to "come back if the child becomes unwell, even if the rash looks the same or fades."<sup>237</sup>

## NON-BLANCHING – SORT THE CAUSE

| Cause                                     | Key features                                    |
|-------------------------------------------|-------------------------------------------------|
| <b>Meningococcal / invasive bacterial</b> | Ill, febrile — emergency (identified at Step 1) |

|                                              |                                                                                                                                                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IgA vasculitis<br/>(Henoch–Schönlein)</b> | Palpable purpura on buttocks and legs in a <i>well</i> child; abdominal pain, arthralgia/arthritis; check urine for blood and protein (renal involvement) |
| <b>Immune thrombocytopenia (ITP)</b>         | Well child, isolated low platelets, bruising; often post-viral; watch for mucosal bleeding (epistaxis, gums)                                              |
| <b>Mechanical purpura</b>                    | Forceful cough, vomiting or breath-holding; classically above the nipple line (head and neck, SVC territory) — benign                                     |
| <b>Leukemia</b>                              | Pallor, hepatosplenomegaly, fatigue, bone pain, unexplained bruising                                                                                      |

## BLANCHING – SORT THE CAUSE

| Cause                  | Key features                                                                                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Viral exanthems</b> | Roseola, measles, rubella, slapped-cheek (parvovirus B19); hand-foot-and-mouth and varicella are vesicular, not purely maculopapular |
| <b>Scarlet fever</b>   | Sandpaper rash + strawberry tongue → penicillin                                                                                      |

## Urticaria

Transient wheals → antihistamine;  
airway or systemic features → see  
[Anaphylaxis](#)

## Kawasaki disease

Fever  $\geq 5$  days + rash +  
non-purulent conjunctivitis +  
mucositis/strawberry tongue +  
extremity changes + cervical node  
— easily mistaken early for scarlet  
fever or a viral exanthem (see  
[Kawasaki Disease](#))

### ⚠ ANY UNWELL CHILD IS THE EMERGENCY

- Escalate immediately — with or without a rash, and whatever the rash looks like.
- This alone is sufficient.

### ▶ ESCALATE URGENTLY – REGARDLESS OF BLANCHING – ALSO IF

- Non-blanching rash with fever, ill appearance, or rapid spread.
- **Retiform purpura** (branching, angulated, stellate) rather than round petechiae/purpura — suggests vascular occlusion (DIC, severe sepsis, vasculitis) and is higher-acuity than ordinary purpura.<sup>119</sup>

- **Mucosal involvement** (eyes, mouth, genitals) with blistering or skin peeling → Stevens–Johnson syndrome / TEN or toxic shock syndrome. Superficial peeling with **perioral crusting and radial fissures but mucous membranes spared**, especially in young children → **staphylococcal scalded skin syndrome (SSSS)** — mucosal sparing is the classic clue that separates SSSS from SJS/TEN. The **Nikolsky sign** (gentle shearing pressure separates the epidermis) is positive in SJS/TEN, SSSS, and pemphigus.
- **Vesicles in a neonate** → herpes simplex. Skin vesicles are **absent in roughly 40–50% of neonatal HSV with CNS or disseminated disease** — keep the threshold low for a full sepsis + HSV workup (and empiric acyclovir) in any unwell neonate, even with no rash.<sup>120</sup>
- Fever  $\geq$  5 days with rash plus any Kawasaki features.<sup>35</sup>

#### ◆ BEDSIDE TIPS

**Faded or changed rash?** Ask whether a parent photographed it — useful for evanescent eruptions like urticaria.

**Stroke test:** a wheal along a tongue-blade stroke = dermatographism (common, benign); a wheal raised specifically over a pigmented lesion = **Darier's sign** (mastocytosis — worth recognizing in recurrent unexplained wheals). **Herpes zoster** in a dermatomal band is usually benign and immunocompetent in children, especially after early-life varicella or vaccination; consider underlying immunodeficiency only if it is multidermatomal, sacral, recurrent, prolonged or treatment-refractory, or has invasive features.

## Preseptal & Orbital Cellulitis

HOURS

CPS 2026

The orbital septum is the firewall — eye signs separate sight-threatening orbital disease from benign preseptal swelling.

### ► THE DISTINCTION THAT DRIVES EVERYTHING

- The **orbital septum** is the anatomic firewall. **Preseptal (periorbital)** = infection anterior to it — usually from skin trauma or sinusitis; mostly benign. **Orbital** = posterior to it — a

complication of **ethmoid sinusitis**  
eroding the lamina papyracea;  
threatens vision and brain.<sup>89</sup>

- The distinction drives etiology, imaging, antibiotics, and disposition.<sup>89</sup>

## DIFFERENTIATE AT THE BEDSIDE

| Clinical feature                      | Preseptal      | Orbital               |
|---------------------------------------|----------------|-----------------------|
| Eyelid swelling & erythema            | Present        | Present               |
| Pain <b>with</b> eye movement         | Absent         | Present               |
| Extraocular movements                 | Normal         | Impaired / restricted |
| Proptosis                             | Absent         | Often present         |
| Chemosis / diplopia / ↓ acuity / RAPD | Absent         | May be present        |
| Fever / toxic appearance              | May be present | Usually present       |



## ◁ ETIOLOGY & MICROBIOLOGY

- **Preseptal — traumatic:** skin flora after a bite/scratch → *S. pyogenes*, *S. aureus*.
- **Preseptal — atraumatic:** a complication of bacterial rhinosinusitis.
- **Orbital:** mirrors acute sinusitis — *Streptococcus (anginosus, pyogenes, pneumoniae* groups); *S. aureus* and oral anaerobes in chronic sinusitis / older children; *H. influenzae* if under-immunized.<sup>89</sup>
- **MRSA:** generally low but rising — risk factors do not reliably predict it<sup>89</sup>; a **negative nasal MRSA swab reliably excludes** it and guides the oral step-down.

## ◆ IMAGING – CONTRAST CT ORBITS + SINUSES

- Well children with uncomplicated **PRESEPTAL** disease need **NO IMAGING**.<sup>89</sup>
- Contrast CT is first-line; MRI preferred if intracranial extension or cavernous sinus thrombosis is suspected.
- Its job: find who needs surgery or has a complication.

- Image **IF** toxic appearance, optic-nerve signs (abnormal vision, RAPD), intracranial signs (focal deficit, seizure), prominent forehead swelling, or high-risk labs (CRP > 120 mg/L, WBC > 20,000/ $\mu$ L)<sup>89</sup>.
- Otherwise reasonable to **DELAY CT 24-36 H** to judge response when ENT / ophthalmology are available.<sup>89</sup>

#### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — orbital signs, systemic illness, age < 1, and all true orbital cellulitis → IV antibiotics.
- **Discharge** — mild preseptal in a well child on oral antibiotics with 24 h review.
- **Refer** — ophthalmology and ENT for orbital involvement; surgery for an abscess.

## Preseptal & Orbital – Management

HOURS

RISK-BASED

Preseptal is mostly oral outpatient; orbital is admitted, IV, and shared with ENT + ophthalmology.

### DECISION

#### Preseptal or orbital?

#### PRESEPTAL – MOSTLY OUTPATIENT

Well, immunized child, eyelid erythema/edema with **normal painless eye movements, normal vision, no proptosis** → oral antibiotics + review. Admit the very young, severe, or not-improving.

#### ORBITAL – ADMIT, EMERGENCY

Pain/restriction on eye movement, **proptosis, chemosis, diplopia, reduced acuity, or RAPD** → admit: CBC/CRP, blood cultures, IV antibiotics, **same-day ENT + ophthalmology**, CT per criteria. Do not swab the conjunctiva (contamination).

## WHAT TO SEND

- Uncomplicated preseptal, well child: no work-up.
- Severe / uncertain / young child:  
**CBC + CRP**. Severe disease: **blood cultures before antibiotics**. Drained pus / abscess → culture ± PCR.  
**MRSA nasal swab** where available — a negative result reliably excludes MRSA and guides the oral step-down.

## EMPIRIC ANTIMICROBIAL THERAPY

| Scenario                                 | First-line                    | Dose (max) · MRSA / notes                                                                                                                                                                |
|------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Traumatic preseptal (bite / broken skin) | PO cephalexin<br>IV cefazolin | Cephalexin 25 mg/kg/dose q6h (100 mg/kg/day, max 4 g/day); cefazolin 33 mg/kg/dose q8h (100 mg/kg/day). MRSA suspected → TMP-SMX 4–6 mg/kg/dose (TMP) q12h or clindamycin. <sup>89</sup> |
| Atraumatic preseptal (from sinusitis)    | PO cephalexin<br>IV cefazolin | Doses as above.<br><b>Escalate to / add amoxicillin-clavulanate</b> (co-amoxiclav 30 mg/kg [amoxicillin] q8h) if no response by 24–48 h or the child is under-immunized. <sup>89</sup>   |

| Scenario                                    | First-line                                                  | Dose (max) · MRSA / notes                                                                                                                                                                                                              |
|---------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Orbital cellulitis (admitted)               | IV ceftriaxone + vancomycin                                 | Ceftriaxone 50 mg/kg q12h (max 4 g/day) + vancomycin 15 mg/kg q6h (target levels).<br>Alternative: ceftriaxone + clindamycin (adds anaerobe/MSSA cover) or amoxicillin-clavulanate. PO step-down by culture + MRSA swab. <sup>89</sup> |
| Severe · toxic / intracranial / optic-nerve | IV ceftriaxone (or cefotaxime) + vancomycin + metronidazole | Ceftriaxone 50 mg/kg q12h + vancomycin 15 mg/kg q6h + metronidazole 10 mg/kg q8h.<br>CNS-penetrating triple cover; consult ID; assess for surgery.                                                                                     |

#### ◆ WHY THESE CHOICES – ADAPTED TO LOCAL EPIDEMIOLOGY

- In Kuwait, near-universal Hib and pneumococcal conjugate vaccination reduces the *H. influenzae* / *S. pneumoniae* contribution, so a first-generation cephalosporin (cephalexin/cefazolin — strong streptococcal + MSSA cover) is a reasonable narrow empiric choice for skin- and

sinus-origin preseptal disease; escalate to amoxicillin-clavulanate on non-response or under-immunization.

- For orbital disease, **VANCOMYCIN IS INCLUDED EMPIRICALLY** given community-associated MRSA prevalence higher than North America; ceftriaxone covers sinusitis streptococci with good tissue/CNS penetration; metronidazole covers anaerobes / intracranial spread.<sup>89</sup>
- Always tailor to immunization status, prior antibiotics, and the **LOCAL ANTIBIOGRAM**.

| Condition                                                   | Duration   |
|-------------------------------------------------------------|------------|
| Preseptal cellulitis                                        | 5–7 days   |
| Orbital cellulitis (uncomplicated) <sup>89</sup>            | 10–14 days |
| Orbital + subperiosteal / orbital abscess (no intracranial) | 2–3 weeks  |

#### ✦ SURGERY, ADJUNCTS & STEP-DOWN

- **Surgery:** subperiosteal or orbital abscess → prompt ENT + ophthalmology for possible drainage (most small

subperiosteal collections settle medically)<sup>89</sup>; intracranial complications → neurosurgery.

- **IV → oral** once clearly improving, narrowed by cultures / MRSA swab.
- **Adjuncts:** intranasal saline/steroids only with subspecialty input; systemic corticosteroids are **not** routinely recommended.<sup>91</sup> Escalate to ID for systemic illness, non-response, large abscess, or intracranial extension.

#### ⚠ **SIGHT & BRAIN AT RISK – RED FLAGS**

- Falling visual acuity, a relative afferent pupillary defect, severe or bilateral proptosis, ophthalmoplegia, cranial-nerve palsies, or altered consciousness → cavernous sinus thrombosis or intracranial spread.
- Image urgently and escalate immediately.
- Bilateral involvement or rapid progression despite IV therapy is an emergency.

◆ **DIFFICULT IV ACCESS?** Reassess for an early IV→oral switch at every line failure rather

than re-sticking — see [IV → Oral Switch & the DIVA Patient](#) (Quick Reference).

## BRUE (Brief Resolved Unexplained Event)

ASSESS

< 1 YR • AAP 2016

A frightening but now-resolved episode in a well-appearing infant — a diagnosis of exclusion.

+ Calculator: [Corrected age](#)

### DEFINE

- An infant < 1 year, when an observer reports a **sudden, brief, now-resolved** episode of one or more of: cyanosis or pallor; absent, decreased, or irregular breathing; a marked change in tone (hyper- or hypotonia); or altered responsiveness.<sup>32</sup>
- Diagnose a BRUE **only when no explanation is found** after appropriate history and examination.<sup>32</sup>



### ▶ LOWER-RISK – ALL OF

- Age > 60 days; gestation  $\geq$  32 wk and corrected (postmenstrual) age<sup>32</sup>  $\geq$  45 wk.
- First event (no prior BRUE, not in clusters); duration < 1 min; no CPR by a trained provider; no concerning history or examination findings.<sup>32</sup>

### ▶ LOWER-RISK – DO

- **Should** — educate caregivers about BRUEs and use shared decision-making to guide evaluation, disposition, and follow-up; offer resources for caregiver CPR training.
- **May** — obtain pertussis testing and a 12-lead ECG; briefly monitor with continuous pulse oximetry and serial observations.<sup>32</sup>

### × LOWER-RISK – DO NOT / NEED NOT

**Should not:** blood counts or cultures, CSF, electrolytes/renal panel/calcium, ammonia, blood gases, or metabolic screen (urine organic acids, plasma amino acids, acylcarnitine); chest X-ray, echocardiogram, EEG, or studies for reflux; home cardiorespiratory monitoring; acid-suppression or

antiepileptic drugs. **Need not:** viral respiratory testing, urinalysis, glucose, bicarbonate, lactate, anemia work-up, or neuroimaging; admission **solely** for cardiorespiratory monitoring.<sup>32</sup>

**⚠ CONCERNING FEATURES → NOT LOWER-RISK, EVALUATE**

- Toxic appearance, lethargy, vomiting, or respiratory distress at assessment; significant compromise / CPR during the event; recurrent or prior events; family history of sudden cardiac death, long-QT syndrome, unexplained drowning or motor-vehicle death, SIDS, or sibling BRUE; history suggesting seizure; paroxysmal cough or pertussis exposure; any trauma or concern for maltreatment; unexplained sibling death or dysmorphic features.
- These mandate **targeted** evaluation — the testing restrictions above apply to lower-risk infants only.

#### ✦ PEARL – “LOWER-RISK” IS NOT “NO-RISK”

- In the multicenter BRUE network, about **4%** of infants had a serious underlying condition (most often a seizure disorder or an airway abnormality) and events recurred in roughly **7%** discharged from the ED and **11%** discharged from hospital.<sup>33</sup>
- The AAP criteria reliably identify low-risk infants but over-classify many as higher-risk.<sup>33</sup>
- Safety-net every family, offer CPR training, and consider asking carers to video any recurrence for diagnostic clarity.

#### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — higher-risk BRUE (age < 2 months, prematurity, repeated events, or concerning features) for monitoring.
- **Discharge** — lower-risk BRUE with education, CPR resources, and follow-up.
- **Refer** — by cause (cardiology, neurology, safeguarding).

# Adolescent Emergencies – Kuwait

## Context

HOURS

ADOLESCENT

Mental-health crises and menstrual emergencies — within the Kuwait consent and confidentiality framework.

### ▶ MENTAL HEALTH & SELF-HARM

- Screen the psychosocial context in a calm, private setting — Home, Education, Eating, Activities, Drugs, Sexuality, Suicide/safety (HEEADSSS).<sup>225</sup>
- **Acute self-harm, suicidal ideation, or intentional ingestion → manage per the Self-Harm / Intentional Ingestion card** (continuous one-to-one observation, remove the means, treat the ingestion, mental-health assessment before discharge). This card adds the **adolescent consent, confidentiality, and safeguarding** framework below.

### ▶ MENSTRUAL EMERGENCIES

- **Heavy menstrual bleeding:** assess for hemodynamic compromise and anemia. Check CBC, ferritin,

coagulation screen and **von Willebrand studies** (especially at menarche)<sup>239</sup>, thyroid function, and a pregnancy test — performed within the local legal/ethical and safeguarding framework, and discussed with a senior clinician where culturally sensitive.

- **Severe dysmenorrhea:** first-line is a scheduled NSAID; escalate to hormonal therapy for a menstrual indication if refractory.<sup>239</sup>
- **Amenorrhea:** investigate primary amenorrhea if no menarche by age 15 with normal secondary sexual characteristics (or by 13 with none); in secondary amenorrhea, exclude pregnancy first.

| Drug                                           | Dose                                   | Route & notes                                                 |
|------------------------------------------------|----------------------------------------|---------------------------------------------------------------|
| Tranexamic acid                                | 15–25 mg/kg/dose (max 1.5 g) every 8 h | Acute heavy menstrual bleeding; PO or IV <sup>239</sup>       |
| Combined oral contraceptive / oral progestogen | per gynecology protocol                | Hormonal control of acute bleeding or refractory dysmenorrhea |

| Drug                                     | Dose                                                    | Route & notes                                   |
|------------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| NSAID<br>(ibuprofen /<br>mefenamic acid) | ibuprofen 10 mg/kg/dose;<br>mefenamic acid per protocol | Dysmenorrhea; also reduces menstrual blood loss |
| Iron                                     | 3–6 mg/kg/day elemental iron                            | Iron-deficiency anemia from chronic loss        |

### ⚠ SAFEGUARDING OVERRIDES CONFIDENTIALITY

- Confidentiality with an adolescent is real but **not absolute**.
- When there is imminent risk of serious harm — self-harm, suicide, abuse, or exploitation — override it: involve the guardian and escalate through the hospital safeguarding pathway.
- Treat life- or limb-threatening conditions immediately, with or without consent.

### ◆ KUWAIT LEGAL FRAMEWORK

- Under **LAW NO. 70/2020** (Medical Profession & Patients' Rights), informed consent is required before any intervention and, for a minor, is given by the **LEGAL GUARDIAN (WALI)**.<sup>51</sup>

- **CONFIDENTIALITY (ARTICLE 13):** the practitioner must not disclose a patient's secret — whether it came to their knowledge, was discovered through practice, or was entrusted to them — except in defined circumstances: a written order from a court, the public prosecution, or a competent investigating authority; disclosure to prevent or report a crime; notification of a communicable disease to the authority designated by the Ministry; the patient's written consent; information exchange required between the care team for the patient's management; or reporting of child abuse (Law 21/2015) or elder abuse/neglect (Law 18/2016).
- Under the **MENTAL HEALTH LAW NO. 14/2019**, a physician may hold a patient for **UP TO 72 HOURS** for psychiatric assessment when there is danger to self or others or an inability to self-care/consent (Art. 11);<sup>52</sup> involuntary referral is also permitted by judicial order or at the request of a relative (to second kinship), the treating physician, or an investigator. Disclosing a patient's psychological information is a **CRIMINAL OFFENCE**.
- Confirm exact article wording against the official gazette text and

## The Child with Obesity ASSESS

AAP 2023

Obesity reshapes acute care — weight-based dosing, a harder airway, and comorbidities that present as emergencies.

### ▴ DEFINE

- **Obesity** = BMI  $\geq$  95th percentile for age and sex; **severe obesity**  $\geq$  120% of the 95th percentile (or BMI  $\geq$  35). A chronic disease with genetic, physiologic, and environmental contributors — not a failure of willpower.<sup>126</sup>
- Focuses on how obesity **modifies acute care** — drug dosing, the airway, resuscitation, and the comorbidities that present as emergencies.

### ▴ APPROACH & WORK-UP

- Confirm the **BMI percentile**; targeted history (diet, activity, sleep/snoring, family history, medications) and exam for comorbidity signs —



acanthosis nigricans, hypertension (correct-size cuff), hepatomegaly, hip/knee findings, papilledema.

- **Screen  $\geq 10$  years:** fasting glucose/HbA1c (T2DM), ALT (fatty liver), and a lipid panel; blood pressure with an appropriately sized cuff.<sup>126</sup>

## MANAGEMENT – DRUG DOSING IS THE KEY BEDSIDE TRAP

| Drug                                 | Dose                                                   | Route & notes                                                                                                                                                                                                                                                                 |
|--------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resuscitation drugs · defibrillation | Actual (total) body weight – but cap at the adult dose | Length-based tapes (Broselow) mis-estimate weight in obesity <sup>127</sup> — verify actual weight.                                                                                                                                                                           |
| General rule                         | No universal scalar – dosing is drug-specific          | There is <b>no valid one-size default</b> ; the correct weight basis depends on each drug's pharmacokinetics. Confirm with clinical pharmacy / the formulary. Never exceed the adult dose; scaling a hydrophilic drug to total body weight can cause overdose. <sup>127</sup> |

| Drug                                            | Dose                                  | Route & notes                                                                   |
|-------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------|
| Lipophilic drugs, large $V_d$                   | Loading dose nearer total body weight | Then maintain on lean/adjusted weight (e.g., some sedatives, induction agents). |
| Cefazolin<br>succinylcholine<br>antineoplastics | Total body weight (TBW)               | TBW is the correct scalar for these. <sup>127</sup>                             |
| Aminoglycosides<br>hydrophilic drugs            | Adjusted or lean body weight          | Volume of distribution altered; use adjusted body weight. <sup>127</sup>        |

◆ **RULE OF THUMB** When total body weight exceeds ~120% of ideal body weight, choose the weight basis deliberately — **PER DRUG, NOT BY A BLANKET RULE. NEVER EXCEED THE RECOMMENDED ADULT DOSE** for the indication, regardless of weight. For any drug not listed above, or with a narrow therapeutic index, **INVOLVE CLINICAL PHARMACY**.

### ⚠ **AIRWAY & VENTILATION**

- Anticipate **difficult bag-mask ventilation and intubation** and **rapid desaturation** (reduced functional residual capacity).

- Pre-oxygenate well, position **ramped/HELP** (ear-to-sternal-notch), and have adjuncts ready.
- Higher risk of OSA and obesity-hypoventilation.

### ▲ **RED FLAGS – DON'T-MISS ACUTE COMORBIDITIES**

- **T2DM → DKA or HHS** — hyperglycemic hyperosmolar state is disproportionately common in adolescents with obesity and T2DM.
- **SCFE** — hip, thigh, or knee pain with a limp in an adolescent<sup>135</sup> (see the **SCFE** card).
- **Idiopathic intracranial hypertension** — headache, papilledema, visual change.
- **Cholelithiasis / pancreatitis**, VTE in adolescents, worsening asthma control.

### ✦ **PEARL – WEIGH THE CHOICE, NOT JUST THE CHILD**

Document both actual weight and ideal body weight, pick the scalar per drug

class, and avoid codeine — the same CYP2D6/respiratory risk applies.<sup>126</sup>

## Pertussis (Whooping Cough)

MINUTES

AAP Red Book 2024 • CDC

In the young infant, the danger is apnea and cyanosis — the “whoop” is often absent.

### RECOGNIZE

- Paroxysmal coughing fits ± inspiratory whoop or post-tussive vomiting, worse at night, with well periods in between; often marked lymphocytosis.
- **Infants < 6 months / unimmunized:** may present with **apnea, cyanosis or bradycardia** and little cough — a low threshold to admit and monitor.<sup>158</sup>

### SEVERE / MALIGNANT PERTUSSIS

Extreme leukocytosis (WBC > 50 ×10<sup>9</sup>/L), pneumonia, pulmonary hypertension and refractory hypoxemia carry high mortality in young infants — escalate early to critical care; exchange transfusion/leukoreduction is considered in specialist units.<sup>158</sup>

## WORKUP

- **Nasopharyngeal PCR** — test of choice in the first 3–4 weeks of cough; culture (< 2 weeks) for confirmation/typing; serology in later/adolescent disease.<sup>158</sup>
- CBC — marked **lymphocytosis**; **extreme leukocytosis** predicts pulmonary hypertension and death in young infants.
- Chest X-ray if respiratory distress.

| Age        | Azithromycin                                                    | Notes                                                                          |
|------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|
| < 6 months | 10 mg/kg once daily for 5 days                                  | Macrolide of choice — lowest risk of infantile pyloric stenosis <sup>158</sup> |
| ≥ 6 months | 10 mg/kg (max 500 mg) day 1, then 5 mg/kg (max 250 mg) days 2–5 | PO                                                                             |

## PUBLIC-HEALTH & CONTACTS

- Pertussis is **NOTIFIABLE** — inform public health.
- Droplet precautions until 5 days of macrolide.<sup>158</sup>
- Offer **POST-EXPOSURE PROPHYLAXIS** (same macrolide dosing) to close contacts, prioritizing households

with infants < 12 months or pregnant women<sup>158</sup>, and check the family's immunization status.

- Reinforce **MATERNAL TDAP** each pregnancy (27–36 weeks) as the main protection for young infants.<sup>158</sup>

#### ▲ ADMIT · REFER

- **Admit** — any infant < 6 months, apnea/cyanosis, feeding difficulty, pneumonia or extreme leukocytosis.
- **Refer/Call** — critical care for severe pertussis.

# Complex & Technology-Dependent Care

Technology-dependent & medically complex children — tracheostomy, feeding tubes, ECMO, post-ICU, transport

## The Child with Medical Complexity

### ASSESS

When a Child with Medical Complexity presents unwell — start from their baseline, and treat the carer as the expert.

#### ▲ WHO – AND WHY THEY DECOMPENSATE FAST

- **Children with Medical Complexity (CMC)** have chronic conditions with **medical fragility**, multiple coexisting problems needing **multidisciplinary care**, and often **medical technology dependence** — a tracheostomy, long-term ventilation, enteral feeding tubes and pumps, a central venous catheter, a ventriculoperitoneal shunt, or an intrathecal baclofen pump. They usually have

**polypharmacy, coordinated care needs, and frequent healthcare use.**<sup>100</sup>

- They have little reserve, present **atypically**, and the **carer knows the baseline** — when they say “this is not normal for him,” believe it and look hard.<sup>101</sup>

#### ▲ **THE UNWELL CHILD – A STRUCTURED APPROACH**

- **Start from baseline:** their usual saturations, tone, secretions, feeds, alertness, and seizure pattern.  
**Deviation from baseline is the alarm**, not the absolute number.
- **Read the emergency care plan first** — diagnosis, baseline, devices and sizes, drugs, allergies, and the agreed goals of care.
- **Think the common six:**
  - infection
  - **device malfunction**
  - constipation
  - pain / dystonia
  - seizures
  - medication problem (including **withdrawal**).



- These cause most acute presentations.<sup>101</sup>

## TECHNOLOGY PROBLEM – WHERE TO LOOK

| Device                         | Red flag → action                                                                                                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tracheostomy / home ventilator | Desaturation, distress, or alarm → blocked or displaced tube, tracheitis → see <a href="#">Tracheostomy Emergencies</a> . Ventilator alarm → think DOPES (displacement, obstruction, pneumothorax, equipment, stacked breaths) |
| Gastrostomy / jejunostomy      | Dislodged or blocked tube, leak, or peritonism → see <a href="#">Gastrostomy (Feeding) Tube Problems</a> ; a fresh stoma can close within hours                                                                                |
| Central venous catheter        | Fever = line infection until proven otherwise – paired cultures (line + peripheral) before antibiotics; do not pull a tunnelled line lightly. Occlusion or breakage → specialist                                               |
| VP shunt                       | Vomiting, headache, drowsiness, irritability, or a bulging fontanelle = shunt malfunction / raised ICP until proven otherwise → urgent imaging + neurosurgery                                                                  |

| Device        | Red flag → action                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baclofen pump | Itch, rigidity, fever, agitation, autonomic instability = <b>acute withdrawal</b> (can be life-threatening) → urgent specialist; bridge with benzodiazepines ± oral baclofen |

### ⚠ **POLYPHARMACY & DO-NOT-STOP DRUGS**

- Do **not** abruptly stop chronic medications — **most should be continued** — including anticonvulsants, baclofen, and long-term steroids, or long-term oxygen / ventilation.<sup>101</sup>
- If on long-term steroids and acutely unwell, give **stress-dose steroids** (see [Adrenal Crisis](#)).
- Reconcile every drug, check interactions, and bring the child's own equipment and ventilator settings with them.

◆ **PLAN BEFORE THE CRISIS** Agree and document the **BASELINE, THE EMERGENCY CARE PLAN, AND THE GOALS OF CARE** with the family and the medical home in advance — and make sure it travels with the child.<sup>101</sup>

# Tracheostomy Emergencies

MINUTES

The deteriorating child with a tracheostomy — oxygenate both ways, then work the algorithm: patent, obstructed, or displaced?

## ▲ FIRST MOVES – CALL FOR HELP, OXYGENATE BOTH WAYS

- **Call for help.** Apply **high-flow oxygen to both the stoma and the face** as soon as it is available — use **two sources if you have them**; with a **single oxygen source, the stoma is the priority** (it is the airway in most tracheostomy patients), moving to the face if the stoma is not the effective route<sup>94</sup>. Attach **pulse oximetry ± waveform capnography** (exhaled CO<sub>2</sub> = a patent or partly patent airway). Bring the **tracheostomy emergency kit**: same-size and one-size-smaller tubes, a suction machine + catheters, water-based lubricant, a cuff syringe, and a bag-valve-mask. Use the **bedside emergency card** (tube type and size, cuff status, **suction depth**); suction with a **premarked catheter to a set depth** — just past the tube tip, no deeper.<sup>94</sup>

Routine **humidification** (HME / “Swedish nose”) prevents the mucus plugs behind most obstructions.

- **Rapid airway assessment:** look, listen, and feel at **BOTH the mouth and the stoma** — assess **chest rise, breath sounds (air entry), airflow, and exhaled CO<sub>2</sub>** (if available).

DECISION

① **Is the tracheostomy tube in place?**

**NO**

**Accidental decannulation.** If trained, **reinsert the tracheostomy tube**. If not trained or reinsertion fails, **ventilate via whichever route achieves chest rise** (upper airway or stoma). Continue ventilation and await expert airway support.<sup>94</sup>

**YES**

Tube is in place — proceed to the steps below.

② REMOVE WHAT MAY BE CAUSING THE PROBLEM

- Remove any **speaking valve or cap**, remove the **inner cannula** (if present), and **check the ventilator / circuit** (if ventilated) — this alone often relieves the obstruction.
- (Some inner cannulas must be reinserted to connect a breathing circuit.)

DECISION

③ Can you pass a suction catheter?

PASSES

Tube is **patent** — suction to clear secretions (a **mucus plug** is the commonest cause; if it will not clear, **change the tube** — do not persist)<sup>94</sup>.  
Ventilate via the tracheostomy if not breathing. Continue ABCDE.

WON'T PASS

Tube is **obstructed or displaced**.  
**Deflate the cuff** (if present) — if this improves things, air is moving around the tube<sup>94</sup> → **remove the tracheostomy tube**.

after tube removal / still not ventilating



④ VENTILATE – VIA WHICHEVER ROUTE ACHIEVES CHEST RISE

- **Upper airway** (if patent): cover the stoma (gauze / finger) and bag-mask via the mouth/nose ( $\pm$  oral or nasal airway), or an LMA.
- **Stoma** (if the upper airway is not patent or oral/nasal ventilation is ineffective — e.g., laryngectomy, subglottic stenosis): ventilate over the stoma with an appropriately sized pediatric face-mask, or an LMA.

when trained help arrives

⑤ RE-ESTABLISH THE AIRWAY

- Re-cannulate with the **same-size tube**; if it will not pass, use **one size smaller**. If it still will not pass, **do not force — stop and keep ventilating** until expert airway support arrives, or **orally intubate** (an uncut / longer ETT, cuff sited **beyond the stoma**)
- Confirm with **capnography**. A **neck roll** (shoulder roll, neck extended) increases the likelihood of successful insertion.<sup>94</sup>

### ▶ DOPE – CAUSES OF DETERIORATION

- **D — Displacement** (accidental decannulation)
- **O — Obstruction** (mucus plug, blood, granulation tissue)
- **P — Pneumothorax**
- **E — Equipment failure** (ventilator / circuit, speaking valve, cap, inner cannula, oxygen failure)

### ✦ KEEP THE OBTURATOR AT THE BEDSIDE

- Keep the tracheostomy **obturator at the bedside at all times.**
- In an emergency it may be needed to **reinsert the child's existing tube** when a new tube is unavailable or there is no time to open one — reinsertion is **difficult or impossible without it.** <sup>203</sup>

### ⚠ NO UPPER AIRWAY? (LARYNGECTOMY / FIXED OBSTRUCTION)

- If the child **cannot be ventilated from above** (laryngectomy, severe subglottic stenosis), the **stoma is the only airway** — do not waste time on face-mask ventilation;

oxygenate and re-cannulate or intubate the stoma.

- Know this from the care plan **before** the emergency.

### **⚠ BLEEDING – EXCLUDE TRACHEO-INNOMINATE FISTULA (TIF)**

- **Massive bleeding — or a brief “sentinel” bleed — 3 days to 6 weeks after tracheostomy is TIF until proven otherwise** (high risk in neurologically impaired children)<sup>203</sup>.
- This is an airway/ENT/surgical emergency — call expert help (ENT, surgery, anesthesia/ICU) immediately.
- As an expert-performed bridge while help arrives: **overinflate the cuff** to tamponade (~85% effective); if it fails, **orally intubate past the stoma** (ETT cuff distal to the bleed) and apply the **Utley maneuver** — a finger through the stoma compressing the artery against the back of the sternum — then **activate massive transfusion, call ENT + surgery, and go to OR.**<sup>203</sup>



- Any tracheostomy bleeding needs ENT review before discharge.

### ▲ TRACHEITIS

- Increasingly **purulent / copious secretions, fever, plugging, rising work of breathing** — can obstruct. Send a **tracheal aspirate**; cover *S. aureus* (including MRSA) and *Pseudomonas* per the local antibiogram; step up humidification and suction.<sup>203</sup>

### ⚠ WHEN NOT TO CHANGE THE TUBE YOURSELF – CALL ENT / THE TRACHEOSTOMY TEAM

As a general pediatrician, **do not attempt an elective tube change** when: it is the **first change** (the surgical / ENT team performs or directly supervises the first change, usually at **day 3–7** post-formation, to confirm a patent tract)<sup>204</sup>; the stoma is **fresh / immature (< 7 days**, especially percutaneous) — high risk of a **false passage** or an unrecoverable airway; you are **not trained in that specific tube**, or the correct-size tubes and suction are not at hand; the child is **unstable, on high ventilator / oxygen support, or has a known difficult airway** — change only

where **ENT / anesthesia and airway rescue are immediately available**; or there is **active bleeding or suspected tracheo-innominate fistula** (an emergency, not a routine change); the tracheostomy was placed for **critical upper airway obstruction** — the upper airway is **not a reliable rescue route**, so a lost or displaced tube can mean a **lost airway** (treat the stoma as the only airway and change only with ENT / anesthesia and airway rescue at the bedside); or the child has had **major airway / tracheal reconstruction** (laryngotracheal reconstruction, slide tracheoplasty, tracheal resection & anastomosis) — the tube may cross or protect a **fresh repair**, so changes are done by the **surgical team only** (risk of disrupting the anastomosis or a false passage). Otherwise **keep the current tube, oxygenate, and get expert help**.

#### ELECTIVE TUBE CHANGE (ROUTINE)

- Two people.
- Supine with a **shoulder roll, neck extended**; pre-oxygenate and suction.
- Load the obturator, lubricate, deflate the cuff.
- Remove the old tube along its curve; insert the new one on the **obturator** (tip toward the posterior wall, then rotate downward) on expiration — **never force. Remove the obturator immediately**, insert the inner cannula, inflate the cuff, and confirm (chest rise, breath sounds, capnography, suction catheter passes)<sup>204</sup>
- Secure ties to a one-finger gap.

#### ⚠ CALL ENT / ANESTHESIA EARLY

- Any bleeding · suspected TIF (emergency) · failed or difficult reinsertion · obstructing granulation · **accidental decannulation with a fresh stoma (< 7 days)** — prefer oral intubation and avoid blind reinsertion.
- An established stoma can still narrow > 50% within ~12 h of decannulation, so if decannulated

> ~8 h and the child is stable, call ENT before repeated attempts (dilation may be needed).<sup>203</sup>

### SOURCE

National Tracheostomy Safety Project (NTSP) pediatric emergency algorithm; Care of Infants and Children with Tracheostomies — American Thoracic Society Clinical Practice Guideline (*Am J Respir Crit Care Med* 2025;211(11):2001–2020).

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — any airway compromise → ED/PICU.
- **Discharge** — after the tube is secure and patent and the caregiver is confident.
- **Refer** — ENT and the home-ventilation/tracheostomy team.

### ◎ LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES (no login required)

- **CINCINNATI CHILDREN'S – TRACHEOTOMY CARE** — caregiver handbook (Arabic available) plus short videos

including an **EMERGENCY TUBE CHANGE**:

[cincinnatichildrens.org](http://cincinnatichildrens.org)



- **ABOUTKIDSHEALTH (SICKKIDS) – TRACHEOSTOMY & VENTILATION** — home care, suctioning and **EMERGENCY MANAGEMENT**:

[aboutkidshealth.ca/trachvent](http://aboutkidshealth.ca/trachvent)



# Tracheostomy Tubes & Cuff Types

## REFERENCE

Know the exact tube before you change or ventilate — same stated size from different manufacturers is not interchangeable.

## TUBE – WHAT IT IS AND WHEN

| Tube          | Material & features                                                                                                                                                          | Notes                                                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| PVC tube      | PVC, firm, thermosensitive, radiopaque line; neonatal / pediatric / pediatric-long; cuffed (soft air cuff) or uncuffed. Pediatric sizes are single-lumen (no inner cannula). | Single-patient, disposable. Sized by <b>inner diameter (ID, mm)</b> . <sup>203</sup>                                  |
| Silicone tube | Silicone, soft and flexible, reusable; cuff options tight-to-shaft (TTS), air, or foam; extended-proximal and adjustable-le                                                  | Silicone is MRI-conditional — <b>wire-reinforced / metal variants may not be</b> ; check. Sized by ID. <sup>203</sup> |

| Tube | Material & features                       | Notes |
|------|-------------------------------------------|-------|
|      | ngth<br>(wire-reinforced) variants exist. |       |

## CUFF TYPES – INFLATE WITH WHAT, WATCH WHAT

| Cuff                 | Inflate with                       | Use / caution                                                                                                                                                                         |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uncuffed             | –                                  | Most children; airflow around the tube (speech), least mucosal injury.                                                                                                                |
| Air cuff             | Air                                | Positive-pressure ventilation or aspiration protection; keep <b>cuff pressure &lt; 20–25 cmH<sub>2</sub>O</b> (minimal-leak) — over-inflation causes mucosal ischemia. <sup>204</sup> |
| Tight-to-shaft (TTS) | Sterile water (never air)          | Deflated, it sits flush like an uncuffed tube (speech / airflow); inflate only when needed. Air diffuses out of silicone <sup>204</sup> — <b>use water.</b>                           |
| Foam cuff            | Self-inflating (pilot open to air) | For a difficult seal or tracheomalacia; actively <b>deflate and cap the pilot</b> to insert / remove, then release to self-inflate.                                                   |

### ⚠ CUFF SAFETY

- Never use a **speaking valve or cap with the cuff inflated** — no route to exhale (can be fatal)<sup>204</sup>; deflate cuff first.
- Deflate cuff whenever a seal is not needed, to protect mucosa and allow speech.

### ▲ BEFORE ANY TUBE CHANGE

- Same-size tubes from **different manufacturers differ in outer diameter and length — not interchangeable**; if you must substitute in an emergency, be ready to **downsize**.<sup>203</sup>
- Keep a **same-size and one-size-smaller** spare at the bedside; know the exact tube from the **bedside emergency card**.
- After insertion, confirm airway — chest rise, **capnography**, suction catheter passes.

### 🕒 LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES (no login required)

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## Gastrostomy (Feeding) Tube Problems

### HOURS

A dislodged tube is time-critical — the stoma can close within hours. Tract age decides who replaces it.

### ▶ **DISLODGED TUBE – THE CLOCK IS THE STOMA**

- Empty tract narrows within hours — per **ESPGHAN / NASPGHAN** it can **close within ~6 h**, usually lost by **24 h**.<sup>102</sup> Keep it open **now**, then replace.
- **Tract age decides the route** — ask when the tube was first placed.

## DECISION

### How old is the stoma tract?

#### IMMATURE < 6 WEEKS

High risk of a **false tract / intraperitoneal placement** → **do NOT insert a tube into the stoma**. Cover stoma with gauze, keep nil-by-stoma, get to **hospital** for guided replacement by **pediatric surgery / gastroenterology / interventional radiology**.<sup>102</sup>

#### MATURE ≥ 6 WEEKS

Epithelialized tract → bedside replacement usually safe. Replace **promptly** with a **same-size balloon tube**; lubricate, advance gently, inflate the balloon with water. **Caveat: a first PEG, or poor wound healing** (malnutrition, chemotherapy), may mature more slowly — up to **~3 months**; if unsure, **treat as immature**.<sup>102</sup>

#### IF THE SAME SIZE WILL NOT PASS

- Place the largest **Foley catheter** that passes atraumatically to hold the stoma open, then **serially dilate in 2-Fr steps** to the target size.
- **Do not inflate a Foley balloon inside the tract to dilate.**<sup>102</sup>
- If > 24 h out, or any resistance / false-passage concern → stop and refer.

#### ◆ CONFIRM BEFORE FEEDING

- Verify intragastric position before use: **ASPIRATE GASTRIC CONTENTS (PH ≤ 5.5) – BEWARE A FALSE PH ON PROTON-PUMP INHIBITORS.**
- After any **DIFFICULT OR UNCERTAIN** replacement, confirm with a **WATER-SOLUBLE CONTRAST STUDY** — the ESPGHAN standard before first use of a replaced balloon tube.<sup>102</sup> **NEVER USE AIR INSUFFLATION** to confirm.
- Document tube type, French size, and balloon volume.

#### ▴ BLOCKED TUBE & STOMA ISSUES

- **Blocked:** flush gently with **warm water**; do not force. Hold feeds until patency is confirmed.

- **Buried bumper:** suspect if the tube **will not rotate or advance freely**, or pain, leakage, or resistance on feeding — prevented by avoiding traction and rotating + advancing the tube regularly. Buried bumper or **suspected peritonitis** (pain, feed intolerance, fever, rigidity) → stop feeds, urgent **surgical review**.<sup>102</sup>
- **Granulation, leak, redness, excoriation:** see [Enteral Feeding Tubes — Types & Complications](#).

#### ⚠ ADMIT / ESCALATE

- Peritonitis or systemic infection
- **buried bumper syndrome**
- uncertain or **difficult placement**
- concern of a **false passage**
- any immature-tract dislodgement
- **tube out > 24 h** → see surgery / gastroenterology.

#### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — peritonitis, a dislodged tube with a lost tract, or inability to replace it.
- **Discharge** — a replaced, patent tube with feeds tolerated.

- **Refer** — surgery/gastroenterology for tract problems or early dislodgement (< 4 weeks).

© **LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES** (no login required)

- **CINCINNATI CHILDREN'S – G-TUBE HOME CARE** — caregiver handbook + procedure videos (Arabic available):

[cincinnatichildrens.org](https://cincinnatichildrens.org)



- **TOGETHER BY ST. JUDE – TYPES OF TUBES & DEVICES** — illustrated visual guide to NG / ND / NJ and G / GJ / J tubes and low-profile buttons:

[together.stjude.org](https://together.stjude.org)



# Enteral Feeding Tubes – Types, Delivery & Complications

## REFERENCE

Know the route before you feed or give a drug — the stomach tolerates boluses, the jejunum does not.

## TUBE TYPES – ROUTE, HOW TO FEED, AND WHAT NOT TO GIVE

| Tube      | Route & how to feed                                                                                                                                                                                                                                                                   | Do NOT                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| NGT / OGT | <p>Stomach; short-term; bolus or continuous. <b>Nasogastric vs orogastric route is a unit / clinical choice – evidence favors neither.</b><sup>192</sup> Choose by expected duration, gag strength, and respiratory status: prefer OGT when nasal airflow matters (preterm, CPAP,</p> | <p>Never feed with position unconfirmed. Use <b>OGT</b> (not NGT) in suspected base-of-skull injury.</p> |

| Tube                                      | Route & how to feed                                                                                                                                                                                                           | Do NOT                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                                           | respiratory distress) or the gag is weak; NGT is more secure for a stronger gag or longer use. Confirm gastric aspirate pH $\leq$ 5.5 before each use (x-ray if in doubt; local policy: confirm by x-ray). <sup>192,391</sup> |                                                                                                                |
| NJT                                       | Nose → jejunum (post-pyloric); for gastroparesis or high aspiration risk. Continuous feeds only.                                                                                                                              | <b>No bolus</b> (dumping). pH is unreliable — confirm position by imaging. <sup>102</sup>                      |
| Gastrostomy (PEG / high-profile / button) | Abdominal wall → stomach; long-term. Bolus or continuous.                                                                                                                                                                     | Force or over-rotate a PEG bumper (buried bumper). A low-profile button needs an <b>extension set</b> to feed. |

| Tube                   | Route & how to feed                                                                                                                                           | Do NOT                                                                        |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Gastro-jejunosomy (GJ) | Through the gastric stoma, tip in jejunum; <b>gastric + jejunal ports.</b> Jejunal port = <b>continuous only</b> ; gastric port = <b>venting / medicines.</b> | <b>Blind replacement</b> — needs fluoroscopy. No bolus down the jejunal port. |
| Jejunostomy (J)        | Abdominal wall → jejunum. <b>Continuous feeds only</b> , small volumes.                                                                                       | Bolus feed. Attempt a bedside change — refer (surgical tract).                |

### ⚠ MEDICINES THROUGH A FEEDING TUBE

- Use **liquid formulations** where possible; **never crush enteric-coated or modified / sustained-release** tablets (blocks the tube, alters the drug).<sup>102</sup>
- Give each drug **separately** and **flush with water** before, between, and after. **The route matters:** down a **jejunal (post-pyloric)** tube avoid hypertonic or large-volume drugs (dumping, perforation), and **some medications cannot be**



**given jejunally at all** — check with pharmacy.

- Some **parenteral (injectable) preparations** can be given enterally when no oral liquid exists (e.g., **clonidine, glycopyrrolate**) — confirm suitability and dose with pharmacy first.

#### ▲ DIFFERENTIATE THE TUBE AT THE BEDSIDE

- **Where it enters** — nose / mouth (NGT/OGT) vs the abdomen (gastrostomy / GJ / J).
- **Profile** — an external bumper with a **balloon port** (water valve) = balloon gastrostomy; **flush to the skin** needing an extension = low-profile **button**; a firm internal flange that resists gentle traction = **PEG**.
- **Ports** — **two labelled ports (GASTRIC + JEJUNAL)** = GJ; a single feed port plus a balloon port = gastrostomy.
- Read the **French size, length, and brand** off the tube and the care plan. Unsure of the route? Confirm with **contrast** — do not feed to find out.

## MANAGE THE COMMON STOMA COMPLICATIONS

| Problem                    | What & why                                                 | Management                                                                                                                                                                                                                                                     |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Granulation on (granuloma) | Red, moist, friable overgrowth from friction and moisture. | Stop the movement (secure the tube, correct balloon volume); <b>saline soaks</b> ; topical <b>silver nitrate</b> (protect surrounding skin with a barrier) or a short <b>topical steroid</b> ; silver / foam / hydrocolloid dressing. <sup>102</sup>           |
| Leak                       | Gastric contents track around the tube.                    | <b>Do not just upsize the tube</b> (enlarges the stoma). Correct balloon volume; treat <b>constipation, retching, delayed emptying</b> (vent, prokinetic, slow the feed); treat granulation / infection; barrier cream. Consider buried bumper. <sup>102</sup> |
| Redness                    | Irritation vs infection.                                   | < 5 mm erythema is usually irritation — barrier protection. <b>Spreading redness, pain, swelling, or pus = infection</b> → swab, then topical or systemic antibiotics (cover skin flora).                                                                      |

| Problem     | What & why                                | Management                                                                                                                                     |
|-------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Excoriation | Enzymatic burn from leaked feed and acid. | Stop the leak at source; <b>barrier products</b> (zinc or dimethicone cream, pectin barrier ring / powder); keep dry; involve the stoma nurse. |

### ◆ GOLDEN RULES

- Confirm route before feeding or giving a drug; **GASTRIC TOLERATES BOLUSES, JEJUNAL IS CONTINUOUS ONLY**; flush to keep patent.
- A fresh gastrostomy stoma can close within hours if the tube falls out — see **Gastrostomy (Feeding) Tube Problems.**

### ◎ LEARN MORE – OPEN-ACCESS PEDIATRIC RESOURCES (no login required)

- **TOGETHER BY ST. JUDE – TYPES OF TUBES & DEVICES** — illustrated visual guide to NG / ND / NJ and G / GJ / J tubes and low-profile buttons:

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## ECMO – What the General Pediatrician Needs to Know

ASSESS

KNOW WHEN TO CALL • ELSO

Temporary heart–lung support for severe but **reversible** failure — your job is to recognize the candidate and call **early**.

+ Calculator: [PARDS oxygenation index](#)

### WHAT IT IS

- Temporary machine support that **oxygenates the blood and removes CO<sub>2</sub>** ( $\pm$  supports the circulation) to buy time for **potentially reversible** heart or lung failure. It is a **bridge** — to recovery, a decision, or transplant — **not a cure**, and is run only in a specialized PICU / ECMO center.
- **VV (veno-venous) = lung support** (the heart still works); **VA (veno-arterial) = heart  $\pm$  lung**

**support**, and covers **ECPR** (ECMO during a refractory but reversible cardiac arrest).<sup>240</sup>

#### **⚠ WHEN TO THINK ECMO – CALL EARLY**

Severe **cardiac or respiratory failure not responding to maximal conventional therapy**, in a child with a **reversible cause** (e.g., fulminant myocarditis, ARDS / severe pneumonia, PPHN; neonatal MAS or CDH). **Call the PICU / ECMO center early** — before catastrophic deterioration or irreversible brain injury; timely referral improves outcomes, so do not wait for arrest. **The center decides candidacy, thresholds, and timing.**<sup>240</sup>

#### **⚠ USUALLY NOT A CANDIDATE**

- An **irreversible** condition with no exit strategy · **severe irreversible brain injury** · a **lethal chromosomal / congenital anomaly** (e.g., trisomy 13 or 18 — not 21) · **uncontrolled bleeding**.
- Extreme prematurity, very low birth weight, and many days of prior ventilation lower the chance of benefit.

- The final decision rests with the ECMO team.<sup>240</sup>

#### ✦ YOUR ROLE – BEFORE AND AFTER

- You are **not** expected to run ECMO — you **recognize the failing-yet-reversible child and escalate**.
- Resuscitate and support maximally, **protect the brain and other organs**, avoid irreversible injury, and **call the PICU / ECMO center early**.
- Afterwards, **survivors need long-term multidisciplinary follow-up**: up to about  $\frac{1}{3}$  have some neurodevelopmental impairment and new problems can appear later — track development and arrange **hearing tests** (sensorineural loss can be progressive / late).<sup>240</sup>
- Support the family through a **high-risk, uncertain course** (major complications include bleeding, clots, and stroke).

### ▲ ADMIT · DISCHARGE · REFER

- **Escalate** — refractory cardiac or respiratory failure with a reversible cause → **PICU / ECMO center now.**
- **Discuss early** — call **before** the child is in extremis; deterioration is often rapid.

### SOURCE

- Extracorporeal Life Support Organization (ELSO) general and pediatric / neonatal guidelines.
- Candidacy, thresholds, and timing are set by the PICU / ECMO center.

# Post-PICU Syndrome (PICS-p)

## ASSESS

## AFTER CRITICAL ILLNESS

Surviving the PICU is only the start — new physical, cognitive, emotional, and social problems can follow the child and family for years.

### WHAT IT IS

- **Post-Intensive Care Syndrome — pediatric (PICS-p):** new or worsening impairment across **four domains** after critical illness — **physical, cognitive, emotional, and social** — affecting the **child and the family**. It emerges over the months after discharge and can persist for years.<sup>206</sup>
- You, the general pediatrician, are the one who **sees these children back** — so you screen for it, coordinate care, and reassure.



## THE FOUR DOMAINS – WHAT TO LOOK FOR

| Domain    | Watch for                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Physical  | ICU-acquired weakness, fatigue, pain, poor feeding / growth, reduced exercise tolerance, poor sleep; residual respiratory or organ problems. |
| Cognitive | Attention, memory, and executive difficulties; developmental regression in the young; declining school performance.                          |
| Emotional | PTSD, anxiety, depression, nightmares / delusional memories, fear, irritability or aggression, behavioral change.                            |
| Social    | Trouble reintegrating at school, absenteeism, withdrawal from peers and activities, identity and participation problems.                     |

### ■ HIGHER RISK AFTER...

- **Longer PICU stay, mechanical ventilation, sedation and delirium,** renal replacement, **ECMO**, cardiac arrest, sepsis, ARDS, or major trauma; oncology / neurology diagnoses; an **unscheduled admission**; younger age and pre-existing conditions.<sup>206</sup>

### ⚠️ DON'T FORGET THE FAMILY (PICS-F)

- Parents and siblings are affected too: at PICU discharge up to **~60% of parents are at risk of PTSD and ~75% of depression**, alongside anxiety, exhaustion, financial strain, and impaired family functioning.<sup>206</sup>
- Ask how the **family** is coping, and support them.

### ✦ YOUR ROLE

- At the first post-discharge visits and beyond: **screen every domain** (development, school, mood / sleep, physical function) and the family's wellbeing; **coordinate multidisciplinary follow-up** — physiotherapy / occupational therapy / speech-language therapy, psychology, neurodevelopmental assessment, and **school liaison**; reconcile medications and immunizations; and treat the family as part of the patient.

- Normalize and follow up rather than dismiss — recovery is a trajectory, not an event.

#### ▲ ADMIT · DISCHARGE · REFER

- **Screen** — every child at follow-up after a significant PICU admission.
- **Refer** — neurodevelopment / rehabilitation, psychology / psychiatry, and the relevant specialty for persistent domain-specific problems; social work for family support.
- **Escalate** — new severe mental-health symptoms (e.g., self-harm risk) or acute physical deterioration.

#### SOURCE

- Manning JC, Pinto NP, Rennick JE, Colville G, Curley MAQ.
- Conceptualizing Post Intensive Care Syndrome in Children — the PICS-p Framework. *Pediatr Crit Care Med*. 2018;19(4):298–300; pediatric PICS reviews.

# Aeromedical Transport – Pediatric & Neonatal

## ASSESS

### PRE-FLIGHT ASSESSMENT • MOH A-ADM-014

Match the transport tier to clinical risk — two predictable cabin stresses (hypoxia, gas expansion) drive the pre-flight assessment.

#### + Calculator: Oxygen cylinder duration

#### ▶ TWO CABIN STRESSES – ASSESS BEFORE EVERY FLIGHT

- **Hypobaric hypoxia** — cabins are pressurized to a maximum **8,000 ft** equivalent ( $\approx$  breathing **15% oxygen** at sea level); a healthy  $\text{PaO}_2$  falls to  **$\sim 60\text{--}65$  mmHg** ( $\text{SpO}_2 \sim 89\text{--}94\%$ ). Limited cardiopulmonary reserve desaturates further. **Resting  $\text{SpO}_2$  alone is insensitive. If available, a HAST (hypoxia-altitude simulation test) best predicts in-flight oxygen need in a child with chronic lung or cardiac disease.**<sup>241</sup> **If HAST is not available,** use the practical **alternative:** base the decision on **resting oximetry, the diagnosis and trajectory, and any prior flight tolerance,** and — when in doubt —

**give empiric in-flight oxygen**

(double an existing flow) or choose a **sea-level cabin / air ambulance** for higher-risk children.

- **Trapped-gas expansion** — gas in a closed space expands ~**35%** from sea level to 8,000 ft. Pneumothorax, recent surgery with retained gas, and intracranial or intraocular gas need a **sea-level cabin (air ambulance)** when the gas cannot be excluded. A **current pneumothorax requires a chest drain before any urgent flight**<sup>241</sup> — even if it is minimal.

### **⚠ NEONATAL & INFANT AIR TRANSPORT – STABILITY, NOT AGE, GOVERNS**

- **Term neonate:** defer non-urgent flying for roughly the **first 7 days** (ideally to ~40 weeks corrected) to confirm feeding, thermoregulation, and no undiagnosed cardiorespiratory problem or silent pneumothorax.<sup>241</sup>
- **Preterm without chronic lung disease:** the “~3 months corrected age” figure is a **soft guide, not a fixed cutoff** — fitness depends on **gestational maturity + demonstrated cardiorespiratory**

**stability + being apnea-free**, judged individually. Avoid flying during a respiratory infection until ~6 months corrected age (apnea risk). Former preterms with **BPD / neonatal chronic lung disease**: discuss with a respiratory pediatrician and, where available, a **HAST** before flying (~70% desaturate on hypoxic challenge).<sup>241</sup>

- **An incubator is not mandatory for a stable infant** — a physiologically stable infant may travel in an aircraft **bassinet / in-arms**. A **transport incubator** is needed for thermal instability, continuous monitoring, controlled oxygen/humidity, or mechanical ventilation.
- **Respiratory support changes the route, not by age**: a ventilated or monitoring-dependent infant goes by **dedicated air ambulance** (incubator/transport platform, continuous SpO<sub>2</sub> + ECG, capnography if ventilated, altitude-appropriate settings, sea-level-cabin option). A **stable, monitored, ventilated infant can still be flown** for urgent surgery with a specialist team.

- **Tier by stability:** stable, self-ventilating, thermally stable → **bassinet/in-arms commercial** (after the term deferral / individualized preterm assessment); oxygen-dependent or apnea-risk → **commercial + medical clearance + in-flight O<sub>2</sub>** (roughly double the flow at cruise); unstable/ventilated/monitoring-dependent → **dedicated air ambulance**.

## IN-FLIGHT OXYGEN – INFANTS & YOUNG CHILDREN

| Current support / status    | Action                                                                                             |
|-----------------------------|----------------------------------------------------------------------------------------------------|
| Already on oxygen           | <b>Double the flow rate in flight</b> as a starting point; confirm it is deliverable in the cabin. |
| Needs > 4 L/min at baseline | Commercial flight <b>not advised</b> → escalate to <b>air ambulance</b> . <sup>241</sup>           |

### 🕒 OXYGEN LOGISTICS

- Carriers supply oxygen at fixed flows (often 2–4 L/min, set rates or none).
- A need for precise or higher flow, or a corrected requirement > ~4 L/min, cannot be reliably managed on a scheduled flight.

- Carry an airline-approved **PORTABLE OXYGEN CONCENTRATOR (POC)** with battery for  $\geq 150\%$  of journey time; the US prohibits passenger oxygen cylinders (use a POC or carrier oxygen).<sup>241</sup>

## TRANSPORT TIER – ASSIGN THE LOWEST TIER THAT SAFELY COVERS THE WHOLE JOURNEY

| Tier | Mode & escort                                                                                                                               | Typical pediatric patient                                                                                                  |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 0    | Commercial seat, no escort                                                                                                                  | Stable, self-caring; PEWS low.                                                                                             |
| 1    | Commercial seat + non-medical attendant                                                                                                     | Minor / vulnerable; needs help with mobility, not clinical care.                                                           |
| 2    | Commercial seat + a competent trained caregiver; a nurse/RT is required only if oxygen- or ventilation-dependent, or other risk is expected | Stable <b>tracheostomy on room air</b> (with a trained caregiver competent in the device), stable oxygen, or pump feeding. |
| 3    | Commercial seat or stretcher + physician + nurse (add RT for home-vent)                                                                     | Stable established home ventilation, or a fragile child with deterioration potential.                                      |



| Tier | Mode & escort                                | Typical pediatric patient                                                       |
|------|----------------------------------------------|---------------------------------------------------------------------------------|
| 4    | Dedicated air ambulance + critical-care team | Critically ill or unstable (meets air-ambulance criteria below). <sup>242</sup> |

### ◎ SCORING & PERSONNEL (PER POLICY)

- The policy stratifies patients by **LEVEL OF CARE 0-4** and a **NEWS-BASED RISK LEVEL** (Low = 1-4 · Medium = 5-6 · High =  $\geq 7$ ). In children, substitute an age-appropriate **PEWS** (NEWS2 is validated only  $\geq 16$  yr).<sup>242</sup>
- If findings fall between tiers, choose the **HIGHER TIER**. A low score with **HIGH DETERIORATION POTENTIAL** (end-stage lung disease, potential airway/hemodynamic compromise) is classified up by the accompanying physician.
- **AT LEAST TWO COMPETENT PERSONNEL** escort any patient; for a child, include a **PARENT/LEGAL GUARDIAN** where possible.

◎ **LOCAL AVAILABILITY (KUWAIT)** A **DEDICATED PEDIATRIC / NEONATAL AEROMEDICAL TRANSPORT TEAM IS NOT YET ESTABLISHED** — transfers are currently staffed by the **ACCOMPANYING PEDIATRIC / TREATING TEAM**. Confirm current MOH capability early and coordinate

through the A-ADM-014 pathway (Directorate of Treatment Abroad).<sup>242</sup>

### **⚠ DEDICATED AIR AMBULANCE (SEA-LEVEL CABIN) – WHEN COMMERCIAL CARRIAGE WILL NOT DO**

Any one absolute criterion mandates it: **acute / unstable ventilation** or a credible need to intubate or escalate in transit; **hemodynamic instability** needing vasoactive / inotropic infusion; a **trapped-gas condition** needing sea-level cabin (unresolved pneumothorax, recent intracranial / intraocular gas, significant bowel obstruction); need for **invasive monitoring or mechanical support** (arterial-line titration, ICP monitoring, IABP, **ECMO**, VAD, external pacing); **refractory hypoxemia** or oxygen need beyond carrier capability; **status epilepticus** / uncontrolled seizures; or a communicable disease needing **isolation**. Under the policy, **Medevac (air ambulance)** also applies to **priority 1–2 cases**, expedited transfer of an unstable patient, brain-death repatriation, or when the commercial route/port is not cleared. **Absolute contraindications to flight until treated:** untreated pneumothorax, hemothorax or a compromising effusion, cardiac tamponade, recent abdominal

surgery / immediate post-op, or a penetrated eye injury.<sup>242</sup>

#### ✦ AIRWAY & CUFFS AT ALTITUDE

- The cabin is a poor place to rescue a failing airway — **secure a difficult or unstable airway definitively before departure**, not aloft; do not defer a foreseeable intubation to the aircraft.
- Endotracheal / tracheostomy **cuff gas expands  $\sim\frac{1}{3}$**  as cabin pressure falls — **check for a leak and adjust the cuff volume as needed**; monitor cuff pressure during ascent and descent.<sup>241</sup>
- Confirm the ventilator and oxygen are airline-approved with battery for  $\geq 150\%$  of journey time.

#### ▲ TEAM, CLEARANCE & HANDOVER

- **Escort competence:** current **pediatric / neonatal resuscitation** (PALS / NRP) and competence with the child's devices; the accompanying physician is the **MRP** en route.
- **Clearance:** pre-flight medical report + **risk assessment** reviewed and approved by the **MRP** (senior

registrar or above) **7–14 days and again 48–72 h** before departure; airline **MEDIF** for commercial carriage. A **return/repatriation reassessment is mandatory** before flying home.<sup>242</sup>

- **Handover:** document status before, during and on arrival, and give a structured handover to the receiving team using the policy's **handover tools**.

#### ▲ **ADMIT · DISCHARGE · REFER**

- **Escalate** — any air-ambulance criterion, or a deteriorating trajectory → critical-care transport team + treatment-abroad / transport committee.
- **Proceed commercial** — stable child at Tier 0–3 with a completed pre-flight clinical assessment (HAST where available) and confirmed oxygen/device support.
- **Refer** — the MOH Directorate of Treatment Abroad (DTA) decides transport type; involve the relevant clinical council for team and equipment.

## SOURCE

- State of Kuwait — MOH Aeromedical Evacuation & Transport Policy **A-ADM-014** (effective Oct 2025; review 2027) — the transport forms and checklists are in the policy on the MOH website.
- Pre-flight assessment thresholds align with the British Thoracic Society clinical statement and pediatric air-travel evidence. Neonatal/infant guidance: BTS air-travel recommendations (*Thorax* 2011); Fitness to fly in the paediatric population (*Eur J Pediatr* 2018); Khanna et al., in-flight hypoxia in healthy infants (*Thorax* 2013).

## Diabetic Ketoacidosis

**HOURS**

ISPAD 2022 • TREKK/CPEG 2023

Glucose > 11 mmol/L (200 mg/dL) + pH < 7.3  
or bicarbonate < 18 + ketonemia ( $\beta$ -  
hydroxybutyrate  $\geq$  3 mmol/L).

**+ Calculators: DKA fluids & insulin ·  
Corrected Na · anion gap · Ca · osmolality**



rehydration and 0.45% versus 0.9% saline did not change neurological outcomes in the 1,389-episode PECARN FLUID trial.<sup>7,344</sup>

#### ◆ SEVERITY & PICU

- **For DKA managed in the PICU, follow the latest Kuwait PICU Task Force DKA guideline** — the local task-force protocol governs fluid, insulin and electrolyte targets in the unit.<sup>172</sup>
- Consider PICU for severe DKA, young children (age < 5 y), reduced consciousness, long symptom duration, compromised circulation, or high-risk features (pH < 7.1, pCO<sub>2</sub> < 21 mmHg, urea > 7 mmol/L).<sup>7</sup>

#### ⚠ CEREBRAL INJURY – FURTHER DETAIL

- If focal neurological deficits develop, obtain an urgent CT to exclude cerebral venous thrombosis or intracranial hemorrhage. **Do not postpone treatment to obtain a CT.** Perform hourly GCS so cerebral injury is



caught early — it is the leading cause of death in pediatric DKA.<sup>7</sup>

◆ **AVOID WHERE POSSIBLE**

Intubation (ventilation removes the child's respiratory compensation for the acidosis), central venous catheters (thrombosis risk; a large dead space can make insulin delivery erratic if insulin runs through the CVC), and routine urinary catheterization.<sup>7</sup>

✦ **PEARL – THE MASQUERADES OF NEW-ONSET DKA**

- First-presentation diabetes is missed when DKA wears a disguise: **abdominal pain and vomiting** taken for gastroenteritis or appendicitis, **Kussmaul breathing** taken for pneumonia or an asthma attack, and **polyuria** taken for a urinary tract infection or, in a younger child, secondary enuresis.
- A bedside glucose and ketones settle it in seconds.

## ADMIT · DISCHARGE · REFER

- **Admit** — all children with DKA are admitted, but the **level of care follows severity**<sup>7</sup>: **mild DKA** can be managed on a general pediatric ward with hourly monitoring; **moderate/severe**, young age (< 5 y), reduced consciousness, or high-risk features → high-dependency/PICU.
- **Discharge** — not from DKA; transition to subcutaneous insulin with diabetes-team education.
- **Refer** — pediatric endocrinology/diabetes team for all; PICU for cerebral-injury risk.

## Hyperglycemic Hyperosmolar State (HHS)

MINUTES

ISPAD 2022

Not “severe DKA” — very high glucose, very high osmolality, little ketosis, and higher mortality.

+ Calculators: Corrected Na · anion gap · Ca · osmolality · DKA fluids & insulin

### ▴ DEFINE (DIFFERS FROM DKA)

- Glucose > **33 mmol/L (600 mg/dL)**, serum osmolality > **320 mOsm/kg**, **pH > 7.3 and bicarbonate > 15**, only mild ketosis, with altered consciousness/profound dehydration. Increasingly seen in adolescents with obesity/type-2 diabetes.<sup>7</sup>

### ⚠ MANAGE IT DIFFERENTLY FROM DKA

- Fluid deficits are larger (~12–15%) and **fluid — not insulin — is the priority first.**<sup>7</sup>
- Start insulin **later and lower** (only once glucose stops falling with fluids alone<sup>7</sup>, ~0.025–0.05 unit/kg/h).
- Watch for **rhabdomyolysis, malignant-hyperthermia-like fever, thrombosis and arrhythmia from potassium/phosphate shifts.**<sup>7</sup>
- Mortality is higher than DKA.<sup>7</sup>

### ✦ MIXED PICTURES EXIST

- Some children have features of both HHS and DKA.

- When ketoacidosis coexists, follow DKA fluid/insulin caution but expect a **bigger fluid requirement** and monitor osmolality and creatine kinase closely.

#### **WORKUP**

- Glucose, **venous blood gas**, beta-hydroxybutyrate/ketones.
- Electrolytes with **corrected sodium**, potassium, **phosphate, magnesium, calcium**; BUN/creatinine.
- **Calculated effective osmolality**; ECG + continuous cardiac monitoring.
- **Creatine kinase** — screen for rhabdomyolysis.

| Step              | Approach                                                        | Notes                                                     |
|-------------------|-----------------------------------------------------------------|-----------------------------------------------------------|
| Fluids first      | 0.9% saline; restore circulation, then replace deficit steadily | Aim to lower glucose gradually with fluid alone initially |
| Insulin (delayed) | 0.025–0.05 unit/kg/h once glucose plateaus on fluids            | No bolus; avoid rapid osmolar shifts                      |

| Step    | Approach                                                                                                      | Notes                                            |
|---------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Monitor | Osmolality,<br>K <sup>+</sup> , PO <sub>4</sub> ,<br>Ca <sup>2+</sup> ,<br>creatinine<br>kinase, neuro<br>obs | Anticipate<br>rhabdomyolysis and<br>hyperthermia |

#### ▲ ADMIT · REFER

- **Admit** to high-dependency/PICU.
- **Call** pediatric endocrinology and critical care.

## Adrenal Crisis / Salt-Wasting CAH

MINUTES

BSPED 2023

Shock + low sodium + high potassium + low glucose; collapse with illness.

+ Calculators: [Hyponatremia 3% saline](#) · [Maintenance fluids & bolus](#)

#### ▲ RECOGNIZE

- Known adrenal insufficiency/CAH under stress, or a neonate (1–4 weeks) with vomiting, poor feeding, weight loss, hyperpigmentation, ambiguous genitalia (in females).<sup>15</sup>

- Labs: low sodium, high potassium, metabolic acidosis, low glucose.

### ① STEROID – DON'T WAIT

Hydrocortisone immediately (draw 17-OHP/cortisol first only if no delay).<sup>243</sup>



### ② FLUIDS + GLUCOSE

Saline bolus, repeat; treat low glucose; maintenance with dextrose-saline.



### ③ ELECTROLYTES & CAUSE

High potassium usually corrects with fluids + steroid (ECG for peaked T waves). Treat the trigger (often infection); cover empirically for sepsis if suspected. Endocrine + PICU.

| Drug / fluid         | Dose                                          | Route & notes                                                              |
|----------------------|-----------------------------------------------|----------------------------------------------------------------------------|
| Hydrocortisone       | < 1 yr 25 mg<br>1–5 yr 50 mg<br>> 5 yr 100 mg | IV/IM, then 50–100 mg/m <sup>2</sup> /day divided every 6 h <sup>243</sup> |
| Sodium chloride 0.9% | 20 mL/kg                                      | IV bolus, repeat                                                           |
| Dextrose 10%         | 5 mL/kg                                       | IV, for low glucose                                                        |

| Drug / fluid     | Dose              | Route & notes                                     |
|------------------|-------------------|---------------------------------------------------|
| Fludro-cortisone | 0.1 mg once daily | Oral — maintenance only, not for the acute crisis |

### ⚠ DO NOT USE DEXAMETHASONE FOR SALT-WASTING CRISIS

- It has **no mineralocorticoid effect**.<sup>15</sup>
- Give hydrocortisone immediately — do not delay it for lab results.
- High-dose hydrocortisone (> 50 mg) covers the mineralocorticoid need acutely; fludrocortisone is added later for maintenance.<sup>15</sup>

### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — all → monitored bed; hydrocortisone before labs, treat hypoglycemia/hyponatremia.
- **Discharge** — only after stabilization, endocrine review, and stress-dosing/emergency-injection education.
- **Refer** — pediatric endocrinology.

# Hypoglycemia (Beyond the Newborn)

MINUTES

PES 2015

Plasma glucose < 3.9 mmol/L (70 mg/dL) — treat now; draw the critical sample first only when it will not delay treatment.<sup>81</sup>

+ Calculator: Glucose infusion rate

## RECOGNIZE

- **Autonomic:** sweating, tremor, tachycardia, hunger, pallor, anxiety.  
**Neuroglycopenic:** lethargy, irritability, poor feeding, behavior change, confusion, seizures, coma.
- **Whipple triad** confirms it: low plasma glucose + consistent symptoms + resolution with glucose. Confirm any bedside reading with a **laboratory plasma glucose** (glucometers are unreliable when low).<sup>81</sup>
- Red-flag context: hepatomegaly (glycogen storage / fatty-acid oxidation defect), midline defects or micropenis (hypopituitarism), hyperpigmentation (adrenal



insufficiency), fasting- or illness-triggered episodes, or a high glucose-infusion requirement.

**⚠ DRAW THE CRITICAL SAMPLE – BUT NEVER DELAY TREATING SEVERE OR SYMPTOMATIC HYPOGLYCEMIA**

Obtain the sample before glucose **whenever it can be drawn immediately**. If the child is **symptomatic or severely hypoglycemic, treat first** and take the sample as soon as access allows — do not delay treatment for the sample. At the moment of hypoglycemia (ideally glucose < 2.8 mmol/L): **lab glucose, venous gas/bicarbonate, insulin + C-peptide,  $\beta$ -hydroxybutyrate + free fatty acids** (the pivotal branch point), **lactate, ammonia, cortisol + growth hormone, acylcarnitine profile + carnitine, plasma amino acids**; and the **first-void urine** for ketones, organic acids  $\pm$  toxin/sulfonylurea screen. One missed sample can cost the diagnosis.<sup>81</sup>

**① CORRECT NOW**

**IV D10 2 mL/kg** (200 mg/kg) slow push — avoid large hyperosmolar boluses. If alert with a safe airway: oral rapid carbohydrate ~0.3 g/kg.<sup>81</sup>



## ② MAINTAIN

- Continue **D10 at maintenance** and titrate the **glucose-infusion rate** to keep glucose > 3.9 mmol/L (central line for > D12.5)
- A requirement > **8–10 mg/kg/min strongly suggests hyperinsulinism** (a hallmark<sup>81</sup> — confirm on the critical sample); maintain euglycemia with escalating dextrose as an initial measure until the endocrinology team advises.



**Glucagon 0.5–1 mg IM/IV/SC (or 0.03 mg/kg, max 1 mg) when there is no IV access or hyperinsulinism is suspected — a rise > 1.7 mmol/L (30 mg/dL) suggests hyperinsulinism. It fails in glycogen-depleted states (ketotic hypoglycemia, fatty-acid oxidation defects).<sup>81</sup>**

## ▴ READ THE KETONES/FREE-FATTY-ACIDS BRANCH

- **Low ketones + low free fatty acids** (non-ketotic) with detectable insulin → **hyperinsulinism** (congenital  $K_{ATP}$ , Beckwith-Wiedemann, exogenous insulin/sulfonylurea, dumping).<sup>81</sup>

- **Low ketones + high free fatty acids** (hypoketotic) → **fatty-acid oxidation defect** (e.g. VLCADD or LCHADD) — abnormal acylcarnitines ± hyperammonemia/cardiomyopathy.
- **High ketones + high free fatty acids** (appropriate) → **ketotic hypoglycemia of childhood** (commonest, 1–5 yr, illness-triggered, diagnosis of exclusion), **GH/cortisol deficiency**, or **glycogen storage disease** (hepatomegaly, high lactate).<sup>81</sup>

#### ▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — unexplained symptomatic or persistent hypoglycemia → until fasting safety is documented and a cause established.
- **Discharge** — only after a documented safe fast and a sick-day/emergency plan (home glucagon where indicated).
- **Refer** — pediatric endocrinology/metabolic; hyperinsulinism or a glucose infusion

rate > 8–10 mg/kg/min → **consult endocrinology with congenital hyperinsulinism (CHI) expertise.**

## Acute Electrolyte Emergencies

### MINUTES

Stabilize the heart, stop the seizures, then correct at a safe rate.

✚ Calculators: Hyponatremia 3% saline · Corrected Na · anion gap · Ca · osmolality

### ▲ HYPERKALEMIA – GET AN ECG FIRST

- **Hyperkalemia** = serum  $K^+$  > **5.5 mmol/L** (mild 5.5–6.0 · moderate 6.0–6.5 · severe > **6.5** or ECG changes); **hypokalemia** =  $K^+$  < **3.5 mmol/L** (severe < 2.5).<sup>244</sup>
- ECG progression: peaked T waves → flattened P / prolonged PR → widened QRS → sine wave → arrest. Treat urgently if there are **ECG changes or potassium > 6.5 mmol/L**.<sup>244</sup>
- If the child is well with a normal ECG, repeat a **free-flowing venous sample** first — hemolysis causes

pseudohyperkalemia.<sup>244</sup> **Stop all potassium intake** and potassium-raising drugs.

## HYPERKALEMIA – STABILIZE, SHIFT, REMOVE

| Step / drug                            | Dose                                                                         | Route & notes                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STABILIZE ·<br>Calcium gluconate 10%   | 0.5 mL/kg (max 20 mL) IV over 5–10 min                                       | Cardiac monitor; protects the heart, does <b>not</b> lower potassium. Repeat after 15 min if ECG still abnormal. <b>Not in the same line as bicarbonate</b> (precipitates); caution in digoxin toxicity <sup>244</sup> |
| SHIFT ·<br>Insulin (soluble) + glucose | 0.1 unit/kg (max 10 units) + glucose 0.5 g/kg IV (= 5 mL/kg of dextrose 10%) | Onset 10–20 min; <b>monitor glucose</b> closely — early and late hypoglycemia <sup>244</sup>                                                                                                                           |
| SHIFT ·<br>Salbutamol (nebulized)      | 2.5–5 mg (high dose 10–20 mg)                                                | Additive to insulin; may cause tachycardia <sup>244</sup>                                                                                                                                                              |
| SHIFT ·<br>Sodium bicarbonate          | 1 mmol/kg IV                                                                 | Only if acidotic; modest, adjunctive effect                                                                                                                                                                            |

| Step / drug                  | Dose                                      | Route & notes                                                                                                                                                         |
|------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REMOVE<br>• Potassium binder | 0.5–1 g/kg<br>(max 30 g)<br>oral / rectal | Sodium/calcium polystyrene sulfonate; slow onset — not for immediately life-threatening hyperkalemia; avoid with ileus, bowel obstruction, or NEC risk <sup>244</sup> |
| REMOVE<br>• Furosemide       | 1 mg/kg IV                                | If making urine                                                                                                                                                       |
| REMOVE<br>• Dialysis         | –                                         | Refractory or severe; involve renal / PICU early                                                                                                                      |

#### ◆ THE LOGIC

- Calcium and the shifters (insulin, salbutamol, bicarbonate) only **BUY TIME**; only binders, loop diuretics, and dialysis actually remove potassium from the body.<sup>244</sup>
- Recheck potassium and glucose after treating.

## SODIUM – CORRECT AT A SAFE RATE

◎ **DEFINITIONS** Hyponatremia =  $\text{Na}^+ < 135$  **MMOL/L** (severe  $< 125$ ); hypernatremia =

$\text{Na}^+ > 145 \text{ MMOL/L}$  ( $> 150$  in hypernatremic dehydration).<sup>99</sup>

| Problem                                     | Treatment                                                                                  | Notes                                                                                                                                                                                                   |
|---------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hyponatremia with seizures / encephalopathy | 3% saline 2 mL/kg (max 100 mL) IV over 10 min; repeat once or twice until symptoms improve | Each bolus raises sodium by about 2 mmol/L; aim for a <b>5–6 mmol/L</b> rise in the first 1–2 h, then stop boluses — no more than 10 mmol/L in the first 5 h and 15–20 mmol/L in 48 h <sup>347,99</sup> |
| Hyponatremia (no severe symptoms)           | Treat the cause; correct slowly                                                            | Do <b>not</b> exceed <b>8 mmol/L per 24 h</b> (osmotic demyelination). Volume depletion → 0.9% saline; SIADH → fluid restriction <sup>99</sup>                                                          |
| Hypernatremic dehydration                   | Resuscitate shock with isotonic fluid first, then rehydrate over ~48 h                     | Lower sodium $\leq 0.5 \text{ mmol/L per hour}$ ( $\leq 10\text{--}12 \text{ per 24 h}$ ) — faster risks cerebral edema <sup>99</sup>                                                                   |

## CALCIUM & MAGNESIUM

⊙ **DEFINITIONS** Hypocalcemia = total calcium  $< 2.1 \text{ MMOL/L}$  or ionized  $< 1.1 \text{ MMOL/L}$  (neonate  $< 2.0 \text{ mmol/L}$ ; correct total calcium for albumin). Hypomagnesemia =  $\text{Mg}^{2+} < 0.7 \text{ MMOL/L}$  (blocks correction of low  $\text{K}^+$  and  $\text{Ca}^{2+}$ ).

| Problem                                  | Treatment                                                                 | Notes                                                                                                          |
|------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Symptomatic hypocalcaemia                | Calcium gluconate 10% 0.5–1 mL/kg slow IV, cardiac monitor; then infusion | Tetany, seizures, prolonged QT, Chvostek / Trousseau signs. Avoid extravasation (tissue necrosis) <sup>1</sup> |
| Refractory hypocalcaemia or hypokalaemia | Check and replace magnesium                                               | Low magnesium blocks correction of both — replace it first <sup>244</sup>                                      |

### ⚠ THE SAFE-RATE RULE

Move fast to **stabilize the heart** (calcium) and **stop seizures** (3% saline), but correct chronic sodium **slowly**: overcorrecting low sodium causes osmotic demyelination, and dropping high sodium too fast causes cerebral edema.<sup>99</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic or severe derangements, or those needing infusions/monitoring → high-dependency/PICU.
- **Discharge** — mild, asymptomatic, correctable causes, with a follow-up recheck.



- **Refer** — nephrology/endocrinology by cause; PICU for arrhythmia risk or rapid shifts.

## Hypercalcemia HOURS

Corrected calcium > 2.6 mmol/L (10.5 mg/dL);  
> 3.5 mmol/L (14 mg/dL) is a crisis.<sup>245</sup>

**+ Calculator:** Corrected Na · anion gap · Ca  
· osmolality

### RECOGNIZE

- **Stones, bones, groans & psychiatric moans:** nephrocalcinosis/stones; bone pain; constipation, nausea, anorexia, pancreatitis; lethargy, confusion, weakness, hypotonia, coma. Infants: poor feeding, failure to thrive, irritability.<sup>245</sup>
- **Polyuria → dehydration** (nephrogenic diabetes insipidus) — vicious cycle worsening the calcium. Cardiac: **short QT**, bradyarrhythmia; monitor ECG when severe/symptomatic.<sup>245</sup>
- **Correct for albumin:** add 0.8 mg/dL per 1 g/dL albumin below 4.0; confirm with **ionized calcium**.<sup>245</sup>

- Mild 2.6–3.0
- moderate 3.0–3.5
- severe > 3.5 mmol/L.

#### **FIND THE CAUSE – PTH IS THE BRANCH POINT**

- **High or inappropriately normal PTH**  
→ primary hyperparathyroidism (think MEN) or **familial hypocalciuric hypercalcemia** (low urine Ca/Cr — benign, do **not** operate).<sup>245</sup>
- **Suppressed PTH** → malignancy (PTHrP/osteolysis), vitamin D intoxication, granulomatous disease, immobilization, thiazides, milk-alkali. Send PTHrP, 25-OH & 1,25-(OH)<sup>245</sup><sub>2</sub> vitamin D, phosphate, magnesium, urine Ca/Cr.
- **Infants:** subcutaneous fat necrosis, Williams syndrome, idiopathic infantile hypercalcemia, neonatal severe hyperparathyroidism.

## LOWER THE CALCIUM – HYDRATE, THEN REMOVE

| Step / drug               | Dose                                                                     | Route & notes                                                                                                                                                 |
|---------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stop intake               | –                                                                        | Stop all calcium & vitamin D (diet, fluids, supplements); stop thiazides, lithium, vitamin A/D                                                                |
| 0.9% saline (cornerstone) | 10–20 mL/kg bolus, then ~2–3× maintenance (~3000 mL/m <sup>2</sup> /day) | IV; restores volume then drives calciuresis; target brisk urine output <sup>245</sup>                                                                         |
| Furosemide                | 1 mg/kg IV                                                               | <b>Only after volume repletion</b> ; for fluid overload; monitor K <sup>+</sup> /Mg <sup>2+</sup>                                                             |
| Calcitonin (salmon)       | 4 IU/kg SC/IM every 6–12 h (up to 8 IU/kg q6h)                           | Rapid onset (2–6 h), modest; <b>tachyphylaxis by 48 h<sup>245</sup></b> — a bridge. <i>Salmon</i> is the potent form used clinically (≈40× human calcitonin). |

| Step / drug                        | Dose                                                                                                      | Route & notes                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Zoledronic acid (preferred)</b> | <b>0.025–0.05 mg/kg/day IV (max 4 mg); 4 mg/5 mL vial, dilute 1:20 in 0.9% saline, infuse over 15 min</b> | Preferred bisphosphonate for high bone turnover (malignancy, immobilization) — higher response rate (complete response ~88% vs ~70%) and a much shorter infusion (15 min vs 2–4 h) than pamidronate (Major, <i>J Clin Oncol</i> 2001). <sup>207</sup> Onset 24–48 h; ensure volume replete; watch for hypocalcemia; reduce/withhold in renal impairment |
| <b>Pamidronate (alternative)</b>   | <b>0.5–1 mg/kg IV over 2–4 h</b>                                                                          | Alternative where zoledronic acid is unavailable; same indications <sup>207</sup> ; onset 24–48 h; ensure volume replete; watch for hypocalcemia                                                                                                                                                                                                        |
| <b>Glucocorticoid</b>              | <b>Prednisone 1–2 mg/kg/day</b>                                                                           | For vitamin D-mediated, granulomatous disease, or lymphoma/leukemia <sup>245</sup>                                                                                                                                                                                                                                                                      |
| <b>Dialysis</b>                    | <b>–</b>                                                                                                  | Severe/refractory, renal failure, or cannot tolerate a saline load                                                                                                                                                                                                                                                                                      |

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — moderate–severe, symptomatic, or  $> 3.5$  mmol/L → monitored bed/PICU with ECG.
- **Discharge** — mild & asymptomatic with an outpatient workup and offending agents stopped.
- **Refer** — pediatric endocrinology ± oncology/nephrology; parathyroidectomy for primary hyperparathyroidism (not for FHH).

## Thyroid Storm (Thyrotoxic Crisis)

MINUTES

ATA 2016 · BWPS

Decompensated hyperthyroidism: fever, tachyarrhythmia and agitation, often with a trigger.

### ▲ RECOGNIZE

- Known/undiagnosed Graves' disease with **hyperthermia**, marked **tachycardia or tachyarrhythmia** (atrial fibrillation/heart failure), agitation or reduced consciousness,

and GI upset — usually with a trigger (infection, surgery, DKA, medication non-adherence).

### ⚠ **TREAT ON CLINICAL SUSPICION**

- Do not wait for thyroid function results.
- Provide supportive care (cooling, fluids, treat the trigger) and block the thyroid axis in the right **order** —  $\beta$ -blockade and a thionamide first, and give **iodine only after** the thionamide to avoid fuelling hormone synthesis.<sup>246</sup>

### ▴ **WORKUP**

- **TSH (suppressed), free T4, free T3, TRAb/TSI.**
- CBC, LFTs, glucose, electrolytes, calcium; **ECG.**
- Diagnosis is **clinical** (Burch-Wartofsky<sup>246</sup>  $\geq 45$  highly suggestive) — do not delay treatment for labs.

| Step / drug                                   | Dose                                                                | Route & notes                                                                                                                                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 · Propranolol (β-block)                     | 0.25–0.5 mg/kg/dose PO/NG q6h or 0.01–0.05 mg/kg IV (max 1 mg/dose) | Controls heart rate, tremor and agitation; slows peripheral T4→T3. Caution in heart failure — treat failure alongside <sup>246</sup>                                                                    |
| 2 · Methimazole (thionamide)                  | 0.5–1 mg/kg/day PO/NG/PR divided q8h (max 30 mg/day)                | Blocks new hormone synthesis; give the thionamide <b>before</b> iodine. Carbimazole is the equivalent prodrug <sup>246</sup>                                                                            |
| 2 · PTU (thionamide — alternative)            | 5–10 mg/kg/day PO/NG divided q8h                                    | Also blocks peripheral T4→T3. <b>Either thionamide is acceptable</b> — no difference in mortality (Lee, <i>JAMA Netw Open</i> 2023); ATA 2016 has historically listed PTU first in storm <sup>208</sup> |
| 3 · Potassium iodide (strong iodine solution) | 1–5 drops PO/NG q8h                                                 | Blocks hormone <b>release</b> <sup>246</sup> ; give ≥ 1 h <b>after</b> the thionamide                                                                                                                   |
| 4 · Hydrocortisone                            | 1–2 mg/kg/dose IV q6h (or 50–100 mg/m <sup>2</sup> /day)            | Reduces peripheral T4→T3 conversion and covers relative adrenal insufficiency <sup>246</sup>                                                                                                            |

### ▴ ADMIT · REFER

- **Admit** to PICU/HDU.
- **Call** pediatric endocrinology and critical care.

## Metabolic Crisis (Inborn Error)

MINUTES

ANY AGE

Suspected inborn error of metabolism —  
"sepsis that doesn't fit."

+ Calculators: Glucose infusion rate ·  
Corrected Na · anion gap · Ca · osmolality

### ▴ RECOGNIZE (THINK INBORN ERROR)

- Encephalopathy, vomiting, seizures, or apnea after a symptom-free interval or a catabolic trigger (fasting, intercurrent illness, surgery); poor feeding; tachypnea. Family history or parental consanguinity.<sup>247</sup>
- Mimics sepsis — "sepsis that doesn't fit." Presents at **any age**, not only the newborn. Not every metabolic crisis is hyperammonemic — the pattern points to the group:<sup>247</sup>



| Pattern                           | Acid-base · ammonia                                                   | Pointers                                                              |
|-----------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Urea-cycle (hyperammonemia)       | Respiratory alkalosis; very high ammonia; low urea <sup>247</sup>     | Little or no acidosis; encephalopathy ± seizures                      |
| Organic acidemia (MMA / PA / IVA) | High-anion-gap metabolic acidosis + ketosis; ammonia high (secondary) | Hypoglycemia; offending acid on urine organic acids <sup>247,13</sup> |
| MSUD                              | Ketosis with normal / near-normal ammonia; little or no acidosis      | Maple-syrup odor (urine/cerumen); high plasma leucine <sup>247</sup>  |

#### ① STOP CATABOLISM

**Stop all protein — a temporary emergency measure (first ~24–48 h) until the metabolic team advises;** prolonged protein restriction itself worsens catabolism, so protein is reintroduced on their guidance. Run high-dextrose fluids; avoid hypotonic fluids and lactated Ringer's.<sup>247</sup> Treat low glucose. **This card is initial stabilization pending specialist consultation.**



② IMMEDIATE LABS + CALL METABOLIC TEAM

- Ammonia (free-flowing sample, **on ice, run STAT and process immediately**)<sup>247</sup>, blood gas, glucose, lactate, ketones, electrolytes/anion gap.
- Send as **confirmatory** (not routine/STAT): plasma amino acids, urine organic acids, acylcarnitines.

high ammonia

③ NITROGEN SCAVENGERS + COFACTORS

See drug table — coordinate with the metabolic service.

ammonia > ~500 / rising / coma

**Urgent hemodialysis or CRRT. Transfer to a metabolic center.**<sup>247</sup>

| Drug / measure                         | Dose                                                                 | Route & notes                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dextrose 10%                           | 1.5× maintenance; glucose infusion rate 8–10 (up to 15–20) mg/kg/min | IV — reverses catabolism                                                                                                                                                                                                                                                                                                       |
| Sodium benzoate + sodium phenylacetate | 250 mg/kg load over 90–120 min, then 250 mg/kg/day                   | IV; central line preferred, <b>peripheral acceptable when life-saving</b> — ask pharmacy for an appropriately diluted preparation. <sup>247</sup> In <b>MMA / PA</b> , phenylacetate/phenylbutyrate is <b>not</b> the primary scavenger (depletes glutamine) — benzoate ± carglumic acid per the metabolic team. <sup>13</sup> |
| L-arginine hydrochloride               | 200–600 mg/kg                                                        | IV load then per day; omit in arginase deficiency <sup>247</sup>                                                                                                                                                                                                                                                               |

| Drug / measure | Dose                                                                                                                                      | Route & notes                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Carglumic acid | Acute 100–250 mg/kg/day (2–4 divided doses), then maintenance ~50 mg/kg/day (range 10–100) in 2 divided doses – titrate to plasma ammonia | Oral/NG; for NAGS deficiency & some organic acidemias <sup>247,13</sup> |
| L-carnitine    | 100 mg/kg                                                                                                                                 | IV / oral; for suspected organic acidemia                               |

#### ◆ AVAILABILITY

- Nitrogen scavengers, carglumic acid and L-carnitine are **SPECIALIST STOCK** held in tertiary metabolic / PICU units, not general wards.
- Phone the metabolic on-call **EARLY** — high-dextrose fluids, protein cessation and dialysis access can start anywhere while drugs are sourced. **AVOID VALPROATE.**<sup>247</sup>

#### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected → PICU; stop protein, high-dose dextrose, treat hyperammonemia.

- **Discharge** — not acutely; under the metabolic team with an emergency regimen.
- **Refer** — metabolic/genetics urgently and a center capable of ammonia scavengers/hemofiltration.

## Focused Specialty

07

## Emergencies

Focused eye, dental, dermatologic, ENT & neurosurgical emergencies

### Pediatric Eye Emergencies

MINUTES

#### SIGHT-THREAT

Chemical burn — irrigate before anything else. Leukocoria, orbital cellulitis, and a ruptured globe are the sight- and life-threats not to miss.

#### ⚠ **CHEMICAL INJURY – IRRIGATE FIRST**

- Immediate, copious irrigation with **saline (or clean water)** for **≥ 15–30 min BEFORE** examination or imaging; evert the lids, sweep out particulate matter, recheck pH

until neutral ( $\approx 7.0-7.5$ ). **Alkali** (drain/oven cleaner, cement, wet button batteries) penetrates deeper than acid.<sup>188</sup>

- Then urgent ophthalmology.

### ▶ THE CAN'T-MISS DIAGNOSES

- **Orbital (post-septal) cellulitis** — **proptosis, pain on or limited eye movement, reduced acuity or color vision, a relative afferent pupillary defect**, chemosis → emergency. **CT orbits + sinuses, IV antibiotics, ophthalmology + ENT**. Distinguish from **preseptal** (lid swelling only, with **normal** movement, vision, and pupil)<sup>188</sup> (see [Preseptal & Orbital Cellulitis](#)).
- **Ruptured / penetrating globe** — after trauma with a **distorted (teardrop) pupil**, prolapsed dark uveal tissue, or marked subconjunctival hemorrhage → cover with a **rigid shield (never a pad or any pressure)**, keep **NPO**, give analgesia + antiemetic, update tetanus, and refer urgently. **No drops, no ointment.**<sup>188</sup>

- **Leukocoria (white pupil reflex)**, a new squint, or an **absent red reflex** → **retinoblastoma** until proven otherwise → urgent ophthalmology (a life-threatening tumor).<sup>188</sup>
- **Ophthalmia neonatorum** — hyperacute purulent conjunctivitis in the **first days of life** → gonococcal risk (corneal perforation) → emergency: **systemic ceftriaxone or cefotaxime is the main treatment** (adequate systemic therapy makes topical antibiotics unnecessary; prefer cefotaxime in the jaundiced neonate or with calcium-containing IV fluids), frequent saline irrigation of discharge, admit and swab; **test for both *N. gonorrhoeae* and *C. trachomatis***.<sup>158</sup>
- **Acute red painful eye** with photophobia/reduced vision, or **hyphema** (blood layering in the anterior chamber after trauma) → same-day ophthalmology.
- **Corneal ulcer (microbial keratitis)** — red, painful, photophobic eye with a **white/opaque corneal infiltrate or haze** ± discharge; high risk in **contact-lens** wearers and after

trauma → sight-threatening,  
same-day ophthalmology. **Do not  
patch**; stop contact-lens wear.<sup>188</sup>

- **Congenital / infantile glaucoma** —  
the infant triad of **watering  
(epiphora), photophobia, and  
blepharospasm** with an **enlarged or  
cloudy cornea (buphthalmos)** →  
urgent ophthalmology (needs  
surgery); do not dismiss as a  
blocked tear duct.<sup>188</sup>
- **New anisocoria (unequal pupils)** —  
may signal a **third-nerve palsy,  
Horner syndrome, or raised ICP /  
herniation** (see [Raised ICP](#)); with  
reduced consciousness, ptosis, or  
trauma it is a red flag → urgent  
assessment. Check pupils in **both  
light and dark** to localize.<sup>188</sup>

#### **⚠ SAFEGUARDING – RETINAL HEMORRHAGES**

Multilayered retinal hemorrhages  
extending to the periphery in an infant  
with encephalopathy suggest **abusive  
head trauma** — involve safeguarding and  
ophthalmology (see [Fractures —  
Non-Accidental Red Flags](#)).

✦ **PEARL – CHECK THE RED REFLEX IN EVERY**



## CHILD

A symmetrical red reflex helps exclude retinoblastoma, cataract, and vitreous pathology; any **white, asymmetric, or absent** reflex needs urgent ophthalmology.<sup>188</sup>

### ✦ YOUR ROLE – THE GENERAL PEDIATRICIAN

- You are the first responder, not the eye surgeon.
- Three time-critical moves are yours: **(1) irrigate immediately** for any chemical splash — don't wait for ophthalmology; **(2) shield** (never pad or press) a suspected ruptured globe, keep **NPO**, give analgesia + antiemetic; **(3) check the red reflex in every child** and treat leukocoria / an absent reflex as retinoblastoma.
- Document **visual acuity, pupils, and eye movements** before referral. **Avoid drops, ointment, or any pressure** on a possibly open globe, and never patch a suspected corneal ulcer.
- Then refer to ophthalmology (add ENT for orbital cellulitis) within the timeframe the diagnosis demands.

## REFER

- **Ophthalmology urgently** — chemical injury, suspected globe rupture, leukocoria/absent red reflex, orbital cellulitis (add **ENT**), acute vision loss, or hyphema.

## Dental & Oral Emergencies

MINUTES

### TIME-CRITICAL

An avulsed permanent tooth is a true emergency — reimplant within minutes. Never reimplant a primary tooth. Watch the swelling that threatens the airway.

## DENTAL TRAUMA

- **Luxation** (displaced or mobile tooth) — **do not attempt to reposition it**; analgesia and same-day dental referral.<sup>178</sup>
- **Crown fracture** — preserve any fragment and keep it moist (milk or saline); analgesia; prompt dental review — pulp exposure needs same-day care.
- **Primary tooth avulsion** — **do NOT reimplant** (risks of damaging the permanent successor); reassure and

arrange dental review.<sup>178</sup>

- **Post-extraction bleeding** — firm gauze pressure 15–20 min ± tranexamic acid; if it persists, consider an undiagnosed **bleeding disorder**.

### ⚠ **AVULSED PERMANENT TOOTH – MINUTES MATTER**

- **Reimplant as soon as possible.** Hold the tooth by the **crown** — **never touch or scrub the root** (protect the periodontal ligament).
- If dirty, rinse ~10 s in **cold milk or saline**, reinsert, have the child bite on gauze.
- If reimplantation isn't possible, transport the tooth in **Hank's solution or cold milk** — **never water**.<sup>178</sup>
- Then urgent dental care for splinting; **assess tetanus prophylaxis** — indicated for a contaminated avulsion and guided by both the child's immunization status and the nature of the wound; consider antibiotics.
- Prognosis falls sharply once **extra-oral dry time exceeds ~60 min**, and periodontal-ligament

cells are then considered non-viable — **reimplant anyway.**<sup>178</sup>

#### ✦ PEARL – MILK BEATS WATER

If Hank's solution isn't to hand, **cold milk** is the best readily-available storage medium; tap water is hypotonic and kills periodontal-ligament cells, worsening the outcome.<sup>178</sup> **Time is tooth.**

#### ▴ PRIMARY HERPETIC GINGIVOSTOMATITIS (HSV-1)

- The commonest cause of a painful mouth with fever in the **1–5-year-old: fever** (usually **3–5 days**, occasionally up to a week), **swollen, friable, bleeding gingiva**, and crops of **painful vesicles → ulcers** on the gingiva, tongue, lips, and buccal mucosa (often with perioral spread), drooling, and **refusal to eat or drink**. Tender **cervical lymphadenopathy** is usual and, being reactive, resolves as the illness settles. Self-limited — the illness runs **~10–14 days.**<sup>195</sup>
- **Management is supportive** — the priority is **hydration and analgesia** (acetaminophen ± ibuprofen); offer cold, bland fluids. **Dehydration from**

**poor intake is the main reason to admit.** Avoid topical lidocaine/anesthetic gels in young children (toxicity).<sup>194</sup>

- **Oral acyclovir — 15–20 mg/kg/dose 4×/day for 5–7 days** (max single dose 800 mg); immunocompromised children **20 mg/kg/dose 4×/day for 7–10 days**; immunocompromised adolescents 400 mg 3×/day for 10 days or until resolution.<sup>194</sup>

**Significant or disseminated disease in an immunocompromised child needs IV acyclovir** (5–10 mg/kg/dose q8h) rather than a prolonged oral course. Shortens fever, oral lesions, and viral shedding **only if started within 72 h of symptom onset**; not routine for the well child presenting late.

- Distinguish from **herpangina** (coxsackievirus — ulcers on the soft palate/tonsillar pillars, **gingiva spared**) and hand-foot-and-mouth disease.<sup>195</sup> **Admit/refer** the dehydrated, the immunocompromised, or suspected **eczema herpeticum** / disseminated HSV.

## ▶ ORAL SOFT-TISSUE INJURIES

- **Tongue lacerations** — most heal without sutures (rich blood supply). **Repair** if the wound gapes at rest, is large ( $> \sim 1\text{--}2\text{ cm}$ ), bleeds persistently, is through-and-through, or splits the free edge/tip (a bisecting flap) — these risk a notched tongue.
- **Lip lacerations** — a wound that **crosses the vermilion border** needs precise realignment of the border first (a 1 mm step is visible); through-and-through wounds need layered closure. Palpate and image for an **embedded tooth fragment** or foreign body after a fall onto the teeth.
- **Buccal & gingival mucosa** — most intraoral tears heal spontaneously; suture only large, gaping, or food-trapping wounds. A deep cheek wound can injure the **parotid duct or facial nerve** — refer.
- **Embedded foreign bodies** — account for **every tooth fragment**: a missing piece may be buried in the lip/tongue (soft-tissue radiograph or ultrasound) or aspirated (**chest**

**X-ray if unaccounted for).** Remove grit/glass, irrigate, and assess tetanus status.

- **Intruded tooth** (driven into the socket) — **refer to a pediatric dentist.** A **primary** intruded tooth is usually left to **re-erupt spontaneously** (do not reposition — protect the permanent tooth bud); a **permanent** intruded tooth needs dental assessment for repositioning.<sup>178</sup>
- Repair intraoral and lip wounds within ~24 h with analgesia; give antibiotics for through-and-through lip wounds, bites, or grossly contaminated wounds.

#### ▀ **DENTOALVEOLAR ABSCESS / FACIAL SWELLING**

- Analgesia and **dental source control** — **incision and drainage, root canal (pulp) treatment, or extraction of the causative tooth** — with a **mandatory dental referral.** For a **localized** abscess without systemic involvement or spreading, definitive dental treatment is the primary management and **systemic antibiotics are generally not required; reserve antibiotics for**

**facial cellulitis, systemic signs, significant spreading, or suspected deep-space infection** (per clinical judgment and MOH protocols). These commonly present **directly to dental clinics**, where the dentist identifies the source and delivers timely definitive treatment.<sup>194</sup> **Watch for spread to the deep neck, floor of mouth, or orbit** — a spreading facial-space infection can progress to Ludwig's angina (next).

#### ⚠ **AIRWAY – LUDWIG'S ANGINA**

Firm submandibular swelling — in children often **unilateral at onset** before becoming bilateral — with a **raised (elevated) tongue**, drooling, or trismus is a deep-space infection that can obstruct the airway. **Call for airway help early — anesthesia, oral & maxillofacial surgery, ENT, or pediatric surgery** (all are equipped to manage these cases), give IV antibiotics, and do **not** lay the child flat.<sup>196</sup> **Time-critical — urgently call or refer.**

#### ✦ **YOUR ROLE – THE GENERAL PEDIATRICIAN**

- Most of what reaches you is yours to start.



- For an **avulsed permanent tooth**, reimplant or store it in **cold milk** and send urgently — minutes decide the tooth; **never reimplant a primary tooth**.
- Give analgesia, control post-extraction bleeding with pressure ± tranexamic acid, **assess tetanus** for contaminated wounds, and suture only the soft-tissue lacerations that need it — account for **every tooth fragment**.
- Treat herpetic gingivostomatitis supportively, watching hydration.
- Escalate the two true emergencies fast: a spreading facial-space infection or **Ludwig's angina** (airway — call/refer anesthesia, oral & maxillofacial surgery, ENT, or pediatric surgery), and refer to dental / maxillofacial for avulsion, pulp exposure, or abscess.

#### **REFER**

- **Dental / maxillofacial urgently** — avulsion, fractures with pulp exposure, facial fractures, or a facial-space infection; refer an **intruded tooth to a pediatric dentist**;

for any airway threat call **anesthesia, oral & maxillofacial surgery, ENT, or pediatric surgery** — all are equipped to manage these cases.

## Dermatologic Emergencies HOURS

### CAN'T-MISS

Fever + rash — find the killers:  
meningococcemia / purpura fulminans,  
SJS/TEN, staph scalded skin, and eczema  
herpeticum.

#### ⚠️ **NON-BLANCHING RASH + FEVER**

- Petechiae or purpura that **do not blanch** in an unwell/febrile child = **invasive meningococcal disease / sepsis / purpura fulminans** until proven otherwise — take blood cultures, give **empiric IV antibiotics immediately**, follow the **Septic Shock** pathway.<sup>119</sup>
- Do not wait for investigations.

## ▶ THE CAN'T-MISS RASHES

- **SJS / TEN** — dusky/targetoid macules with skin detachment and **erosions of  $\geq 2$  mucosal sites** (eyes, mouth, genitals), 1–3 weeks after a **drug** (antibiotics, anticonvulsants).<sup>181</sup>  
**Stop the culprit**; fluids, skin + eye care, urgent dermatology + ophthalmology, transfer to a **burns/PICU** unit. **Avoid the culprit drug and relevant cross-reacting agents** — broad beta-lactam avoidance applies mainly when a beta-lactam is the suspected culprit (see [Antibiotic Allergy](#)).
- **Staphylococcal scalded skin syndrome (SSSS)** — young child, fever, **tender diffuse erythema with superficial peeling (Nikolsky +)** and perioral crusting; **mucosae spared**.  
Toxin-mediated: IV anti-staphylococcal antibiotic (**cefazolin** first-line in Kuwait, as flucloxacillin is not available locally) + analgesia + fluids.<sup>184</sup>
- **Eczema herpeticum** — clusters of **monomorphic punched-out vesicles/erosions on eczematous skin** in an unwell child → start **acyclovir** urgently; ophthalmology if periocular. (HSV superinfection.)<sup>185</sup>

- **DRESS** — fever, widespread morbilliform rash, **facial edema**, lymphadenopathy, **eosinophilia**, and organ (liver) involvement 2–8 weeks after a drug → stop the drug, supportive care ± steroids.<sup>186</sup>
- **Necrotizing fasciitis** — **pain out of proportion**, rapidly spreading dusky/edematous skin, systemic toxicity → surgical emergency + broad-spectrum antibiotics.<sup>187</sup>

◆ SJS / TEN – FOCUSED WORKUP & MANAGEMENT

- **Confirm the signs** — dusky/targetoid macules with **skin detachment** and **erosions of  $\geq 2$  mucosal sites**; **Nikolsky +** (lateral shearing pressure detaches epidermis on non-blistered skin), **Asboe–Hansen +** (a blister extends sideways with slight tangential pressure), and **widespread skin tenderness** — typically 1–3 weeks after a culprit drug.<sup>181</sup>
- **Prognosis** — calculate **SCORTEN** (best at day 1 and day 3). **In children it over-estimates mortality** (pediatric death rates are far lower), so use it for guidance, not as a ceiling.<sup>181</sup>
- **Investigations** — CBC with differential, urea & electrolytes, LFTs, glucose, bicarbonate, coagulation profile, CRP/ESR, magnesium, phosphate, lactate, urinalysis. **Sepsis is the leading cause of death** — infective screen (blood cultures; skin swabs for **HSV/VZV and bacteria**) and treat superadded infection. **Chest X-ray**;

***Mycoplasma* testing guided by the clinical picture** (in children, **reactive infectious mucocutaneous eruption [RIME]** — the current term, which subsumes the former *Mycoplasma*-induced rash and mucositis [MIRM] — is a distinct, milder mimic with **prominent mucositis and little skin loss**; RIME can also follow other respiratory pathogens, so test by presentation rather than routinely).<sup>209</sup> **Skin biopsy** if the diagnosis is unclear.

- **Act now — stop the culprit drug immediately** (earlier withdrawal improves survival); **transfer to a burns unit or PICU**.<sup>182</sup>
- **Supportive care is the mainstay** — IV fluids & electrolytes, **thermoregulation**, nutrition, **analgesia; gentle handling, minimize friction/shear; non-adherent dressings** (petrolatum-impregnated gauze or silicone/advanced dressings); avoid **prophylactic** systemic antibiotics — treat proven infection. Early multidisciplinary team: **dermatology**,

**ophthalmology** (prevent ocular scarring; amniotic-membrane grafting in severe eye disease), **urology/gynecology** (prevent genital adhesions), wound care, pain, psychology.<sup>182</sup>

- **Systemic immunomodulation — specialist decision, low-certainty evidence.** Drug withdrawal + supportive care remain the priority. If an agent is used, **ciclosporin** and **etanercept (anti-TNF)** carry the most promising (still low-certainty) mortality signals; **corticosteroids** are controversial; **IVIG monotherapy is not supported** (used, if at all, combined with corticosteroids). Individualized by the specialist team.<sup>183</sup>
- **After recovery** — screen for **ocular, cutaneous, urogenital and psychological** sequelae; document the culprit, give a clear **allergy alert** (avoid re-exposure and cross-reacting agents), and consider **HLA-guided avoidance** where

relevant (e.g. HLA-B\*15:02 & carbamazepine, HLA-B\*58:01 & allopurinol).<sup>181</sup>

**Evidence:** BAD children & young people  
SJS/TEN guideline 2018; European S3 guideline 2024; RIME/MIRM terminology per Damari et al., *J Hosp Med* 2025. Culprit-drug withdrawal + supportive care have the clearest benefit; all systemic immunomodulators remain low-certainty.



◆ SSSS (STAPHYLOCOCCAL SCALDED SKIN)

– FOCUSED WORKUP & MANAGEMENT

- **Recognize** — young child, fever, tender diffuse erythema with **superficial peeling (Nikolsky +); desquamation of the flexures and radial fissuring/crusting around the eyes, mouth and nose; mucosae are spared.**

Toxin-mediated (exfoliative toxin from a distant staphylococcal focus).<sup>184</sup>

- **Cultures** — **swab the focus, not the blisters.**

Nasopharynx/nares, conjunctivae, throat, umbilicus, urine, blood, and any suspected primary site; in a **neonate** include the **umbilical stump and circumcision site. Bulla fluid is sterile** (unlike bullous impetigo) — do not culture it. **Skin biopsy is usually unnecessary** — reserve it (or a frozen section) for when SJS/TEN cannot be excluded (SSSS splits high in the granular layer; TEN is full-thickness).<sup>184</sup>

- **Antibiotics** — start an **IV anti-staphylococcal** agent: **cefazolin** is first-line in Kuwait

(the penicillinase-resistant penicillins — flucloxacillin, nafcillin, oxacillin — are the international first-line but are not available locally), switching **IV** → **oral** once improving; **clindamycin** only as an **anti-toxin** adjunct where susceptible (added benefit uncertain in children, resistance rising); **vancomycin** if MRSA is suspected/prevalent or first-line fails. Doses below — de-escalate once afebrile and per sensitivities.<sup>184</sup>

- **Supportive care** — treat like a **superficial burn**: fluid & electrolyte balance, thermoregulation, nutrition, **analgesia**, and **minimal handling**. **Wound care** with **petrolatum/Vaseline-impregnated gauze** (apply to the dressing — do not smear emollient directly onto raw skin); **warm compresses** to facial crusting. Watch for fluid loss and secondary sepsis.
- **Prognosis** — good in children with prompt treatment (low mortality); higher risk in

neonates and with extensive involvement.<sup>184</sup>

| Antibiotic                                                      | Dose (pediatric)                                     | Route & notes                                                                                                               |
|-----------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Cefazolin</b> (first-line IV)                                | 25–33 mg/kg/dose q8h (max 1 g/dose)                  | IV; first-generation cephalosporin — anti-staphylococcal first-line <b>in Kuwait (flucloxacillin not available locally)</b> |
| <b>Flucloxacillin / nafcillin / oxacillin</b> (where available) | 25–50 mg/kg/dose q6h (max 2 g/dose)                  | IV; penicillinase-resistant anti-staphylococcal — the international first-line, <b>not currently available in Kuwait</b>    |
| <b>Clindamycin</b> (anti-toxin adjunct)                         | 10 mg/kg/dose q8h (max 900 mg/dose)                  | IV/PO; optional — suppresses toxin, only where susceptible; benefit uncertain in children                                   |
| <b>Vanco- mycin</b> (if MRSA)                                   | 15 mg/kg/dose q6h (max 1 g/dose)                     | IV; if MRSA suspected/prevalent or first-line fails                                                                         |
| <b>Oral step-down</b>                                           | <b>Cephalexin</b> 25 mg/kg/dose q6–8h (max 1 g/dose) | PO once afebrile & improving; complete ~7-day course (dicloxacillin/flucloxacillin where available)                         |

**Evidence:** pediatric SSSS systematic review (Pediatr Dermatol 2025); Merck Manual (professional). Bullae are sterile — culture the primary focus; anti-staphylococcal therapy + supportive care is the mainstay.

◆ ECZEMA HERPETICUM — FOCUSED WORKUP  
& MANAGEMENT

- **Recognize** — clusters of **monomorphic punched-out vesicles/erosions** on eczematous skin in an unwell child; **honey-colored crusting** means the lesions are **impetiginized** (secondary *S. aureus*).<sup>185</sup>
- **Isolate** — contact precautions: **gloves and gown** until lesions are dry and crusted.
- **Confirm (don't delay treatment)** — **deroof an intact vesicle** with a sterile needle and swab the clear fluid for **HSV-1/2 & VZV PCR** ( $\pm$  Tzanck smear as a rapid bedside adjunct); take a **bacterial swab (C&S)** for secondary infection. Diagnosis is clinical — start antivirals empirically.<sup>185</sup>
- **Baseline labs** — CBC and a metabolic/renal panel, and check hydration, **before IV acyclovir** (renal safety).
- **Antivirals early — they save lives** — **IV acyclovir 5–10 mg/kg/dose q8h** (dose by ideal

body weight; ensure hydration) if **systemically unwell, severe/disseminated, periocular, or poor oral intake; oral acyclovir per local protocol** for milder disease. Continue ~10–14 days / until lesions crust; step IV → oral on improvement.<sup>185</sup>

- **Cover secondary infection** — if impetiginized, add an **oral anti-staphylococcal antibiotic** for *S. aureus*.
- **Eyes** — facial/periocular involvement warrants **urgent ophthalmology** to exclude **corneal disease (herpetic keratitis)**; start topical ophthalmic antiviral if corneal lesions are suspected.
- **Skin / eczema care** — once antivirals are started (many **pause topical steroids for ~48 h, then restart the appropriate-potency topical corticosteroid** as the child improves — steroids must never delay antivirals).<sup>185</sup> **Bland ointment emollients** (avoid creams — they sting); **non-soap cleansers**; lukewarm **5-min**

**baths, pat (never rub) dry, no soap;** warm compresses to crusted areas; loose cotton clothing; avoid friction.

- **Supportive & referral** — fluids/electrolytes, analgesia, nutrition; admit if extensive/systemic/dehydrated. **Neurology** if CNS involvement is suspected; **dermatology at discharge** for ongoing eczema management.

**Evidence:** StatPearls (Eczema Herpeticum); pediatric emergency guidance. Early antivirals are the priority; topical steroids treat the underlying eczema but must not delay antivirals.

◆ DRESS / DIHS — FOCUSED WORKUP & MANAGEMENT

- **Recognize** — **morbilliform rash with facial edema** starting on the **face** → **upper trunk** → **extremities** (can **worsen even after the drug is stopped**), plus fever, lymphadenopathy, **eosinophilia** and atypical lymphocytes. Onset is **late: 2–8 weeks** after starting the culprit — but within **~24 h on re-challenge**. Commonest culprits: **anticonvulsants** (carbamazepine, phenytoin, lamotrigine, phenobarbital), **allopurinol**, and **antibiotics** (sulfonamides, vancomycin, minocycline). First-degree relatives carry up to **~25% risk** with the same drug (HLA-linked).<sup>186</sup>
- **Organ involvement** — **liver first** (hepatitis), then lung, kidney, **heart (myocarditis)**, spleen, GI and neurologic; **HHV-6/7, EBV, CMV reactivation** is characteristic. **Late endocrine autoimmunity** (hypothyroidism, type 1 diabetes) can appear months afterwards.<sup>186</sup>



- **Diagnose** — clinical, scored with the **RegiSCAR** or **J-SCAR** criteria. **Labs:** CBC + differential (eosinophilia, atypical lymphocytes), **LFTs** (may stay abnormal for months), renal panel + urinalysis (proteinuria/hematuria), CK, lipase, CRP/ESR/**ferritin**, baseline **TFTs + glucose**, viral PCR (HHV-6/7, EBV, CMV), **ECG + troponin + baseline echo**, and CXR if respiratory symptoms; extend the work-up by clinical suspicion. **Skin biopsy** to exclude mimics.<sup>186</sup>
- **Manage** — **stop the culprit immediately** and **label the patient hypersensitive**; supportive care; **high-potency topical corticosteroids**. For organ involvement, **systemic corticosteroids** — prednisolone **0.5–1 mg/kg/day (max 60 mg/day)** with a **slow taper over weeks–months** to avoid relapse; adjuncts/alternatives (specialist): **ciclosporin**, IVIG (limited evidence), JAK inhibitors (anecdotal).<sup>186</sup>

- **Monitor & avoid** — screen for **autoimmune sequelae** (thyroiditis, type 1 diabetes): **TSH, free T4 and glucose for up to 2 years.** <sup>186</sup> **Strict lifelong avoidance** of the culprit and cross-reacting drugs; document the allergy clearly.

**Evidence:** EAACI position paper on DRESS in children 2025; RegiSCAR / J-SCAR criteria. Prompt drug withdrawal + staged (skin-only vs organ) therapy; steroids taper slowly to prevent relapse.

◆ **NECROTIZING FASCIITIS – FOCUSED  
WORKUP & MANAGEMENT**

- **Recognize — pain out of proportion and edema extending beyond the visible erythema;** rapidly spreading dusky skin, fever, **crepitus**, bullae, necrosis or ecchymosis, and a **hard "wooden" feel** of the subcutaneous tissue; **cutaneous anesthesia is a late sign.** Lower > upper extremities. Special forms: **perineum (Fournier gangrene)**, head & neck, and neonatal infection (e.g. from omphalitis).<sup>187</sup>
- **It is a surgical emergency — call the surgeon now.** Early aggressive debridement is the single biggest determinant of survival; **do not let the LRINEC score or imaging delay surgery.**<sup>187</sup>
- **Scoring / labs — the LRINEC score** can help distinguish necrotizing fasciitis from other soft-tissue infection, **but a low score does not exclude it** (poor sensitivity in children).<sup>187</sup> Send CBC + differential, basic metabolic panel, glucose, CRP,

**serum lactate**, coagulation profile, and **blood cultures (2 sets)** — then start antibiotics without waiting.

- **Imaging** — CT (or MRI) may support the diagnosis but **must never delay surgical exploration**.
- **Empiric antibiotics** (after cultures) — broad-spectrum covering **gram-positive incl. MRSA + gram-negative + anaerobes**, plus **clindamycin for its antitoxin effect**: e.g. **vancomycin 15 mg/kg/dose q6h (max 1 g/dose) + piperacillin-tazobactam 100 mg/kg/dose (piperacillin) q6–8h (or meropenem 20 mg/kg/dose q8h, max 1 g/dose) + clindamycin 10–13 mg/kg/dose q8h (max 900 mg/dose);** de-escalate on sensitivities.<sup>187</sup>
- **Surgery = the gold standard** — surgical exploration is both the **definitive diagnosis and the treatment** (expect repeat looks/debridement); send intra-operative specimens for **Gram stain, culture, and histology — alert microbiology**

**in advance:** processing is **time-critical once tissue reaches the lab** (expedite the Gram stain and keep direct communication).

- **Who leads — shared surgical responsibility.** The operative lead follows the **anatomical site: orthopedics** (limbs), **ENT or maxillofacial** (head & neck), **plastics** (soft-tissue coverage), **urology** (perineum / Fournier), **general/pediatric surgery** (trunk / abdomen). In complex or multi-region cases, **several surgical subspecialties may need to collaborate together in the ED.**
- **Support & referral —** aggressive hemodynamic resuscitation / PICU; **IVIG** as an adjuvant in selected cases (streptococcal toxic shock).<sup>187</sup> Refer to **infectious diseases and plastic/reconstructive surgery** (with pediatric/general surgery).

**Evidence:** IDSA skin & soft-tissue infection guidelines 2014. Surgery is time-critical — neither the LRINEC score nor imaging should delay operative exploration.

### ⚠ **NEONATE**

Vesicles, or an unwell neonate with a rash, may be **neonatal HSV** — start **high-dose IV acyclovir 20 mg/kg/dose every 8 h** and a full sepsis work-up without delay. Treat for **14 days** (skin-eye-mouth disease) or **21 days** (CNS or disseminated disease); for **CNS disease**, repeat CSF HSV PCR near the end of therapy and continue until it is negative, then give **oral acyclovir suppression — 300 mg/m<sup>2</sup>/dose 3×/day for 6 months.** <sup>158</sup>

### ✦ **PEARL – READ THE PATTERN**

**Non-blanching + unwell** = sepsis;  
**monomorphic vesicles on eczema** = eczema herpeticum; widespread peeling with **mucosal erosions** = SJS/TEN, versus **mucosa spared / superficial** = SSSS.

### ▴ **REFER**

- **Dermatology + the relevant team urgently; PICU/burns** for extensive TEN; **ophthalmology** for any

mucosal/ocular involvement;  
**urology/gynecology** for genital involvement.

## Pediatric ENT Emergencies

MINUTES

### AIRWAY / ENT

The airway threats, bleeds, and batteries that can't wait — stabilize first, then call ENT.

#### ⚠ **BUTTON BATTERY IN THE NOSE OR EAR – HOURS MATTER**

- A disc battery lodged in the nose or ear causes **liquefactive necrosis and septal/ostial perforation within hours; unilateral foul or bloody discharge** is the clue.
- ENT emergency — **urgent removal**; do **not irrigate** a nasal battery.<sup>190</sup> (Swallowed batteries → **Button Battery Ingestion.**)

#### ▶ **RECOGNIZE & ACT**

- **Acute upper-airway obstruction** — stridor, drooling, tripod posture → epiglottitis / bacterial tracheitis /

severe croup: keep the child **calm and upright, minimal handling**, call **anesthesia + ENT**, and **do not examine the throat**<sup>1</sup> (see [Croup](#), [Difficult Airway](#), [Retropharyngeal Abscess](#)).

- **Post-tonsillectomy bleeding** — any bleed (classically **day 5–10**) is an emergency: sit up, suction, **IV access + fluids/blood**, tranexamic acid, call ENT + anesthesia, prepare for the operating room. Swallowed blood **hides the true loss**.
- **Deep-neck / peritonsillar / retropharyngeal abscess** — trismus, muffled “hot-potato” voice, neck stiffness or torticollis → IV antibiotics, imaging, ENT; watch the airway.<sup>164</sup>
- **Acute mastoiditis** — post-auricular swelling/erythema with a **protruding pinna** after otitis media → IV antibiotics + ENT (risk of intracranial spread / venous sinus thrombosis).<sup>158</sup>
- **Severe epistaxis** — sit forward, **pinch the soft part 10–15 min**, topical vasoconstrictor; if uncontrolled or recurrent, ENT — consider a **bleeding disorder**, or juvenile angiofibroma in an adolescent boy.



- **Ear/nose foreign body** — refer for removal; for a live insect, instill lidocaine or oil first.

### ⚠ **AIRWAY**

Do **not** try to visualize the airway or lay the child flat when epiglottitis or a deep-neck abscess is suspected — summon **anesthesia/ENT**, keep the child calm and upright.

### ✦ **PEARL – UNILATERAL DISCHARGE = FOREIGN BODY / BATTERY**

Unilateral foul or bloody nasal discharge in a toddler is a **foreign body — and a battery until excluded**; examine, image if needed, and remove urgently.<sup>190</sup>

### ✦ **YOUR ROLE – THE GENERAL PEDIATRICIAN**

- Secure the child first, not the diagnosis.
- In suspected epiglottitis or a deep-neck abscess, keep the child **calm and upright**, give oxygen, do **not** examine the throat or lay them flat, and summon **anesthesia + ENT**.
- For post-tonsillectomy bleeding, get **IV access, fluids/blood, and tranexamic acid** while ENT is

called — assume swallowed blood hides the true loss.

- Recognize the emergencies that can't wait: a **nasal/aural button battery** and **unilateral bloody discharge** need same-hour ENT.
- Control epistaxis with **10–15 min of firm pressure** before escalating.
- Your job is early recognition, first stabilization, and a timely ENT / anesthesia call.

#### ▲ REFER

- **ENT urgently** — nasal/aural battery, post-tonsillectomy bleed, deep-neck abscess, mastoiditis, or uncontrolled epistaxis; **anesthesia** for any airway threat.

# Pediatric Neurosurgical Emergencies

MINUTES

TIME-CRITICAL

Shunt failure, rising ICP, and cord compression — recognize early, control the pressure, and call neurosurgery.

## ⚠ VP SHUNT MALFUNCTION – TREAT AS RAISED ICP

- In a child with a shunt, **headache, vomiting, drowsiness, irritability, a bulging fontanelle, or any change from baseline = a blocked shunt until proven otherwise.**<sup>189</sup>
- Assess, image (**CT or rapid-sequence MRI + shunt series**), call **neurosurgery urgently**.
- Shunt **infection** (fever, redness/swelling along the tract, or a new malfunction) carries the same urgency — manage on a defined pathway: **sample CSF from the shunt/reservoir plus blood cultures**; start empiric antimicrobials covering **staphylococci (including MRSA) and Gram-negative organisms** — e.g. **vancomycin + an**

**anti-pseudomonal beta-lactam (cefepime, ceftazidime, or meropenem); plan removal or externalization of the infected hardware; and involve neurosurgery and infectious diseases early.**<sup>210</sup>

- Controlling raised ICP is the priority while awaiting neurosurgery.

#### ▀ **RECOGNIZE & ACT**

- **Raised ICP / impending herniation** — falling GCS, an unequal or fixed pupil, **Cushing's triad** (hypertension, bradycardia, irregular breathing), or abnormal posturing → **head up 30°, midline neck, normocapnia, treat seizures, hyperosmolar therapy** (hypertonic saline or mannitol), urgent neurosurgery (see<sup>80</sup> [Raised ICP](#)).
- **Non-traumatic intracranial hemorrhage** — sudden severe headache, focal deficit, or coma (AVM, aneurysm, tumor) → urgent CT + neurosurgery.
- **Depressed / open skull fracture or penetrating head injury** → neurosurgery + antibiotics + tetanus

(see [Traumatic Brain Injury](#)).

- **Acute spinal cord compression — neck or back pain with arm or leg weakness**, a sensory level, or bladder/bowel dysfunction (tumor, abscess, trauma) → **urgent MRI**, dexamethasone if tumor, and **urgent spine-surgical referral — neurosurgery or pediatric orthopedic spine surgery** (either may be consulted for extradural pathology: epidural abscess, vertebral/extradural tumor, or bony trauma) ± oncology — time-critical for recovery. In **traumatic spinal cord injury**, also involve **PICU for hemodynamic (MAP) support** — maintain mean arterial pressure and avoid hypotension to preserve cord perfusion; watch for **neurogenic shock** (hypotension with bradycardia).
- **CSF leak** — clear rhinorrhea/otorrhea or a tense pseudomeningocele after surgery/trauma → neurosurgery (meningitis risk).

**⚠ DON'T DELAY IMAGING**

A shunted child who is drowsy or vomiting needs **urgent imaging and a**

**neurosurgical call** — deterioration can be rapid; do not attribute it to a viral illness.

✦ **PEARL – ASK THE PARENTS**

Parents of a shunted or hydrocephalic child usually recognize their own child's **“shunt-failure pattern”** — take that history seriously and image.

✦ **PEARL – PROTECT THE PUPIL EXAM**

Have **ophthalmology look for papilledema**, but **avoid dilating the pupils in a drowsy or deteriorating child** — mydriatic drops abolish the pupillary response, a key sign of herniation.

✦ **YOUR ROLE – THE GENERAL PEDIATRICIAN**

- Recognize, control the pressure, and call — you are not expected to operate.
- In any shunted child, treat headache / vomiting / drowsiness / any change from baseline as a **blocked shunt until proven otherwise**: image (CT or rapid-sequence MRI + shunt series) and call neurosurgery.
- While waiting, deliver the **ICP first-aid that is yours** — head up

30°, midline neck, normocapnia, treat seizures and fever, and give hyperosmolar therapy for impending herniation.

- Take a parent's **"this is his shunt-failure pattern"** seriously.
- Don't anchor on a viral illness, and refer **time-critically** for cord compression (urgent MRI) and open / depressed skull fractures.

#### REFER

- **Neurosurgery urgently** — shunt malfunction/infection, raised ICP, intracranial hemorrhage, or a depressed/open skull fracture.  
**Acute spinal cord compression** — neurosurgery **or** pediatric orthopedic spine surgery (either for extradural pathology); add **PICU** for traumatic cord injury (MAP support).

## Gastroenterology & Hepatology

08

GI bleeding · liver failure · pancreatitis · hepatitis · neonatal cholestasis · ingestion

## Upper Gastrointestinal Bleeding

**MINUTES**

Hematemesis or melena — resuscitate first, then find and treat the source.

**+ Calculator:** [Blood products](#)

### RECOGNIZE & STRATIFY

- Bleeding proximal to the ligament of Treitz: **hematemesis** (fresh or coffee-ground) or **melena**. Confirm true blood — swallowed maternal or epistaxis blood and red foods mislead.<sup>54</sup>
- **Clinically significant** = hemodynamic instability, a hemoglobin fall  $\geq 2$  g/dL, or a transfusion requirement — most need PICU.<sup>54</sup>
- **Age-based causes:** neonate — swallowed maternal blood, vitamin K deficiency bleeding, stress gastritis; infant/child — reflux esophagitis, gastritis, peptic ulcer, Mallory-Weiss tear; any age with liver disease —



**varices.** Consider an **ingested button battery** (urgent chest X-ray; see [Button Battery Ingestion](#)).<sup>257</sup>

○ **Investigate — order → look for → what it means**

| Order                                  | Look for                                    | What it tells you                                                                                 |
|----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------|
| CBC + <b>serial</b> hemoglobin         | Hb fall $\geq 2$ g/dL; thrombocytopenia     | Severity & transfusion need <sup>54</sup>                                                         |
| Blood group + <b>crossmatch</b>        | —                                           | Prepare blood early in any significant bleed                                                      |
| Coagulation — PT/INR, aPTT, fibrinogen | Coagulopathy                                | Correct before/with endoscopy; a prolonged INR also flags underlying liver disease <sup>257</sup> |
| LFTs + <b>albumin</b>                  | High bilirubin, low albumin                 | Chronic liver disease → suspect a <b>variceal</b> source <sup>257</sup>                           |
| Urea & creatinine                      | Urea raised out of proportion to creatinine | Supports an <b>upper</b> (vs lower) GI                                                            |

source —  
absorbed  
blood-protein  
load<sup>257</sup>

Venous  
blood gas +  
lactate

Metabolic  
acidosis, rising  
lactate

Hypoperfusion  
→ the bleed is  
severe;  
escalate<sup>257</sup>

#### ① RESUSCITATE FIRST

ABC, two large-bore IVs, crystalloid;  
send crossmatch; correct  
coagulopathy/thrombocytopenia.  
Protect the airway in brisk  
hematemesis.



#### DECISION

**Hemodynamically unstable?**

#### UNSTABLE / MASSIVE

Resuscitate to **hemodynamics** and  
activate the **massive transfusion  
protocol** — **not** a fixed Hb target. PICU;  
urgent endoscopy once stabilized.<sup>22</sup>

## STABLE

**Restrictive transfusion** — transfuse at **Hb < 7 g/dL**, target 7–9 g/dL.

Over-transfusion raises portal pressure and can worsen variceal bleeding.<sup>22</sup>



## DECISION

**Variceal suspected?**

(known/suspected liver disease, splenomegaly, umbilical-catheter history)

## VARICEAL

**Octreotide** infusion (start early, after senior/GI discussion). In **cirrhotic** portal hypertension, add **broad-spectrum IV antibiotic prophylaxis** (per local susceptibility) for 5–7 days (e.g., ceftriaxone); role in non-cirrhotic PH unclear.<sup>216</sup> Urgent endoscopy — band ligation/sclerotherapy.

## NON-VARICEAL

High-dose IV **proton pump inhibitor**.  
Endoscopy for hemostasis — injection  
+ cautery/clips ( $\pm$  hemostatic  
powder)<sup>23</sup> — **early ( $\leq 24$  h)** once  
resuscitated; emergent/urgent ( $\leq 6$ – $12$   
h) endoscopy only if still unstable  
despite resuscitation.<sup>288</sup>



**Uncontrolled bleeding  $\rightarrow$  balloon  
tamponade (Sengstaken-Blakemore)  
as a bridge in massive variceal  
bleeding; interventional radiology  
(TIPS) or surgery — keep the massive  
transfusion protocol running.**<sup>257</sup>

| Drug                      | Dose                                                                                                                         | Route & notes                                                                                                                                                                                                       |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octreotide                | bolus 1–2 mcg/kg (max 50 mcg), then 1–2 mcg/kg/h infusion (titrate up to ~5 mcg/kg/h if bleeding continues)                  | IV; variceal bleeding. Once bleeding stops, wean gradually (rebound risk); monitor glucose <sup>21</sup>                                                                                                            |
| Pantoprazole / omeprazole | 1–2 mg/kg (max 40 mg) IV bolus, then 0.1–0.2 mg/kg/h infusion (max 8 mg/h) for severe bleeding, or intermittent high-dose IV | Non-variceal; also stress-ulcer prophylaxis <sup>23</sup>                                                                                                                                                           |
| Ceftriaxone               | 50 mg/kg/dose once daily (max 2 g)                                                                                           | Broad-spectrum IV prophylaxis <b>per local susceptibility, 5–7 days</b> , in <b>cirrhotic</b> portal hypertension with acute variceal bleeding; benefit in non-cirrhotic portal hypertension unclear <sup>216</sup> |
| PRBC transfusion          | restrictive: transfuse at Hb < 7 g/dL, target 7–9 g/dL                                                                       | Especially variceal/cirrhosis; transfuse promptly for hemodynamic instability or ongoing massive bleeding regardless of Hb <sup>22</sup>                                                                            |

### ⚠️ AT THE BEDSIDE

**Do not routinely place a nasogastric tube** — it does not improve outcomes, is complicated in about a third of patients, and endoscopy is the diagnostic and therapeutic tool.<sup>279,280</sup> **If varices are suspected** (known chronic liver disease or portal hypertension, splenomegaly, prior varices), **avoid blind nasogastric tube insertion** — it can traumatize varices and provoke bleeding, and gains nothing before endoscopy; leave gastric access to the endoscopy/anesthesia team.<sup>279</sup> **Protect the airway** in brisk hematemesis; do not over-transfuse a stable variceal bleed.<sup>22</sup>

### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — all significant bleeds → resuscitation and endoscopy.
- **Discharge** — a minor self-limited bleed after observation, with follow-up.
- **Refer** — gastroenterology (endoscopy); surgery/interventional radiology for uncontrolled bleeding.

# Lower Gastrointestinal Bleeding

HOURS

AGE-STRATIFIED

Hematochezia — most are benign, but resuscitate first and never miss the surgical causes.

+ Calculator: Blood products

## RECOGNIZE & STRATIFY

- Bleeding distal to the ligament of Treitz: **hematochezia** (fresh or maroon blood/clots per rectum), or melena from a brisk right-colonic/small-bowel source. Confirm it is blood — red foods, drugs, and bismuth mislead.<sup>257</sup>
- **Separate the sick from the well first:** a pale, tachycardic child, or blood with bilious vomiting or severe pain, points to a **surgical emergency** (intussusception, malrotation with volvulus, or NEC in a neonate) — see the red lines below.<sup>257</sup>
- **Age-based commonest causes:**
  - **Neonate** — swallowed maternal blood, anal fissure, cow's-milk protein allergy, **NEC**, vitamin K deficiency bleeding, malrotation/volvulus.

- **Infant / toddler** — anal fissure, cow's-milk protein colitis, **intussusception**, **Meckel diverticulum** (painless, brisk), infectious colitis.
- **Child / adolescent** — infectious colitis, **juvenile polyp** (painless), anal fissure, inflammatory bowel disease, Henoch-Schönlein purpura, hemorrhoids.

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### ○ Investigate — order → look for → what it means

| Order                      | Look for                                        | What it means                                                     |
|----------------------------|-------------------------------------------------|-------------------------------------------------------------------|
| CBC + serial hemoglobin    | Anemia, Hb fall, thrombocytopenia; microcytosis | Severe chronic (microcytic) anemia                                |
| Coagulation — PT/INR, aPTT | Coagulopathy                                    | Bleeding diathesis, vitamin K deficiency, liver failure, neonatal |
| Blood group + crossmatch   | —                                               | Preparation for an transfusion, significant bleeding              |



|                                                                          |                                       |                                  |
|--------------------------------------------------------------------------|---------------------------------------|----------------------------------|
| Urea & creatinine                                                        | A <b>raised</b> urea:creatinine ratio | Sugg <b>upper</b> instead record |
| Stool — culture, <i>C. difficile</i> , ova/parasites; fecal calprotectin | Pathogen; raised calprotectin         | Infect vs inf bowe disea         |
| Abdominal ± Doppler ultrasound                                           | Target / pseudo-kidney sign; free air | Intuss NEC / perfor              |
| <sup>99m</sup> Tc-pertechnetate (Meckel) scan                            | Ectopic gastric mucosa                | Meckel divert painle bleed       |

#### ① RESUSCITATE FIRST

ABC, IV access, crystalloid, crossmatch; transfuse by the **restrictive** strategy (Hb < 7 g/dL) unless unstable or massively bleeding.<sup>22</sup>



#### DECISION

**Sick child, or surgical features?**

## SURGICAL / UNSTABLE

Bilious vomiting, peritonism, a palpable mass or **currant-jelly** stool, or a distended tender neonatal abdomen → **urgent surgical review + imaging** (ultrasound for intussusception; contrast study for malrotation). Do not delay.<sup>257</sup>

## WELL / STABLE

Treat by the likely cause — fissure care and address constipation; trial a cow's-milk-protein-free diet in a well infant with colitis; supportive care for infectious colitis.<sup>257</sup>



Persistent, significant, or unexplained bleeding → endoscopy (colonoscopy ± upper endoscopy) by pediatric gastroenterology; a Meckel scan for painless brisk bleeding.<sup>257</sup>

## ⚠ RED LINES – DO NOT MISS

- **Intussusception** — colicky pain, drawing up the legs, currant-jelly stool, a sausage-shaped mass — ultrasound first, then air/contrast enema reduction.

- **Malrotation with volvulus** — bilious vomiting with rectal bleeding is a surgical emergency until excluded.
- **Meckel diverticulum** — massive **painless** rectal bleeding in a young child.
- **NEC** in a preterm/neonate — distension, bloody stools, systemic upset (see **Necrotizing Enterocolitis**).

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#### ▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — any significant bleed, hemodynamic compromise, or a suspected surgical cause.
- **Discharge** — a well child with a minor self-limited cause (fissure, small polyp bleed) and follow-up.
- **Refer** — pediatric surgery for intussusception/volvulus/Meckel/NEC; gastroenterology for endoscopy, suspected IBD, or recurrent/unexplained bleeding.

## Acute Liver Failure HOURS

Coagulopathy not corrected by vitamin K — transfer to a transplant center early.

### + Calculator: Paracetamol & iron

#### ▴ DEFINE (PALF)

- Biochemical liver injury with a **coagulopathy not corrected by vitamin K**:  $\text{INR} \geq 1.5$  **with** hepatic encephalopathy, or  $\text{INR} \geq 2.0$  **without** it — and no known chronic liver disease.<sup>55</sup>
- Encephalopathy is hard to read in infants (irritability, poor feeding, reversed sleep). Look for hypoglycemia, jaundice, hyperammonemia.
- Seek a treatable cause: acetaminophen, herpes simplex virus, autoimmune hepatitis, Wilson disease, metabolic (galactosemia, tyrosinemia), gestational alloimmune liver disease.<sup>55</sup>

○ Investigate — order → look for → what it means (NASPGHAN)

Order

Look for

What

|                                                                                                                                                                                                                                     |                                                                                           |                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>PT / INR</b>                                                                                                                                                                                                                     | INR $\geq$ 1.5 with encephalopathy, or $\geq$ 2.0 without it — not corrected by vitamin K | <b>Meed</b><br><b>diag</b><br>the t<br>prog     |
| Ammonia, glucose, venous gas/bicarbonate, lactate ( $\pm$ pyruvate)                                                                                                                                                                 | Hyperammonemia, hypoglycemia, metabolic acidosis, high lactate                            | <b>Com</b><br><b>seve</b><br>PICU<br>esca       |
| Bilirubin, albumin/total protein, AST/ALT/GGT                                                                                                                                                                                       | Rising bilirubin, low albumin                                                             | Degr<br>hepa<br>& syn                           |
| CBC, electrolytes, Ca/Mg/phosphate                                                                                                                                                                                                  | Cytopenias, electrolyte derangement                                                       | Supp<br>targe<br>risk <sup>2</sup>              |
| <b>Etiology panel:</b><br>acetaminophen level,<br>HSV PCR + viral serology,<br>autoimmune markers,<br>ceruloplasmin/Wilson studies, metabolic screen<br>(galactosemia, tyrosinemia),<br>gestational alloimmune liver disease workup |                                                                                           | Direc<br>thera<br>acyc<br>immu<br>chela<br>exch |

A **treatable** cause

**Suspect it → contact a pediatric liver transplant center / PICU immediately**



#### **SUPPORT**

- Treat hypoglycemia (continuous dextrose, glucose infusion rate 6–8 mg/kg/min)
- Give one dose of parenteral vitamin K. **Do not routinely correct the INR without bleeding or a pre-invasive procedure** — it is the prognostic marker the transplant team needs.<sup>55</sup>
- Manage hyperammonemia/encephalopathy with transplant-hepatology/PICU guidance (lactulose/ammonia-lowering, nurse head-up, avoid sedatives, transfer early); cover empirically for sepsis if unwell.
- Consider adding acyclovir in a neonate with suspected HSV infection, and N-acetylcysteine if acetaminophen toxicity is suspected.

| Drug                  | Dose                                            | Route & notes                                                                                                                                                                        |
|-----------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10% dextrose infusion | glucose infusion rate 6–8 mg/kg/min, titrate    | Keep glucose normal; check frequently                                                                                                                                                |
| Vitamin K             | single parenteral dose                          | To exclude vitamin K-responsive coagulopathy                                                                                                                                         |
| N-acetylcysteine      | per acetaminophen protocol                      | Acetaminophen toxicity; not routine in non-acetaminophen disease — the pediatric RCT showed no survival benefit, so any use is a transplant-center/hepatology decision <sup>20</sup> |
| Acyclovir             | consider in a neonate/infant with suspected HSV | HSV is a treatable cause <sup>158</sup>                                                                                                                                              |

### ⚠️ AVOID HARM

- Avoid sedatives/benzodiazepines (not cleared by a failing liver; they mask encephalopathy) and hepatotoxic drugs. **Do not chase a normal INR with plasma** in a non-bleeding child — it erases the key trend.

- Watch for cerebral edema (rising intracranial pressure) as encephalopathy advances.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → PICU/transplant center.
- **Discharge** — not acutely.
- **Refer** — hepatology/transplant urgently; PICU for encephalopathy/cerebral edema.

## Acute Pancreatitis HOURS

NASPGHAN 2018

Diagnose on two of three — then fluids, analgesia, and early feeding.

#### ▲ DIAGNOSE (≥ 2 OF 3)

- Abdominal pain compatible with acute pancreatitis (epigastric, may radiate to the back).
- Amylase and/or **lipase**  $\geq 3\times$  the **upper limit of normal** (lipase is more specific).<sup>53</sup>



- Imaging consistent with acute pancreatitis (ultrasound first; CT/MRI if needed).
- Look for a cause in every case: structural/biliary, trauma (including inflicted injury), drugs, infection, metabolic, systemic illness, genetic.

### ○ Investigate — order → look for → what it means

| Order                               | Look for                                            | What it means                                      |
|-------------------------------------|-----------------------------------------------------|----------------------------------------------------|
| <b>Lipase</b> (± amylase)           | <b>≥ 3× the upper limit of normal</b>               | Diagnostic criteria: more than 3× amylase          |
| CBC, CRP, hematocrit                | Leukocytosis, rising CRP, hemoconcentration         | Inflammation, burden, severe                       |
| <b>BUN, creatinine</b>              | Rising BUN, acute kidney injury                     | Key predictor of escalation, fluid                 |
| Glucose, calcium                    | Hyperglycemia, hypocalcemia                         | Severe markers                                     |
| ALT, bilirubin, GGT ± triglycerides | Raised ALT/bilirubin; high triglycerides or calcium | Biliary (gallbladder), metabolic causes, suspicion |

obst  
**MRC**  
then  
whe  
thera  
proc  
need

Abdominal  
ultrasound first-line;  
**MRCP** for suspected  
biliary or  
pancreatic-duct  
obstruction/anomaly;  
CT for suspected  
complications

Gallstones,  
necrosis, ductal  
anomaly,  
choledocholithiasis

Etiol  
com

#### ① FLUIDS

Crystalloid **1.5–2× maintenance** first  
24–48 h (10–20 mL/kg bolus for  
hypovolemia). Target urine output >  
0.5–1 mL/kg/h; monitor cardiac,  
pulmonary, and renal status closely for  
48 h. <sup>53</sup>



## ② ANALGESIA & FEEDING

- Adequate analgesia, including opioids when needed. **Start enteral nutrition early** (within 48–72 h, as tolerated — oral/gastric/jejunal); enteral is preferred over parenteral.
- Prolonged gut rest is no longer recommended.<sup>53</sup>



**No routine prophylactic antibiotics — reserve them for proven infection (e.g., infected necrosis).**<sup>53</sup>

| Therapy                                                | Dose / setting                                                                                                                                        | Notes                                                   |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Crystalloid (balanced, or saline + 5% dextrose)</b> | <b>1.5–2× maintenance, first 24–48 h; avoid overly aggressive fluids – reassess for pulmonary edema, AKI, and shock, especially in severe disease</b> | Most important early intervention                       |
| <b>Morphine</b>                                        | <b>0.05–0.1 mg/kg IV, titrate</b>                                                                                                                     | Opioid analgesia when needed                            |
| <b>Early enteral nutrition</b>                         | <b>within 48–72 h</b>                                                                                                                                 | Oral/gastric/jejunal; enteral preferred over parenteral |

### ⚠ **WATCH FOR SEVERE DISEASE**

- Most complications declare within 48 h.
- Persistent organ dysfunction (respiratory, cardiovascular, renal) defines severe acute pancreatitis — escalate to PICU.
- Gallstone pancreatitis with suspected obstruction: **MRCP first** (first-line and prerequisite), then **ERCP** for stone clearance if obstruction is confirmed; involve surgery and GI.

### ▴ **ADMIT · DISCHARGE · REFER**

- **Admit** — essentially all confirmed cases — for IV fluids, analgesia, and monitoring; ICU for organ dysfunction or predicted severe disease.
- **Discharge** — mild disease once pain is controlled, feeding is re-established, and the cause is being addressed.
- **Refer** — surgical or gastroenterology review for biliary/structural causes, necrosis, or complications.

## Acute Hepatitis HOURS

Most recover — the job is to catch the few progressing to liver failure.

### ASSESS

- Raised transaminases ± jaundice, nausea, anorexia, right-upper-quadrant pain, dark urine, pale stools.
- Causes: viral (hepatitis A/B/C/E, EBV, CMV, adenovirus, enterovirus, herpes simplex virus, SARS-CoV-2), drug-induced (including acetaminophen), autoimmune hepatitis, Wilson disease, metabolic, ischemic.<sup>158</sup>
- **Check synthetic function and a coagulation profile (INR) at presentation** — it separates uncomplicated hepatitis from evolving liver failure. In a sick or deteriorating child, also send **ammonia, lactate, glucose**, and a **toxicology (acetaminophen) screen**.<sup>55</sup>

○ **Investigate — order → look for → what it means**

| Order                                                                          | Look for                                  | What it tells                                                                                |
|--------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|
| AST/ALT, bilirubin                                                             | Markedly raised transaminases ± jaundice  | Confirms hepatocellular injury                                                               |
| PT/INR, albumin                                                                | Prolonged INR, low albumin                | <b>Synthetic → evolving liver failure</b><br>separates uncomplicated hepatitis <sup>55</sup> |
| Ammonia, glucose, lactate, venous gas                                          | Hyperammonemia, hypoglycemia, acidosis    | Complicates severity in sick/deteriorating child <sup>55</sup>                               |
| Acetaminophen level; viral panel — hepatitis A/B/C/E, EBV, CMV, <b>HSV PCR</b> | A treatable/infectious cause              | Directs specific therapy (N acyclovir) public-health notification                            |
| Autoimmune markers, ceruloplasmin, ferritin                                    | Autoimmune hepatitis, Wilson disease, HLH | Specific etiologies missed <sup>55</sup>                                                     |

#### MANAGE

- Mostly supportive: hydration, antiemetics, avoid hepatotoxins, review every drug.
- Test for **viral hepatitis** as indicated; send autoimmune markers, ceruloplasmin, and **ferritin**. Notify per public-health requirements for notifiable viral hepatitis.



**Red flags → treat as acute liver failure:** a rising INR not corrected by vitamin K, hypoglycemia, or any encephalopathy. Refer urgently.

#### ⚠ **DON'T MISS THE TURN**

- A child with hepatitis who develops coagulopathy (INR  $\geq$  1.5–2.0), hypoglycemia, or altered behavior is no longer “just hepatitis” — escalate to the Acute Liver Failure pathway (see [Acute Liver Failure](#)) and contact a transplant center.
- Consider acetaminophen toxicity and Wilson disease in any severe presentation.<sup>55</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — any coagulopathy, hypoglycemia, encephalopathy, severe vomiting or dehydration, or diagnostic uncertainty.
- **Discharge** — well child with stable or improving enzymes, normal INR, tolerating oral intake, with follow-up.
- **Refer** — urgent transplant-center/hepatology referral if it evolves to acute liver failure (rising INR, encephalopathy).

## Neonatal Cholestasis ASSESS

NASPGHAN/ESPGHAN 2025

Conjugated jaundice is never physiological — the biliary-atresia clock is ticking.

### ▲ THE RULE

- **Any infant still jaundiced at 2 weeks** (3 weeks if breastfed and well) needs a **fractionated (total and direct) bilirubin** — do not assume breast-milk or physiological jaundice.<sup>191</sup>



- **Cholestasis** = conjugated/direct bilirubin  $> 1.0 \text{ mg/dL}$  ( $> 17 \text{ }\mu\text{mol/L}$ ), **regardless of the total bilirubin** — always pathological (NASPGHAN/ESPGHAN 2017; the older “ $> 20\%$  of total” rule is no longer used).<sup>308,191</sup>
- Pale (acholic) stools and dark urine point to biliary obstruction — use a stool-color reference card.
- **Prioritize the treatable causes** — exclude biliary atresia (time-critical), sepsis, urinary tract infection, hypothyroidism, hypopituitarism, galactosemia, and tyrosinemia.<sup>191</sup>

### ○ Investigate — order → look for → what it means

| Order                                          | Look for                                                                                                                  | What it tells you                                             |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Fractionated (total + direct) bilirubin</b> | <b>Direct <math>&gt; 1.0 \text{ mg/dL}</math> (<math>&gt; 17 \text{ }\mu\text{mol/L}</math>), regardless of the total</b> | = cholestasis — <b>always pathological</b> <sup>308,191</sup> |
| GGT, ALT/AST, albumin                          | Pattern of injury (high- vs low-GGT)                                                                                      | Narrows the differential <sup>191</sup>                       |
| <b>Coagulation (PT/INR)</b>                    | Coagulopathy                                                                                                              | Fat-soluble vitamin                                           |

deficiency → give  
**vitamin K**<sup>191</sup>

Abdominal  
ultrasound  
(fasting),  
first-line

Absent/small  
gallbladder,  
triangular-cord  
sign

Clue to **biliary  
atresia**  
(time-critical)<sup>191</sup>

Septic +  
metabolic  
screen —  
glucose,  
ammonia,  
lactate,  
blood gas,  
urine  
reducing  
substances,  
thyroid,  
cultures

A treatable  
cause

Exclude  
sepsis/UTI,  
hypothyroidism,  
galactosemia,  
tyrosinemia<sup>191</sup>

**Neonatal conjugated  
hyperbilirubinemia = a medical  
emergency → urgent pediatric  
gastroenterology consultation**



**ACT NOW**

Start the workup above the **same day**;  
give **vitamin K** (cholestasis impairs  
fat-soluble vitamin absorption and  
risks bleeding); do not wait for an  
outpatient appointment.<sup>191</sup>

### ⚠ **BILIARY ATRESIA IS TIME-CRITICAL**

- Kasai portoenterostomy works best **within the first 60 days of life** — outcomes fall sharply after.<sup>191</sup>
- A delayed referral for “prolonged jaundice” can cost the native liver.
- When in doubt, fractionate the bilirubin and refer the same day.

### ▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — for workup and any decompensation — a medical emergency for evaluation.
- **Discharge** — a stable infant with an expedited outpatient workup.
- **Refer** — pediatric hepatology urgently (exclude biliary atresia before 6 weeks).

# Foreign Body Ingestion HOURS

ESPGHAN 2026

X-ray to locate it first; then the object and where it sits decide the timing.

## ▲ FIRST STEP

- Ask **what, when, and how many**.
- **Plain radiographs** — a **chest X-ray (PA or AP frontal + lateral**, including the neck / upper esophagus) and an **abdominal X-ray**; together these cover neck-to-pelvis. The **lateral CXR** locates and characterizes an esophageal object and shows the button-battery **double-ring / halo** sign (vs a coin); the **abdominal film** tracks an object that has passed into the stomach or bowel.
- **Radiolucent objects** (most plastic, wood, glass, thin aluminum, fish/food bones) may not show — **a normal film does not exclude them**; consider further imaging or endoscopy if suspicion is high.<sup>24</sup>
- Keep child fasting (nothing by mouth) until disposition decided.<sup>24</sup>

Timing

Which swallowed objects

**Now**  
endoscopy  
≤ 2 h

Button battery in the esophagus;  
**two or more magnets**, or a  
magnet plus a metal object (any  
level)<sup>24</sup>; sharp or pointed object in  
the esophagus; esophageal  
blockage — drooling, unable to  
swallow saliva; airway distress or  
signs of perforation; **any**  
**symptomatic child**.

---

**Within 24 h**  
next day at  
the latest

Any other object lodged in the  
esophagus — never leave one  
there beyond 24 h. Sharp object,  
or magnets, sitting in the stomach  
or duodenum. Object ≥ 2.5 cm  
(25 mm, ~1 inch) wide or ≥ 5–6  
cm (~2 inches) long — too big to  
pass the pylorus or duodenum.

---

**Observe**  
at home,  
elective

A single blunt object (e.g., a coin)  
already past the esophagus into  
the stomach, < 2.5 cm wide and <  
5 cm long, with no battery or  
magnet. 80–90% pass within 1–2  
weeks. Re-image only if  
symptoms appear; remove if it is  
still in the stomach at ~ 2–4  
weeks.<sup>24</sup>

---

**Water**  
**beads**  
admit &  
observe;  
surgery, not  
endoscopy

**Superabsorbent (water) beads**  
are radiolucent — plain films miss  
them; use **abdominal ultrasound**.  
Small ingestions (a few beads)  
usually pass: admit, serial  
abdominal examination, watch for  
bilious vomiting, distension or  
pain. **Endoscopy rarely retrieves**  
**them** and is not the pathway.  
**Involve pediatric surgery early**

for large ingestions or any sign of obstruction — most symptomatic children need an operation.<sup>310,311</sup>

### ⚠ **RED LINES**

- Batteries and magnets are never simply observed — an esophageal battery or two or more magnets is an emergency at any level (see **Button Battery Ingestion**).
- An esophageal foreign body left beyond 24 h risks pressure necrosis, perforation, and aorto-/tracheo-esophageal fistula.<sup>24</sup>
- Return for pain, vomiting, fever, drooling, bleeding, or black stool.

◆ **WHO REMOVES IT** A foreign body lodged in the **UPPER (CERVICAL) ESOPHAGUS, AT THE LEVEL OF THE CLAVICLES (T2)**, typically removed by ENT with a **RIGID ESOPHAGOSCOPE**; objects lower (esophagus/stomach) usually managed by GI with flexible endoscopy.

◆ **SEE ALSO** **Button (Lithium) Battery Ingestion**.

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic, an esophageal object, or high-risk (magnets, sharp, long).
- **Discharge** — asymptomatic gastric blunt object with return advice.
- **Refer** — GI/surgery for endoscopy (esophageal, magnets, sharp objects).

## Button (Lithium) Battery Ingestion

MINUTES

ESPGHAN 2026

An esophageal battery burns within 2 hours — move fast.

### LOCALIZE

Urgent **chest X-ray (PA or AP frontal + lateral)** covering neck to upper esophagus (add an **abdominal X-ray** if the battery is not seen). A "double-ring / halo" sign distinguishes battery from coin. Esophageal location = emergency.<sup>25</sup>



## DECISION

### Battery in the esophagus?

#### YES — ESOPHAGEAL

#### Endoscopic removal within 2 hours.

See mitigation doses below — do not let them delay removal.<sup>25</sup>

#### PAST ESOPHAGUS, asymptomatic

Often expectant with follow-up imaging; remove if symptomatic, co-ingested magnet, or fails to progress.

| Mitigation | Dose                              | Route & notes                                                                                                 |
|------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------|
| Honey      | 10 mL every 10 min, up to 6 doses | Oral; within 12 h. Pre-hospital. <b>Only if ≥ 12 months — never give honey under 12 months.</b> <sup>25</sup> |
| Sucralfate | 1 g every 10 min, up to 3 doses   | Oral; in-hospital, before removal <sup>25</sup>                                                               |

#### ⚠ AFTER REMOVAL

Injury continues — watch for delayed strictures and tracheo-/aorto-esophageal fistula.<sup>25</sup>



### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all esophageal batteries → emergency.
- **Discharge** — after confirmed passage past the esophagus in an asymptomatic child, per protocol.
- **Refer** — GI/ENT/surgery IMMEDIATELY for esophageal lodgement — do not delay.

## Functional Constipation & Fecal Impaction

### ASSESS

ESPGHAN/NASPGHAN · NICE

Common, benign, and clinical — but disimpact before maintenance and screen for red flags.

### ▲ RECOGNIZE (ROME IV)

- **< 4 years** —  $\geq 2$  of the following for  $\geq 1$  month;  **$\geq 4$  years** —  $\geq 2$  at least once a week for  $\geq 1$  month (with insufficient criteria for IBS):
  - $\leq 2$  defecations/week
  - painful or hard stools
  - large-diameter stools
  - a large fecal mass in the rectum

- excessive stool retention
- (if toilet-trained)  $\geq 1$  episode of fecal incontinence/week or a history of large stools that may obstruct the toilet.<sup>150</sup>
- Retentive posturing / withholding — often presents as **overflow soiling** mistaken for diarrhea.
- Acute presentations: abdominal pain, a palpable fecal mass, urinary symptoms, or a distressed withholding child.

#### ▴ **DIAGNOSE – CLINICAL**

- History + examination make the diagnosis. **Without a red flag, no routine investigations** — no bloods (celiac, thyroid, calcium, lead), imaging, allergy testing, or transit studies.<sup>150</sup>
- **A digital rectal exam is not needed when  $\geq 2$  Rome IV criteria are met** — reserve it for a single criterion, a red flag, or to confirm impaction when the abdominal exam is unclear. A plain abdominal film is not used to diagnose constipation.<sup>150</sup>

**⚠ RED FLAGS – THINK ORGANIC DISEASE, DO NOT JUST TREAT**

- Onset from birth or delayed meconium > 48 h (Hirschsprung)
- ribbon stools
- failure to thrive
- unintentional weight loss or slow growth
- bilious vomiting
- severe distension
- blood without a fissure
- abnormal anal position or tone, absent anal/cremasteric reflex, sacral dimple, hair tuft or lower-limb weakness (spinal)
- perianal scarring or extreme fear on exam (safeguarding)
- family history of autoimmune disease or cystic fibrosis.
- Examine the perineum and back.<sup>149</sup>

## TREAT

- Disimpact first if a fecal mass is present — **maintenance alone will worsen overflow soiling**. Then start maintenance the same day, add behavioral measures, and review at 2 weeks.<sup>149</sup>
- **Goal — one soft, painless stool daily with no withholding or soiling**. Continue maintenance at goal for  $\geq 2$  months and until **symptom-free for  $\geq 1$  month** — in a toilet-training child, continue until trained — then **wean gradually rather than stopping abruptly**; do not stop while symptoms persist.<sup>150</sup>
- **Break the pain-withholding cycle** — an anal fissure or painful defecation drives retention; keeping stool soft (with topical fissure care) is what lets maintenance work.<sup>258</sup>

## POLYETHYLENE GLYCOL (PEG 3350/4000) – FIRST-LINE FOR BOTH STEPS

| Step                        | Dose                                                                                 | Notes                                                                                                                                                   |
|-----------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Disimpaction</b>         | PEG 1–1.5 g/kg once daily (oral) for 3–6 consecutive days                            | Oral is as effective as enemas and less invasive; expect a few messy days. A rectal enema is an alternative if oral is refused or fails. <sup>149</sup> |
| <b>Maintenance</b>          | PEG 0.4–1.5 g/kg once daily (start ~0.4 g/kg, titrate up to ~1.5 g/kg/day as needed) | First-line and <b>superior to lactulose</b> . Titrate to the goal above; reassess at 2 weeks if inadequate. <sup>150</sup>                              |
| <b>Add-on / alternative</b> | Stimulant (senna or bisacodyl, age-appropriate); lactulose if PEG unavailable        | Second-line adjunct when PEG alone is insufficient — not a substitute for disimpaction.                                                                 |

### ◆ PEG – WITH OR WITHOUT ELECTROLYTES

- Same drug (macrogol); the only difference is added balancing salts. **With electrolytes** for high-volume **disimpaction**; **without** (more palatable) is an equal option for **maintenance**. The electrolyte-containing macrogol stocked locally covers

both; improve palatability by chilling and mixing it into a flavored drink.<sup>277,278</sup>

#### ◆ SIDE-EFFECTS & PRECAUTIONS

- **PEG (osmotic)** — dose-related loose stools, bloating, flatulence and nausea; over-titration can cause diarrhea or overflow soiling — **reduce the dose, do not stop**. Maintain adequate fluid intake. Clinically significant electrolyte disturbance is rare (the electrolyte form guards against it further), and there is **no tolerance or dependence** with long-term use.<sup>150</sup>
- **Stimulants (senna, bisacodyl)** — abdominal cramps and diarrhea; in nappy-wearing or incontinent children, prolonged skin contact with senna can cause a blistering perianal ("senna") dermatitis, so keep the skin clean and barrier-protected. No good evidence for a "lazy bowel" at therapeutic doses.<sup>150</sup>
- **Lactulose** — flatulence, bloating and cramps, with more adverse events than PEG and inferior

efficacy; reserve it for where PEG is unavailable or unlicensed.<sup>276</sup>

- **General** — titrate to one soft, painless stool daily; do not stop abruptly (relapse); recheck adherence and dose before escalating.

#### ◆ INFANTS < 1 YEAR

- PEG is effective and first-line; **LACTULOSE IS THE ALTERNATIVE** where PEG is unavailable or unlicensed.
- In a **REFRACTORY** infant, consider a **2-4 WEEK TRIAL OF A COW'S-MILK-PROTEIN-FREE DIET**.
- **INFANT DYSCHIZIA** — straining then passing a *soft* stool, < 9 months — is **NOT** constipation and needs no laxative.<sup>150</sup>

#### ✦ WHEN MAINTENANCE ISN'T WORKING

- **Check adherence and dose first** — under-dosing and stopping too early are the commonest reasons for relapse.
- Re-check for a **retained impaction**; re-disimpact if present.

- Add a **stimulant** (senna or bisacodyl); then reconsider the diagnosis / red flags.
- Prucalopride, lubiprostone and linaclotide are **not established in children** — specialist use only.<sup>149</sup>

#### ✦ DIET, BEHAVIOR & COUNSELING

- Normal fiber and fluid; **fiber supplements, extra fluid, and probiotics are not proven** and do not replace a laxative.<sup>150</sup>
- Demystify and remove blame; scheduled toilet sits ~5 min after meals with foot support; a stool diary and reward system.
- **Long-term PEG is safe and not habit-forming** — reassure families to keep the dose adequate.<sup>150</sup>
- Relapse is common and treatment often runs months; about half are off laxatives by 6–12 months — set expectations and follow up.<sup>149</sup>

#### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — rarely — failed disimpaction or severe symptoms.



- **Discharge** — most, with disimpaction, maintenance laxatives, and education.
- **Refer** — pediatric **gastroenterology** for red flags / suspected organic disease, failure of optimized maintenance after ~3 months, diagnostic uncertainty, or a child needing ongoing second-line therapy; **surgery for suspected Hirschsprung** — rectal suction biopsy is the gold standard (contrast enema and anorectal manometry are adjuncts).<sup>150</sup>

## Reflux & GERD ASSESS

INFANT-FOCUSED • 2018

Distinguish the thriving "happy spitter" from true reflux disease.

### ▶ REFLUX VS DISEASE

- **GER (physiological)**: effortless regurgitation in a comfortable, thriving infant; peaks ~4 months, resolves by 12–18 months. Not a disease<sup>28</sup> — reassurance.

- **GERD:** reflux causing troublesome symptoms or complications — feeding refusal, faltering growth, esophagitis, recurrent aspiration.<sup>28</sup>

① REASSURE + CONSERVATIVE

- Parental education; avoid overfeeding; smaller, more frequent feeds; thickened feeds; hold upright after feeds; continue breastfeeding.
- Do **not** use prone/side positioning for sleep (SIDS risk).<sup>28</sup>



② TRIAL IF NOT IMPROVING (INFANT)

2–4 weeks of extensively hydrolyzed (or amino acid) formula, or maternal dairy exclusion if breastfed — overlaps with cow's milk protein allergy.<sup>28</sup>



**INFANT • still not responding**

**Refer to gastroenterology** — do **not** start acid suppression in an infant yourself. The gastroenterologist assesses and decides the next step: a **PPI / H2 trial, or upper-GI endoscopy or a contrast study.**<sup>28</sup>

Time-limited **4–8 week** trial of acid suppression (proton-pump inhibitor or H<sub>2</sub>-antagonist), then **wean and reassess**.<sup>28</sup>

**⚠ DO NOT ACID-SUPPRESS SIMPLE REFLUX**

- Proton-pump inhibitors are **not** recommended for unexplained crying, distress, or visible regurgitation in an otherwise well infant — no benefit and real harms.<sup>28</sup>
- Avoid acid suppression for isolated cough/wheeze. **Red flags** (investigate / refer): bilious or projectile vomiting, blood in vomit or stool, onset after 6 months or persisting beyond 12–18 months, faltering growth, fever, lethargy, or a bulging fontanelle.

**▴ ADMIT • DISCHARGE • REFER**

- **Admit** — rarely — only for complications (apnea, faltering growth, aspiration) or diagnostic uncertainty.

- **Discharge** — uncomplicated reflux with reassurance and feeding advice.
- **Refer** — gastroenterology for red flags, faltering growth, or treatment failure.

## Ingestion Injuries – See Also

Corrosive ingestion is covered in the Toxicology section.

### ◆ CROSS-REFERENCE CAUSTIC (CORROSIVE)

**INGESTION** (Toxicology & Environmental) — acid/alkali esophageal injury; no charcoal, no emesis, airway first, endoscopy at 12–24 h.

## Acute IBD Flare & Toxic Megacolon

**MINUTES**

ESPGHAN-ECCO 2025

Bloody diarrhea with systemic upset — and the colon that dilates and can perforate.

+ Calculator: [PUCAI](#)

### ► RECOGNIZE

- Known or new IBD with **frequent bloody stools**, abdominal pain, fever, tachycardia and raised inflammatory markers; grade severity (e.g. **PUCAI** in ulcerative colitis).
- **Toxic megacolon** — colonic dilatation with systemic toxicity (fever, tachycardia, tenderness/distension); a surgical emergency.<sup>248</sup>

### ⚠ WHAT MAKES IT WORSE

- **Avoid antimotility agents, opioids and anticholinergics** — they can precipitate toxic megacolon.<sup>248</sup>
- Exclude infection (including **C. difficile** and enteric pathogens) before/while escalating immunosuppression.
- Rising distension, worsening toxicity, or free air means **surgery**.

### ► WORKUP

- **PUCAI at admission and daily** (score it below, or use the **PUCAI calculator**) — acute severe colitis =  $\text{PUCAI} \geq 65$ . **Day 3 > 45** → plan

second-line therapy (screen, discuss); **day 5  $\geq$  65** → start second-line; **35–65 on day 5** → continue IV steroids 2–5 more days with daily scoring; **< 35** → responding, continue and transition to oral.<sup>248</sup>

- CBC, CRP/ESR, albumin, electrolytes, renal and liver function.
- **Stool infection panel including *C. difficile*** (consider CMV on biopsy if steroid-refractory).
- **Abdominal X-ray** for colonic dilatation / toxic megacolon; limited unprepped sigmoidoscopy where needed.

#### ◆ PUCAI – SIX CLINICAL ITEMS, NO LABS (0–85)

| Item                                   | Score                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Abdominal pain</b>                  | none <b>0</b> · can be ignored <b>5</b><br>· cannot be ignored <b>10</b>                                                                               |
| <b>Rectal bleeding</b>                 | none <b>0</b> · small amount, in < 50% of stools <b>10</b> · small amount with most stools <b>20</b> · large amount (> 50% of stool content) <b>30</b> |
| <b>Stool consistency (most stools)</b> | formed <b>0</b> · partially formed <b>5</b> · completely unformed <b>10</b>                                                                            |

| Item                                | Score                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------|
| Stools per 24 h                     | 0–2 <b>0</b> · 3–5 <b>5</b> · 6–8 <b>10</b> · > 8 <b>15</b>                             |
| Nocturnal stools (waking the child) | no <b>0</b> · yes <b>10</b>                                                             |
| Activity level                      | no limitation <b>0</b> · occasional limitation <b>5</b> · severely restricted <b>10</b> |

- **< 10** remission · **10–34** mild · **35–64** moderate · **≥ 65** severe (acute severe colitis — admit for IV steroids).<sup>309,248</sup>

| Step            | Approach                                                                    | Notes                                                                                                                                                                              |
|-----------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resuscitate     | IV fluids, correct electrolytes (esp. K <sup>+</sup> ), transfuse if needed | Monitor for toxic megacolon                                                                                                                                                        |
| Treat the flare | IV corticosteroids for severe colitis                                       | Gastroenterology-directed; <b>escalate to infliximab or cyclosporine by day 5</b> if inadequate response; pharmacologic <b>VTE prophylaxis for all hospitalized</b> <sup>248</sup> |
| Toxic megacolon | Bowel rest, decompression, urgent surgical review                           | Colectomy if deteriorating/perforation                                                                                                                                             |

### ▲ ADMIT · REFER

- **Admit** all acute severe colitis.
- **Call/Refer** — pediatric gastroenterology and surgery (early joint care for toxic megacolon).

09

## Hematologic & Oncologic

Neutropenia · tumor lysis · sickle cell · bleeding

### Chemotherapy-induced Febrile Neutropenia

MINUTES

Fever in a child on chemotherapy — a medical emergency until proven otherwise.

#### ◆ NOT ON CHEMOTHERAPY?

- This card is for the **oncology / chemotherapy (or post-transplant) patient**. For a **well, previously healthy, immunocompetent child** with an incidental fever + neutropenia, use [Febrile Neutropenia — Otherwise](#)



Healthy Child instead — empiric antibiotics are usually **not** required.<sup>154,60</sup>

▴ **DEFINE**

- Fever — single oral temp  $\geq 38.3$  °C, or  $\geq 38.0$  °C sustained over 1 h between checks, **without antipyretic use** (even if first reported at home) — **plus** absolute neutrophil count  $< 500$  cells/ $\mu$ L (or expected to fall below).<sup>36</sup>

**Empiric broad-spectrum antibiotics within 1 hour — do not wait for the count<sup>36</sup>**



## WORKUP

- Blood cultures (peripheral + each central line lumen — at least one must be drawn, central preferred; do not delay antibiotics waiting for both), full blood count with differential, renal panel (electrolytes, creatinine, glucose), lactate, urinalysis ± culture, focused exam (skin, perianal, line site)
- Start antibiotics immediately if unstable — even without labs collected.<sup>36</sup>

| Drug                    | Dose                                                                   | Route & notes                                                                                                            |
|-------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Piperacillin-tazobactam | 75–80 mg/kg/dose (piperacillin component) every 6 h (max 4000 mg/dose) | IV; first-line anti-pseudomonal monotherapy <sup>36,282</sup>                                                            |
| Cefepime                | 50 mg/kg/dose every 8 h (max 2000 mg/dose)                             | IV; alternative monotherapy if piperacillin-tazobactam is unavailable <sup>282</sup>                                     |
| Meropenem               | 20 mg/kg/dose every 8 h (max 1 g/dose; 2 g/dose if CNS/severe)         | IV; for penicillin / $\beta$ -lactam allergy; also for resistant organisms, CNS infection, or instability <sup>282</sup> |

| Drug                               | Dose                                                                                  | Route & notes                                                                                                    |
|------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Ceftazidime (+ vancomycin)         | 50 mg/kg/dose every 8 h (max 2000 mg/dose)                                            | IV; if piperacillin-tazobactam is unavailable — give with vancomycin for gram-positive cover <sup>282</sup>      |
| Vancomycin (add only if indicated) | 15 mg/kg/dose every 6 h if < 12 yr; every 8 h if ≥ 12 yr (max 1 g/dose; level-guided) | IV; for instability, line/skin infection, known resistant organism — not routine; requires drug-level monitoring |

#### ◆ LOCAL CHOICE

- Follow your local antibiogram — the empiric agent should match local resistance.
- Add an **AMINOGLYCOSIDE** (amikacin or gentamicin) **ONLY** for hemodynamic instability or known/suspected multidrug-resistant gram-negative rods, and until culture results are available.<sup>282</sup>
- **SELECTED LOW-RISK** (including but not limited to: anticipated neutropenia ≤ 7 days, no comorbidity, stable, tolerating oral intake, reliable caregiver & transport) may be managed as an outpatient as per local protocol: day-1 IV **CEFTRIAXONE** (100 mg/kg, max 4 g) + gentamicin, stepping down to oral ciprofloxacin

+ amoxicillin-clavulanate; review daily.<sup>282</sup>

- Do not routinely add vancomycin (other than as the gram-positive companion to ceftazidime above); granulocyte colony-stimulating factor only in consultation with hematology/oncology. **AGE < 2 MONTHS** (rare): ampicillin + ceftazidime, in consultation with infectious diseases / pediatric hematology-oncology.<sup>36</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — high-risk febrile neutropenia (oncology/chemotherapy) → immediate empiric antibiotics.
- **Discharge** — selected low-risk after risk assessment with close follow-up (the well-appearing non-oncologic child).
- **Refer** — oncology/hematology.

# Febrile Neutropenia – Otherwise Healthy Child

## ASSESS

## IMMUNOCOMPETENT CHILD

The well child with fever, commonly a viral illness, found to have an incidental low ANC — usually transient and benign, unlike chemotherapy-induced neutropenia from myelosuppression. Some malignancies (e.g., a localized brain tumor) do not suppress the marrow and so do not cause neutropenia.

### ◆ THE KEY QUESTION – WHICH CAME FIRST?

- **INFECTION → NEUTROPENIA** (a virus transiently suppresses or consumes neutrophils in a healthy child) behaves very differently from **NEUTROPENIA → INFECTION** (marrow failure, malignancy, post-transplant).<sup>154</sup>
- Most well, febrile children with incidental neutropenia are the former and do **NOT** need empiric febrile-neutropenia antibiotics.<sup>154</sup>

### ▴ DEFINITIONS & MECHANISM

- **Neutropenia = absolute neutrophil count (ANC) < 1500/μL.**
- Mild 1000–1500

- moderate 500–1000
- **severe < 500**; very severe < 200.
- Normal ANC varies by age and ethnicity (benign ethnic neutropenia).<sup>154</sup>
- **Fever (neutropenic definition):**  
single oral temp  $\geq 38.3$  °C, or  $\geq 38$  °C sustained > 1 h, or twice > 38 °C in 12 h. Avoid rectal temperatures (mucosal trauma).<sup>154</sup>
- **Common viral triggers:** RSV, influenza A/B, parvovirus B19, Epstein-Barr virus (EBV), cytomegalovirus (CMV), human herpesvirus 6 (HHV-6), varicella-zoster. Other unidentified viruses may also cause neutropenia or other transient cytopenias. Mechanisms: transient marrow suppression, peripheral destruction/margination, or an autoimmune process.<sup>154</sup>

#### **WHAT MAKES IT LOW-RISK (USUALLY PRESUMED VIRAL)**

- **Well-appearing**, immunocompetent, clear viral context, no features of sepsis or primary immunodeficiency.

- **Short anticipated duration** — viral neutropenia typically resolves within ~2–4 weeks, rarely beyond 6 weeks; risk rises mainly when neutropenia is prolonged<sup>154</sup> (> 30 days).
- No prior neutropenia, recurrent/severe infections, failure to thrive, or family history of a congenital syndrome; not on a neutropenia-inducing drug; not immunosuppressed.

#### DECISION

**Ill-appearing or any high-risk feature?**

#### LOW-RISK • WELL CHILD

Well-appearing, presumed viral, short-duration, no clue to an underlying cause. **Low risk of serious infection** — manage supportively with close follow-up. A history of a previously normal ANC is reassuring here.

## HIGH-RISK

**Ill-appearing** (hypothermia, hypotension, lethargy), or known/suspected **severe congenital neutropenia** (no prior normal ANC on record; e.g., Kostmann), aplastic anemia, or other **marrow-failure/immunodeficiency** → medical emergency.

Most transient neutropenia is viral — it does not require antimicrobial therapy.<sup>154</sup>



## LOW-RISK – EVALUATION & DISPOSITION

| Step                      | What to do                                                                                                                                                                          | Notes                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Cultures</b>           | Blood culture stratified by severity: not required in mild (ANC 1000–1500), consider in moderate (500–1000), obtain in severe (< 500); consider urine culture (esp. young / female) | Canadian guidance leans toward routine culture. Don't rely on urinalysis — pyuria is often absent in neutropenia. <sup>60</sup> |
| <b>Blood film</b>         | Peripheral blood film in all newly discovered neutropenia (if not already done)                                                                                                     | Look for blasts, dysplasia or a left shift — helps exclude malignancy / marrow failure.                                         |
| <b>Identify the virus</b> | Consider a respiratory multiplex PCR (not just influenza/RSV)                                                                                                                       | Confirming a viral cause supports a benign, transient course; may not always identify a specific viral species.                 |
| <b>Targeted tests</b>     | As indicated: chest radiograph, abdominal imaging, lumbar puncture, stool testing                                                                                                   | A normal radiograph can mislead — infiltrates may appear only as the ANC recovers.                                              |

| Step        | What to do                                                        | Notes                                                                              |
|-------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Disposition | Discharge home with close follow-up if well and follow-up assured | Recheck CBC to confirm recovery <sup>154</sup> ; most normalize within ~2–4 weeks. |

## HIGH-RISK – MANAGEMENT

| Empiric therapy | Agent / approach                                                                                     | Notes                                                                                         |
|-----------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Act promptly    | Full infectious work-up, hospitalize, start empiric antimicrobials without waiting for the ANC trend | Involve hematology ± infectious diseases early.                                               |
| First-line      | Piperacillin-tazobactam (broad-spectrum, anti-pseudomonal)                                           | Backbone of high-risk empiric cover, as in chemotherapy-related FN. <sup>36</sup>             |
| Tailor by risk  | Adjust for known colonization (MDRO, MRSA), prior prophylaxis, local antibiogram                     | Add MRSA cover (vancomycin) for hemodynamic instability, line, or skin/soft-tissue infection. |

### ⚠ DON'T OVER-TREAT – OR UNDER-TREAT

- The well child with an incidental low ANC during a viral illness is **not** chemotherapy-induced myelosuppression — reflexive broad-spectrum antibiotics add resistance and harm with little benefit.
- Conversely, never anchor on “just a virus”: an **ill-appearing** child, **prolonged** neutropenia (> 30 days), recurrent/severe or unusual infections, oral ulcers, or failure to thrive warrant a congenital/marrow-failure work-up and prompt treatment.<sup>154</sup>

### ▶ ADMIT · DISCHARGE · REFER

- **Admit** — only if febrile with concerning features or a suspected/confirmed bacterial infection.
- **Discharge** — well child — expectant management with a scheduled recheck of counts.

- **Refer** — hematology for persistent (> 6 weeks), profound (< 500 consistently), or recurrent neutropenia.

## Tumor Lysis Syndrome MINUTES

Massive cell breakdown — after starting chemotherapy, or spontaneously in high-burden leukemia/lymphoma.

✚ Calculators: BSA & hyperhydration · eGFR (Schwartz)

### ▲ THE BIOCHEMICAL QUARTET

- **High** potassium, **high** phosphate, **high** uric acid; **low** calcium → acute kidney injury (see Acute Kidney Injury), arrhythmia, seizures.<sup>37</sup>

① HYDRATION IS THE BACKBONE OF CARE

- Vigorous IV fluids (up to ~3–4 L/m<sup>2</sup>/day per renal status; if **body surface area** unknown or unmeasurable, use weight-based hydration — 4-2-1 rule at 1.5× maintenance). **No potassium** in fluids.
- Routine urinary alkalinization not recommended.<sup>37</sup>



DECISION

**Risk level?**

**INTERMEDIATE**

Allopurinol + hydration.<sup>37</sup>

**HIGH / ESTABLISHED**

Rasburicase + hydration. Do not give allopurinol on a rasburicase day. High risk includes (not limited to) white cell count > 50,000, elevated or rapidly rising uric acid, or suspected lymphoma with bulky disease (mediastinal mass, multiple significantly enlarged nodes).<sup>37</sup>



## ② TREAT THE ELECTROLYTES

- Manage hyperkalemia (see [Electrolyte Emergencies](#)).
- Treat symptomatic hypocalcemia with appropriate replacement.
- For refractory derangement or oliguric kidney injury, escalate urgently to nephrology/PICU — dialysis is a specialist decision (AEIOU indications).

| Drug        | Dose                                                                                                              | Route & notes                                                                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Allopurinol | 10 mg/kg/day<br>PO divided<br>every 8 h<br>(max 800 mg/day)                                                       | PO; start before chemotherapy; prevents new urate <sup>37</sup>                                                                                                |
| Rasburicase | Fixed by weight, 1 dose/24 h:<br>< 15 kg → 1.5 mg<br>15–25 kg → 3 mg<br>26–35 kg → 4.5 mg<br>> 35 kg → 6 mg (max) | IV over 30 min; breaks down existing urate. Reassess daily by uric acid — often 1 dose is adequate. <sup>67</sup> <b>Do NOT give in known G6PD deficiency.</b> |

### ⚠ RASBURICASE + G6PD DEFICIENCY

- Rasburicase is **contraindicated** in glucose-6-phosphate dehydrogenase deficiency (causes hemolysis and

methemoglobinemia) — screen before giving — **quantitative G6PD level** if status unknown (e.g., new diagnosis not yet screened); especially relevant here given high prevalence.<sup>67</sup>

- Send urate samples on ice.<sup>37</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all at-risk or established TLS → hyper-hydration and monitoring (allopurinol for intermediate risk; rasburicase for high risk or rapidly rising uric acid); severe/refractory → PICU with the dialysis team (nephrology).
- **Discharge** — after resolution of TLS and confirmed stability; not acutely; after resolution.
- **Refer** — oncology/nephrology (dialysis for refractory).

## Sickle Cell Crises

MINUTES

Pain crisis, acute chest syndrome, splenic sequestration, and stroke — plus every patient is functionally asplenic.

+ Calculator: Blood products

### VASO-OCCLUSIVE (PAIN) CRISIS

| Step             | Action                                                                     | Notes                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analgesia — fast | Opioid within 30–60 min; add NSAID/acetaminophen                           | Use a pain protocol; reassess every 15–30 min. <b>Do not delay analgesia</b> — splinting (shallow, guarded breathing from pain) can progress to acute chest syndrome. <sup>38,69</sup> |
| Hydration        | Maintenance fluids (0.75× maintenance if concern for acute chest syndrome) | Avoid over-hydration (acute chest risk) <sup>38</sup>                                                                                                                                  |
| Oxygen + warmth  | If SpO <sub>2</sub> < 95% or below baseline                                | Keep warm; treat the trigger <sup>38</sup>                                                                                                                                             |

#### ⚠️ ACUTE CHEST SYNDROME

- **Definition:** new pulmonary infiltrate + respiratory symptoms / fever / chest pain — the leading



cause of death in patients with sickle cell disease.

- **Treatment:** oxygen (keep  $\text{SpO}_2 \geq 95\%$ ), broad-spectrum antibiotics covering atypicals (cefotaxime + azithromycin/clarithromycin), bronchodilator if wheezing, incentive spirometry (10 breaths every 2 h while awake).
- **Escalation:** early hematology/PICU discussion for **simple vs exchange transfusion** if severe hypoxia or falling hemoglobin.<sup>38</sup>

#### **⚠ SPLENIC SEQUESTRATION**

- Rapidly enlarging spleen + falling hemoglobin **and/or platelet count** → hypovolemic shock. Give fluids for shock and consult hematology immediately.
- Transfuse **cautiously** (~5 mL/kg toward partial correction) — over-transfusion causes rebound hyperviscosity as the spleen releases trapped cells.<sup>38</sup>

## ⚠️ ACUTE STROKE

- Sudden focal deficit, seizure, or new altered consciousness → **urgent neuroimaging** with immediate hematology and neurology involvement.
- **Ischemic stroke** — transfuse urgently, do not delay. **Exchange transfusion is the definitive treatment** (goal HbS < 30%; keep total Hb  $\leq$  10 g/dL to avoid hyperviscosity), under hematology and neurology. If the child is significantly anemic (**Hb < 9 g/dL**) or exchange is not immediately available, give a **simple (top-up) transfusion as an immediate bridge** — but do not raise Hb above ~10 g/dL. Aim to start exchange **within 6 h.**<sup>281</sup>
- **Hemorrhagic stroke** is managed case by case, in tandem with neurosurgery.

## ◆ FEVER IN SICKLE CELL DISEASE IS AN EMERGENCY

All patients functionally asplenic, at risk of overwhelming infection by encapsulated organisms — give prompt empiric antibiotics — **CEFTRIAXONE** is the agent of choice (once-daily; enables outpatient

management of low-risk cases). Use **CEFOTAXIME IN INFANTS < 1 MONTH**, in whom ceftriaxone is contraindicated (bilirubin displacement); another broad-spectrum agent covering encapsulated bacteria may be used if allergic.<sup>70,38</sup> Assess for the crises above. **CONSIDER ORAL ANTIBIOTICS ONLY IF STRICTLY LOW-RISK (PER LOCAL PROTOCOL):** age > 1 year and clinically well with normal vitals ( $\text{SpO}_2 \geq 95\%$ ); no focal source, severe pain, acute chest, or sequestration; reassuring labs ( $\text{WBC } 5\text{--}30 \times 10^9/\text{L}$ , reticulocytes  $> 100 \times 10^9/\text{L}$ , platelets  $> 100 \times 10^9/\text{L}$ ); a first parenteral antibiotic dose given in the ED with 2–4 h observation; and a reliable home situation with guaranteed 24-h follow-up.

#### ▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — severe uncontrolled pain, acute chest syndrome, sequestration, stroke, priapism, or sepsis.
- **Discharge** — pain controlled on oral analgesia, well and hydrated, with follow-up.
- **Refer** — hematology; PICU for acute chest/stroke; exchange-transfusion decisions.

# Acute Bleeding in Hemophilia

## MINUTES

"Factor First" — replacing clotting factor is always the priority; give factor before imaging or detailed assessment so there is no delay to life/limb-saving treatment.

### + Calculator: Blood products

#### PRINCIPLE

- In any serious bleed — especially head injury — give factor **immediately**, before the CT scan or full work-up. Each unit/kg of factor VIII (factor 8) raises the level by ~2%; each unit/kg of factor IX (factor 9) by ~1%.<sup>39</sup>

| Scenario / agent                    | Dose                        | Target & notes                              |
|-------------------------------------|-----------------------------|---------------------------------------------|
| Hemophilia A — severe / head injury | Factor VIII<br>~50 units/kg | IV; aim factor level 80 <sup>39</sup> –100% |
| Hemophilia A — minor bleed          | Factor VIII<br>~25 units/kg | IV; aim ~50% <sup>39</sup>                  |

| Scenario / agent                        | Dose                                                      | Target & notes                                                                                                                                             |
|-----------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hemophilia B — severe                   | Factor IX<br>100–140<br>units/kg                          | IV; aim 80 <sup>39</sup> –100%                                                                                                                             |
| Mild hemophilia A                       | Desmopressin (DDAVP)                                      | Only if a documented prior response; not for major bleeds <sup>39</sup>                                                                                    |
| Known inhibitor — life/limb threat      | rFVIIa 90 mcg/kg, or FEIBA 75–100 units/kg                | IV; bypassing agents — call hematology urgently for use and guidance <sup>39</sup>                                                                         |
| Tranexamic acid (mucosal / oral bleeds) | 25 mg/kg/dose PO every 8 h, or 10 mg/kg/dose IV every 8 h | Excellent adjunct for oral/mucosal bleeds.<br><b>Avoid with FEIBA</b> (thrombosis risk) and <b>in hematuria</b> (ureteric clot obstruction). <sup>39</sup> |

### ⚠ HEAD INJURY

- Treat to 100% factor level immediately and **before** the CT.
- If weight unknown, give roughly three times the usual joint-bleed dose.
- Contact the on-call hematologist.
- Do not hesitate to give factor; use the Breslow tape to estimate weight if unknown.<sup>72</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — major or life-threatening bleeds (head, neck, gut, compartment).
- **Discharge** — minor bleeds controlled after factor, with a home plan.
- **Refer** — hematology for all; neurosurgery/surgery as needed — give factor before imaging.

## Immune Thrombocytopenia (ITP)

ASSESS

ASH 2019

Isolated thrombocytopenia in a well child — treat the bleeding, not the number.

+ Calculator: Blood products

### ▲ RECOGNIZE

- Sudden bruising, petechiae or mucosal bleeding in an otherwise **well** child (often 1–4 weeks after a viral illness), **isolated** platelets  $< 100 \times 10^9/L$  with normal hemoglobin, white count and film.<sup>249</sup>

- Grade the bleeding, not just the count: **"dry" purpura** (skin only) vs **"wet" purpura** (mucosal — oral blood blisters, epistaxis, GI/urinary) — wet bleeding drives treatment.<sup>249</sup>

#### ⚠ **RED FLAGS – THIS IS NOT SIMPLE ITP**

- Hepatosplenomegaly, lymphadenopathy, bone/joint pain, pallor or fatigue, **any other cytopenia**, or an atypical film → suspect **leukemia, aplastic anemia, HUS or NAI** and involve hematology before steroids.
- Steroids can partially treat and mask leukemia — **never start empiric steroids without a film reviewed by hematology**<sup>249</sup>.

#### ✦ **FIRST DECISION – OBSERVE OR TREAT**

- For a child with **no or minor (dry) bleeding**, observation is preferred **regardless of platelet count** — most childhood ITP is self-limiting.<sup>249</sup>
- Drug therapy is for **significant mucosal (wet) bleeding** or bleeding that affects daily life, not for a low number alone.<sup>249</sup>

Actively bleeding toddlers with counts  $< 10 \times 10^9/L$  may need a short oral-steroid course — decided case by case.

#### ▲ WORKUP

- **CBC + peripheral blood film** — confirm **isolated** thrombocytopenia and look for blasts, schistocytes or other cytopenias.
- **Blood group**, and a **direct antiglobulin test** (Coombs test) before anti-D.<sup>249</sup>
- **No routine bone marrow** in typical ITP — reserve for atypical features (other cytopenias, organomegaly, constitutional symptoms) or before splenectomy.<sup>249</sup>



| Option                      | Dose                                                   | Route & notes                                                                                                                                                               |
|-----------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prednisolone / prednisolone | 2–4 mg/kg/day (max 120 mg/day) for 5–7 days, then stop | PO; short courses need no taper (taper only if prolonged/recurrent); ideally the film is seen first; cost-effective with tolerable short-course side-effects <sup>249</sup> |
| IV immunoglobulin (IVIG)    | 0.8–1 g/kg as a single dose (may repeat once)          | IV; fastest platelet rise — first-line when a rapid rise is needed; expensive and requires special access <sup>249</sup>                                                    |
| Anti-D immunoglobulin       | 50–75 µg/kg once                                       | IV; only if Rh(D)-positive, non-splenectomized, no autoimmune hemolysis <sup>249</sup> — watch for hemolysis                                                                |
| Tranexamic acid             | 10–15 mg/kg/dose (max 1000 mg) every 8 h               | PO/IV; adjunct for mucosal bleeding — <b>avoid in hematuria</b> <sup>249</sup>                                                                                              |
| Platelet transfusion        | Only for life-threatening bleeding                     | Give <b>with</b> IVIG + steroid; consumed rapidly — not for the count alone <sup>249</sup>                                                                                  |

#### ◆ BRIDGE TO DEFINITIVE CARE

- For life-threatening bleeding (intracranial, massive GI): platelets + IVIG 1 g/kg + IV methylprednisolone

+ tranexamic acid together, and call hematology/critical care.<sup>249</sup>

- Avoid IM injections, aspirin/NSAIDs and contact sport while counts are low.

#### ▲ ADMIT · REFER

- **Admit** — wet purpura, severe bleeding, or any diagnostic doubt.
- **Refer/Call** — hematology for all new severe thrombocytopenia; urgently if red flags for malignancy — do not hesitate to reach out for guidance.

## New-Diagnosis Oncologic Emergencies

MINUTES

COG · AAP 2026

The child who arrives before anyone knows they have cancer — new malignancies can present with emergencies that are lethal in hours.

➕ Calculators: BSA & hyperhydration · Blood products

## ▶ PATTERNS TO RECOGNIZE

- **Hematologic malignancy (5):** febrile neutropenia / sepsis (functionally neutropenic despite a high WBC), hyperleukocytosis, mediastinal mass, tumor lysis syndrome, and severe coagulopathy (a subset of leukemia — APML).
- **Solid tumors (3):** tumor lysis syndrome, mediastinal mass (by origin/location), and local effects (spinal cord compression, raised ICP).

## ▶ MEDIASTINAL MASS (SVC COMPRESSION SYNDROME)

- Orthopnea, stridor, facial/neck swelling and plethora, distended neck veins, cough worse when supine — often T-cell lymphoma/leukemia.<sup>250</sup>
- **Do NOT sedate or lie the child flat** and avoid general anesthesia — the airway and great vessels can collapse irreversibly. Keep **upright**, give oxygen, get an **erect** chest film / echo, and call oncology + anesthesia/critical care.<sup>250</sup> **Always consult hematology-oncology once a mediastinal mass is identified.**

- **Getting tissue safely:** obtain the diagnosis by the least-invasive route that avoids general anesthesia — peripheral node, bone marrow, or pleural/pericardial fluid. If a procedure is unavoidable in a cooperative child, use light sedation with local/regional anesthesia and **preserve spontaneous ventilation** (avoid muscle relaxants and positive-pressure ventilation). When general anesthesia, rigid bronchoscopy, or ECMO might be needed, give **steroids** and — if the child is stable — postpone to let the mass shrink; if general anesthesia is unavoidable in moderate-to-high-risk disease, plan to escalate fast: rigid bronchoscopy, airway stent, and ECMO on standby.<sup>284</sup>

#### ■ **HYPERLEUKOCYTOSIS / LEUKOSTASIS**

- WBC > 100 ×10<sup>9</sup>/L with headache, hypoxia, confusion, priapism or visual change.<sup>250</sup>
- Hydrate (2–3× maintenance), start **tumor-lysis** prophylaxis (see [Tumor Lysis Syndrome](#)), and **avoid red-cell transfusion** unless critically anemic

— it raises viscosity. Keep platelets  $\geq 20\text{--}30 \times 10^9/\text{L}$ . Urgent oncology  $\pm$  leukapheresis.<sup>250</sup>

### ▀ COAGULOPATHY (SUSPECT APML)

- Deranged coagulation studies in suspected leukemia are a medical emergency. Inform hematology/oncology and the hematopathologist promptly to identify **acute promyelocytic leukemia (APML)** on morphology.<sup>283</sup>
- If APML is identified, **All-Trans Retinoic Acid (ATRA)** is life-saving and given without delay — under hematology/oncology guidance after hematopathology confirmation.<sup>283</sup>

### ▀ MALIGNANT SPINAL CORD COMPRESSION

- Back pain (esp. night/waking) — or **neck pain with arm weakness or numbness** when the compression is cervical — leg weakness, a sensory level, urinary retention or constipation — a neurosurgical-oncological emergency.<sup>250</sup>

- Give **dexamethasone** immediately, arrange **whole-spine MRI**, and call oncology plus the **pediatric spine service — orthopedics or neurosurgery, depending on local coverage** — outcome depends on how quickly power is preserved.<sup>250</sup>

#### ■ WORKUP

- **All:** CBC + differential, **peripheral blood smear**, electrolytes, renal & liver profile, **uric acid, LDH, phosphate, potassium, calcium**, and coagulation screen (tumor-lysis / DIC).<sup>250</sup>
- **Mediastinal mass / SVC:** **erect + supine chest X-ray**, CT chest and echocardiogram; obtain the **least-invasive** diagnostic tissue (peripheral blood, effusion, marrow).
- **Leukostasis:** WBC, coagulation, type & crossmatch; neuroimaging if neurological signs.
- **Cord compression: whole-spine MRI with contrast** — do not wait for it to start dexamethasone if the deficit is progressing.<sup>250</sup>

| Drug                                           | Dose                                                                                                                     | Route & notes                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dexamethasone (cord compression)               | 1–2 mg/kg IV once (max 10 mg), then per oncology                                                                         | IV; do not delay for imaging if deficit is progressing — but confirm no undiagnosed leukemia first where possible <sup>250</sup>                                                                                                                                                                              |
| Methylprednisolone (anterior mediastinal mass) | 16 mg/m <sup>2</sup> /dose IV every 12 h; if IV access is difficult, PO prednisone 20 mg/m <sup>2</sup> /dose every 12 h | IV/PO; <b>obtain a biopsy first if the patient's condition allows</b> — steroids can rapidly shrink lymphoma and obscure the diagnosis. If a malignant effusion is present, it may be aspirated and sent for flow cytometry. <b>Assess airway security</b> before any sedation or positioning. <sup>250</sup> |
| IV fluids (leukostasis / TLS)                  | 2–3× maintenance, no potassium initially                                                                                 | IV; with rasburicase/allopurinol per <a href="#">Tumor Lysis Syndrome</a>                                                                                                                                                                                                                                     |

### ⚠ TWO TRAPS

- Sedating a child with an anterior mediastinal mass for a “quick CT” or line has caused fatal airway collapse — imaging is **erect and unsedated**.
- And giving steroids before a firm in possible hematological malignancy

can obscure the diagnosis —  
coordinate with oncology.<sup>250</sup>

#### REFER

- **Call** pediatric oncology urgently for all of the above; involve critical care, anesthesia, neurosurgery and hematopathology as required.

10

## Neonatal

NRP 2025 · AAP hyperbilirubinemia 2022

### Neonatal Resuscitation

MINUTES

NRP 2025

Effective ventilation is the single most important step.

+ Calculator: APGAR

DECISION

**Term? Good tone? Breathing or crying?**



## YES

Delayed cord clamping  $\geq 60$  s, dry, skin-to-skin, keep warm, monitor.<sup>4</sup>

## NO

Warm, dry, stimulate, position airway, clear secretions if needed. SpO<sub>2</sub> (right hand) + ECG.

apnea / gasping / HR < 100

### GOLDEN MINUTE

Start PPV within 60 s. Term 21% oxygen; preterm 21–30%. Assess for chest rise after 15 s; if none, perform MR SOPA corrective steps.<sup>4</sup>

HR < 60 after 30 s effective PPV

### COMPRESSIONS

Coordinate 3:1 (90 compressions + 30 breaths/min), increase to 100% oxygen, ideally via a secured airway. Reassess after 60 s. **Compressions are effective only after establishing good chest rise.**<sup>4</sup>

HR still < 60

Give epinephrine; treat hypovolemia; reconsider airway, pneumothorax, dose, technique.<sup>4</sup>

| Drug / fluid             | Dose                                                   | Route & notes                                                                                                |
|--------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Epinephrine (adrenaline) | IV/IO 0.02 mg/kg (0.2 mL/kg)<br>ET 0.1 mg/kg (1 mL/kg) | 0.1 mg/mL (1:10,000). IV/IO preferred; ET only while establishing access. <sup>4</sup> Repeat every 3–5 min. |
| Volume (saline or blood) | 10 mL/kg                                               | IV/IO, for hypovolemia <sup>4</sup>                                                                          |

◆ **ALGORITHM** Full AHA/AAP neonatal resuscitation algorithm — [cpr.heart.org](https://cpr.heart.org)



# Critical CHD – Pulse Oximetry Screening

ASSESS

≥ 24 H • AAP 2025

Screen every well newborn before discharge — catches duct-dependent lesions before they collapse.

## WHO & HOW TO SCREEN

- **Who:** asymptomatic infants at **≥ 24 hours** of age (or shortly before discharge if earlier), in the well-baby nursery, **not on supplemental oxygen.**<sup>5</sup>
- **How:** measure SpO<sub>2</sub> in the **right hand (pre-ductal)** and **either foot (post-ductal).**<sup>5</sup>

## DECISION

Result of the screen?

PASS

SpO<sub>2</sub> **≥ 95% in BOTH limbs and ≤ 3%** difference between them. Routine newborn care.<sup>5</sup>

## IMMEDIATE FAIL

Any reading **< 90%** in either limb → straight to echocardiography.<sup>5</sup>

### borderline

#### RETEST ONCE

90–94% in either/both limbs, **or** > 3% hand–foot difference → repeat **once after 1 hour** (one retest, not two).<sup>5</sup> Still not passing → fail.

### any fail

**Urgent echocardiography — and rule out sepsis, PPHN, and respiratory causes (4–5× more common than CCHD)<sup>5</sup>**

## ⚠ SCREENING DOES NOT REPLACE EXAMINATION

- It can **miss left-sided obstruction**, especially coarctation of the aorta.
- Check **femoral pulses** and four-limb saturations/blood pressure on every newborn.
- Some CCHD is not hypoxemic in the first hours — a normal screen does not exclude it.<sup>5</sup>
- For the lesions, presentation, and emergency management, see

## Neonatal & Infant CHD (Cardiac & Shock).

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — a failed screen → keep for echo; cyanosis/shock → NICU + alprostadil.
- **Discharge** — a passed screen with a normal exam.
- **Refer** — pediatric cardiology for any failed screen (echo before discharge).

◆ **SEE ALSO** Lesion recognition & management: Neonatal & Infant CHD and the Hypercyanotic (Tet) Spell (Cardiac & Shock).

# Persistent Pulmonary Hypertension (PPHN)

MINUTES

TERM / LATE PRETERM

Labile, severe hypoxemia out of proportion to the lung disease — the circulation never made the switch.

## RECOGNIZE

- Severe cyanosis/desaturation **poorly responsive to oxygen**, often labile (large swings with handling).<sup>50</sup>
- **Differential cyanosis:** pre-ductal SpO<sub>2</sub> (right hand) higher than post-ductal (foot) by **> 5–10%** — right-to-left shunt across the duct.<sup>50</sup>
- Common settings: meconium aspiration, sepsis/pneumonia, birth asphyxia, congenital diaphragmatic hernia, or idiopathic.<sup>50</sup>
- Confirm by **echocardiography** — and exclude duct-dependent congenital heart disease before committing to therapy.<sup>50</sup>

### ① OPTIMIZE LUNGS, AVOID SWINGS

- **Intubate and ventilate** — recruit the lung (gentle ventilation; surfactant if parenchymal disease — meconium aspiration / RDS). **Minimal handling**, cluster cares, sedation/analgesia.
- Keep temperature, glucose, calcium, and pH normal.<sup>50</sup>



### ② TARGETS

Pre-ductal SpO<sub>2</sub> low-to-mid 90s, PaO<sub>2</sub> 55–80 mmHg (7.3–10.6 kPa). Support blood pressure and cardiac output with fluids/inotropes<sup>50</sup> so systemic pressure  $\geq$  pulmonary pressure.



### ③ PULMONARY VASODILATOR

- Suspected PPHN needs urgent neonatology/cardiology/PICU care; iNO, milrinone, noradrenaline, and ECMO referral are specialist-directed.<sup>50</sup>
- Start **inhaled nitric oxide** once the oxygenation index is  $\geq$  20–25 (after lung recruitment)
- Response is a clear rise in oxygenation within ~30 min.<sup>152</sup>



No response with OI rising toward ~40, or refractory shock → ECMO — refer to an ECMO center early.<sup>50</sup>

⚠ **ECHOCARDIOGRAM – ALWAYS**

- A neonatal echocardiogram should **always** be performed when PPHN is suspected — to confirm the diagnosis and exclude underlying congenital heart disease; iNO can be detrimental in some CHDs.<sup>50</sup>

| Therapy               | Dose / setting      | Notes                                                                                                                                                               |
|-----------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhaled nitric oxide  | start 20 ppm        | Above 20 ppm rarely adds benefit and raises methemoglobin/NO <sub>2</sub> — monitor. Assess response after 20–30 min. Do not stop abruptly (rebound) <sup>152</sup> |
| Milrinone / inotropes | specialist-directed | Support cardiac output; milrinone for left-ventricular dysfunction, in addition to its pulmonary vasodilator properties                                             |



| Therapy       | Dose / setting      | Notes                                                                                                                                                       |
|---------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noradrenaline | specialist-directed | Add-on to milrinone when blood pressure drops — raises systemic vascular resistance and further lowers the pulmonary/systemic pressure ratio <sup>286</sup> |

#### ◆ NOTE

- The approach above is suggested when the cause of pulmonary hypertension is **INCREASED PULMONARY VASCULAR RESISTANCE (PVR) AND** evidence of an ischemic hit is absent. Other scenarios of pulmonary hypertension are approached differently.
- A trial of **ORAL SILDENAFIL** can be attempted where iNO is not available, after specialist consultation.<sup>285</sup>

#### ⚠ OXYGENATION INDEX & A KEY CONTRAINDICATION

- $OI = (\text{mean airway pressure} \times FiO_2 \times 100) \div PaO_2 \text{ in mmHg}$ . **Do not give inhaled nitric oxide in duct-dependent congenital heart**

**disease** — confirm the heart on echo first.<sup>50</sup>

- Wean  $\text{FiO}_2$  and settings slowly; sudden small drops can trigger prolonged desaturation.<sup>50</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → NICU/PICU.
- **Discharge** — not acutely; after weaning support, with follow-up.
- **Refer** — neonatology/cardiology; an ECMO center if refractory.

◆ **SEE ALSO** Neonatal & Infant CHD — exclude duct-dependent CHD before attributing cyanosis to PPHN.

## Neonatal Hypoglycemia HOURS

CPS 2019

Often asymptomatic — screen the at-risk, treat early, find the persistent few.

+ Calculator: Glucose infusion rate

## ▲ RECOGNIZE & WHO TO SCREEN

- Signs are non-specific: jitteriness, poor feeding, lethargy, hypotonia, temperature instability, apnea — and, when severe, seizures. **Many affected newborns are asymptomatic, so at-risk infants must be screened.**<sup>81</sup>
- Screen at-risk infants from ~2 h of age, pre-feed: infant of a diabetic mother (IDM), large- or small-for-gestational-age (LGA / SGA-IUGR), late-preterm / preterm, perinatal stress or asphyxia, hypothermia or sepsis, maternal beta-blocker, and congenital syndromes (e.g. Beckwith-Wiedemann) or midline / microphallus features (suggesting hypopituitarism).<sup>81</sup>
- Confirm a low point-of-care (POC) reading with a **laboratory plasma glucose** — whole-blood / POC reads ~15% lower and falls further with processing delay — but **treat first; do not wait for the lab**<sup>81</sup>.

| Phase                     | Threshold                    | Meaning                                                                                                                                   |
|---------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Transitional — first 72 h | < 2.6 mmol/L (47 mg/dL)      | The physiologic nadir is normal — catch the at-risk or persistently low infant, don't over-treat healthy transitional dips. <sup>81</sup> |
| Persistent — beyond 72 h  | < 3.3 mmol/L (60 mg/dL)      | Also applies to hypoglycemia first presenting after 72 h — below this warrants investigation for a hypoglycemia disorder. <sup>81</sup>   |
| On IV dextrose            | hold > 2.6 mmol/L (47 mg/dL) | Suspect a disorder if glucose cannot be held <sup>81</sup> > 2.6 mmol/L on <u>GIR</u> 5–8 mg/kg/min.                                      |

#### DECISION

**First 72 h — symptomatic (any level), or asymptomatic glucose < 2.6 mmol/L (47 mg/dL)?**

**SYMPTOMATIC, or < 1.8 mmol/L (32 mg/dL)**

IV D10 mini-bolus 2 mL/kg, then infusion at GIR 5–8 mg/kg/min. Recheck in 20–30 min; escalate GIR in 2 mg/kg/min steps.<sup>81</sup> No IV access → glucagon IM.

**ASYMPTOMATIC 1.8–2.5 mmol/L (32–45 mg/dL)**

Feed + buccal 40% dextrose gel 0.5 mL/kg; recheck in 30 min.<sup>83</sup>

still 1.8–2.5 mmol/L

**REPEAT ONCE**

Feed ± a second buccal gel dose; recheck in 30 min.

second repeat, or symptoms appear

**Start IV D10 — mini-bolus 2 mL/kg then GIR 5–8 mg/kg/min; escalate as needed.**

| Drug             | Dose                         | Route & notes                                                                                                             |
|------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Dextrose 10%     | 2 mL/kg (200 mg/kg) bolus    | IV/IO; then infusion at GIR 5–8 mg/kg/min, escalate by 2 mg/kg/min as needed. Concentrations > 12.5% need central access. |
| Dextrose gel 40% | 0.5 mL/kg (200 mg/kg) buccal | First-line for asymptomatic; massage into buccal mucosa, then feed. May repeat once. <sup>83</sup>                        |

| Drug      | Dose                                    | Route & notes                                                                                                                                                   |
|-----------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Glucagon  | 0.2 mg/kg<br>(200 mcg/kg)<br>– max 1 mg | IM/IV as a bridge when IV access is not yet available; less effective in hyperinsulinism; watch for hyponatremia. <sup>81</sup>                                 |
| Diazoxide | 1.7–5 mg/kg/dose PO q8h                 | Endocrinology-led, for confirmed hyperinsulinism; usually with chlorothiazide. Pre-treatment echo (fluid retention, pulmonary hypertension risk). <sup>81</sup> |

### ⚠️ PERSISTENT / REFRACTORY – THINK HYPERINSULINISM

Persisting beyond 48–72 h, recurring, or needing **GIR > 8–10 mg/kg/min** to stay euglycemic warrants a **critical (hypoglycemia) sample**, drawn at glucose < 2.6–2.8 mmol/L (best < 50 mg/dL) *before* correcting: plasma glucose, insulin, C-peptide, beta-hydroxybutyrate (ketones), free fatty acids, cortisol, growth hormone — plus ammonia, lactate, and an acylcarnitine profile / urine organic acids. Hyperinsulinism = inappropriately low ketones and free fatty acids with detectable insulin, a glycemic response

to glucagon, and a high glucose requirement. Refer to endocrinology / metabolic.<sup>81</sup>

#### ✦ PEARLS

- Glucose is normally lowest in the first 24–72 h (transitional); the goal is to catch the at-risk or persistently low infant, not to over-treat healthy transitional dips.
- Do not discharge an at-risk infant until persistent hypoglycemia is excluded — i.e. pre-feed glucose stays above target<sup>81</sup> through a normal feed–fast cycle.

#### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic, persistent, or needing IV dextrose/high glucose-infusion rate.
- **Discharge** — transient, resolved, feeding well with stable pre-feed glucose.
- **Refer** — endocrinology/metabolic for persistent or recurrent hypoglycemia (hyperinsulinism workup).

# Neonatal Hyperbilirubinemia

HOURS

≥ 35 WK • AAP 2022

Use total serum bilirubin against the hour- and gestation-specific nomogram.

+ Calculators: Blood products • Corrected age

## ASSESS & DECIDE

- Any jaundice under 24 h, or visible jaundice → measure bilirubin.

**Transcutaneous bilirubin (TcB)** is a valid screen; **confirm with a total serum bilirubin (TSB)** when the TcB is within 3 mg/dL (50 μmol/L) of the phototherapy threshold, is ≥ 15 mg/dL (250 μmol/L), or once phototherapy has started.<sup>301</sup> Use the **total** value against the threshold (do not subtract the direct fraction).<sup>12</sup>

**Severe hyperbilirubinemia** = peak TSB > 425 μmol/L (25 mg/dL) or need for exchange transfusion.<sup>301</sup>

- Thresholds for phototherapy and exchange depend on **gestational age (35–40 wk), age in hours, and neurotoxicity risk factors** — read the AAP nomogram or app each time, never a single fixed number.<sup>12</sup>



- **Neurotoxicity risk factors:** gestation < 38 weeks, albumin < 3.0 g/dL (30 g/L), isoimmune hemolysis (positive DAT), G6PD deficiency or other hemolysis, sepsis, clinical instability in the prior 24 h — these **lower** the threshold.<sup>12</sup>
- **Suspect hemolysis** when the TSB rises  $\geq 5 \mu\text{mol/L/h}$  (0.3 mg/dL/h) in the first 24 h or  $\geq 3.5 \mu\text{mol/L/h}$  (0.2 mg/dL/h) thereafter, or fails to respond to intensive phototherapy → CBC, reticulocytes, smear, DAT, G6PD/pyruvate-kinase assay; sepsis screen if clinically indicated.<sup>301</sup>
- **Find the cause** when the level is high or rising — hemolytic: ABO/Rh isoimmunization (DAT), G6PD deficiency, hereditary spherocytosis; non-hemolytic: cephalohematoma or bruising, polycythemia, sepsis, hypothyroidism, and breast-milk vs inadequate-breastfeeding jaundice.<sup>12</sup>
- **Conjugated (direct) hyperbilirubinemia is a different problem** — for neonatal cholestasis see [Neonatal Cholestasis](#) in the GI & Hepatology section.

## TREAT

At/above the phototherapy line → phototherapy + feeding/hydration + find the cause.<sup>12</sup>

At 2 mg/dL (30 μmol/L) of the exchange line – pre-exchange

## ESCALATION OF CARE

Intensive phototherapy + IV fluids without delay; withhold enteral feeds; urgent TSB, direct bilirubin, CBC, albumin, chemistries, type & crossmatch, hemolysis work-up; **repeat TSB every 2–3 h**; consult neonatology and prepare transfer to a unit that can perform exchange. IVIG is not routine: give 0.5–1 g/kg over 2 h (may repeat once at 12 h) for **isoimmune** hemolytic disease at the pre-exchange threshold when exchange cannot be performed promptly.<sup>12,301</sup>

At/above exchange line, or encephalopathy

**Immediate exchange transfusion for any sign of acute bilirubin encephalopathy (high tone, arching, high-pitched cry, lethargy). NICU.**<sup>12</sup>

## ◆ NOMOGRAM & FOLLOW-UP

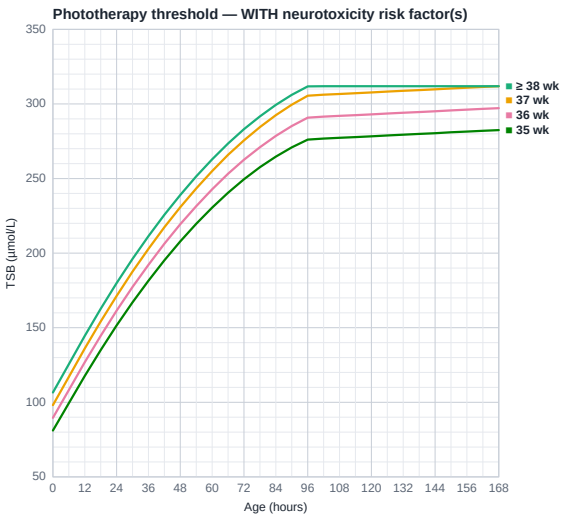
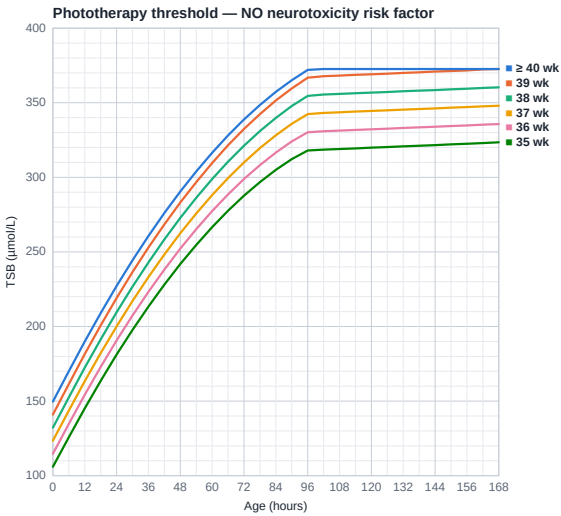
- Kuwait laboratories report bilirubin in **MMOL/L**. Plot the hour-specific total serum bilirubin on the **AAP 2022 THRESHOLD CURVES** — in  $\mu\text{mol/L}$  using the **CPS 2026 FIGURES** (Figures 2–7, AAP data reproduced with permission; cps.ca)<sup>301,12</sup> or the online calculators below, which accept either unit (+ **17.1** converts  $\mu\text{mol/L}$  to mg/dL). Decision values on this card are given in both units.
- Calculators: [peditools.org/bili2022](https://peditools.org/bili2022)  
 or [bilitool.org](https://bilitool.org)  

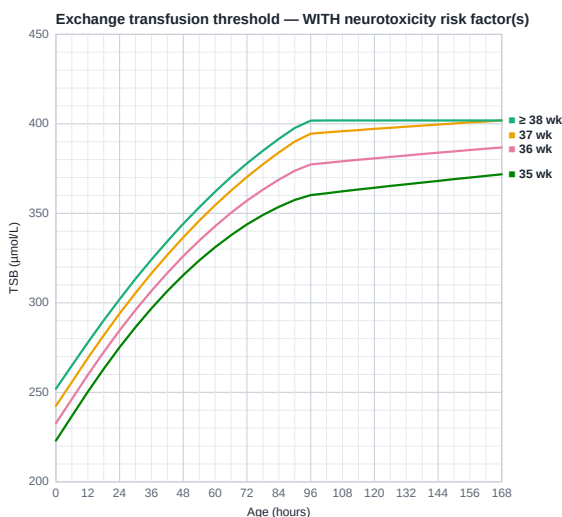
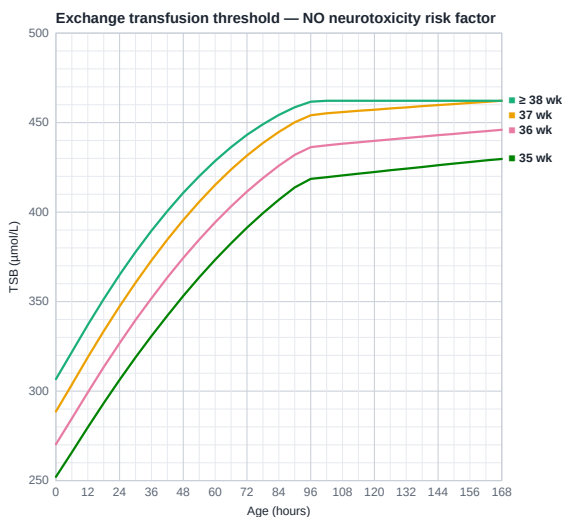
- Before discharge, set the follow-up interval by the **GAP BETWEEN THE TSB AND THE PHOTOTHERAPY THRESHOLD**: the smaller the gap, the sooner the recheck (often within 24 hours).<sup>12</sup>

## Treatment thresholds in $\mu\text{mol/L}$ – AAP 2022 curves, 35 to $\geq 40$ weeks

Redrawn from the AAP 2022 hour-specific thresholds as published in  $\mu\text{mol/L}$  by the Canadian Paediatric Society (2026), gridlines every 10  $\mu\text{mol/L}$  and 6 h, 0–168 h; beyond 168 h the 35–37-week lines rise by  $\leq 20 \mu\text{mol/L}$  and the  $\geq 38$ -week lines are flat – use the calculator for days 8–14. Full text: [CPS 2026 guideline \(doi:10.1093/pch/pxaf034\)](https://doi.org/10.1093/pch/pxaf034) · [AAP 2022 guideline \(doi:10.1542/peds.2022-058859\)](https://doi.org/10.1542/peds.2022-058859).

Neurotoxicity risk factors: albumin < 30 g/L, isoimmune or other hemolytic disease, sepsis, significant clinical instability in the preceding 24 h. <sup>301,12</sup>





### ◆ $\Delta$ TSB – THE “RULES OF 30” (CPS 2026)

- **$\Delta$ TSB** = hour-specific phototherapy threshold – measured TSB or TcB (screen at  $\geq 12$  h of age, before discharge;

TcB valid only if not on phototherapy).<sup>301</sup>

- **$\Delta$ TSB  $\leq 30 \mu\text{mol/L}$**  → delay discharge: < 24 h old, consider starting phototherapy;  $\geq 24$  h old, repeat TSB in 4–12 h and start phototherapy if rising. **31–60** → repeat TSB in 12–24 h. **61–90** → repeat in 24–48 h. **> 90** → routine follow-up with the primary provider.<sup>301</sup>
- With neurotoxicity risk factors, phototherapy may be started when the TSB is within  $30 \mu\text{mol/L}$  of the threshold.<sup>301</sup>
- **Stop phototherapy** once the TSB is  $> 30 \mu\text{mol/L}$  below the threshold ( $\geq 38$  wk) or  $> 60 \mu\text{mol/L}$  below (35–37 wk). Re-check TSB 12–24 h after stopping for rebound (re-crossing the threshold within 72–96 h); with hemolytic disease re-check at 6–12 h, then every 24 h until the TSB is falling and  $\Delta$ TSB  $> 60$ . TcB may be used  $\geq 18$  h after phototherapy stops.<sup>301</sup>

**⚠ DON'T RELY ON A SINGLE VALUE**

- Plot against time.

- A rapidly rising level, early jaundice, or hemolysis can cross thresholds fast — recheck on the interval the nomogram specifies.<sup>12</sup>
- TcB should **not** be used for follow-up when the baby is on phototherapy — use serum bilirubin.<sup>12</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — levels at/above phototherapy or exchange thresholds, or a rapid rise.
- **Discharge** — below threshold, feeding well, with a bilirubin-recheck plan.
- **Refer** — hematology for hemolysis; an exchange-capable unit near exchange thresholds; **audiology** (formal hearing assessment) if peak TSB > 400  $\mu\text{mol/L}$ , the pre-exchange threshold was reached, or exchange was performed;  
**neurodevelopmental follow-up** if the pre-exchange threshold was reached, exchange was performed, or neurological signs occurred; after

immune hemolytic disease, check hemoglobin at 4–10 weeks for late anemia.<sup>301</sup>

## Neonatal Sepsis – Early-Onset (rapid approach)

### MINUTES

Non-specific: temperature instability, poor feeding, lethargy, apnea, distress.

### ▲ ACT ON SUSPICION

- Septic screen (blood culture ± CSF/urine, blood count, CRP), check glucose. Start empiric antibiotics promptly — do not wait for results. Support airway/breathing/circulation; NICU.<sup>251</sup>

### ◆ EARLY- VS LATE-ONSET

**Early-onset ( $\leq 72$  h)** is vertically acquired — group B *Streptococcus*, *E. coli*, *Listeria*; empiric **ampicillin + gentamicin** (below). **Late-onset ( $> 72$  h)** is usually nosocomial (coagulase-negative staphylococci, *S. aureus*, Gram-negatives) and needs



different empiric cover per your unit protocol.

| Drug       | Dose                                      | Route & notes                                                                                                 |
|------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Ampicillin | 50 mg/kg/dose (meningitis 100 mg/kg/dose) | IV; interval by postnatal age. Use the meningitic dose if CSF not obtained or CNS involvement. <sup>251</sup> |
| Gentamicin | 4–5 mg/kg                                 | IV (interval by gestation/age) <sup>251</sup>                                                                 |
| Cefotaxime | 50 mg/kg                                  | IV, add if meningitis suspected <sup>251</sup>                                                                |
| Acyclovir  | 20 mg/kg every 8 h                        | IV, add if herpes simplex suspected <sup>251</sup>                                                            |

### ⚠ BILIOUS VOMITING IN A NEONATE

Green vomiting is **malrotation with volvulus until proven otherwise** — see Surgical.

### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — all with risk factors or clinical signs → empiric antibiotics after cultures.
- **Discharge** — not until sepsis is excluded/treated; a reassuring 36–48 h course with negative cultures may stop antibiotics.

- **Refer** — neonatology/PICU for instability.

◆ **SEE ALSO** The neonate ( $\leq 28$  days) is always high-risk in the febrile-infant pathway — use [Fever in the Young Infant](#) for the full age-stratified workup; this card is the neonatal-specific rapid approach.

## Omphalitis & Delayed Umbilical Separation

HOURS

NEONATE

Red or discharging umbilical stump — treat as neonatal sepsis; watch for necrotizing fasciitis.

### RECOGNIZE

- Periumbilical **erythema, induration, tenderness**; purulent or malodorous discharge from the stump.<sup>159</sup>
- Spreading abdominal-wall cellulitis, or systemic signs (poor feeding, lethargy, temperature instability, tachycardia) = **invasive infection / sepsis**.<sup>108</sup>

- Uncommon with dry cord care (~1 in 1000) but can progress rapidly<sup>108</sup> — treat as a neonatal emergency.
- **Risk factors:** low birth weight or prematurity, prolonged rupture of membranes or maternal infection, non-sterile or home delivery, umbilical catheterization, and unclean cord practices (e.g., cow-dung application — also a tetanus risk). Onset is typically around day 3.<sup>159</sup>

#### ⚠ **RED FLAGS → NECROTIZING FASCIITIS / MYONECROSIS (SURGICAL EMERGENCY)**

- Pain or tenderness out of proportion, rapidly spreading or **dusky / bullous** skin, crepitus, peau-d'orange edema, systemic toxicity.
- It can deteriorate fast despite mild initial findings — reassess often and mark the margins.
- Any of these → **urgent pediatric surgery for debridement**; do not delay for imaging.<sup>108</sup>
- Debridement resects the umbilical vein, both arteries, and any urachal remnant (even if they look normal); reported mortality is high

(~60–85%), so err toward early surgical exploration in the critically ill infant.<sup>108</sup>

### INVESTIGATE & TREAT

- Blood culture (+ discharge swab), CBC, CRP; full neonatal sepsis workup.
- Start IV antibiotics **immediately** — cover staphylococci (incl. MRSA), group A Streptococcus, Gram-negatives (E. coli, Klebsiella, Proteus) and anaerobes (Bacteroides, Clostridium); the exact regimen follows the local antibiogram and neonatal formulary.<sup>159</sup>
- If systemic: keep nil by mouth (septic ileus), IV fluids / parenteral nutrition; manage as neonatal sepsis.

**EMPIRIC IV ANTIBIOTICS – DOSES SHOWN ARE GENERAL PEDIATRIC (PER DOSE · INTERVAL); ADJUST FOR NEONATES BY GESTATIONAL/POSTNATAL AGE AND FOR HEPATIC/RENAL IMPAIRMENT; CONFIRM AGAINST THE LOCAL FORMULARY AND ANTIBIOGRAM**

| Drug            | Dose (per dose · interval)       | Role                                                                                             |
|-----------------|----------------------------------|--------------------------------------------------------------------------------------------------|
| Cefotaxime      | 50 mg/kg/dose q6h                | Gram-negative cover                                                                              |
| + Cefazolin     | 33–50 mg/kg/dose q8h             | Gram-positive (MSSA) — use <b>vancomycin</b> where MRSA is prevalent                             |
| ± Metronidazole | 7.5–10 mg/kg/dose q6–8h          | Add for anaerobes — foul discharge or maternal chorioamnionitis                                  |
| Meropenem       | 40 mg/kg/dose q8h                | <b>OR (severe cases)</b> — alternative for necrotizing fasciitis (with vancomycin + clindamycin) |
| ± Vancomycin    | 15 mg/kg/dose q6h (max 1 g/dose) | Where MRSA is prevalent; monitor levels                                                          |
| ± Clindamycin   | 5–10 mg/kg/dose q6–8h            | Necrotizing fasciitis (antitoxin effect)                                                         |

**◆ DURATION**

- About 7 days for uncomplicated periumbilical cellulitis; 10–14 days

IV with bacteremia or sepsis; 14–21 days if meningitis; prolonged and surgically-guided in necrotizing fasciitis.

- Tailor to cultures, susceptibilities, and clinical response — older infants may step down to oral once clearly improving.<sup>108</sup>

#### ◆ DELAYED UMBILICAL SEPARATION

- Normal separation is by ~2 weeks (range 3–45 days, mean ~10 days); antiseptic cord care delays it versus dry care, so a **well** infant is usually benign.
- Not off by **3–4 weeks** = delayed (up to ~10% of newborns). ⚠ **Suspect leukocyte adhesion deficiency (LAD)** if delayed separation + omphalitis or soft-tissue infection that forms **no pus** + persistent marked leukocytosis without infection (often  $40\text{--}60 \times 10^9/\text{L}$ ) + poor wound healing → **consult immunology whenever LAD is suspected**. There is a growing group of LAD syndromes with variable defects; normal expression of CD18 alone is not sufficient to exclude LAD-1 —

functional assays must be performed if clinical suspicion is high.<sup>159</sup>

- Most isolated delay in a well infant with a normal white count is **not** LAD.<sup>159</sup>
- Other causes: a patent urachus or omphalomesenteric duct (see below), factor XIII deficiency (stump bleeding).

### UMBILICAL GRANULOMA

- Moist pink granulation after separation → silver-nitrate cautery (protect surrounding skin) or topical salt.
- Refer if persistent discharge or a polyp suggests a duct remnant.<sup>159</sup>

### ⚠ DON'T MISS – PATENT DUCT REMNANTS

- **Patent omphalomesenteric (vitelline) duct** = persistent ileum-to-umbilicus connection;  
**patent urachus** = bladder-to-umbilicus. Both present after cord separation with intermittent umbilical drainage:  
**small-bowel contents** (omphalomesenteric duct) or **clear**

**urine-like fluid, more with crying or voiding** (urachus). <sup>302,303</sup>

- A sinus or small opening may be visible at the umbilical base; **ultrasound** may show a tubular structure extending deep to the umbilicus. If an opening is present it can be cannulated for a **sinogram/fistulogram** to demonstrate communication with the ileum or bladder. <sup>303,304</sup>
- Other remnants that mimic granuloma: **umbilical polyp** (bright-red, mucosa-covered, does not respond to silver nitrate — ectopic ileal/gastric mucosa) and urachal or omphalomesenteric **cyst/sinus** (persistent wetness, infected periumbilical mass). <sup>305</sup>
- ► **Refer all to pediatric surgery** — prolapsed bowel through a patent omphalomesenteric duct is a surgical emergency; small urachal remnants often close spontaneously in infancy, so excision is decided by the surgeon. <sup>303,304</sup>

#### ▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → IV antibiotics (neonatal-sepsis pathway).



- **Discharge** — not acutely; complete the IV course — older infants may step down to oral.
- **Refer** — pediatric surgery for necrotizing fasciitis or a suspected duct remnant; immunology for suspected leukocyte adhesion deficiency.

## Necrotizing Enterocolitis

HOURS

### Modified Bell

Preterm or stressed neonate: feeding intolerance, distension, bloody stools.

### RECOGNIZE

- Feeding intolerance, bilious or rising gastric residuals, **abdominal distension and tenderness**, bloody stools.<sup>84</sup>
- Systemic: lethargy, apnea, temperature instability, shock. Mostly preterm (onset often the 2nd–4th week, later at lower gestation); term infants with congenital heart disease or asphyxia are also at risk.<sup>84</sup>

- Risk rises with prematurity and formula feeding; human (breast) milk and standardized feeding protocols are protective.<sup>84</sup>

#### CONFIRM

- Abdominal X-ray (supine + left-lateral decubitus): **pneumatosis intestinalis** (intramural gas — the radiologic hallmark), portal venous gas, a persistent fixed dilated loop, or free air.<sup>85</sup>
- Bloods: CBC, CRP, blood gas, lactate, blood culture, coagulation, electrolytes.
- Bowel ultrasound (wall thickness/perfusion, free fluid) is a useful adjunct.<sup>85</sup>
- Repeat X-ray every 6 h while acute.<sup>85</sup>

| Modified Bell stage       | Findings                                                                                                                            | Bowel rest + antibiotics                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| I —<br>Suspected          | Nonspecific signs (apnea, temperature instability), distension, occult or gross blood; X-ray normal or ileus                        | NPO + NG drainage ~3 days; stop antibiotics if NEC is excluded <sup>85</sup> |
| II —<br>Definite (proven) | Above + absent bowel sounds ± tenderness; pneumatosis ± portal venous gas (IIB: metabolic acidosis, thrombocytopenia) <sup>85</sup> | NPO + NG ~7–10 days; antibiotics 7–10 days                                   |
| III —<br>Advanced         | Critically ill: shock, acidosis, DIC, peritonitis. IIIA bowel intact; IIB perforated (pneumoperitoneum) <sup>85</sup>               | NPO + NG ~14 days; antibiotics 14 days; surgery                              |

## TREAT

- Stop enteral feeds (NPO), nasogastric tube on free drainage, IV fluids and parenteral nutrition.
- Broad-spectrum antibiotics covering Gram-negatives and anaerobes — e.g. ampicillin + gentamicin + metronidazole (or piperacillin-tazobactam); add vancomycin if a central line or a resistant organism is suspected.<sup>85</sup>
- Correct acidosis, thrombocytopenia, and coagulopathy; support blood pressure.
- Serial exams and X-rays.
- **Surgical consultation** if no improvement or signs of perforation are present.

## ⚠ SURGERY NOW

Pneumoperitoneum (perforation) is an absolute indication. Also: deterioration despite maximal medical therapy, a fixed dilated loop, abdominal-wall erythema, a palpable mass, or a positive paracentesis (stool / bile / pus). Options: laparotomy with resection, or a primary peritoneal drain in the unstable extremely-low-birth-weight infant.<sup>85</sup>

## ✦ PEARLS

- **Pneumatosis intestinalis is the radiologic hallmark**; portal venous gas signals more extensive disease and a worse prognosis.<sup>84</sup>
- **Spontaneous intestinal perforation (SIP) is a distinct entity** — earlier onset (first week), often after indomethacin or steroids, a focal perforation with *no* pneumatosis; treat with anaerobe-covering antibiotics (~7 days) plus drainage or laparotomy.<sup>84</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → NICU, nil by mouth, gut decompression, antibiotics.
- **Discharge** — not acutely; after recovery, with feeding advice.
- **Refer** — pediatric surgery (perforation or deterioration).

## Neonatal Seizures

MINUTES

ILAE 2023

Often subtle — and often a treatable cause behind them.

### ▲ RECOGNIZE & CONFIRM

- Subtle signs: lip-smacking, cycling or pedaling, eye deviation, apnea, focal clonic or tonic movements.<sup>30</sup>

- **Only focal clonic and focal tonic seizures can be diagnosed clinically** — all other types need EEG or amplitude-integrated EEG (aEEG) to confirm. Electrographic-only seizures are common, especially after the first drug (electroclinical dissociation), so monitor with EEG and **treat electrographic seizures.**<sup>30</sup>
- Common causes: hypoxic-ischemic encephalopathy (most common), stroke, intracranial hemorrhage, infection / meningitis, metabolic (low glucose, calcium, sodium, magnesium), inborn errors, and genetic or structural epilepsies.<sup>30</sup>

#### FIND THE CAUSE

- Glucose, calcium, magnesium, sodium; blood gas, lactate, ammonia.
- Sepsis screen including lumbar puncture, with **CSF HSV PCR** (start empiric IV acyclovir if neonatal HSV is possible)<sup>30</sup>
- Cranial ultrasound, then MRI. If unexplained: metabolic work-up (plasma amino acids, urine organic acids, CSF glucose / lactate / glycine) and genetic testing for early-onset epilepsy (e.g. KCNQ2, SCN2A). **Correct reversible causes first** — refractory hypocalcemia is often driven by low magnesium.<sup>30</sup>

| Drug          | Dose                                                      | Route & notes                                                                                                                                                                 |
|---------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenobarbital | 20 mg/kg IV load (further 10–20 mg/kg to ~40 mg/kg total) | First-line regardless of etiology (superior to levetiracetam, NEOLEV2). Use the shortest course; discontinue early once acute provoked seizures are controlled. <sup>86</sup> |
| Levetiracetam | 40–60 mg/kg (max 4.5 g)                                   | IV; second-line — preferred when there is cardiac disease. <sup>30</sup>                                                                                                      |

| Drug                        | Dose                                    | Route & notes                                                                                                                                       |
|-----------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenytoin / fosphenytoin    | 20 mg/kg (PE)                           | IV; second-line; favored for suspected sodium-channelopathy (e.g. KCNQ2, SCN2A). <sup>30</sup>                                                      |
| Midazolam infusion          | 0.05–0.4 mg/kg/h (after 0.2 mg/kg load) | IV; refractory seizures. <sup>30</sup>                                                                                                              |
| Lidocaine (lignocaine)      | specialist infusion (load, then taper)  | IV; refractory. <b>Avoid with phenytoin and in cardiac disease</b> (proarrhythmic). <sup>30</sup>                                                   |
| Pyridoxine                  | 100 mg IV                               | Trial under EEG + cardiorespiratory monitoring (apnea / cerebral depression); for vitamin B6-dependent epilepsy or unexplained refractory seizures. |
| Pyridoxal-5-phosphate (PLP) | 30 mg/kg/day ÷ doses                    | If pyridoxine fails and PNPO deficiency is suspected.                                                                                               |
| Folinic acid                | 3–5 mg/kg/day                           | Add-on for pyridoxine-dependent epilepsy with incomplete response.                                                                                  |



| Drug                                                     | Dose                                         | Route & notes                                           |
|----------------------------------------------------------|----------------------------------------------|---------------------------------------------------------|
| Dextrose 10% / calcium gluconate 10% / magnesium sulfate | 2 mL/kg / 0.5–1 mL/kg slow / 0.1–0.2 mmol/kg | IV; correct hypoglycemia, hypocalcemia, hypomagnesemia. |

### ✦ PEARLS

- **Phenobarbital remains first-line** (superior to levetiracetam for electrographic control, NEOLEV2 RCT), but use the shortest course and **stop early** once acute provoked seizures are controlled — often before discharge.<sup>86</sup>
- Refractory with no cause found: trial **pyridoxine**, then **pyridoxal-5-phosphate** if PNPO is suspected, then add **folinic acid** — these treat vitamin-responsive epilepsies that anticonvulsants miss.<sup>30</sup>

### ⚠ JITTERY IS NOT SEIZING

- Jitteriness stops with gentle holding and has no eye or autonomic features; true seizures do not stop with restraint.<sup>30</sup>
- Don't miss glucose, calcium, magnesium, or infection.

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → NICU with monitoring/aEEG.
- **Discharge** — not acutely; under neurology with a plan.
- **Refer** — neurology and metabolic/genetics for the cause.

## Vitamin K Deficiency Bleeding (VKDB)

MINUTES

AAP 2022

Preventable with one injection at birth — the late form presents as catastrophic intracranial hemorrhage.

+ Calculator: Blood products

### ▲ RECOGNIZE (THREE WINDOWS)

- **Early** (0–24 h) — usually maternal enzyme-inducing drugs (anticonvulsants, rifampicin, warfarin).<sup>252</sup>
- **Classic** (day 2–7) — bruising, GI or umbilical bleeding in a baby who missed prophylaxis.<sup>252</sup>

- **Late** (2–12 weeks) — exclusively breastfed, no/oral-only prophylaxis, or malabsorption/cholestasis; up to half present with **intracranial hemorrhage** (bulging fontanelle, seizures, pallor, poor feeding).<sup>252</sup>

#### ⚠ **ANY BLEEDING NEONATE**

- Check a coagulation screen (prolonged PT/aPTT with normal platelets and fibrinogen suggests VKDB), give vitamin K without waiting for confirmation, and image the brain if any neurological sign.<sup>252</sup>
- Investigate for **cholestasis/malabsorption** in late VKDB.<sup>252</sup>

#### ▴ **WORKUP**

- **Coagulation screen — prolonged PT** (most sensitive) and aPTT with **normal platelets and fibrinogen** (distinguishes from DIC); rapid correction after vitamin K confirms.<sup>252</sup>
- CBC / film.

- **Neuroimaging** (cranial ultrasound / CT) if any neurological sign — late VKDB commonly presents as intracranial hemorrhage.<sup>252</sup>

| Setting              | Dose                                                                | Route & notes                                                                                       |
|----------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Prophylaxis at birth | Phytonadione (vitamin K <sub>1</sub> )<br>1 mg (0.5 mg if < 1500 g) | IM — single dose; the reliable route <sup>252</sup>                                                 |
| Active bleeding      | Vitamin K <sub>1</sub> 1–2 mg slow IV                               | Plus <b>FFP</b> or prothrombin complex concentrate for serious/intracranial bleeding <sup>252</sup> |

#### ◆ COUNSELING

- IM prophylaxis at birth prevents both classic and late VKDB; purely oral regimens are less protective, especially in breastfed infants and cholestasis.<sup>252</sup>
- Reinforce with families who decline the injection.

#### ▲ ADMIT · REFER

- **Admit** any bleeding neonate; PICU/neurosurgery for intracranial hemorrhage.

- **Refer** — gastroenterology if underlying cholestasis/malabsorption.

## Neonatal Abstinence Syndrome

ASSESS

ESC-NOW • NEJM 2023

The jittery, inconsolable, poorly feeding newborn after in-utero drug exposure.

### RECOGNIZE

- Tremors/jitteriness, irritability and a high-pitched cry, hypertonia, poor feeding/uncoordinated suck, vomiting/loose stools, sweating, sneezing, and disturbed sleep — typically within the first days after opioid exposure (later for long-acting agents).<sup>253</sup>

### ⚠ EXCLUDE MIMICS; WATCH FOR SEIZURES

Hypoglycemia, sepsis, hypocalcemia, hyperthyroidism and hypoxic-ischemic encephalopathy can look similar — check them. **Seizures** occur in severe withdrawal and warrant pharmacological treatment and escalation.<sup>253</sup>

## ✦ NON-PHARMACOLOGICAL CARE FIRST

- The **Eat-Sleep-Console** approach — rooming-in, breastfeeding where appropriate, swaddling, low stimulation, skin-to-skin — reduces the need for medication and shortens stay.<sup>253</sup>
- Reserve opioids (morphine or methadone) for infants who cannot feed, sleep or be consoled despite optimal supportive care.<sup>253</sup>

## ▲ WORKUP

- Primarily **clinical** — the **Eat-Sleep-Console** assessment (feeds  $\geq 1$  oz / breastfeeds · sleeps  $\geq 1$  h · consoled  $\leq 10$  min).
- Confirm exposure with **maternal history + maternal/infant toxicology** (urine, meconium, or umbilical cord).<sup>253</sup>
- Exclude mimics: glucose, calcium, sepsis screen and thyroid studies as indicated.

### ▲ ADMIT · REFER

- **Admit**/observe for scoring; escalate if seizures or failure of supportive care.
- **Refer** — neonatology, social work and safeguarding as indicated.

## Neonatal Dosing by Corrected Age (PMA)

### DOSING

NeoFax / Lexicomp / BNFc • confirm against unit for

The dose per kg barely changes — corrected age changes the **interval**.

+ Calculator: Corrected age

### ▲ THE ONE IDEA

- **Postmenstrual age (PMA) = gestational age at birth + weeks since birth.** Immature kidneys clear renally-eliminated drugs slowly, so the more premature the baby and the younger the postnatal age, the **longer the interval** between doses.
- For aminoglycosides and vancomycin, the dose stays roughly the same per kg while the gap

between doses stretches.<sup>255</sup>

### ⚠ LEVELS GOVERN, NOT A FIXED SCHEDULE

- Any published regimen is a **starting point** only.
- After the first doses, the **drug level** sets the interval for gentamicin, amikacin and vancomycin — draw a trough just before the next dose and do not give it until the level is acceptable. **Extend the interval (do not cut the dose)** for renal impairment / oliguria, perinatal asphyxia or therapeutic cooling, and concurrent indomethacin / ibuprofen or vancomycin.<sup>255</sup>
- Confirm every value against your unit's formulary and a live NeoFax / BNF for Children.

### ▶ DOSES – USE A NEONATAL REFERENCE

- Neonatal and premature dosing is **PMA- and postnatal-age-dependent**, and aminoglycoside/vancomycin regimens are titrated to **drug levels** — so specific doses and intervals are **not reproduced here**.



- Use a maintained neonatal dosing reference — **NeoFax**, **Lexicomp**, or the **BNF for Children** — for gentamicin, amikacin, vancomycin and  $\beta$ -lactam dosing by PMA and postnatal age, and confirm every value against your **unit protocol / NICU formulary**.

◆ **EMPIRIC EARLY-ONSET SEPSIS — AMPICILLIN + GENTAMICIN** (ampicillin covers group B *Streptococcus* and *Listeria*; gentamicin adds Gram-negative cover and synergy). Avoid ceftriaxone in neonates (bilirubin displacement, calcium precipitation).<sup>251</sup>

NeoFax • AAP Red Book

#### ✦ **CAFFEINE CITRATE – THE QUINTESSENTIAL NEONATAL DRUG**

- Apnea of prematurity: **load 20 mg/kg**, then **5 mg/kg once daily** maintenance (up to 10 mg/kg/day for refractory apnea).<sup>254</sup>
- Always prescribe as **citrate** — caffeine base is half the number (20 mg citrate = 10 mg base).<sup>254</sup>

Caffeine • CAP trial / FDA label

#### ◆ **SOURCES**

- NeoFax / Micromedex (gentamicin, amikacin, vancomycin,  $\beta$ -lactams); vancomycin AUC target — ASHP/IDSA/PIDS consensus 2020; ANMF Neonatal Medicines consensus 2024; AAP Red Book (empiric regimens); CAP trial (Schmidt, NEJM 2006) and FDA Cafcit label (caffeine).
- Neonatal dosing varies by unit and drug-monitoring protocol — this is a starting point, not a substitute for your local formulary.

## 11

# Neurologic

AES / ESETT · febrile seizure

## Convulsive Status Epilepticus (> 1 month of age)

**MINUTES**

Seizure  $\geq$  5 minutes or repeated without recovery — treat by the clock.

**+ Calculator: Infusion dose  $\leftrightarrow$  rate**

0-5 MIN • STABILIZE (CLOCK STARTS AT SEIZURE ONSET)

- **Time is counted from the start of the seizure, not from arrival.** A child who reaches you already seizing  $\geq 5$  min gets the benzodiazepine **immediately** — airway, oxygen and IV/IO access proceed **in parallel**, never before the drug.<sup>11,293</sup>
- Airway + oxygen 10-15 L/min via non-rebreather mask • IV/IO • monitor.
- **Check glucose** (treat if low); send sodium, calcium, magnesium, glucose.<sup>11</sup>

5 min from onset • 1st line

BENZODIAZEPINE — MAY REPEAT ONCE AFTER 5 MIN

Lorazepam IV/IO, or diazepam IV/IO;  
**no IV access → do not wait for one:**  
intramuscular midazolam (preferred),  
buccal/intranasal midazolam, or **rectal diazepam**.<sup>11,293</sup>

~15-20 min • 2nd line

ESETT: THE THREE ARE EQUALLY EFFECTIVE

**Levetiracetam** (preferred — fewer side effects and easy administration), valproate, or fosphenytoin/phenytoin (phenobarbital is an alternative).<sup>10</sup>

~40 min · refractory

**3RD LINE · PICU** Refractory status epilepticus: urgent  
PICU/anesthesia/neurology (anesthetic infusion, continuous EEG, find the cause). Intubation is a team decision whenever needed — not reserved for this step.<sup>205</sup>

## FIRST LINE – BENZODIAZEPINES

| Drug      | Dose                                                                               | Route & notes                                                                                  |
|-----------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Lorazepam | 0.1 mg/kg<br>(max 4 mg)                                                            | IV/IO                                                                                          |
| Diazepam  | 0.2–0.3 mg/kg<br>(max 10 mg)<br>IV/IO ·<br>rectal 0.5<br>mg/kg (max 20<br>mg)      | IV/IO; rectal when<br>there is no IV access<br>— an accepted<br>first-line route <sup>11</sup> |
| Midazolam | 0.1 mg/kg IV<br>(max 4 mg)<br>0.2 mg/kg<br>IM/buccal/int<br>ranasal (max<br>10 mg) | IM/buccal/intranasal<br>preferred when no IV<br>access <sup>11</sup>                           |

## SECOND LINE – CHOOSE ONE

| Drug          | Dose                    | Route & notes                                                                                                                                                                       |
|---------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Levetiracetam | 40–60 mg/kg (max 4.5 g) | IV; often preferred for safety <sup>10</sup>                                                                                                                                        |
| Valproate     | 40 mg/kg (max 3 g)      | IV; avoid if metabolic/mitochondrial disease or liver failure suspected. Avoid in suspected mitochondrial/POLG disease, significant liver disease, or thrombocytopenia/coagulopathy |
| Fosphenytoin  | 20 mg PE/kg (max 1.5 g) | IV (PE = phenytoin equivalents)                                                                                                                                                     |
| Phenytoin     | 20 mg/kg (max 1.5 g)    | IV slow, cardiac monitoring — use if fosphenytoin unstocked                                                                                                                         |
| Phenobarbital | 15–20 mg/kg IV          | Alternative second-line; monitor for respiratory depression and hypotension                                                                                                         |

### ◆ DRAVET SYNDROME

- Known/suspected Dravet syndrome (prolonged febrile hemiclonic seizures in infancy): prefer **LEVETIRACETAM** or **VALPROATE**; sodium-channel blockers (phenytoin/fosphenytoin) are avoided as **MAINTENANCE** but are **NOT**

contraindicated for acute status epilepticus.<sup>205</sup>

- Phenobarbital is an alternative.

### THIRD LINE – REFRACTORY (PICU)

| Drug                             | Dose                              | Route & notes                                                                                                           |
|----------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Midazolam infusion               | 0.2 mg/kg load → 0.05–0.4 mg/kg/h | IV; titrate to EEG <sup>205</sup>                                                                                       |
| Ketamine / thiopental / propofol | specialist dosing                 | IV infusion titrated to EEG; <b>propofol short-term only</b> (propofol infusion syndrome — monitor labs) <sup>205</sup> |
| Pyridoxine                       | 100 mg IV                         | Trial in refractory seizures in infants <sup>205</sup> < 2 yr                                                           |
| Dextrose 10% (if low glucose)    | 2 mL/kg                           | IV/IO                                                                                                                   |

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → monitored bed; refractory or needing infusions → PICU.
- **Discharge** — not from the acute episode; a first seizure that stops may go home after workup, neurology follow-up.

- **Refer** — neurology (EEG, cause);  
**PICU for refractory status, or**  
**airway/hemodynamic compromise.**

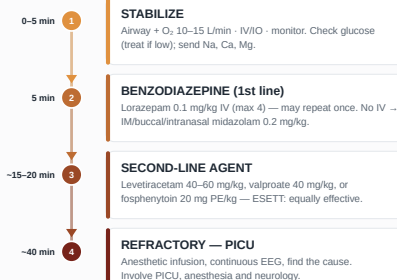
## Convulsive status epilepticus – treat by the clock

Stabilize → benzodiazepine at 5 min → second-line at ~20 min → refractory (PICU) at ~40 min.



### SEIZURE ≥ 5 MIN = STATUS EPILEPTICUS

Two benzodiazepine doses maximum, then move to second-line — don't keep repeating.



## Febrile Seizure

HOURS

6 MO–5 YR

Seizure with fever, no CNS infection or prior afebrile seizure, and no pre-existing neurologic abnormality.

### ▴ SIMPLE VS COMPLEX

- **Simple:** generalized, under 15 minutes, once in 24 hours — benign; reassure, treat the fever source,

antipyretics for comfort; no routine EEG, imaging, or LP.<sup>169</sup>

- **Complex:** focal, over 15 minutes, or recurs within 24 hours — evaluate more carefully.<sup>169</sup>

#### ✦ PEARL – NOT A SIMPLE FEBRILE SEIZURE

- Not a simple febrile seizure if any are present: **focal** features, duration **> 15 min** or recurrence within 24 h (complex), age **< 6 months or > 6 years**, incomplete recovery, or a child unwell between events.<sup>169</sup>
- A febrile seizure is, by definition, a seizure with fever in a child **without** CNS infection, metabolic disturbance, or prior afebrile seizures.<sup>292</sup> When any of the features above is present, **exclude those first** — meningitis/encephalitis (see **Bacterial Meningitis & Encephalitis**), hypoglycemia, electrolyte disturbance, and an underlying (non-febrile) epilepsy — before labelling it a *complex febrile seizure*.



## ⚠ LOWER THE THRESHOLD FOR LUMBAR PUNCTURE

- Consider meningitis with any meningism or history/exam suggesting CNS infection; in an infant 6–12 months **under-immunized against Hib or pneumococcus** (or of unknown immunization status); or in a child **pretreated with antibiotics** (may mask meningitis).<sup>169</sup>
- Age < 12 months alone is no longer an indication — AAP 2011 replaced the 1996 advice to “strongly consider” LP in every infant under 12 months; LP is now an *option* only for the under-immunized 6–12-month-old or the antibiotic-pretreated child.<sup>169</sup>
- A seizure lasting  $\geq 5$  minutes is status epilepticus (see [Convulsive Status Epilepticus](#)) — treat as above.

## ▀ ADMIT · DISCHARGE · REFER

- **Admit** — complex features, prolonged/focal, incomplete recovery, or a source needing admission.

- **Discharge** — simple febrile seizure, well child, normal exam, reliable carers with safety-net advice.
- **Refer** — neurology only for complex, recurrent, or atypical seizures — not routine for simple febrile seizures.

## Bacterial Meningitis & Encephalitis

MINUTES

> 2 MO

Fever plus altered consciousness, neck stiffness, or a bulging fontanelle — do the LP first, but never delay antibiotics for it.

### RECOGNIZE & ACT

- **Infants (non-specific):** fever, poor feeding, lethargy or decreased interaction, vomiting, irritability, ± rash; inconsolable or worsening irritability with progressive lethargy are red flags. **Nuchal rigidity is uncommon in infants.**<sup>103</sup>
- **Older children:** headache, neck pain/stiffness, photophobia, vomiting, impaired consciousness, seizures, **non-blanching rash.**<sup>119</sup>

- Take **blood cultures** (weight-based volumes), then give **empiric IV antibiotics immediately** — do not delay if LP is deferred.<sup>390</sup>
- Add **acyclovir** if encephalitis features (behavior change, focal neurology, seizures). **Encephalitis criteria:** altered mental status  $\geq 24$  h plus  $\geq 2$  of — fever, new seizures, focal neurology, CSF pleocytosis, abnormal neuroimaging, or abnormal EEG.<sup>262</sup>
- Watch for **SIADH** and **raised ICP** (see [Raised ICP](#)).

## EMPIRIC & TARGETED THERAPY – MENINGITIC DOSES

| Drug        | Dose                                                                              | Route & notes                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Ceftriaxone | 50 mg/kg/dose q12h (or 100 mg/kg/dose once daily; max 4 g/day (2 g/dose if q12h)) | IV (IM if IV not immediate); first-line empiric. Covers <i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>H. influenzae</i> . <sup>390</sup> |
| Cefotaxime  | 50–75 mg/kg/dose q6h (max 3 g/dose)                                               | IV; alternative to ceftriaxone — <b>preferred in neonates</b> . <sup>103</sup>                                                                |

| Drug          | Dose                                                                                             | Route & notes                                                                                     |
|---------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Vancomycin    | 15–20 mg/kg/dose q6h; target trough 15–20 mg/L (AUC/MIC-guided) for CNS infection (max 1 g/dose) | IV; add empirically for possible cephalosporin-resistant pneumococcus. <sup>390</sup>             |
| Ampicillin    | 75 mg/kg/dose q6h (or 50 mg/kg/dose q4h; max 12 g/day)                                           | IV; add for <i>Listeria</i> in neonates / immunocompromised / at risk. <sup>103</sup>             |
| Meropenem     | 40 mg/kg/dose q8h (max 2 g/dose)                                                                 | IV; alternative if a 3rd-gen cephalosporin is contraindicated — get ID advice.                    |
| Dexamethasone | 0.15 mg/kg/dose q6h (max 10 mg/dose)                                                             | IV; within 4 h of the first antibiotic — see steroid note. <b>Not</b> in neonates. <sup>390</sup> |

| Drug      | Dose                                                                                                                                                                                                                            | Route & notes                                                                                                                                                                                                                                            |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acyclovir | <p>≤ 4 mo: use neonatal HSV dosing (20 mg/kg/dose every 8 h). &gt; 4 mo–&lt; 12 y: 15–20 mg/kg/dose every 8 h (Red Book target 20; balance against nephrotoxicity). ≥ 12 y: 10 mg/kg/dose every 8 h. Duration 21 days (CNS)</p> | <p>IV, infuse over 1 h with adequate hydration + renal monitoring; add if HSV encephalitis suspected. Beyond the neonatal period the higher 20 mg/kg dose is <b>not</b> routine — increased neurotoxicity / nephrotoxicity (Lexicomp).<sup>158</sup></p> |

#### ◆ DURATION BY PATHOGEN

- N. meningitidis and H. influenzae 7 days; S. pneumoniae 10–14 days; group B Streptococcus 14–21 days; Listeria ≥ 21 days; aerobic Gram-negative bacilli 21 days (or ≥ 2 weeks after the first sterile CSF in neonates) — confirm the final duration with infectious diseases.<sup>390</sup>
- Confirm the pathogen and tailor to response.

## ◆ STEROIDS – DEXAMETHASONE

- Give 0.15 mg/kg/dose q6h just before, with, or **within 4 h** of the first antibiotic (earlier is better).  
**Continue 4 days if Hib or *S. pneumoniae*** is confirmed.
- Reduces hearing loss, especially in Hib. <sup>390</sup>

## DURATION, TARGETED THERAPY & FOLLOW-UP

- When susceptibilities return, narrow therapy (penicillin-susceptible pneumococcus or meningococcus → penicillin G or ampicillin; GBS → penicillin/ampicillin + gentamicin). **IV duration:** *S. pneumoniae* 10–14 d
- Hib 7–10 d
- *N. meningitidis* 5–7 d
- GBS 14–21 d (longer if cerebritis/ventriculitis). **Formal audiology before discharge** for every child. <sup>103</sup>
- Repeat LP is not routine — reserve it for failure to improve, resistant pneumococcus, or selected neonates/complex cases after ID advice; also repeat the LP at the end of therapy for confirmed HSV encephalitis. <sup>390</sup>

- Arrange **chemoprophylaxis** for close contacts of meningococcal (and Hib per criteria) via public health.<sup>390</sup>

**⚠ DEFER THE LP (BUT NEVER THE ANTIBIOTICS) IF**

- reduced or fluctuating consciousness, focal signs, signs of raised ICP, coagulopathy, or skin infection over the site (stabilize a cardiorespiratorily unstable child before any imaging) — image (contrast CT and/or MRI) first.<sup>390</sup>
- A purpuric / non-blanching rash with fever is meningococcal sepsis until proven otherwise<sup>119</sup> (see [The Rash](#), [Septic Shock](#)).

**▲ ADMIT · DISCHARGE · REFER**

- **Admit** — all → hospital; reduced consciousness, shock, seizures, or raised ICP → PICU.
- **Discharge** — not from the acute illness; complete the course with audiology and neurodevelopmental follow-up.

- **Refer** — infectious diseases/neurology; PICU for complications; public health for notification and contact prophylaxis.

## Brain Tumor – New Presentation

**HOURS**

HeadSmart / RCPCH

Persistent morning headache and vomiting with a visual, coordination, or behavioral sign — image the whole neuraxis, don't LP, and call neurosurgery.

### ▶ **RECOGNIZE – HEADSMART RED FLAGS**

- **Raised ICP:** recurrent **headache worse on waking or lying down**, **persistent vomiting (classically early-morning)**, and **papilledema** or a falling conscious level. <sup>259</sup>
- **Visual / eye signs:** new squint or double vision, **nystagmus**, reduced acuity, an **abnormal head tilt**, or abnormal eye movements. <sup>259</sup>
- **Motor / coordination:** **ataxia** or unsteady gait, new focal weakness or facial droop, and in a young child **regression / loss of milestones**. <sup>259</sup>



- **Age-specific:** infants — **enlarging head, bulging fontanelle, “sunsetting” eyes, developmental regression**; older children / teens — **behavior or school change, fatigue, delayed or arrested puberty, polyuria/polydipsia** (diabetes insipidus), and **precocious puberty** — associated with intracranial (pineal / suprasellar) germ-cell tumors.<sup>259</sup>
- A new **focal seizure**, or any of the above that is **persistent or progressive**.<sup>259</sup>

### ⚠ **DO NOT LP**

With a mass or **obstructive hydrocephalus**, an LP can precipitate **herniation** — **image first**. Don't dismiss persistent morning vomiting as gastroenteritis.<sup>259</sup>

### ◆ **DIAGNOSE**

- **MRI brain and whole spine with contrast** is the test of choice (spine imaging stages leptomeningeal / “drop” metastases); **contrast CT** if MRI is not immediately available or to

exclude acute hydrocephalus / hemorrhage.

- Look actively for **obstructive hydrocephalus** — it is a neurosurgical emergency.<sup>259</sup>

#### ▲ ACUTE MANAGEMENT

- **Position first:** head of bed up **30°**, neck midline (no venous obstruction), avoid hypotonic fluids and hyperthermia; treat seizures — the same first-line measures as for raised ICP (see **Raised ICP**).
- **Dexamethasone** for symptomatic peritumoral (vasogenic) edema / raised ICP — specialist-directed; commonly **0.25–0.5 mg/kg/dose**, and for severe edema or impending herniation a **1–2 mg/kg IV load then 1–1.5 mg/kg/day divided q6h** (max **16 mg/day**). Add a **PPI (e.g., omeprazole)** for gastric protection.
- **Call neurosurgery urgently**; treat hydrocephalus (EVD / CSF diversion) as directed.<sup>259</sup>
- **Transfer to PICU** if drowsy or dropping conscious level, and manage as **Raised ICP** in the interim.

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — any suspected tumor with raised ICP, hydrocephalus, a focal deficit, or a reduced conscious level.
- **Discharge** — not during the acute presentation; expedite imaging and specialist review.
- **Refer** — pediatric **neurosurgery / neuro-oncology** at a specialist center for resection and hydrocephalus management; involve oncology.

## Brain Abscess

HOURS

NEUROSURGICAL

Fever, headache, focal deficit, or new seizures — image with contrast MRI, treat by source, don't LP.

### ▲ RECOGNIZE

- **Subacute:** headache (most common), fever, **focal neurological deficit**, new-onset seizures, and features of **raised ICP** (vomiting, altered consciousness, papilledema). The classic triad (fever + headache + focal deficit) occurs in a **minority**.<sup>104</sup>

- **The source predicts the organisms:**  
contiguous — **otitis/mastoiditis, sinusitis, dental**, meningitis;  
hematogenous — **cyanotic congenital heart disease** (right-to-left shunt), endocarditis, pulmonary; direct — penetrating trauma or neurosurgery;  
**immunocompromise** (add fungi, *Nocardia*, *Toxoplasma*).<sup>104</sup>

#### ◆ **DIAGNOSE**

- **Contrast MRI with diffusion-weighting** is the test of choice — abscess shows **restricted diffusion** (a tumor usually does not); contrast CT if MRI unavailable.<sup>105</sup>
- Take **blood cultures. LP only after neurosurgical evaluation** — low yield, herniation risk with a focal mass; abscess drainage and culture is best.<sup>105</sup>
- Contrast MRI + neurosurgery + IV antibiotics by source.
- Urgent neurosurgery + ID; empiric therapy follows source and local resistance.

## DECISION – WHAT IS THE LIKELY SOURCE?

OTOGENIC / SINUS / DENTAL /  
HEMATOGENOUS

Ceftriaxone (or cefotaxime) +  
metronidazole + vancomycin. Covers  
the *Streptococcus anginosus* (milleri)  
group and anaerobes.<sup>105</sup>

POST-TRAUMA / POST-NEUROSURGICAL

Vancomycin + an antipseudomonal  
beta-lactam (cefepime, ceftazidime, or  
meropenem) — covers *S. aureus* (incl.  
MRSA), Gram-negatives and  
*Pseudomonas*.<sup>105</sup>

## EMPIRIC AGENTS – PEDIATRIC DOSES

| Drug                       | Dose                                                                                                | Route & notes                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Ceftriaxone</b>         | 50 mg/kg/dose<br>q12h (or 100<br>mg/kg/dose<br>once daily;<br>max 4 g/day<br>(2 g/dose if<br>q12h)) | IV; streptococci +<br>Gram-negatives.<br>Cefotaxime is an<br>alternative. |
| <b>+<br/>Metronidazole</b> | 7.5<br>mg/kg/dose<br>q6h (max 500<br>mg)                                                            | IV; anaerobic cover for<br>otogenic / sinus /<br>dental sources.          |
| <b>+<br/>Vancomycin</b>    | 15 mg/kg/dose<br>q6h; trough<br>15–20 mg/L<br>(max 1<br>g/dose)                                     | IV; add for <i>S. aureus</i> /<br>MRSA or severe<br>disease.              |

| Drug                            | Dose                                | Route & notes                                                                       |
|---------------------------------|-------------------------------------|-------------------------------------------------------------------------------------|
| Meropenem (OR —<br>alternative) | 40 mg/kg/dose<br>q8h (max 2 g)      | IV; cephalosporin allergy, resistant Gram-negatives, or suspected <i>Nocardia</i> . |
| ±<br>Vancomycin                 | 15 mg/kg/dose<br>q6h (max 1 g/dose) | Add to meropenem where <i>S. aureus</i> / MRSA is likely.                           |

## SURGERY, DURATION & ADJUNCTS

- **Neurosurgical aspiration or excision** for diagnosis and for potential abscess drainage, or if not responding — send pus for aerobic, anaerobic, fungal and mycobacterial studies. **IV antibiotics ≈ 4–8 weeks**, guided by clinical response and serial imaging; involve **infectious diseases**.<sup>105</sup> **Treat the source** (mastoid, sinus, dental). **Urgent PICU consult for ICP management**; give **anticonvulsants** for seizures.
- Reserve **steroids** for significant edema / mass effect — they can reduce antibiotic penetration.<sup>105</sup>

## ⚠ DON'T LP RIGHT AWAY – WATCH FOR HERNIATION

- With a focal mass, an LP can precipitate herniation and rarely

changes management<sup>104</sup> — image first.

- Rapid deterioration, a large abscess with midline shift, or impending herniation → emergency neurosurgery & PICU consult (see [Raised ICP / The Comatose Child](#)).

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → for treatment and monitoring; potential abscess drainage.
- **Discharge** — after prolonged IV therapy and imaging response, with follow-up.
- **Refer** — neurosurgery (drainage) and infectious diseases; find the source (ENT/dental/cardiac).

## Raised ICP / The Comatose Child

### MINUTES

Protect the brain: airway, glucose, and reverse herniation fast.

+ Calculator: [Glasgow Coma Scale](#)

## RECOGNIZE HERNIATION

- Falling conscious level (GCS / AVPU), unequal or fixed pupils, abnormal posturing, and the **Cushing triad** (hypertension + bradycardia + irregular breathing) = impending herniation.<sup>80</sup>
- Coma work-up — **DEFG: "Don't Ever Forget Glucose."** Sort causes with **AEIOU-TIPS**: alcohol/toxins, epilepsy, insulin/glucose, opioids, uremia/metabolic, trauma, infection, psychiatric, stroke/shock.

## STABILIZE

- Airway (add C-spine control if trauma; use an appropriately sized collar), oxygen, ventilation.
- Check glucose; treat seizures.
- Position the head midline and elevate the head of the bed to 30<sup>80°</sup>.
- Keep normothermic and normoglycemic; avoid hyponatremia — keep sodium in the normal-to-high range.<sup>80</sup>
- **Call PICU.**





## LOWER ICP

Maintain oxygenation and **normocapnia** (avoid routine hyperventilation); give hyperosmolar therapy; urgent CT + neurosurgery.<sup>80</sup>

| Drug                             | Dose                                      | Route & notes                                                                                                        |
|----------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Hypertonic saline 3%             | 3–5 mL/kg over 10–15 min                  | IV; for raised ICP / herniation <sup>80</sup>                                                                        |
| Mannitol                         | 0.5–1 g/kg over 10–15 min                 | IV; alternative (caution if hypovolemic/hypotensive) <sup>80</sup>                                                   |
| Dextrose 10% — hypoglycemia only | 2–5 mL/kg (0.2–0.5 g/kg); recheck glucose | IV/IO <b>only to correct documented hypoglycemia</b> (“don’t ever forget glucose”); use the smallest effective dose. |

## ⚠ FLUIDS – AVOID FREE WATER

In suspected raised ICP, **avoid dextrose-containing and hypotonic fluids** as a bolus or maintenance — once the glucose is metabolized the leftover free water worsens cerebral edema. Maintain with **isotonic 0.9% saline**.<sup>80</sup> The one exception is **documented hypoglycemia**, which injures the brain and must be corrected.

### ⚠ **DO NOT DELAY TRANSFER**

Do not delay transfer to PICU/neurosurgery; repeat hyperosmolar therapy is specialist-guided.

### ⚠ **HYPERVENTILATION**

Brief hyperventilation **only** as a bridge for impending herniation; sustained hypocapnia causes cerebral ischemia.<sup>80</sup>

### ▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → PICU.
- **Discharge** — not from the acute episode.
- **Refer** — neurosurgery/neurology and the primary-cause specialty (tumor, infection, metabolic, trauma).

# Guillain-Barré Syndrome / Acute Flaccid Paralysis

MINUTES

EAN/PNS 2023

Ascending weakness that can reach the diaphragm — the emergency is respiratory failure and autonomic instability.

## RECOGNIZE

- Progressive, fairly **symmetric ascending weakness** with **reduced/absent reflexes**, often after a GI or respiratory infection; a refusal to walk in a young child. <sup>260</sup>
- CSF (if done) shows **high protein with normal cell count** (albuminocytologic dissociation) — may be normal in the first week; MRI may show nerve-root enhancement. <sup>260</sup>

## THE TWO THINGS THAT KILL

- **Respiratory failure** — track vital capacity/effort; rising CO<sub>2</sub> and a falling forced vital capacity (roughly < 15–20 mL/kg) mean elective intubation, not a trial of NIV. **Autonomic instability** — labile blood pressure, arrhythmia,

ileus; put the child on a cardiac monitor.<sup>260</sup>

- Rapid progression or bulbar weakness (choking, weak cough) = ICU.<sup>260</sup>

#### ✦ DON'T BE REASSURED BY A NORMAL EARLY TAP

- Weakness can outrun the CSF and the reflexes early on.
- If the picture fits, admit and monitor breathing serially rather than discharging on a single normal test.
- Always exclude a compressive cord lesion (a sensory level, back pain or sphincter loss points to the cord, not GBS).<sup>260</sup>

#### ▀ WORKUP

- **Lumbar puncture** — albuminocytologic dissociation (high protein, normal cells); **may be normal in the first week.**<sup>260</sup>
- **Nerve conduction studies / EMG** (subtype; may be normal early — repeat).

- **Serial bedside spirometry** — FVC (and NIF/MIP) **every 4–6 h** in the progressive phase, plus bulbar assessment;<sup>260</sup> act on the **"20/30/40" thresholds** (FVC < 20 mL/kg, MIP < 30, MEP < 40 cmH<sub>2</sub>O) or a **> 30% fall** from baseline — these predict the need for ventilation.<sup>287</sup>
- **Spirometry is unreliable under ~5–6 years** or with facial/bulbar weakness (poor mouth seal) — there, rely on **serial clinical assessment**: single-breath count, cough strength, paradoxical breathing, ability to lie flat, and the blood-gas/CO<sub>2</sub> trend.<sup>260</sup>
- **Continuous ECG/BP** for dysautonomia.<sup>260</sup>
- **Spinal MRI with contrast** (nerve-root enhancement; excludes a compressive cord lesion).
- Anti-ganglioside antibodies (GQ1b for Miller-Fisher); **stool for poliovirus/enterovirus** in acute flaccid paralysis.<sup>260</sup>

| Treatment         | Dose                        | Notes                                                                                                                            |
|-------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| IV immunoglobulin | 2 g/kg total, over 2–5 days | First-line; equivalent to plasma exchange — a <b>second course is not recommended</b> in non-responders (SID-GBS) <sup>260</sup> |
| Plasma exchange   | Per center protocol         | Alternative — not combined with IVIG <sup>260</sup>                                                                              |
| Corticosteroids   | Not recommended             | No benefit in GBS <sup>260</sup>                                                                                                 |

#### ▲ ADMIT · REFER

- **Admit** all suspected cases for respiratory and cardiac monitoring.
- **Call** neurology; critical care early if bulbar/respiratory involvement or rapid progression. Acute flaccid paralysis is **notifiable** (polio surveillance) — send stool.

## The Child with Neuromuscular Disease

**MINUTES**

DMD Care 2018 · CHEST 2023

A known neuromuscular disease rewrites every emergency — the weak cough, the anesthetic that can kill, the heart, and the

steroids.

### ▶ RESPIRATORY – THE USUAL KILLER

- It is the **weak cough, not the lungs**: a “simple” chest infection tips into **hypercapnic respiratory failure**. Support ventilation with **NIV (bilevel)** and clear secretions with **mechanical insufflation–exsufflation (cough assist)** — not oxygen alone.<sup>261</sup>
- **The oxygen saturation lies** — it can be normal while CO<sub>2</sub> climbs. Track **capnography / blood-gas pCO<sub>2</sub>**, not SpO<sub>2</sub>.

### ⚠ ANESTHETIC / RSI HAZARD – FLAG BEFORE ANY SEDATION

- **Succinylcholine is contraindicated** in dystrophinopathies and many myopathies (hyperkalemic cardiac arrest), and **volatile agents** (sevoflurane, isoflurane, desflurane) risk **anesthesia-induced rhabdomyolysis / hyperkalemia** (see [Rhabdomyolysis](#)). **Safe approach**: total IV anesthesia (propofol, opioid, benzodiazepine)

+ a **non-depolarizing** blocker (rocuronium) reversible with sugammadex.<sup>261</sup>

- Check ionized K<sup>+</sup>, ECG and CK after any procedure.<sup>261</sup>

### ⚠ DO NOT GIVE UNCONTROLLED OXYGEN

- High-flow O<sub>2</sub> masks a rising CO<sub>2</sub> and can abolish respiratory drive<sup>261</sup> → **CO<sub>2</sub> narcosis and arrest.**
- Titrate O<sub>2</sub> only **with** ventilatory support and CO<sub>2</sub> monitoring — correct the hypoventilation, not the number.<sup>261</sup>

### ▀ CARDIAC

- **Duchenne / Becker** → dilated cardiomyopathy and arrhythmia (may be silent — reduced mobility hides symptoms). **Emery-Dreifuss / LGMD1B** → progressive **AV / conduction block** and sudden death (may need pacing).<sup>261</sup>
- Any acute deterioration: **12-lead ECG + echocardiogram**; troponin and BNP/NT-proBNP if acute myocardial injury or heart failure is suspected.<sup>261</sup>



### ▀ ADRENAL – STRESS-DOSE STEROIDS

- A child on **chronic daily steroids** (prednisone/deflazacort for Duchenne) **cannot mount a stress response**. Acute illness, vomiting, trauma, surgery or shock needs **stress-dose steroids** to prevent adrenal crisis — **never stop steroids abruptly**<sup>261</sup> (see [Adrenal Crisis](#)).

### ▀ MYASTHENIC CRISIS

- Worsening bulbar/respiratory weakness → treat with **IVIG or plasma exchange** and support the airway.<sup>261</sup> Follow **FVC and NIF**, not SpO<sub>2</sub>.
- **Avoid drugs that worsen myasthenia**: aminoglycosides, macrolides, fluoroquinolones, **IV magnesium**, beta-blockers, and procainamide.<sup>261</sup>

### ✦ TWO NUMBERS THAT MISLEAD, ONE THAT DOESN'T

- In neuromuscular weakness a normal **SpO<sub>2</sub>** and a normal **respiratory rate** are falsely reassuring — the child fails from

## **CO<sub>2</sub> retention and a failing cough.**

- The honest bedside numbers are **pCO<sub>2</sub> (capnography/gas)** and **FVC / NIF**.
- A metabolic/mitochondrial myopathy adds a fourth: start **IV dextrose (D10) early** — fasting drives catabolism, hypoglycemia and rhabdomyolysis.<sup>261</sup>

### ▲ **WORKUP**

- **VBG/ABG with pCO<sub>2</sub> + continuous capnography** — the key respiratory measure; **FVC and NIF** where the child can cooperate.
- Electrolytes incl. **potassium, magnesium, calcium; creatine kinase**; renal function; **glucose**; lactate.
- **12-lead ECG**; troponin + BNP/NT-proBNP; echocardiogram if cardiac concern.
- Chest X-ray ± sputum culture if infective; swallow/bulbar assessment.<sup>261</sup>

| Situation                                                  | Dose                                                                                | Route & notes                                                    |
|------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Stress<br>steroids<br>—<br>toleratin<br>g oral             | ≈ triple the<br>maintenance<br>steroid dose<br>during the<br>illness                | PO; for the child on<br>chronic steroids<br>(Duchenne)           |
| Stress<br>steroids<br>— unwell<br>/<br>vomiting<br>/ shock | Hydrocortison<br>e ≈ 2 mg/kg<br>(or 50 mg/m <sup>2</sup> )<br>IV, then<br>continued | IV/IM; treat as adrenal<br>crisis <sup>261</sup> — do not wait   |
| Myasthe<br>nic crisis                                      | IVIg 2 g/kg<br>over 2–5 days<br>or plasma<br>exchange                               | Plus airway support;<br>treat the<br>trigger/infection           |
| Metaboli<br>c<br>myopath<br>y —<br>catabolic<br>stress     | IV dextrose<br>(D10) early                                                          | Prevents<br>fasting-driven<br>decompensation /<br>rhabdomyolysis |

#### ◆ LOCAL & ANTICIPATORY

- Ask for the child's **EMERGENCY / ANESTHESIA ALERT CARD** and neuromuscular care plan on arrival, and involve the **NEUROMUSCULAR TEAM** early.
- Keep succinylcholine and volatile agents off the plan for dystrophinopathies, and flag the diagnosis to anesthesia/critical care

before any transfer, sedation or intubation.<sup>261</sup>

▲ **ADMIT · REFER**

- **Admit** — any respiratory decompensation, myasthenic/adrenal crisis, new cardiac signs, or a child who needs assisted airway clearance.
- **Call** neuromuscular/neurology, cardiology and critical care/anesthesia; new acute flaccid weakness without a known diagnosis → see **Guillain-Barré / Acute Flaccid Paralysis**.

# Orthopedic &

12

## Musculoskeletal

The limping child · fractures & NAI · SCFE · compartment syndrome · bone & joint infection

### Orthopedic Emergencies – Call Now

MINUTES

Immediate orthopedic ( $\pm$  vascular/spinal) consultation — do not wait for imaging.

#### ⚠ IMMEDIATE ORTHOPEDIC CONSULTATION

- Septic arthritis
- Compartment syndrome
- Open fracture
- SCFE
- Joint dislocation
- Fracture with pulseless limb
- Unstable pelvic injury
- Spine injury with neurological deficit<sup>133</sup>

# Osteoarticular Infections

HOURS

## RECOGNIZE

Acute hematogenous osteomyelitis (AHO) and septic arthritis (SA) — consider in any limp, refusal to use a limb, or pseudoparalysis.

### CLINICAL PRESENTATION

- **Pseudoparalysis / limp / refusal to weight-bear**, or focal limb pain — often the only symptom. Fever supports diagnosis but may be absent.<sup>61</sup>
- **Osteomyelitis**: point tenderness over the **metaphysis** of a long bone ± localized swelling/erythema; fluctuance suggests a periosteal abscess.<sup>61</sup>
- **Septic arthritis**: joint effusion, swelling, and pain on any movement of the isolated joint; the limb is held in a position of comfort.<sup>61</sup>
- *S. aureus* → acute, rapidly progressive, higher CRP (MRSA more severe). *Kingella kingae* (young children) → milder, indolent, lower markers.<sup>61</sup>
- **Chronic recurrent multifocal osteomyelitis (CRMO/CNO)** — consider with recurrent bone pain,

multifocal lesions, sterile cultures, and poor response to antibiotics.<sup>61</sup>

### ▶ MOST COMMON SITES & PATHOGENS

- **Bone:** metaphysis of long bones — **femur, tibia, humerus**. Pelvic bones are easily missed (vague pain, mimics abdomen).<sup>61</sup>
- **Joint: hip and knee** most common, then ankle, elbow, shoulder. SA + adjacent osteomyelitis is common ( $\approx$  32% of SA), especially with *S. aureus* and longer symptom duration.<sup>61</sup>
- **Pathogens:** *S. aureus* overall most common; *K. kingae* dominates under  $\sim$ 4 yr (inoculate fluid into blood-culture bottles). Add *Salmonella* in sickle cell disease; *H. influenzae* type b if under-immunized.<sup>61</sup>

### ◆ KOCHER CRITERIA – HIP SA VS TRANSIENT SYNOVITIS

- Four predictors, one point each:  
**NON-WEIGHT-BEARING; FEVER > 38.5 °C;  
ESR > 40 MM/H; WHITE-CELL COUNT >  
12,000/ML.**

- Many add **CRP > 20 MG/L** as a strong independent predictor.
- A risk-stratification aid for the hip, not a rule-out.<sup>64</sup>
- More positive criteria substantially raise septic-arthritis probability, but always interpret alongside CRP, ultrasound, and clinical judgment.<sup>64</sup>

## KOCHER – PROBABILITY OF SEPTIC ARTHRITIS

| Predictors present | Approx. probability of SA |
|--------------------|---------------------------|
| 0 of 4             | < 0.2 %                   |
| 1 of 4             | ~ 3 %                     |
| 2 of 4             | ~ 40 %                    |
| 3 of 4             | ~ 93 %                    |
| 4 of 4             | ~ 99 %                    |

### ⚠ WORK-UP

- Blood cultures **before** antibiotics.  
**CRP** at baseline (sensitive, falls fast — best for monitoring; ESR less useful).<sup>61</sup>
- Plain films to exclude mimics;  
**ultrasound** for hip/shoulder effusion; **MRI is the imaging modality of choice** (add contrast if abscess or soft-tissue extension is suspected); **MRI with gadolinium** is the most sensitive/specific test,



defining abscess, adjacent osteomyelitis, and soft-tissue extent.<sup>61</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected septic arthritis/osteomyelitis → IV antibiotics after cultures/aspiration. **Delay antibiotics until a joint aspirate is obtained whenever clinically feasible.**
- **Discharge** — not acutely; step down to oral with monitoring and orthopedic follow-up.
- **Refer** — orthopedics (aspiration/washout) and infectious diseases.

# Osteoarticular Infections — Management

HOURS

CULTURES FIRST

Cultures first, then empiric IV cefazolin; involve orthopedics early for any septic joint or bony abscess.

DECISION

**Osteomyelitis or septic arthritis?**

## OSTEOMYELITIS (AHO)

Most respond to **antibiotics alone**. Surgery (drainage/debridement) for a **subperiosteal/intraosseous abscess**, or if not improving in the first few days — repeat imaging and reconsider debridement.<sup>61</sup>

## SEPTIC ARTHRITIS

**Orthopedic emergency** — urgent **aspiration ± arthroscopic/open drainage & irrigation**. Hip and shoulder are especially time-critical (avascular-necrosis risk). Send synovial fluid for cell count, Gram stain, culture (blood-culture bottle) + PCR.<sup>61</sup>

**Empiric therapy after cultures: IV cefazolin — covers MSSA and *K. kingae*.**<sup>61</sup>

## ANTIBIOTIC THERAPY

| Scenario                          | Drug & dose                                                                             | Notes                                                                                                                              |
|-----------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Empiric — first-line IV</b>    | Cefazolin 37.5–50 mg/kg/dose IV q6–8h (max 2 g/dose)                                    | Covers MSSA <b>and</b> <i>K. kingae</i> — the two commonest pathogens. Start after blood cultures. <sup>61</sup>                   |
| <b>Under 4 yr / unimmunized</b>   | Cefuroxime 50 mg/kg/dose IV q8h (max 1.5 g/dose)                                        | Broadens to cover <i>H. influenzae</i> in unimmunized young children. <sup>61</sup>                                                |
| <b>MSSA — confirmed</b>           | Continue cefazolin, or cloxacillin 37.5–50 mg/kg/dose IV q6h (cloxacillin max 2 g/dose) | Cloxacillin narrows further but irritates veins. <sup>61</sup>                                                                     |
| <b>MRSA — suspected/confirmed</b> | Add/use vancomycin 15 mg/kg/dose IV q6h (max 1 g/dose; level-guided)                    | Where MRSA prevalent or known carrier. Note: <i>K. kingae</i> is resistant to vancomycin, clindamycin & cloxacillin. <sup>61</sup> |

| Scenario                                          | Drug & dose                                                  | Notes                                                                                                                          |
|---------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Oral<br>step-down —<br>MSSA /<br>culture-negative | Cephalexin<br>40–50<br>mg/kg/dose PO<br>TID (max 6<br>g/day) | First-line oral.<br>Cloxacillin 25<br>mg/kg/dose QID is an<br>alternative (poor<br>taste). <sup>76</sup>                       |
| Oral<br>step-down —<br>MRSA                       | Clindamycin,<br>TMP-SMX, or<br>linezolid                     | Per susceptibilities,<br>with<br>infectious-diseases<br>input; MRSA usually<br>needs longer before<br>switching. <sup>76</sup> |

#### ◆ IV → ORAL SWITCH & DURATION

- Switch to oral when **CLINICALLY IMPROVED** (afebrile, using the limb) and **CRP FALLING** ( $\approx$  50% drop, often < 20–30 mg/L) with assured adherence — even after prior bacteremia.<sup>76</sup> **DURATION (CRP-GUIDED): AHO 3–4 WEEKS**
- isolated SA **2–3 WEEKS**
- SA with adjacent osteomyelitis **4 WEEKS**
- hip/shoulder SA<sup>61</sup> **4–6 WEEKS**.
- Randomized data (Peltola) support short courses once response is good and CRP normalizes early.<sup>63</sup>

### ⚠ FOLLOW-UP

- Confirm clinical resolution and a **normal/normalizing CRP** before stopping.
- End-of-therapy radiographs mainly if the **growth plate** was involved or a large lytic lesion present — these and any physeal involvement need **orthopedic follow-up**.
- Routine follow-up MRI/CT is not needed — reserve for complications or non-improvement.<sup>61</sup>

◆ **DIFFICULT IV ACCESS?** Reassess for an early IV→oral switch at every line failure rather than re-sticking — see [IV → Oral Switch & the DIVA Patient](#) (Quick Reference).

# The Limping Child

ASSESS

APPROACH

A limp is a symptom — use age and red flags to separate the benign from septic joint, SCFE, malignancy, and abuse.

## DEFINE

- A **limp** is any deviation from normal age-appropriate gait; pain present in ~80%. Broad differential — separate benign causes from the **can't-miss** ones: septic joint, osteomyelitis, SCFE, malignancy, and inflicted injury.<sup>133</sup>

## APPROACH – AGE NARROWS THE DIFFERENTIAL

- **Toddler (1–3 y):** toddler's fracture, transient synovitis, developmental dysplasia of the hip, inflicted injury.
- **Child (4–10 y):** transient synovitis, Legg-Calvé-Perthes, septic arthritis, reactive/juvenile idiopathic arthritis.
- **Adolescent (> 10 y):** SCFE, apophysitis/overuse, sports injury.<sup>133</sup>
- **Spinal:** consider discitis, vertebral osteomyelitis, or a spinal tumor in a child refusing to walk with a **normal**

**hip exam** — especially with back pain or spinal tenderness.<sup>133</sup>

- **Malignancy:** persistent night pain, pain out of proportion to the exam, constitutional symptoms, or failure to improve<sup>133</sup> → **urgent investigation.**
- History: pain, trauma, fever, systemic or night symptoms. Examine gait (antalgic vs Trendelenburg), hip, knee, spine, neurology — **knee pain often means hip disease.**<sup>133</sup>
- **Any age — look beyond the hip:** consider a **first presentation of juvenile idiopathic arthritis (JIA)**, which occurs across **all age groups**, and **examine every joint** — not only the painful one (a commonly missed step that reveals oligo- or polyarticular disease).<sup>133</sup>

#### ⚠ **WORK-UP**

Plain radiographs of the involved site — **both hips (add a frog-leg view for SCFE)** if the exam does not localize. **FBC, CRP/ESR, blood culture** if febrile or systemically unwell; ultrasound for a hip effusion; MRI if osteomyelitis or malignancy is suspected.<sup>133</sup>

## ▀ RED FLAGS

- **Fever + refusal to weight-bear + raised inflammatory markers** → septic joint/osteomyelitis. Normal early CRP/ESR/white count do **not** exclude osteoarticular infection — reassess if symptoms persist or worsen. <sup>133</sup>
- **Night pain, weight loss, pallor, bruising, or hepatosplenomegaly** <sup>133</sup> → malignancy/leukemia.
- **Multiple or unexplained fractures, or a fracture in a non-ambulatory infant** <sup>136</sup> → inflicted injury.
- **Adolescent hip or knee pain with a limp** <sup>135</sup> → SCFE until excluded.

## ✦ PEARL – POINT TO THE HIP

Isolated knee or thigh pain in a child is hip pathology until proven otherwise — SCFE and Perthes both refer pain to the knee. <sup>133</sup>

## ▀ ADMIT · DISCHARGE · REFER

- **Admit** — septic arthritis, osteomyelitis, fracture, or systemic illness; unable to weight-bear with red flags.



- **Discharge** — transient synovitis in a well child with safety-net advice and review.
- **Refer** — orthopedics for red flags; oncology/rheumatology as indicated.

## Transient Synovitis vs Septic Arthritis

HOURS

KOCHER

The commonest atraumatic limp — but rule out the septic joint first; fever + non-weight-bearing + high CRP means aspirate.

### ▀ DEFINE

- **Transient synovitis** — self-limiting, usually post-viral hip inflammation and the commonest cause of an atraumatic limp (3–8 years); symptoms usually resolve within **7–10 days**. Must be separated from **septic arthritis**, a surgical emergency.<sup>64</sup>

## ▶ APPROACH – READ THE CHILD

- **Transient synovitis:** a well child, low-grade or no fever, **weight-bears with discomfort**, mildly raised markers, often a recent upper-respiratory infection.
- **Septic arthritis:** an unwell/febrile child who **refuses to move the joint**, severe pain on any movement, the limb held in flexion/abduction/external rotation.<sup>64</sup>

## ◆ KOCHER CRITERIA – HIP: SEPTIC ARTHRITIS VS TRANSIENT SYNOVITIS

- One point each: **NON-WEIGHT-BEARING; FEVER > 38.5 °C; ESR > 40 MM/H; WBC > 12,000/ML.**
- Many add **CRP > 20 MG/L** as a strong independent predictor.
- Probability of septic arthritis: 0/4 < 0.2%
- 1/4 ≈ 3%
- 2/4 ≈ 40%
- 3/4 ≈ 93%
- 4/4 ≈ 99%.
- A risk aid, not a rule-out.<sup>64</sup>

### ⚠ WORK-UP

- FBC, **CRP**, ESR, blood culture; hip **ultrasound** (an effusion is seen in both and does not distinguish).<sup>64</sup>
- If septic arthritis is suspected → **urgent ultrasound-guided aspiration** (cell count, Gram stain, culture into a blood-culture bottle).<sup>64</sup>

### ▀ MANAGEMENT

- **Septic arthritis** — orthopedic emergency: urgent joint drainage plus IV antibiotics after cultures (see [Osteoarticular Infections](#)).
- **Transient synovitis** — rest, NSAIDs, review in 24–48 h; settles over days. Re-evaluate promptly if it does not improve or markers rise.<sup>64</sup>

### ✦ PEARL – WHEN IN DOUBT, ASPIRATE

Fever + refusal to weight-bear + a raised CRP is a septic joint until an aspirate proves otherwise — do not label it transient synovitis by default.<sup>64</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — any suspicion of septic arthritis (Kocher criteria, raised markers) → aspiration.
- **Discharge** — transient synovitis, afebrile, mobilizing, with review.
- **Refer** — orthopedics for aspiration/washout if septic.

## Nursemaid's Elbow

ASSESS

1–4 YR

A pulled, pronated arm the child will not use — reduce by hyperpronation; no film needed if the story is classic.

### ▲ DEFINE

- **Nursemaid's elbow (radial head subluxation, 'pulled elbow')** — the annular ligament slips over the radial head after axial traction on an extended, pronated arm.<sup>134</sup>  
Commonest elbow injury in children aged 1–4 years.

## ▲ APPROACH & WORK-UP

- Typical story: sudden **pull or lift** on the arm. Child holds the arm slightly flexed and **pronated**, refuses to use it, with **no swelling, bruising, or deformity** and minimal tenderness.<sup>134</sup>
- **No imaging needed** with a classic history and a normal-looking arm. Image if the story is atypical, or there is swelling, focal bony tenderness, or deformity.<sup>134</sup>

## ⚠ RED FLAGS

- Swelling, bruising, point bony tenderness, or deformity → suspect a fracture and radiograph.
- An inconsistent or implausible history should prompt consideration of inflicted injury.

## ▲ MANAGEMENT – REDUCE IT

- **Hyperpronation** (support the elbow, firmly pronate the forearm) — a meta-analysis shows it is **more effective and less painful** than supination-flexion — do it first.<sup>134</sup>

- Alternative: **supination-flexion** (supinate the forearm, then fully flex the elbow). A palpable click often signals success.<sup>134</sup>
- Child usually resumes normal use within 5–30 minutes. If reduction fails, try the other maneuver; if still unsuccessful, immobilize, image, reassess.<sup>134</sup>

#### ✦ PEARL – NO FILM NEEDED IF IT WORKS

- With a classic pull mechanism and no swelling, reduce without imaging — if the child uses the arm normally afterwards, diagnosis is confirmed and no radiograph is required.<sup>134</sup>
- Do not attempt repeated reductions if there is swelling, bruising, focal tenderness, or an atypical history — image and consider safeguarding.<sup>134</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — not required.
- **Discharge** — after successful reduction and return of normal arm use.

- **Refer** — orthopedics only if reduction fails, or for swelling/deformity suggesting fracture (image first).

## Slipped Capital Femoral Epiphysis (SCFE)

HOURS

CAN'T MISS

Peri-pubertal limp with hip or knee pain — stop weight-bearing and get frog-leg views of both hips.

### DEFINE

- **Slipped capital femoral epiphysis (SCFE)** — the proximal femoral epiphysis slips through the physis. Peri-pubertal; **obesity is the strongest risk factor**; up to  $\frac{1}{3}$  bilateral. Consider endocrinopathy if the child is young (< 10 years), underweight, or atypical.<sup>135</sup>

### APPROACH & WORK-UP

- Limp with **hip, groin, thigh, or knee pain**; the leg lies **externally rotated** with limited internal rotation and

**obligate external rotation on hip flexion** (Drehmann sign).<sup>135</sup>

- **Stable** (able to weight-bear, with or without walking aids) vs **unstable** (cannot bear weight — high avascular-necrosis risk).<sup>135</sup>
- **AP + frog-leg lateral radiographs of BOTH hips** for a stable slip; for an **unstable** slip avoid forced frog-leg positioning and use AP + **cross-table lateral** to prevent further displacement. Klein's line fails to intersect the epiphysis on the affected side. The frog-leg view is most sensitive.<sup>135</sup>

#### ⚠ **RED FLAGS**

- An **unstable slip** (non-weight-bearing) carries high avascular-necrosis risk.
- Missed or delayed diagnosis → progression and permanent deformity.<sup>135</sup>
- Assess both hips.

#### ▀ **MANAGEMENT**

- **Make the child non-weight-bearing immediately** and refer urgently to orthopedics for **in-situ screw**



## **fixation.**<sup>135</sup>

- **Do not attempt reduction** — forceful manipulation increases avascular-necrosis risk.<sup>135</sup>
- Consider prophylactic contralateral fixation in high-risk patients (young age, obesity, endocrinopathy).<sup>135</sup>

### ✦ **PEARL – FROG-LEG BOTH HIP**

Any obese adolescent with a limp or knee pain gets frog-leg views of both hips and is kept off the leg until SCFE is excluded — knee pain is the classic decoy.<sup>135</sup>

### ▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → non-weight-bearing, urgent surgery.
- **Discharge** — post-operative.
- **Refer** — orthopedics urgently (surgical fixation); endocrinology if bilateral or atypical.

# Fractures – Patterns & Non-Accidental Red Flags

## ASSESS

## SAFEGUARDING

Know the common breaks — and the fractures that mandate a child-protection assessment.

### DEFINE

- Common pediatric fracture patterns and — critically — the fractures that trigger a **child-protection** assessment. Children's growing bone breaks differently from adults'.<sup>136</sup>

### APPROACH – COMMON ACCIDENTAL PATTERNS

- **Buckle/torus** (distal radius, stable), **greenstick**, and **toddler's fracture** (spiral distal tibia in an ambulant toddler).<sup>136</sup>
- **Physeal (Salter-Harris)** fractures cross the growth plate (SH-II commonest) and risk growth arrest — always classify and follow up.<sup>136</sup>
- Radiograph the **joint above and below**; obtain comparison views when needed.<sup>136</sup>

- **Supracondylar humeral fractures** need careful neurovascular assessment — document radial pulse, capillary refill, and **median (including anterior interosseous), radial, and ulnar** nerve function before and after splinting/reduction. A pulseless or poorly perfused hand needs **immediate orthopedic consultation**.<sup>136</sup>

#### **⚠ RED FLAGS – INFLICTED INJURY**

- Escalate to safeguarding when the **injury is inconsistent** with the stated mechanism or the child's development — especially a **long-bone (femur/tibia) fracture in a non-ambulatory infant**, delayed presentation, multiple fractures of different ages, or **rib, metaphyseal 'corner', scapular, or spinous-process** fractures.<sup>136</sup>
- In children < 2 years with a suspicious fracture, obtain a **skeletal survey**.<sup>136</sup>
- Associated **bruising, patterned burns, retinal hemorrhage**, or other unexplained injuries further raise suspicion of abuse.<sup>136</sup>

## ▀ MANAGEMENT

- Immobilize, provide analgesia, and refer to orthopedics for displaced, physeal, or unstable fractures.
- Where abuse is suspected, follow the mandatory **child-protection pathway** — evaluation and reporting are not optional.<sup>136</sup>

### ◆ KUWAIT – REPORTING IS MANDATORY

- Under the **CHILD RIGHTS LAW NO. 21 OF 2015**, children are protected from all forms of violence, abuse, exploitation, and neglect, and **PHYSICIANS ARE LEGALLY OBLIGED TO REPORT** suspected child maltreatment.<sup>144</sup>
- Refer through the **KUWAIT NATIONAL CHILD PROTECTION PROGRAM (KNCPP), MINISTRY OF HEALTH**, via the national **CHILD PROTECTION HOTLINE 147**.
- Ensure the child's immediate safety, document findings objectively, and involve the hospital child-protection team.

✦ **PEARL – IF THEY CAN'T WALK, BE SUSPICIOUS**

A fracture in a child who is not yet walking is highly suspicious for abuse until proven otherwise — do the skeletal survey and involve safeguarding.<sup>136</sup>

✦ **PEARL – DOCUMENT THE NEUROVASCULAR EXAM**

Every **supracondylar fracture** needs documented radial pulse, capillary refill, and median (including anterior interosseous), radial, and ulnar nerve function **before and after** splinting or reduction. **A pulseless hand is an orthopedic emergency, even if it remains pink.**<sup>136</sup>

## Compartment Syndrome

MINUTES

### LIMB-THREATENING

Rising compartment pressure starves muscle and nerve — in children the 3 A's beat the 5 P's; fasciotomy is the fix.

#### DEFINE

- **Acute compartment syndrome** — rising pressure in a fascial compartment cuts perfusion →

muscle/nerve ischemia.  
Limb-threatening; **clinical** diagnosis.  
Most follow fractures (tibia, forearm),  
tight casts, crush, or reperfusion.<sup>137</sup>

### ⚠ **RED FLAGS – THE 3 A'S IN CHILDREN**

- The adult '5 Ps' (pain out of proportion, pain on passive stretch, paresthesia, pallor, pulselessness) are **unreliable in children**.<sup>137</sup>
- Suspect from the **3 A's — rising Analgesia requirement, Anxiety, and Agitation** — appearing on average ~7 hours before any neurovascular change.<sup>137</sup>
- Tensely swollen compartment = most consistent sign.  
**Pulselessness and paralysis are late.**<sup>137</sup>

### ⚠ **WORK-UP**

- Clinical diagnosis.
- Reserve compartment-pressure measurement for the obtunded or equivocal patient ( $\Delta P < 30$  mmHg from diastolic is concerning). **Do**

**not delay fasciotomy to obtain pressures.**<sup>137</sup>

## ▀ MANAGEMENT

- **Remove or split all constricting casts and dressings down to skin;** keep the limb at heart level (do not elevate).<sup>137</sup>
- Analgesia, oxygen, correct hypotension. Dense regional/nerve blocks can mask ischemic pain — in high-risk limb injuries prefer low-concentration blocks and stay vigilant (breakthrough or escalating pain, rising analgesic needs).<sup>137</sup>
- **Emergency surgical fasciotomy without delay** — the definitive treatment.<sup>137</sup>

## ✦ PEARL – ESCALATING PAIN = CALL SURGERY

A child with a limb injury whose analgesia needs keep climbing has compartment syndrome until proven otherwise — split the cast and call surgery.<sup>137</sup>

## ▀ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected → emergency.

- **Discharge** — post-operative.
- **Refer** — orthopedic (lower limb) / plastic surgery (upper limb) IMMEDIATELY (fasciotomy); do not delay for pressure measurement in a clear case.

## Open Fracture HOURS

### ANTIBIOTICS EARLY

Skin breach over bone — early IV antibiotics, tetanus cover, saline dressing, and urgent orthopedics.

### DEFINE

- **Open (compound) fracture** — fracture with an overlying skin breach communicating with bone; high infection risk. Graded Gustilo-Anderson I–III by wound size, contamination, and soft-tissue injury. <sup>138</sup>

### APPROACH & WORK-UP

- Visible wound/bleeding over the fracture, exposed bone, or gross deformity. **Assess neurovascular status distal to the injury.** <sup>138</sup>



- Remove gross contamination, photograph the wound once, then **cover with a saline-soaked sterile dressing**; radiograph; check **tetanus status**.<sup>138</sup>

### ⚠ RED FLAGS

- A **pulseless, pale, cold limb** needs urgent reduction/vascular assessment.
- Gross contamination, type-III wounds, and evolving **compartment syndrome** raise the stakes.<sup>138</sup>

## MANAGEMENT

| Drug                 | Dose                                                               | Route & notes                                                        |
|----------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|
| Antibiotics — timing | First IV dose as soon as possible (ideally < 1 h, at latest < 3 h) | The clock starts at injury, not arrival. <sup>138</sup>              |
| Type I-II            | Cefazolin 30 mg/kg/dose IV (max 2 g) every 8 h                     | First-generation cephalosporin — gram-positive cover. <sup>138</sup> |

| Drug                                | Dose                                                                                                                                        | Route & notes                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Type III                            | Add<br>gentamicin 5–7 mg/kg/dose<br>IV once<br>daily; add<br>benzylpenicillin for<br>farm/soil<br>contamination<br>(anti-clostridial cover) | Add gram-negative<br>cover for<br>high-grade/contaminated<br>wounds. <sup>138</sup> |
| Farm /<br>soil<br>contamin<br>ation | Add<br>penicillin                                                                                                                           | Anti-clostridial cover.                                                             |

#### THEN

- **Tetanus prophylaxis** per immunization status; analgesia; realign and splint the limb. <sup>138</sup>
- **Urgent orthopedic referral** for formal debridement, irrigation, and stabilization in the operating room. <sup>138</sup>

#### PEARL – ANTIBIOTICS ON THE CLOCK

Give the first cephalosporin within an hour of injury, update tetanus, cover with a saline dressing, call orthopedics — early antibiotics drive the infection rate down. <sup>138</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all open fractures and significant limb injuries.
- **Discharge** — not acutely; post-operative/reduction.
- **Refer** — orthopedics (early antibiotics, tetanus, debridement); vascular surgery if perfusion is threatened.

## Fracture with Pulseless Limb

MINUTES

LIMB-THREATENING

An absent pulse needs urgent assessment; a pale, cool, poorly perfused limb needs emergent restoration of circulation.

### ▲ DEFINE

- A fracture with an absent distal pulse may indicate arterial compression, entrapment, or injury.<sup>294,295</sup>

### ▲ APPROACH

- Obtain urgent radiographs.

- Record injury time, pulses, capillary refill, skin temperature and color, and individual peripheral-nerve function; compare with the opposite limb.
- Reassess and document after realignment, splinting, reduction, and surgery.
- Persistent suspected arterial injury requires consultant-directed vascular imaging, usually **CT angiography**, coordinated so that emergency revascularization is not delayed.<sup>295</sup>

#### **RED FLAGS**

- Pallor, coolness, delayed refill, worsening pain, progressive sensory or motor deficit, or deteriorating perfusion → **immediate orthopedic and vascular escalation.**
- Increasing pain or swelling also warrants reassessment for **compartment syndrome.**

#### **MANAGEMENT**

- Splint in the position of deformity in the emergency room.
- Emergent reduction and pinning in the operating room.

- Reassess circulation immediately post-reduction; vascular exploration if the limb remains cold and pulseless.<sup>294</sup>
- **Supracondylar fracture with a pink, warm hand** — arrange urgent reduction and fixation. If the hand remains well perfused afterwards, an absent radial pulse alone does not mandate brachial-artery exploration; the orthopedic consultant documents the decision and arranges close inpatient neurovascular observation.<sup>294</sup>

◆ **PEARL – THE PINK HAND IS STILL AN EMERGENCY**

A pink pulseless hand remains an orthopedic emergency; its perfusion after reduction determines the need for immediate vascular exploration.<sup>294</sup>

▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — all fractures presenting with a pulseless limb.
- **Discharge** — only after definitive management and consultant confirmation that perfusion is restored.

- **Refer** — orthopedics immediately; vascular surgery immediately for ischemia or suspected persistent arterial injury.

## Joint Dislocation MINUTES

### LIMB-THREATENING

Reduce traumatic dislocations promptly; compromised perfusion or threatened skin needs immediate intervention.

#### ▀ DEFINE

- Traumatic loss of joint alignment, with or without an associated fracture.

#### ▀ APPROACH

- Assess the mechanism, associated injuries, deformity, skin condition, and distal neurovascular function; document findings before and after reduction.
- Obtain joint radiographs to identify associated fractures — unless imaging would delay treatment of vascular compromise or threatened skin.

- Obtain post-reduction imaging and repeat the neurovascular examination.

### ⚠ RED FLAGS

- Absent or diminished pulses, impaired perfusion, neurological deficit, an open injury, or threatened skin → **immediate orthopedic assessment.**

### ▀ MANAGEMENT

- Analgesia, then prompt reduction by an orthopedic surgeon under appropriate sedation or anesthesia with monitoring — emergency room or operating theatre depending on hospital capability.
- Immobilize after reduction and obtain post-reduction radiographs.
- Definitive management depends on the joint, injury pattern, and any associated fracture.

### ✦ PEARL – THE HIP CLOCK

A delay in reduction > 6 hours greatly increases the risk of avascular necrosis in traumatic hip dislocation.<sup>296</sup>

## ▲ ADMIT · DISCHARGE · REFER

- **Admit** — hip or true knee (not patellar) dislocations; open, irreducible, unstable, or neurovascularly compromised injuries; associated fractures needing inpatient treatment.
- **Discharge** — selected uncomplicated injuries after stable reduction, satisfactory imaging, normal neurovascular findings, and arranged follow-up; advise return for increasing pain, swelling, or numbness.
- **Refer** — orthopedics immediately; vascular surgery immediately for ischemia or suspected persistent arterial injury.

## Unstable Pelvic Injury MINUTES

### LIFE-THREATENING

Treat a suspected unstable pelvic injury as major trauma — stabilization and hemorrhage control begin before radiological confirmation.

➕ **Calculators:** SIPA shock index · Blood products



## ▀ DEFINE

- A pelvic-ring injury with mechanical instability, usually anterior and posterior ring disruption. Mechanical and hemodynamic instability are distinct: a child may initially maintain circulation despite an unstable fracture.<sup>297</sup>

## ▀ APPROACH

- Pediatric trauma primary survey (see [Major Trauma — Primary Survey](#)).
- Look for pelvic pain, deformity, limb-length asymmetry, and shock.
- Avoid pelvic springing and repeated stability testing; minimize unnecessary movement.
- Apply a pelvic binder as soon as possible; arrange targeted CT once sufficiently stable, with the trauma team — imaging must not delay stabilization or hemorrhage control.

## ⚠ RED FLAGS

- Persistent tachycardia, poor perfusion, hypotension, altered consciousness, or deterioration

despite resuscitation → major hemorrhage.

- Open injury, blood at the urethral meatus, or lower-limb neurological abnormality → immediate specialist assessment.

## ▀ MANAGEMENT

- Activate the pediatric trauma and major-hemorrhage pathways; resuscitate while arranging urgent orthopedic, pediatric-surgical, anesthetic, and intensive-care support.
- A sheet applied over the greater trochanters serves as a pelvic binder when an unstable pelvic fracture is suspected — do not delay application for imaging.
- Ongoing hemodynamic instability despite resuscitation and binding needs immediate hemorrhage control — operative intervention or angioembolization; orthopedics determines temporary and definitive pelvic stabilization.
- Seek urological or pediatric-surgical advice before urethral catheterization if urethral injury is suspected.

- Open injuries require antibiotics and tetanus assessment.

#### ◆ PEARLS

- An apparently reduced pelvis on imaging obtained with a binder on does not establish fracture stability.
- A negative FAST (Focused Assessment with Sonography for Trauma) examination does not exclude pelvic bleeding.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected or confirmed unstable pelvic injuries; may need intensive care.
- **Discharge** — only after definitive stabilization, mobility, and a follow-up plan.
- **Refer** — triage to a pediatric-capable major trauma service; coordinate transfer and hemorrhage control with orthopedics and pediatric surgery.

# Spine Fracture with Neurological Deficit

## MINUTES

Protect spinal alignment, support oxygenation and circulation, and obtain immediate spinal-surgical advice.

### DEFINE

- Traumatic cervical, thoracic, or lumbar fracture or dislocation with new motor, sensory, or sphincter dysfunction suggesting spinal cord, conus medullaris, or cauda equina injury. The same emergency pathway applies when neurological findings occur without a visible fracture (SCIWORA — spinal cord injury without radiographic abnormality).

### APPROACH

- Pediatric trauma primary survey with strict cervical-spine immobilization and log-roll precautions (see [Major Trauma — Primary Survey](#)).
- Document the time and evolution of weakness, sensory loss, and bladder or bowel symptoms; repeat the

neurological assessment after transfers and any deterioration.

- Urgent radiographs and/or CT for injury characterization; urgent MRI for suspected cord or ligamentous injury.
- Do not delay referral to pediatric orthopedic spine or neurosurgery in order to complete imaging.

### **RED FLAGS**

- Progressive weakness, a sensory level, saddle sensory loss, new urinary retention or incontinence, or respiratory compromise → immediate escalation.
- Hypotension with bradycardia may be neurogenic shock — but hemorrhage must also be excluded.

### **MANAGEMENT**

- Maintain age-appropriate spine immobilization and minimize unnecessary movement.
- Analgesia and monitored resuscitation; correct hypoxemia and hypotension promptly.

- Involve intensive care early for mean arterial pressure support — in older children a lower limit of **75–80 mmHg** and upper limit of **90–95 mmHg**, maintained for **3–7 days** according to injury severity and neurological recovery; there is no established MAP target for young children.<sup>298</sup>
- The spine surgeon urgently determines the role of surgical decompression and stabilization, aiming for intervention **within 24 hours** of injury when clinically feasible to optimize neurological recovery.<sup>299</sup>
- Do not give high-dose methylprednisolone routinely for traumatic spinal cord injury.<sup>300</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all; PICU for cord injury, neurogenic shock, or respiratory compromise.
- **Discharge** — not acutely.

- **Refer** — pediatric orthopedic spine or neurosurgery immediately; pediatric-capable major trauma service.

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## Pediatric Trauma & TBI

Primary survey & resuscitation · secondary survey & decisions · traumatic brain injury

### Trauma Team & Roles MINUTES

Know who leads before the patient arrives — role confusion costs the golden hour.

#### ▴ WHO LEADS — SET LOCALLY

- **Trauma team leader (TTL)** — identified by each hospital's own trauma pathway; the designation differs between hospitals and this guide does not prescribe it. Whoever leads runs the primary survey and resuscitation and keeps the overview; the airway is managed by the most skilled airway operator present.<sup>306</sup>

- **Pediatric surgery** (where available; otherwise general surgery) — the surgical survey, hemorrhage source control, and performance of any operation. **The decision to operate, to image first, or to transfer is a team decision**, based on shared situational awareness of the child's physiology and trajectory — not a single-specialty call.<sup>306,98</sup>
- **No trauma team on site (private hospital, peripheral facility):** stabilize with the primary survey (cABCDE), control hemorrhage, secure oxygenation and access, keep the child warm, and **transfer to the nearest trauma-capable center while resuscitating** — call the receiving team early. Do not delay transfer for imaging or specialist review that cannot change management locally.<sup>98</sup>

**⚠️ AGREE THE PATHWAY IN ADVANCE**

Confirm the trauma call, role allocation, and transfer route locally before the patient arrives — front-door role confusion costs time in the golden hour.



# Major Trauma – Primary Survey & Resuscitation

## MINUTES

Stop the bleeding, protect the brain, keep them warm — and read the physiology, because children compensate then crash.

✚ Calculators: [SIPA shock index](#) · [Glasgow Coma Scale](#) · [Blood products](#)

### ▲ HOW CHILDREN DIFFER – ANTICIPATE IT

- **Large head, thin scalp:** head injury is common; scalp-wound blood loss can be significant in a small child.<sup>98</sup>
- **Compliant chest wall:** serious lung injury (pulmonary contusion) occurs **without** rib fractures — a normal-looking chest wall does not exclude it.<sup>98</sup>
- **Less-protected solid organs:** liver and spleen injure easily in blunt abdominal trauma.<sup>98</sup>
- **Great reserve, then sudden collapse:** children hold blood pressure by tachycardia and vasoconstriction, so **hypotension is a late, pre-terminal sign** — act on

tachycardia, cool peripheries, and prolonged capillary refill long before the BP falls.<sup>98</sup>

- **High surface-area-to-mass:** they cool fast — hypothermia drives coagulopathy. Hypoxia is the commonest cause of traumatic arrest.<sup>98</sup>

#### <C> CATASTROPHIC HEMORRHAGE

Control visible bleeding first: direct pressure, wound packing, a **limb tourniquet (record the exact time applied)** for uncontrolled extremity bleeding, and a **pelvic binder** for a suspected pelvic fracture.<sup>264</sup>

#### A – AIRWAY + C-SPINE

- Manual in-line stabilization; open (using a **jaw-thrust maneuver**) and clear the airway.
- Anticipate a difficult airway (large occiput flexes the neck; large tongue). In younger children — especially **< 2 yr** — a **shoulder roll** helps: maintain C-spine precautions and avoid over-extension or flexion.
- Secure the airway if GCS  $\leq 8$  or it is not protected.<sup>98</sup>

## B — BREATHING

- Oxygen 10–15 L/min via non-rebreather mask. **Tension pneumothorax is a clinical diagnosis — treat before imaging** (needle then finger thoracostomy). Both decompression sites are correct: **4th–5th intercostal space anterior to the mid-axillary line** (ATLS), or the **2nd intercostal space, mid-clavicular line** (PALS) — prefer the 2nd space, mid-clavicular line in **younger children and infants**<sup>98</sup>
- Seal an open pneumothorax; suspect massive hemothorax and pulmonary contusion.<sup>98</sup>

## C — CIRCULATION

- Two large-bore IV lines; go early to **intraosseous** if access fails.
- Find the bleeding — **chest, abdomen, pelvis, long bones, and the floor**. Resuscitate with **blood, not repeated crystalloid**; consider tranexamic acid within 3 h.<sup>266</sup>  
**Activate the massive transfusion protocol when required.**<sup>98</sup>

## D – DISABILITY

GCS or AVPU (alert / voice / pain / unresponsive), pupils, and a **glucose** (do not miss hypoglycemia). Manage raised ICP with **urgent neurosurgery consult** — see the Traumatic Brain Injury card.<sup>80</sup>

## E – EXPOSURE / ENVIRONMENT

Fully undress to examine and log-roll, then **cover and actively warm** — every degree of cooling worsens clotting.<sup>264</sup>

## RESUSCITATION – BLOOD, NOT CRYSTALLOID

| Drug / product       | Dose                                                  | Notes                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Packed red cells     | 10 mL/kg boluses, reassess after each                 | First-line for hemorrhagic shock; balanced 1:1:1 (RBC:plasma:platelets) in massive hemorrhage; activate the hospital massive transfusion protocol (pre-agreed with the blood bank) <sup>265</sup> |
| Crystalloid (warmed) | ≤ one 20 mL/kg bolus, only if blood not yet available | Each further crystalloid bolus → more coagulopathy, ventilator and ICU days <sup>98</sup>                                                                                                         |

| Drug / product        | Dose                                                                               | Notes                               |
|-----------------------|------------------------------------------------------------------------------------|-------------------------------------|
| Tranexamic acid (TXA) | 15 mg/kg (max 1 g) IV over 10 min, then 2 mg/kg/h; > 40 kg → 1 g then 1 g over 8 h | Within 3 h of injury <sup>266</sup> |

#### ◆ RESUSCITATION REASONING

- Estimated blood volume ~70 mL/kg in a child (80–90 in an infant), so small losses matter.<sup>98</sup>
- Move early to blood products and minimize crystalloid. **DURING ACTIVE HEMORRHAGE, TRANSFUSE ACCORDING TO HEMORRHAGIC-SHOCK AND HEMOSTATIC-RESUSCITATION NEEDS – NOT TO A FIXED HEMOGLOBIN TARGET.** Once bleeding is controlled and the child is hemodynamically stable, use a restrictive red-cell strategy, generally around **HB 7 G/DL.**<sup>98,339</sup> **TARGET A SYSTOLIC BP NO LESS THAN 70 + 2 × AGE (YR) MMHG** (1–10 yr band; use the age-specific systolic thresholds in the Shock card for other ages). **ADDRESS ONGOING LOSSES. AVOID HYPOTENSION.**<sup>98,80,98</sup>

### ⚠ THE LETHAL TRIAD – AND REASSESS

- Hypothermia, acidosis, and coagulopathy feed each other: keep the child warm, give blood early, correct coagulopathy.<sup>264</sup>
- If the patient deteriorates at any point, **go back to A** and work through again.<sup>98</sup>

## Major Trauma – Secondary Survey & Decisions

### MINUTES

Once the primary survey is controlled — find the injuries and make the calls.

### ▲ SECONDARY SURVEY – HEAD-TO-TOE + AMPLE HISTORY (ALLERGIES, MEDICATIONS, PAST HISTORY, LAST MEAL, EVENTS)

- Start **only after <C>ABCDE is controlled** and resuscitation underway. Examine front and back (log-roll): scalp and face, neck, chest, abdomen, pelvis and perineum, back, limbs, and a focused neurological exam.<sup>98</sup>

- **AMPLE** history: **A**llergies, **M**edications, **P**ast history, **L**ast meal, **E**vents/mechanism. Mechanism predicts the injuries — ask for it explicitly.<sup>98</sup>

#### DECISION

**Abdomen — does this child need a CT?**

#### **PECARN (Pediatric Emergency Care Applied Research Network)**

**intra-abdominal injury rule — 7**

**features:**

abdominal wall trauma / seatbelt sign

GCS  $\leq$  14

abdominal tenderness

chest-wall trauma

abdominal pain

decreased breath sounds

vomiting.<sup>97</sup>

**NONE present**

Very low risk for injury needing intervention — CT can usually be safely avoided; observe with serial examination.<sup>97</sup>

Not low-risk — image (CT if stable) and involve surgery. A **negative FAST does not exclude** intra-abdominal injury in children (FAST = Focused Assessment with Sonography for Trauma — bedside ultrasound for free fluid in the abdomen and pericardium).<sup>97</sup>

#### CHEST

CXR for hemo/pneumothorax and pulmonary contusion; place a **chest drain** for a hemothorax or significant pneumothorax. (*Tension pneumothorax is already decompressed in the primary survey.*)<sup>98</sup>

#### PELVIS

Confirm the binder is correctly positioned and **image** (pelvic X-ray ± CT); do **not** repeatedly spring the pelvis.<sup>98</sup> (*The binder is applied in the primary survey.*)



### C - SPINE

- Clear with the **PECARN** 3-tier rule (clinical clearance / radiograph / CT).
- **SCIWORA** (spinal cord injury without radiographic abnormality) is commoner in children — a normal X-ray does not exclude cord injury; keep the spine immobilized and call neurosurgery/spine urgently if there is any neurological deficit (weakness, sensory change, or bladder/bowel dysfunction), even with normal imaging.<sup>92</sup>

### LIMBS & ANALGESIA

Splint long-bone fractures and check distal pulses. Give **timely analgesia** (intranasal or IV opioid) — pain is routinely undertreated in injured children.<sup>98</sup>

### ⚠ ALWAYS CONSIDER INFLECTED INJURY

- Be alert when the history is inconsistent with the injury or the child's development, presentation is delayed, or the pattern is suspicious (bruising or fracture in a non-mobile infant, torn frenulum, multiple injuries of different ages).

- Document carefully and escalate through local safeguarding.<sup>136</sup>

#### ◦ **ESCALATE & TRANSFER EARLY**

- With no in-house pediatric surgery, involve the trauma team, anesthesia, intensive care and the receiving **trauma center early** — do not wait for the patient to deteriorate.<sup>98</sup>
- Package for transfer: airway secured, hemorrhage controlled, the child warm, and blood en route.
- Do **not** delay transfer for a CT that will not change what you do before the child leaves.<sup>98</sup>

## **Traumatic Brain Injury** **MINUTES**

Decide who needs a CT; in severe injury, stop the second insult.

**+ Calculator: Glasgow Coma Scale**

#### ▲ **IMAGING DECISION – PECARN**

- Use the **PECARN** rule to select children for CT after blunt head trauma and safely avoid imaging in the very-low-risk group. It is

**age-stratified** — use the correct **age column** below. PECARN was not derived or validated for suspected **abusive (inflicted)** head trauma — keep a low threshold to image regardless of the rule when abuse is suspected.<sup>79</sup>

## PECARN HEAD-CT – BY AGE GROUP

| Risk tier                           | < 2 years                                                                                                                                                                                                                                       | ≥ 2 years                                                                                                                                                |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CT recommended (high risk)</b>   | Any of: <b>GCS ≤ 14</b> , altered mental status / agitation / slow response, or a <b>palpable skull fracture</b> . <sup>79</sup>                                                                                                                | Any of: <b>GCS ≤ 14</b> , altered mental status, or signs of a <b>basilar skull fracture</b> . <sup>79</sup>                                             |
| <b>CT vs observe (intermediate)</b> | Any of: <b>occipital / parietal / temporal scalp hematoma</b> , <b>LOC (loss of consciousness) ≥ 5 s</b> , <b>severe mechanism</b> , or <b>not acting normally</b> per the parent <sup>79</sup> — <b>shared decision-making + observation</b> . | Any of: <b>LOC</b> , <b>vomiting</b> , <b>severe mechanism</b> , or <b>severe headache</b> — <sup>79</sup> <b>shared decision-making + observation</b> . |
| <b>No CT (low risk)</b>             | <b>None</b> of the above.                                                                                                                                                                                                                       | <b>None</b> of the above.                                                                                                                                |

### ◆ DECISION TOOLS

- PECARN pediatric head-CT rule —

[mdcalc.com](https://mdcalc.com)



- Severe-TBI management — Brain Trauma Foundation pediatric

guideline, 3rd ed — [doi.org](https://doi.org)



#### SEVERE TBI (GCS $\leq$ 8)

- Secure the airway.
- Keep **SpO<sub>2</sub>  $\geq$  94%** and normocapnia (PaCO<sub>2</sub> 35–40 mmHg).
- Avoid hypotension — **target SBP no less than 70 + 2 × age mmHg (1–10 yr band; age-specific thresholds otherwise).**<sup>80,307</sup> Higher blood-pressure targets may be beneficial in isolated severe TBI, but the threshold, and whether to target SBP or MAP, are not established — set targets with neurosurgery and the receiving intensive care unit (PICU or ICU, per hospital policy).<sup>307</sup>
- Head midline, **elevate the head of the bed 30°**; treat pain, agitation, glucose, and temperature.<sup>80</sup>



## IMPENDING HERNIATION – BRIDGE TO DEFINITIVE CARE

- **Recognize:** new pupil asymmetry or a fixed dilated pupil, Cushing response (bradycardia + hypertension + irregular breathing), extensor posturing, or a falling GCS.
- **Bridge:** head up 30° and midline; secure oxygenation and normocapnia; avoid hypotension; give **one hyperosmolar bolus** (3% saline 2–5 mL/kg, or mannitol 0.5–1 g/kg if not hypovolemic); brief hyperventilation **only** for signs of herniation, as a temporizing measure.<sup>80</sup>
- **Hand over** for urgent CT and neurosurgical review; ongoing intracranial-pressure and perfusion management is intensive-care therapy (PICU or ICU, per hospital policy) and is beyond the scope of this guide.

| Agent                | Dose                                          | Notes                                                                                                                                                      |
|----------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hypertonic saline 3% | 2–5 mL/kg bolus (impending herniation)        | First-line hyperosmolar bridge; infusion regimens are intensive-care therapy <sup>80</sup>                                                                 |
| Mannitol             | 0.5–1 g/kg over 15–20 min                     | Alternative; caution if hypovolemic or hypotensive <sup>80</sup>                                                                                           |
| Levetiracetam        | 20–55 mg/kg IV (max 4.5 g), then 10 mg/kg BID | Prophylaxis to reduce <b>early</b> post-traumatic seizures in severe TBI — a short course (~7 days); phenytoin / fosphenytoin an alternative <sup>80</sup> |

### ⚠️ PREVENT THE SECOND INSULT

- Hypoxia and hypotension each independently worsen outcome — avoid both.
- Do **not** routinely hyperventilate (vasoconstriction lowers cerebral blood flow).
- Avoid hyperthermia and treat seizures promptly.<sup>80</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — all moderate/severe TBI and any abnormal CT to intensive care (PICU or ICU, according to each

hospital's internal policy) with neurosurgical review. Until the child leaves the department, care is led by the consultant responsible at the patient's location — in the emergency department, the trauma team leader.

- **Discharge** — minor head injury meeting a validated low-risk rule, observed, with head-injury advice.
- **Refer** — neurosurgery for surgical lesions or deterioration; safeguarding if abuse is suspected.

## Child Protection & Safeguarding

ASSESS

AAP • NICE NG76

The injury that doesn't fit the history — recognizing it once may prevent a fatal second event.

### ▶ RECOGNIZE (RED-FLAG PATTERNS)

- **Any bruise in a not-yet-mobile infant** is a sentinel injury until proven otherwise. Use **TEN-4-FACEs**: bruising of **T**orso, **E**ars, **N**eck in a child  $\leq 4$  years, **any** bruising  $< 4$

months, or of the **Frenulum**, **Angle of jaw**, **Cheek**, **Eyelid**, **Sclera**, or **patterned marks**.<sup>263</sup>

- History that is **absent, inconsistent, or doesn't match the developmental stage or the injury**; delayed presentation; injuries of different ages.<sup>136</sup>

### ⚠ **ABUSIVE HEAD TRAUMA**

- An infant with unexplained irritability, vomiting, apnea, seizures or reduced consciousness may have **abusive head trauma** — the triad of subdural hemorrhage, **retinal hemorrhages** and encephalopathy.<sup>136</sup>
- Image the brain and arrange a dilated fundoscopy; it is easily mistaken for sepsis or a feeding problem.<sup>136</sup>

### ✦ **EXAMINE, DOCUMENT, ESCALATE**

- Fully undress and examine; document injuries factually with sizes and a body map (photograph per policy).



- In children < 2 years with suspected physical abuse, arrange a **skeletal survey** and consider coagulation studies to exclude mimics.<sup>136</sup>
- Care for the child first; you do not need to be certain to refer.<sup>136</sup>

◆ **LOCAL & LEGAL** Follow Kuwait MOH child-protection procedures — involve the hospital child-protection team/social work and make the mandatory report; ensure the child is in a safe place and that siblings are considered.<sup>145</sup>

### ▴ **WORKUP**

- **Skeletal survey** — all children < **24 months**; **2–5 years selectively** (radiographs of areas of concern; full survey if multiple or occult injuries are suspected); not useful > 5 years. **Repeat limited survey at ~2 weeks** when the first is negative or equivocal — it reveals healing rib and metaphyseal fractures.<sup>312,136</sup>
- **Neuroimaging** — **immediate non-contrast head CT** for any neurological symptom or sign, scalp swelling, or a low hemoglobin (the strongest predictors of an abnormal

CT: head swelling OR 46.7, neurological impairment OR 20.6, low hemoglobin OR 11.8).<sup>314</sup>

**Neurologically normal: CT head is still usually appropriate < 24**

**months** — occult intracranial injury is found in up to 37% of high-risk infants (rib or multiple fractures, facial injury, age < 6 months).<sup>312,313,315</sup>

Where rapid non-sedated MRI is available and expertly reported, **MRI may replace CT as first-line** in a stable infant whose injuries are unrelated to the head.<sup>314</sup> **MRI brain ± whole spine at day 2–5** whenever CT shows hemorrhage, parenchymal injury or skull fracture, or neurology persists.<sup>313,312</sup>

- **Dilated indirect ophthalmoscopy by ophthalmology — within 24 h, no later than 72 h** (hemorrhages fade): for intracranial hemorrhage or injury on imaging, unexplained coma, seizures or altered consciousness in a child < 5 years, visible eye injury, or coagulopathy. **Not a screening test** in the neurologically normal infant — image the brain first.<sup>316</sup>
- **Bleeding-disorder evaluation** when bruising or intracranial hemorrhage is what raises the concern: **PT,**

**aPTT, CBC with platelets, von Willebrand antigen and activity, factor VIII and IX activity.**

Unnecessary when injuries are patterned (hand marks; ear, neck or genital bruising) or clearly non-hemorrhagic (fractures, burns). After transfusion, delay testing until transfused products have cleared; viscoelastic tests (TEG/ROTEM) are not acceptable for this purpose.<sup>317</sup>

- **Occult abdominal injury — screen every child < 5 years: AST, ALT, lipase, urinalysis.**

**Contrast-enhanced abdominal CT if AST or ALT > 80 IU/L or abdominal bruising, tenderness or distension — a quarter of injuries with raised transaminases have no abdominal signs.**<sup>318,312</sup>

#### ▴ **ADMIT · REFER**

- **Admit** for safety and investigation whenever abuse is a realistic concern.
- **Call/Refer** — child-protection team, social work, and safeguarding lead; police per local law.

# Massive Transfusion & Blood Products

MINUTES

Eur. trauma guideline 2023

Balanced-ratio resuscitation with the per-kg doses — and the two silent killers, ionized hypocalcemia and the cold, acidotic child.

✚ Calculators: Blood products · SIPA shock index

## ▲ ACTIVATE EARLY – ANY OF

- > **40 mL/kg** of blood products in 24 h · ~**1 blood volume** ( $\approx$  70–80 mL/kg) in 24 h · or > **20 mL/kg red cells in 2 h** with ongoing bleeding.<sup>98</sup>
- Uncontrolled hemorrhage with instability, a high pediatric age-adjusted shock index (SIPA), or clinical judgment — **activate the protocol early, not late.**<sup>98</sup>

| Component<br>(aim 1:1:1) | Dose                      | Notes                                                                                   |
|--------------------------|---------------------------|-----------------------------------------------------------------------------------------|
| Packed red cells         | 10–20 mL/kg               | Emergency: uncrossmatched <b>O-negative</b> until type-specific is ready <sup>264</sup> |
| Fresh frozen plasma      | 10–20 mL/kg               | Give in a <b>1:1:1</b> ratio with red cells & platelets <sup>265</sup>                  |
| Platelets                | 10–20 mL/kg<br>(often 10) |                                                                                         |
| Cryo-precipitate         | 5–10 mL/kg                | For fibrinogen < 1.5 g/L (or give fibrinogen concentrate) <sup>264</sup>                |

### ⚠ THE TWO SILENT KILLERS

**Ionized hypocalcemia** — citrate in stored products chelates calcium, worsening coagulopathy and cardiac contractility; keep **ionized Ca**  $\geq$  ~1.1 mmol/L and recheck after every ~30–40 mL/kg. **The lethal triad** — hypothermia + acidosis + coagulopathy: **warm every product and the child**, correct acidosis, and treat coagulopathy early.<sup>264</sup>

| Adjunct               | Dose                                                                | Notes                                                                                                                    |
|-----------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Tranexamic acid (TXA) | 15 mg/kg IV (max 1 g) over 10 min, then 2 mg/kg/h for 8 h (max 1 g) | Give <b>within 3 h</b> of injury; consider in TBI with GCS < 12 before intracranial bleed is excluded <sup>266,392</sup> |
| Calcium gluconate     | 50–100 mg/kg IV (max 2–3 g)                                         | Peripheral-safe                                                                                                          |

### ▴ TARGETS

- **Fibrinogen  $\geq 1.5\text{--}2.0$  g/L · platelets  $> 50 \times 10^9/\text{L}$  ( $> 100$  with TBI or ongoing bleeding) · **INR  $< 1.5$ .****

### ✦ DON'T DROP THE PRESSURE

- In children, and especially with TBI or threatened perfusion, maintain age-appropriate blood pressure throughout resuscitation — **target SBP no less than  $70 + 2 \times \text{age (yr)}$  mmHg**, address ongoing losses, avoid hypotension.<sup>80,98,264</sup>
- Use **viscoelastic testing (TEG — thromboelastography / ROTEM — rotational thromboelastometry; bedside whole-blood clotting tests)** to guide which component to give, where available, and watch for

**hyperkalemia** with large-volume or irradiated red cells.<sup>264</sup>

▴ **WORKUP – SEND AT ACTIVATION, REPEAT EVERY 30-60 MIN**

- CBC (hemoglobin, platelets), **fibrinogen**, PT/INR ± aPTT, **ionized calcium**, blood gas with lactate / base deficit, type & crossmatch; **point-of-care viscoelastic coagulation testing** (thromboelastography, TEG/ROTEM) if available; monitor temperature and potassium.<sup>264</sup>

◆ **LOCAL & EVIDENCE**

- Activate your hospital massive-hemorrhage protocol and follow its ratios and dosing.
- Pediatric ratios and thresholds are largely **EXTRAPOLATED FROM ADULT RCTS** (PROPPR, CRASH-2/CRASH-3); the pediatric TXA signal (PED-TRAX) is observational.<sup>265,266</sup>
- Verify every dose against local protocol.
- Sources: European guideline on major bleeding & coagulopathy after trauma, 6th ed. (Rossaint 2023); ATOMAC pediatric MTP; PED-TRAX;

## Renal & Urology

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AKI · HUS · hypertensive emergency · acute  
scrotum · foreskin emergencies · priapism ·  
hematuria

### Acute Kidney Injury HOURS

Abrupt fall in kidney function — rising  
creatinine and/or falling urine output.

+ Calculator: [eGFR \(Schwartz\)](#)

#### RECOGNIZE & DEFINE

- Suspect in any child with  
oliguria/anuria, rising creatinine,  
edema or fluid overload, or  
unexplained electrolyte  
disturbance.<sup>156</sup>
- **KDIGO (Kidney Disease: Improving  
Global Outcomes) stages AKI 1–3** —  
Stage 1 is the threshold that  
diagnoses AKI (see the table  
below).<sup>156</sup>



## KDIGO STAGING – GRADE BY THE WORSE OF CREATININE OR URINE OUTPUT<sup>156</sup>

| Stage | Serum creatinine                                                                                                                                                     | Urine output                                           |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1     | 1.5–1.9× baseline, or a rise $\geq 26.5$ $\mu\text{mol/L}$ ( $\geq 0.3$ mg/dL)                                                                                       | $< 0.5$ mL/kg/h for 6–12 h                             |
| 2     | 2.0–2.9× baseline                                                                                                                                                    | $< 0.5$ mL/kg/h for $\geq 12$ h                        |
| 3     | $\geq 3.0\times$ baseline, or $\geq 353.6$ $\mu\text{mol/L}$ ( $\geq 4.0$ mg/dL), or dialysis started, or – if age $< 18$ y – eGFR $< 35$ mL/min/1.73 m <sup>2</sup> | $< 0.3$ mL/kg/h for $\geq 24$ h, or anuria $\geq 12$ h |

## FIND THE CATEGORY – IT DRIVES TREATMENT

### PRE-RENAL · impaired perfusion (commonest)

Hypovolemia, shock, sepsis — poor perfusion. → **Restore perfusion:** if volume loss, give fluid; if another cause, treat that cause. Reversible if treated early.<sup>156</sup>

HUS (see [Hemolytic Uremic Syndrome](#)), glomerulonephritis, acute tubular necrosis, nephrotoxins. → **Supportive care; treat the cause.** <sup>156</sup>

Posterior urethral valves, stones, neurogenic bladder. → **Insert a urinary catheter first**, then image. After relief, watch for **post-obstructive diuresis** (urine > 3.5–5 mL/kg/h) — replace fluid/electrolytes with nephrology. <sup>156</sup>

#### ► **MANAGE – VOLUME STATUS DRIVES FLUID THERAPY**

Assess volume status first, then treat by category:

- **Hypovolemic / intravascularly deplete** (cool peripheries, tachycardia, prolonged capillary refill, low BP, weight loss) — resuscitate with **isotonic crystalloid**.
- **Euvolemic** (well-perfused, no edema, no fluid-overload signs) — match intake to **insensible losses plus measured output**; do **not** apply routine 4-2-1 (Holliday–Segar) maintenance during established AKI. <sup>156</sup>

- **Hypervolemic / overloaded** (edema, hypertension, raised JVP, pulmonary crackles or a gallop, weight gain) — **restrict fluids**; diuretics do not aid recovery but may be used for established hypervolemia.<sup>156</sup>

Stop nephrotoxins (NSAIDs, aminoglycosides, contrast) and adjust drug doses. Watch potassium, sodium, acid–base; **avoid potassium in maintenance fluids unless treating hypokalemia.**<sup>156</sup> Involve nephrology for KDIGO stage 2–3 or worsening AKI; arrange structured follow-up (stage 2–3, AKI  $\geq$  7 days, incomplete recovery, pre-existing CKD, or recurrent AKI).<sup>156</sup>

#### ► WHEN TO CALL NEPHROLOGY / PICU FOR DIALYSIS – THE "AEIOU" TRIGGERS

Refer urgently for dialysis / CRRT when any of these will **not** correct with medical treatment:

- **A — Acidosis:** severe metabolic acidosis (pH < 7.1) despite bicarbonate / ventilation.<sup>156</sup>
- **E — Electrolytes:** hyperkalemia that keeps rising or recurs despite full medical therapy.
- **I — Intoxication:** a dialyzable poison / drug (salicylate, methanol, ethylene glycol, lithium, theophylline).<sup>156</sup>

- **O — Overload:** fluid overload / pulmonary edema — **or severe hypertension driven by volume overload** — not responding to fluid restriction + diuretics (especially if it limits feeding or ventilation).
- **U — Uremia:** symptomatic — encephalopathy, pericarditis, or bleeding.<sup>156</sup>

**The clinical picture — not the creatinine or urea number — decides.**

Recognize early and call; the specialist chooses the modality (hemodialysis, CRRT, or peritoneal dialysis).<sup>156</sup>

#### **⚠ CROSS-REFERENCE**

Treat life-threatening hyperkalemia immediately — see [Acute Electrolyte Emergencies](#) (Endocrine & Metabolic).

#### **▀ ADMIT · DISCHARGE · REFER**

- **Admit** — all but mild pre-renal AKI; oliguria, hyperkalemia, fluid overload, or acidosis → high-dependency.
- **Discharge** — resolving pre-renal AKI with normalizing function and follow-up.

- **Refer** — nephrology for KDIGO stage 2–3, no clear cause, or need for dialysis.

## Hemolytic Uremic Syndrome HOURS

Triad: microangiopathic hemolytic anemia + thrombocytopenia + acute kidney injury.

**+ Calculators:** [eGFR \(Schwartz\)](#) · [Blood products](#)

### RECOGNIZE

- **Typical (STEC):** preceded by bloody diarrhea; pallor, reduced urine, edema, petechiae, irritability.<sup>267</sup>
- Blood film: fragmented red cells (schistocytes), low platelets, negative direct antiglobulin test, high lactate dehydrogenase.<sup>267</sup>
- **Atypical (non-diarrheal) HUS:**  
STEC-negative; any age. **Primary** — inherited or antibody-mediated complement dysregulation (low C3, family history, relapsing course).  
**Secondary** — invasive *S. pneumoniae* (pneumonia/empyema, meningitis, bacteremia; **direct antiglobulin test positive; avoid**

**plasma-containing products** — use washed red cells and platelets; treat the infection), HIV, drugs (chemotherapy, calcineurin inhibitors, ionizing radiation), systemic disease (lupus, antiphospholipid syndrome, scleroderma), and cobalamin C defect (methylmalonic aciduria with homocystinuria — check homocysteine and methylmalonic acid in infants). The underlying condition is usually already known; this guide details the complement-mediated form.<sup>267</sup>

#### ▀ WORKUP

- CBC + film, U&E/creatinine, LDH, haptoglobin, direct Coombs (negative).
- **Send C3, C4, CH50 and ADAMTS13** (ADAMTS13 = the von Willebrand factor–cleaving enzyme; **severely low confirms TTP (thrombotic thrombocytopenic purpura), normal points to HUS**)<sup>267</sup> **before** any plasma therapy or eculizumab. **If available, also send anti-factor-H antibodies and complement-gene screening** (*CFH*, *CD46/MCP*, *CFI*, *C3*, *CFB*, *THBD* and *DGKE*) to confirm

complement dysregulation, guide prognosis and estimate recurrence risk, including after transplantation — results take weeks and must not delay treatment. Not currently available in Kuwait: store samples and send out.<sup>267,319</sup>

- Stool culture for *E. coli* O157 and Shiga-toxin PCR/antigen when diarrhea is present.<sup>267</sup>

#### SUPPORTIVE CARE IS THE MAINSTAY

- Careful fluid and electrolyte management — in suspected STEC-HUS, early isotonic volume expansion before oligo-anuria may reduce kidney injury; once oliguric or fluid-overloaded, restrict fluids and involve nephrology.<sup>267</sup>
- Manage the acute kidney injury (see [Acute Kidney Injury](#)); transfuse packed red cells for symptomatic anemia or Hb < 6–7 g/dL (platelets only for active bleeding or a procedure); dialysis as needed — dialysis does not change STEC-HUS prognosis.<sup>267</sup>
- Monitor closely through the **5–14-day peak period**: daily CBC, creatinine, electrolytes and LDH, plus urine output, blood pressure and weight.<sup>267</sup>



#### DECISION

**Typical (STEC) or atypical (complement-mediated)?**



**Supportive care only. Benefit of eculizumab in STEC-HUS is unproven — not routine.** The only placebo-controlled RCT (100 children) showed no improvement in the acute course, dialysis or extrarenal disease; use, if any, is a nephrology decision in severe neurological involvement.<sup>267,321</sup>

Complement inhibition (eculizumab/ravulizumab) — nephrology directed. **Meningococcal risk is > 1,000-fold and vaccination alone does not prevent it:** give **MenACWY + MenB  $\geq$  2 weeks before the first dose;** if treatment cannot wait, vaccinate and start **antibiotic prophylaxis (e.g., penicillin V) for the duration of therapy.** Boosters: MenACWY every 5 years (every 3 years if < 7 years at the last dose); MenB 1 year after the series, then every 2–3 years while on treatment. **Any fever = emergency blood cultures and IV ceftriaxone** — 14 of 16 US cases occurred in vaccinated patients, most from non-groupable strains.<sup>322,323</sup>

### ⚠️ AVOID

- Antibiotics in STEC (may increase toxin release — generally avoided), antimotility agents (loperamide), and NSAIDs.<sup>267</sup>
- Do not transfuse platelets unless active bleeding or a procedure is planned.<sup>267</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — all → monitoring and supportive care; anuria/severe → high-dependency. **PICU** for neurologic symptoms or hemodynamic instability; involve **nephrology** early for atypical HUS.
- **Discharge** — not acutely; after hematologic and renal recovery.
- **Refer** — nephrology (dialysis); atypical HUS → a **nephrology specialist** for complement inhibition / eculizumab.

# Hypertensive Emergency

MINUTES

Severe hypertension **with** new or progressing end-organ damage (encephalopathy, seizures, facial palsy, severe headache; heart failure or chest pain; acute kidney injury; retinopathy or blurred vision).

## ASSESS FOR END-ORGAN DAMAGE

- **Brain:** headache, vomiting, altered mental status, seizures, visual symptoms, focal deficits.
- **Heart:** tachycardia, gallop, pulmonary edema, heart failure, chest pain (older children).
- **Kidney:** AKI, oliguria, hematuria, proteinuria.
- **Eyes:** blurred vision, retinal hemorrhages, papilledema.<sup>40</sup>

## WORKUP

- CBC, electrolytes, urea, creatinine.
- Urinalysis ± urine protein.
- ECG.
- Chest radiograph if heart failure suspected.
- Echocardiography if LV dysfunction suspected.

- Fundoscopy if available.
- Renal ultrasound  $\pm$  Doppler when a renal cause is suspected.
- CT/MRI brain only if neurological symptoms or concern for intracranial pathology.<sup>40</sup>

#### ✦ PEARL

In children, severe hypertension is **secondary until proven otherwise** — renal disease is the commonest cause.<sup>40</sup>

#### LOWER GRADUALLY – CONTROLLED DESCENT

- A true hypertensive emergency needs PICU/high-dependency monitoring with a titratable IV infusion under specialist guidance.
- Reduce blood pressure by no more than ~25% in the first 8 hours, then normalize over 24–48 hours.
- Abrupt drops risk cerebral, renal, and retinal ischemia.<sup>40</sup>



#### TREAT COMPLICATIONS

Seizures  $\rightarrow$  benzodiazepine. Fluid overload (e.g., acute glomerulonephritis)  $\rightarrow$  furosemide. Monitor in PICU.<sup>40</sup>

## MEDICATIONS – TITRATABLE IV AGENTS PREFERRED

| Drug                | Dose                                                   | Route & notes                                                                                          |
|---------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Labetalol           | bolus 0.2–1 mg/kg (max 40 mg); infusion 0.25–3 mg/kg/h | IV; <b>preferred first-line (best option) in Kuwait</b> ; avoid in asthma, heart failure <sup>40</sup> |
| Hydralazine         | 0.1–0.5 mg/kg/dose (max 20 mg)                         | IV/IM every 4 h; <b>second-line</b> ; may cause reflex tachycardia                                     |
| Sodium nitropruside | 0.3–8 mcg/kg/min                                       | IV; reserve — cyanide risk with prolonged use/renal impairment <sup>40</sup>                           |
| Esmolol             | 50–300 mcg/kg/min (after load)                         | IV; short-acting; may cause bradycardia/bronchospasm                                                   |
| Furosemide          | 1–2 mg/kg (max 40–80 mg)                               | IV; if fluid overloaded (e.g., post-streptococcal glomerulonephritis)                                  |

◆ **NOTE** Oral or sublingual nifedipine can cause a precipitous, uncontrolled fall and is **NOT** recommended for a true hypertensive emergency.<sup>40</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all true emergencies → PICU with a titratable infusion.

- **Discharge** — not from the emergency; controlled blood pressure with a plan.
- **Refer** — nephrology/cardiology for cause and long-term control.

## Nephrotic Syndrome HOURS

IPNA 2023

Heavy proteinuria, low albumin, and edema — usually steroid-responsive, but watch the complications. Doses follow the International Pediatric Nephrology Association (IPNA) 2023 guideline.

**+ Calculator: eGFR (Schwartz)**

### ▶ RECOGNIZE

- **The triad:** nephrotic-range proteinuria (urine protein/creatinine > 200 mg/mmol, or  $\geq 3+$  on dipstick), **hypoalbuminemia < 30 g/L (< 3.0 g/dL)**, and edema (periorbital → generalized, ascites, scrotal/labial); usually with hyperlipidemia.<sup>49</sup>
- In a child **1–10 years** without the atypical features below, it is almost always steroid-responsive

minimal-change disease — treat without a biopsy.<sup>49</sup>

#### CONFIRM & ASSESS

- Confirm proteinuria (first-morning-void protein/creatinine ratio preferred over a 24-hour collection); send albumin, renal function and electrolytes, lipids; weigh, record blood pressure, assess volume status.
- Assess clinically for active infection before commencing high-dose corticosteroids; investigate according to symptoms and risk factors rather than performing routine infection screening in an otherwise well child.<sup>49</sup>



## FIRST PRESENTATION

- Oral **prednisolone 60 mg/m<sup>2</sup>/dose (≈ 2 mg/kg/dose, max 60 mg)** once daily for 4–6 weeks, then 40 mg/m<sup>2</sup> (max 40 mg) on alternate days for a further 4–6 weeks (total 8–12 weeks)
- Add a no-added-salt diet (no fluid or protein restriction) and ensure pneumococcal vaccination; routine antibiotic prophylaxis is not recommended — instead treat suspected infection promptly, including spontaneous bacterial peritonitis.<sup>49</sup>

## ⚠️ COMPLICATIONS TO PRE-EMPT

- **Hypovolemia** can hide behind gross edema — abdominal pain, cool peripheries, tachycardia, oliguria; needs urgent senior/nephrology input; treat with **fluid resuscitation if needed and 20% albumin plus a diuretic** (avoid diuretic monotherapy). **Hypervolemia** presents differently — edema with **strong peripheral pulses and warm peripheries**; treat with diuretics and close reassessment.<sup>49</sup>



- Watch for **spontaneous bacterial peritonitis** (pneumococcal) and other sepsis, **venous or arterial thromboembolism** (these children are hypercoagulable), and **acute kidney injury (AKI)**.<sup>49</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — first presentation, hypovolemia, peritonitis or sepsis, thromboembolism, severe edema with respiratory compromise, AKI, or a relapse not responding to oral steroid.
- **Discharge** — steroid responsiveness is **not** required before discharge, provided close OPD follow-up is arranged: a well child with no complications, able to tolerate oral steroids, with the family able to monitor first-morning urine and act on a written plan.
- **Refer** — Nephrology for any atypical feature (age < 1 or > 12 years, persistent hypertension, gross hematuria, low complement (C3), renal impairment), steroid resistance,

or frequent relapse — these may need a biopsy or second-line therapy.<sup>49</sup>

## Circumcision – Kuwait Practice

ASSESS

≥ 1 WK • Ped SURGERY

Near-universal in Kuwait — screen first, always give analgesia, and know the complications.

### ▀ BEFORE YOU CIRCUMCISE – CHECKLIST

- **Examine the genitalia first. Do NOT circumcise** if hypospadias, epispadias, chordee, ambiguous genitalia / suspected DSD, micropenis, or buried / webbed penis — the foreskin may be needed for repair → refer to pediatric surgery / urology.<sup>109</sup>
- Contraindicated in an **unstable or sick infant** — confirm normal feeding, elimination and thermoregulation; ensure vitamin K given.<sup>109</sup>
- Family history of a bleeding disorder → coagulation studies / defer.<sup>109</sup>

- Informed parental consent. In Kuwait usually performed by **pediatric (or general) surgery from ~1 week of age** (and later in infancy / childhood).

#### ◆ ANALGESIA – NEVER OMIT

- **Dorsal penile nerve block** or subcutaneous **ring block** (lidocaine 1% without epinephrine,  $\leq 1$  mL, 3–4 mg/kg) — success ~90–100%, complications <0.1%.<sup>110</sup>
- Add topical EMLA, oral sucrose, non-nutritive sucking and acetaminophen.
- Sucrose or EMLA **alone are not enough**; combination analgesia is best.<sup>110</sup>

## METHODS

Gomco clamp, Mogen clamp, or Plastibell (ring) device — each with its own technique and complication profile.<sup>109</sup>

## WOUND CARE

- Clean gently with warm water at each nappy change; avoid soap and wipes (they sting).
- **Gomco / Mogen:** apply petroleum jelly (or antibiotic ointment) to the glans and on the nappy for ~5–7 days to stop the wound sticking. A moist **yellow coating** (granulation) on the glans is **normal healing**, not infection.<sup>109</sup>
- **Plastibell:** the ring falls off by itself in ~5–10 days — **do not pull it**; sponge-bathe until it separates<sup>109</sup>; seek review if it slips onto the shaft or has not fallen by 10–14 days.
- Heals in ~7–10 days. After healing, gently retract the skin at the groove to prevent adhesions / skin bridges.<sup>129</sup>

## ⚠ **COMPLICATIONS – WHEN TO SEEK URGENT REVIEW**

- **Bleeding** more than a few drops (commonest — direct pressure and a compressive dressing around the penis, then pediatric surgery review + coagulation workup)

- spreading redness up the shaft, pus or fever = **wound infection**
- no urine for ~8 h = **retention**<sup>129</sup>
- Plastibell ring migrated with a **dusky glans** = constriction (emergency)
- trapped / concealed penis, adhesions, skin bridges, or late **meatal stenosis** (deflected stream)<sup>129</sup> → pediatric surgery / urology.

## Acute Scrotum

MINUTES

6-HOUR WINDOW

Sudden scrotal pain is testicular torsion until proven otherwise — a clinical diagnosis; do not wait for imaging.

### ▀ DEFINE

- **Acute scrotum** = sudden scrotal pain/swelling. **Testicular torsion until proven otherwise** — a time-critical surgical emergency. Bimodal peaks: the neonate and the peri-pubertal boy.<sup>128</sup>
- Differential: **torsion of the appendix testis** (blue-dot sign, gradual, milder), **epididymo-orchitis** (dysuria,

gradual, older/sexually active),  
trauma, idiopathic scrotal edema,  
incarcerated inguinal hernia.<sup>128</sup>

#### ▀ **APPROACH – FAVORS TORSION**

- **Sudden severe pain** ±  
nausea/vomiting, **high-riding testis**,  
**horizontal lie**, **absent cremasteric reflex**, diffuse tenderness.<sup>128</sup>
- The **TWIST score** (hard testis, swelling, nausea/vomiting, absent cremasteric reflex, high-riding testis) risk-stratifies but does not replace surgical judgment.<sup>128</sup>

#### ⚠ **WORK-UP**

- **Torsion is a clinical diagnosis — do not delay surgical exploration for imaging.** Color Doppler ultrasound (absent intratesticular flow) is useful only when the diagnosis is equivocal **and** will not delay surgery.<sup>128</sup>
- Urinalysis: pyuria or bacteriuria supports bacterial epididymitis — but a **normal urinalysis does not exclude epididymitis, and an**

**abnormal one does not exclude torsion;** send a culture.<sup>320</sup>

### ⚠ **RED FLAGS – THE CLOCK**

- Testicular salvage is **time-dependent:** near-universal if detorsion occurs within ~6 hours of onset, falling sharply thereafter.<sup>128</sup>
- Neonatal torsion and any delayed presentation need urgent senior/urology input.

### ▀ **MANAGEMENT**

- **Immediate urology/surgical referral** for scrotal exploration, detorsion, and **bilateral orchidopexy.**<sup>128</sup>
- If surgery is delayed, a trained clinician may attempt **manual detorsion** (usually 'opening a book' — medial-to-lateral) **only if experienced** as a temporizing measure while immediate exploration is arranged, guided by pain relief; success does not remove the need for surgery and orchidopexy.<sup>128</sup>
- **Consult surgery/urology at once and confirm the diagnosis.** Torsion → surgery. Once torsion is excluded:

**appendix-testis torsion** → NSAID

analgesia and rest;

**epididymo-orchitis** → NSAIDs, rest and scrotal support, with **antibiotics only if urinalysis or urine culture shows bacterial infection** (send the culture before treating — prepubertal epididymitis is usually non-bacterial).<sup>320</sup>

✦ **PEARL – ABDOMINAL PAIN CAN BE A TORSION**

Examine the testes in every boy with acute lower-abdominal or groin pain — torsion frequently refers pain to the abdomen and is missed when the scrotum is not examined.<sup>128</sup>

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — torsion (confirmed or not excluded) → emergency exploration; incarcerated hernia; epididymo-orchitis only if systemically unwell, an infant, or unable to take oral analgesia.<sup>320</sup>
- **Discharge** — once the surgical/urology team has excluded torsion: appendix-testis torsion and epididymo-orchitis go home on NSAIDs ± antibiotics per urine results, review in 48–72 h;



prepubertal epididymitis warrants outpatient renal-tract ultrasound (underlying urogenital anomaly in ~25–30%).<sup>320</sup>

- **Refer** — surgery/urology IMMEDIATELY to confirm the diagnosis (torsion → exploration within 6 h).<sup>128</sup>

◆ **SEE ALSO** [Testicular Torsion](#) — the time-critical diagnosis to exclude first.

## Testicular Torsion MINUTES

### ADOLESCENT & NEONATE

Salvage window ~6 hours. A boy's abdominal pain → examine the scrotum.

#### ▶ RECOGNIZE

- Sudden severe scrotal/lower-abdominal pain, nausea/vomiting, swelling. High-riding/horizontal testis, **absent cremasteric reflex**, negative Prehn sign.<sup>128</sup>

**Clinical suspicion = urgent exploration.  
Don't delay for imaging.**



**ACT**

- **Immediate urology / pediatric surgery** → exploration, detorsion + bilateral orchidopexy.
- Color Doppler ultrasound **only if it will not delay** surgery.
- Manual detorsion may temporize **only in experienced hands** — do not attempt if untrained (can worsen the torsion), and never let it delay surgical exploration.<sup>128</sup>

▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → emergency exploration.
- **Discharge** — post-operative.
- **Refer** — pediatric surgery/urology IMMEDIATELY (explore within 6 h); do not delay for imaging.

◆ **SEE ALSO** [Acute Scrotum](#) for the full differential.

# Paraphimosis

HOURS

REDUCE PROMPTLY

A retracted foreskin trapped behind the glans — edema then ischemia; reduce promptly.

## ▀ DEFINE

- **Paraphimosis** — a retracted foreskin trapped behind the glans, cannot be reduced. The constricting band causes progressive glans and distal-foreskin edema, then venous → arterial compromise. A urological emergency.<sup>129</sup>
- Commonly **iatrogenic** — the foreskin is left retracted after catheterization, cleaning, or examination.<sup>129</sup>

## ▀ APPROACH & WORK-UP

- Painful, swollen glans with an edematous 'collar' of foreskin proximal to it. Diagnosis is clinical.
- Distinguish from balanoposthitis (foreskin/glans inflammation without a trapped retracted ring).

## ⚠ RED FLAGS

Increasing edema, dusky glans, or pain out of proportion signal arterial

compromise and impending necrosis —  
**reduce promptly; escalate to urology if  
reduction fails.**<sup>129</sup>

## MANAGEMENT

### ▴ REDUCE IT

- **Analgesia first** — penile block or procedural analgesia.
- **Reduce edema** — sustained manual compression of the glans ( $\pm$  a wrap such as saline-soaked gauze, ice, or an osmotic agent such as granulated sugar) for several minutes.<sup>129</sup>
- **Reduce the foreskin** — push the glans back with the thumbs while drawing the foreskin forward over it with the fingers.
- **If manual reduction fails** — urology for a dorsal-slit or emergency reduction.<sup>129</sup>

### ✦ PEARL – PREVENTION IS THE CURE

Always return the foreskin to its natural forward position after any catheterization, cleaning, or examination, and teach carers never to force-retract a child's foreskin.<sup>129</sup>

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — if reduction fails or there is tissue compromise.
- **Discharge** — after successful reduction with follow-up.
- **Refer** — urology for failed reduction or recurrent episodes.

## Balanitis / Balanoposthitis & Phimosis

ASSESS

COMMON

Glans/foreskin inflammation and non-retractile foreskin; physiologic phimosis is normal, needs no treatment.

### ▲ DEFINE

- **Balanitis** = inflammation of the glans; **balanoposthitis** = glans + foreskin. **Phimosis** = non-retractile foreskin — **physiologic** (normal in young boys) or **pathologic** (scarred).
- **Physiologic phimosis is normal:** non-retractile in most infants, becomes retractile with age ( $\approx 99\%$  by 17 years). Needs no treatment.<sup>129</sup>

## ▲ APPROACH & WORK-UP

- Balanitis: red, swollen, tender glans/foreskin ± discharge, dysuria. Usually hygiene-related or candidal; consider sexually transmitted causes in adolescents.
- Distinguish physiologic phimosis (soft, healthy, pink opening) from **pathologic** (scarred white fibrotic ring — balanitis xerotica obliterans/lichen sclerosus).<sup>129</sup>
- Clinical diagnosis; swab if recurrent/purulent; urinalysis if voiding symptoms.

## ▲ RED FLAGS

Pathologic phimosis (white scarred ring / BXO), recurrent balanoposthitis, **ballooning or difficulty voiding**<sup>129</sup>, or **urinary retention** — refer to urology.

## MANAGEMENT

| Drug                                             | Dose                                                          | Route & notes                                                       |
|--------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|
| Balanitis<br>—<br>general                        | Hygiene, warm saline soaks                                    | Retract gently only as far as comfortable; avoid irritants.         |
| Balanitis<br>—<br>bacterial/<br>candidal         | Topical/oral antibiotic or topical antifungal                 | Guided by likely organism; treat/screen for STI in adolescents.     |
| Physiologic<br>phimosis                          | No treatment; do not force-retract                            | Reassure; resolves with age.                                        |
| Symptomatic /<br>early<br>pathologic<br>phimosis | Topical betamethasone 0.05% applied twice daily for 4–6 weeks | ≈ 80% success; avoids or defers circumcision. <sup>129</sup>        |
| Pathologic<br>phimosis<br>/ BXO /<br>recurrent   | Refer for circumcision                                        | Definitive for scarred foreskin or lichen sclerosis. <sup>129</sup> |

### ◆ PEARL – HANDS OFF THE YOUNG FORESKIN

Never forcibly retract a young boy's foreskin — it tears, bleeds, and scars, converting harmless physiologic phimosis into pathologic phimosis.<sup>129</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — rarely — severe infection or urinary retention.

- **Discharge** — most, with hygiene and topical treatment.
- **Refer** — urology for pathological phimosis, recurrent balanitis, or balanitis xerotica obliterans.

## Priapism

MINUTES

> 4 H = EMERGENCY

Painful rigid erection > 4 hours; ischemic priapism is a penile compartment syndrome — relieve fast.

### DEFINE

- **Priapism** — persistent erection > 4 hours unrelated to stimulation.  
**Ischemic** (low-flow, painful, rigid corpora — penile compartment syndrome, a true emergency) versus **non-ischemic** (high-flow, usually post-traumatic, not urgent).<sup>130</sup>
- Most pediatric priapism is **ischemic and associated with sickle cell disease**; short self-limiting 'stuttering' episodes often precede a major event.<sup>130</sup>



## ▲ APPROACH & WORK-UP

- Rigid, painful corpora cavernosa with a **soft glans and corpus spongiosum** → ischemic. Ask about sickle cell disease and drugs.
- **Cavernosal blood gas** distinguishes: ischemic = dark, hypoxic, acidotic blood. **Color Doppler ultrasound** also separates the two (ischemic = absent/low cavernosal flow; non-ischemic = normal/high flow). **Do not delay treatment for imaging.**<sup>130</sup>

## ▲ RED FLAGS – THE CLOCK

- Smooth-muscle edema and atrophy begin as early as ~6 hours; risk of permanent erectile dysfunction rises with duration.<sup>130</sup>
- Ischemic priapism > 4 hours is a time-sensitive emergency needing immediate urology (and hematology in sickle cell disease) — do not delay urologic intervention for sickle-cell measures.<sup>130</sup>

## MANAGEMENT

### ▶ RELIEVE THE ERECTION FIRST

- **In sickle cell disease, do not delay penile intervention for exchange transfusion** — urologic relief comes first; sickle measures (hydration, oxygen, analgesia) run concurrently.<sup>290</sup>
- **Corporal aspiration ± saline irrigation plus intracavernosal phenylephrine** (urology-directed), repeated as needed, with blood-pressure and heart-rate monitoring.<sup>130</sup>
- Surgical shunting only after aspiration + sympathomimetic have failed.<sup>130</sup>

| Drug                            | Dose                                                                 | Notes                                                                          |
|---------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Phenylephrine (intracavernosal) | 100–200 mcg/dose every 3–5 min (dilute to 100–500 mcg/mL); max ~1 mg | Monitor BP and HR; use with caution — reduced dose in children. <sup>130</sup> |

### ✦ PEARL – CALL UROLOGY NOW

Acute ischemic priapism is an emergency; in sickle cell disease relieve

the erection first and give sickle-specific care alongside — not instead.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — ischemic priapism → urgent intervention.
- **Discharge** — after detumescence with follow-up.
- **Refer** — urology (aspiration/irrigation) and hematology in sickle cell disease; do not delay urology.

## Gross Hematuria

ASSESS

APPROACH

Confirm true hematuria, then sort glomerular from non-glomerular — that split directs the entire work-up.

#### ▲ DEFINE

- **Gross hematuria** — visibly bloody or tea/cola-colored urine. Confirm **true hematuria** (dipstick positive **and**  $\geq 5$  RBC/high-power field on microscopy) — exclude mimics:

foods (beetroot), drugs (rifampin), myoglobin, urate, menstrual contamination.<sup>131</sup>

#### ▲ APPROACH – GLOMERULAR VS NON-GLOMERULAR

- **Glomerular** — brown/cola-colored, dysmorphic RBCs or RBC casts, proteinuria, hypertension, edema → glomerulonephritis (post-infectious, IgA nephropathy, Alport). **Recurrent painless gross hematuria** with a fever, viral illness or exercise → most likely **IgA nephropathy**; hematuria with **hearing impairment** (or a family history of kidney failure) → **Alport syndrome** — send genetics first.<sup>131</sup>
- **Non-glomerular** — bright red ± clots → urologic: UTI, hypercalciuria/stones, trauma, benign urethrorrhagia, rarely tumor. **Timing helps:** blood at the **start** of the stream → urethra; at the **end** → bladder. **Benign urethrorrhagia** — prepubertal boys, spots of blood at the end of or between voids, resolves spontaneously.<sup>132</sup>

- Commonest causes overall: UTI and hypercalciuria; gross hematuria without glomerular features is **usually benign**.<sup>132</sup>

#### ⚠ WORK-UP

- Urinalysis with **microscopy** (casts, dysmorphic RBCs), urine culture, **blood pressure**; urine protein:creatinine and calcium:creatinine ratios; serum creatinine/electrolytes; **C3/C4 and anti-DNase B** if glomerulonephritis is suspected; **renal-bladder ultrasound** — stones, polycystic kidney disease, trauma.<sup>131</sup>
- Image for trauma or obstructing clot as indicated.

#### ▀ RED FLAGS

- Hypertension, edema, oliguria, or rising creatinine → glomerulonephritis with acute kidney injury.<sup>131</sup>
- Flank pain with clots (stones), significant trauma, a palpable mass, or systemic symptoms.

## ▀ MANAGEMENT

- Treat the underlying cause.  
**Nephrology referral** for glomerular features (hypertension, proteinuria, casts, reduced GFR); **urology** for structural/urologic causes.<sup>131</sup>
- Isolated gross hematuria with normal blood pressure, no proteinuria, and normal ultrasound can usually be observed.<sup>132</sup>

## ✦ PEARL – DIPSTICK + MICROSCOPY + BLOOD PRESSURE

Those three bedside steps triage most children — color and RBC morphology sort glomerular from non-glomerular and direct the whole work-up.<sup>131</sup>

## ▀ ADMIT · DISCHARGE · REFER

- **Admit** — clot retention, significant bleeding, hypertension, or acute nephritis with impairment.
- **Discharge** — well child with a benign cause and outpatient workup.
- **Refer** — nephrology (glomerular) or urology (structural, stones, or mass).

# Acute Post-infectious Glomerulonephritis (PIGN)

ASSESS

2–12 YR

The commonest acute nephritis in children — cola urine, puffy eyes and a high blood pressure 1–6 weeks after an infection, classically streptococcal.

+ Calculator: [eGFR \(Schwartz\)](#)

## RECOGNIZE

- **Acute nephritic syndrome** — cola/tea-colored urine, periorbital/leg edema, **hypertension**, oliguria; onset **1–3 weeks after pharyngitis** or **3–6 weeks after impetigo** (skin infection is the commoner trigger in hot climates). Group A streptococcus is the classic cause, but other bacteria, viruses, fungi and parasites cause the same immune-mediated nephritis — hence **post-infectious** (infection-related) GN. <sup>324,325</sup>
- Ask about recent sore throat, skin sores or scabies; examine the skin, blood pressure and volume status.

## ▲ WORK-UP

- Urinalysis and microscopy — dysmorphic RBCs, **RBC casts**, proteinuria (usually sub-nephrotic); urine protein:creatinine ratio.
- Creatinine, urea, electrolytes (potassium), albumin, CBC.
- **Low C3 with normal C4** supports the diagnosis; **anti-DNase B** confirms recent streptococcal infection (positive in nearly all; ASO titer adds little); throat or skin swab if there is a current infection.<sup>324,325</sup>
- Chest X-ray if breathless or overloaded; renal ultrasound only if the picture is atypical.



## MANAGE — SUPPORTIVE CARE

- **Manage fluid balance** (strict input/output, daily weight, fluid restriction if oliguric or overloaded) and a **no-added-salt diet**.
- **Treat hypertension** — furosemide 1–2 mg/kg IV/oral for edema, overload or hypertension is usually enough; add a calcium-channel blocker if still hypertensive. Severe hypertension with headache, seizures or visual change → [Hypertensive Emergency](#).<sup>324</sup>
- **Antibiotics only for a current infection** (e.g., active pharyngitis or impetigo) — they do not alter the nephritis and are not given for the nephritis itself.<sup>324</sup>
- **No corticosteroids or immunosuppression** for uncomplicated disease.<sup>324</sup>

## ⚠ RED FLAGS

- **Hypertensive encephalopathy** (headache, vomiting, seizures, altered consciousness) — about 1 in 10 admissions.<sup>325</sup>
- **Pulmonary edema**, hyperkalemia, or a **rapidly rising creatinine** → possible crescentic

glomerulonephritis — nephrology the same day (see [Acute Kidney Injury](#)).<sup>324,325</sup>

#### ◆ EXPECTED COURSE

Diuresis and normal blood pressure within 1–2 weeks; C3 normalizes by 8 weeks; microscopic hematuria may persist up to a year. Recurrence is rare.<sup>324</sup>

#### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — hypertension, oliguria, edema/overload, raised creatinine or hyperkalemia.
- **Discharge** — normal blood pressure and urine output; check blood pressure and urinalysis at 1–2 weeks, C3 at 6–8 weeks.
- **Refer** — nephrology if creatinine keeps rising, nephrotic-range proteinuria, C3 still low at **12 weeks** (consider C3 glomerulopathy), or hematuria/proteinuria beyond 1 year — these may need a biopsy.<sup>324</sup>

Muscle breakdown → tea-colored urine, a very high CK, and the twin threats of hyperkalemia and AKI.

### + Calculator: eGFR (Schwartz)

#### ▀ RECOGNIZE

- Myalgia, weakness and **dark (tea-/cola-colored) urine** that is **dipstick-positive for blood but shows few red cells** (myoglobin), with **creatinine kinase > 5× the upper limit**. Triggers: viral myositis (influenza), trauma/crush, extreme exertion or heat, seizures, toxins/drugs, and metabolic myopathies.<sup>268</sup>

#### ⚠ TWO IMMEDIATE DANGERS

- **Hyperkalemia** — get an ECG and treat at once (see Acute Electrolyte Emergencies). **Acute kidney injury** — start **early, generous IV fluids** to maintain a good urine output.<sup>268</sup>
- Also watch calcium/phosphate and for **compartment syndrome** in a swollen, painful limb. **Routine urinary alkalization and**

**mannitol are not recommended.**<sup>268</sup>

#### ✦ CHASE THE CAUSE

Recurrent or exertion-out-of-proportion rhabdomyolysis, or a family history, suggests an **underlying metabolic myopathy** — refer for metabolic work-up once the acute episode is controlled.<sup>268</sup>

#### ▀ WORKUP

- **Creatine kinase** (diagnostic > 5× the upper limit of normal (ULN); **serial** to track the trend; AKI risk climbs > 5000 U/L)<sup>268</sup>.
- **Potassium + ECG**, calcium, phosphate.
- Renal function, venous blood gas/bicarbonate; urinalysis (**heme-positive with few RBCs** = myoglobinuria) + urine myoglobin.<sup>268</sup>
- Uric acid, LDH, AST; metabolic/genetic screen if recurrent or exertion-disproportionate.<sup>268</sup>

### ▲ ADMIT · REFER

- **Admit** — significant CK elevation, any renal impairment or hyperkalemia.
- **Refer** — nephrology if AKI; surgery for compartment syndrome.

## Respiratory, Allergy & Immunology

15

Airway, breathing, anaphylaxis, allergy & immunology

### Anaphylaxis

MINUTES

Acute multisystem allergic reaction — epinephrine first, always.

+ Calculator: [Infusion dose ↔ rate](#)

### ▲ RECOGNIZE & DIAGNOSE

- **Diagnose anaphylaxis (GA<sup>2</sup>LEN 2024 criteria) when any one applies:**
  - 1 • **No known allergen** — acute skin/mucosal signs **plus** respiratory **or** cardiovascular involvement.

## **2 · Likely or known allergen**

**exposure** — any **two** of:

skin/mucosa · respiratory ·  
cardiovascular · severe GI symptoms  
(severe crampy pain, repeated  
vomiting).

## **3 · Known allergen exposure** —

respiratory involvement **alone**  
(non-inhaled allergen) **or**  
cardiovascular involvement **alone**  
(hypotension, poor perfusion,  
collapse) — skin signs may be  
absent.<sup>327</sup>

- **If in doubt, give epinephrine.**

Epinephrine is given immediately for suspected anaphylaxis and may be given on clinical judgment before the criteria are fully met — the risk of a missed dose exceeds the risk of an unnecessary one.<sup>327,9</sup>

- Skin: urticaria, flushing, angioedema, pruritus (absent in up to 20%).<sup>9</sup>
- Breathing: stridor, wheeze, hoarseness, falling SpO<sub>2</sub>, cyanosis, increased work of breathing, signs of respiratory distress, persistent cough.
- Circulation: poor perfusion, hypotension, collapse — **reduced BP = low systolic for age or > 30% drop**

**from baseline.** Infants: floppiness, drooling, pallor, sudden behavior change.<sup>9</sup>

- GI: severe crampy abdominal pain or repeated vomiting — counts toward the diagnosis, especially after a likely allergen.<sup>9</sup>
- **Not anaphylaxis:** repeated vomiting 1–4 h after a food **without** skin, respiratory or cardiovascular signs → think food protein-induced enterocolitis syndrome (FPIES) — fluids and ondansetron, not epinephrine (see [Acute FPIES](#)). **If the infant is lethargic or hypotensive, treat shock immediately** (20 mL/kg bolus) **and give epinephrine if anaphylaxis cannot be excluded.**<sup>363</sup>

**1ST LINE** Intramuscular epinephrine immediately — there is no contraindication<sup>9</sup>



#### THEN

- **Remove the allergen if possible** — stop the infusion or drug, remove the stinger.<sup>9</sup>
- Lay supine, legs up (sit if breathing is hard; never stand suddenly).
- Oxygen 10–15 L/min via non-rebreather mask.
- Stridor, voice change or tongue/lip swelling = impending airway obstruction → call anesthesia/ENT now for early, controlled intubation (see [Difficult Airway](#)).
- Fluid bolus for hypotension.
- Repeat epinephrine every 5–15 min if needed.<sup>9</sup>

refractory after  $\geq 2$  IM doses

**2ND–3RD LINE** Refractory after repeated IM epinephrine → urgent PICU/anesthesia support; IV epinephrine infusion only by an experienced team with monitoring · IV fluids · glucagon if on a beta-blocker.<sup>9</sup>



## MEDICATIONS – EACH GIVEN SEPARATELY

| Drug                                            | Dose                              | Route & notes                                                                                                                                                                                                                              |
|-------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Epinephrine (adrenaline) — 1st line             | 0.01 mg/kg (max 0.5 mg)           | IM, 1 mg/mL (1:1000), anterolateral thigh; repeat every 5–15 min <sup>9</sup>                                                                                                                                                              |
| Epinephrine autoinjector (limited availability) | 0.1 / 0.15 / 0.3 mg               | Some emergency departments dispense on senior-staff authorization; otherwise via the allergy clinic. 0.1 (7.5–14 kg) · 0.15 (15–25 kg) · 0.3 (>25 kg). If the 0.1 mg device is unavailable, the 0.15 mg device may be used down to 7.5 kg. |
| Epinephrine infusion (refractory)               | 0.1–1 mcg/kg/min (max 10 mcg/min) | IV infusion; titrate to effect under PICU supervision with continuous cardiac monitoring <sup>329</sup>                                                                                                                                    |
| Balanced crystalloid                            | 20 mL/kg                          | IV bolus for hypotension, repeat                                                                                                                                                                                                           |
| Salbutamol (for wheeze)                         | 2.5–5 mg nebulized                | Adjunct only — does not replace epinephrine                                                                                                                                                                                                |

| Drug                                | Dose                          | Route & notes                                                                                                             |
|-------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Diphenhydramine (for skin symptoms) | 1 mg/kg/dose (max 50 mg/dose) | IV/IM; adjunct only, not first-line. May use cetirizine (age-appropriate dose) as a non-sedating alternative <sup>9</sup> |
| Glucagon                            | 20–30 mcg/kg (max 1 mg)       | IV over 5 min, only if on a beta-blocker and refractory                                                                   |

**⚠ OBSERVE EVERY CHILD BEFORE ANY DISCHARGE DECISION – NO EARLY DISCHARGE FROM THE EMERGENCY DEPARTMENT**

- **Minimum 2 h** after full resolution — **only when every criterion is met:** good response to a single IM dose within 5–10 min, symptoms completely resolved, the child leaves with **two in-date autoinjectors** dispensed and demonstrated, and a trained adult supervising. **Observe longer (4–6 h) whenever possible**, and always when any criterion is unmet or the department cannot dispense autoinjectors.<sup>328,9</sup>
- **Minimum 6 h** if two IM doses were needed or there is a history of a biphasic reaction.<sup>328</sup>

- **Minimum 12 h (usually admit)** if more than two doses were needed, **history of severe asthma**, anaphylaxis with **severe respiratory compromise**, possible continuing allergen absorption, presentation at night/out of hours, or the family cannot recognize and respond to deterioration.<sup>328</sup>
- Routine corticosteroids are no longer recommended (consider glucocorticoids for persistent bronchospasm, or severe symptoms requiring more than 2 doses of IM epinephrine or IV epinephrine).<sup>9</sup>

#### ▲ **ADMIT · OBSERVE · DISCHARGE · REFER**

- **Admit** — more than two epinephrine doses or an epinephrine infusion; ongoing or biphasic symptoms; **moderate–severe respiratory distress** or hypotension at any point; airway involvement; history of severe asthma; any child in the 12-h observation tier.<sup>328,9</sup>
- **Observe (emergency department / short stay) then discharge** — resolution after one or two IM doses,

the observation tier above completed, no respiratory distress at review.

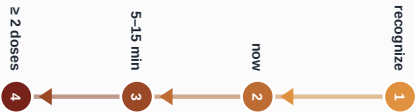
- **Discharge checklist (Kuwait pathway) — Home epinephrine:** where the emergency department stocks autoinjectors, a senior physician dispenses **two devices with hands-on training** before discharge; where it does not, refer to the allergy clinic with the supply pathway (imported device) documented and teach the family the warning signs and to call emergency services at the first sign. Written action plan; allergen-avoidance advice; non-sedating antihistamine for residual urticaria only; no routine steroid.<sup>9</sup>
- **Refer** — Allergy/Immunology for every case; **PICU** for refractory anaphylactic shock despite repeated IM epinephrine requiring an epinephrine infusion, or airway compromise requiring definitive airway management.

# Anaphylaxis – epinephrine first

Recognize (GA<sup>2</sup>LEN criteria) → IM epinephrine immediately →

fluids & repeat IM epinephrine (PICU)

**! IM EPINEPHRINE FIRST, ALWAYS**  
There is no contraindication — give it before antihistamine or steroid.



## RECOGNIZE — GA<sup>2</sup>LEN CRITERIA

- 1 No known allergen: skin/mucosa + respiratory or cardiovascular
  - 2 Likely allergen: any 2 of skin/mucosa · respiratory · cardiovascular · severe GI
  - 3 Known allergen: respiratory alone or cardiovascular alone
- If in doubt, give epinephrine.**

## IM EPINEPHRINE

0.01 mg/kg (max 0.5 mg) of 1:1000 in the anterolateral thigh. Lay supine, legs up; O<sub>2</sub> 10–15 L/min.

## FLUIDS · REPEAT

Fluid bolus for hypotension. Repeat IM epinephrine every 5–15 min as needed.

## REFRACTORY — PICU

Urgent PICU/anesthesia. IV epinephrine infusion by experts only; glucagon if on a beta-blocker.

Grade severity continuously: speech, work of breathing, SpO<sub>2</sub>, silent chest. Doses follow the Global Initiative for Asthma (GINA) 2026 strategy; PICU-level therapies are beyond GINA's scope and follow critical-care practice.

## ▀ GRADE SEVERITY FIRST

- Use a validated score — the **PRAM (Pediatric Respiratory Assessment Measure)**  (validated 2–17 yr) or **PASS (Pediatric Asthma Severity Score)** — with SpO<sub>2</sub> on room air, work of breathing, and ability to speak or feed; reassess and re-score after each step.<sup>77,78</sup>

## ✦ PEARL – WHEN IT ISN'T ASTHMA

- Doubt the diagnosis and look for a mimicker when wheeze is present from the **first weeks of life**, is **monophonic or focal**, fails to respond to adequate bronchodilator and steroid, or comes with faltering growth, clubbing, or chronic wet cough.
- Think: inhaled **foreign body**, tracheo-/bronchomalacia or a

**vascular ring, cystic fibrosis** or primary ciliary dyskinesia, bronchiolitis in the infant, a **cardiac** cause (heart failure), aspiration/gastroesophageal reflux disease (GERD), and — in adolescents — inducible laryngeal obstruction (vocal cord dysfunction).<sup>8</sup>

#### 1ST LINE

- **Oxygen** — start it when SpO<sub>2</sub> is **below 92%**. Then titrate: children **6 years and older** — keep SpO<sub>2</sub> **92 to 95%**; children **5 years and younger** — keep SpO<sub>2</sub> **at or above 92%**. Do not push saturations higher (avoid over-oxygenation).<sup>8</sup>
- **Inhaled salbutamol** by spacer every 20 min × 3 in the first hour, reassessing before each repeat, then titrate to response (dose by age in the table).<sup>8</sup>
- **Systemic corticosteroid** for a **moderate–severe** exacerbation or inadequate response to salbutamol (oral — prednisolone or dexamethasone — as effective as IV). **Not routine for mild**; avoid routine oral steroids for mild viral wheeze in preschoolers.<sup>8</sup>

**2ND LINE****MODERATE-SEVERE**

- Add **ipratropium** × 3.
- Severe or poor response → **IV magnesium sulfate**.
- **Call PICU.**



**life-threatening / not responding**

**3RD LINE • PICU**

**Not responding / life-threatening → call PICU urgently. Continuous nebulized salbutamol, IV salbutamol, non-invasive ventilation and intubation are PICU-directed (see "Beyond GINA" below).<sup>8</sup>**



## BRONCHODILATORS

| Drug                          | Dose                                                                                                                                    | Route & notes                                                                                                                                                                                                        |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Salbutamol (inhaler + spacer) | <p>≤ 5 yr: mild 2–4 · moderate 4 · severe 6 puffs</p> <p>≥ 6 yr: mild 4 · moderate 4–6 · severe 6–10 puffs</p>                          | 100 mcg/puff; <b>shake before each puff</b> , one puff at a time (4–5 breaths each); every 20 min × 3 in the first hour, then <b>titrate to response</b> (hourly → every 1–4 h). GINA sets no fixed cap <sup>8</sup> |
| Salbutamol (nebulized)        | <p>≤ 5 yr: 2.5 mg (any severity)</p> <p>≥ 6 yr: mild 2.5 mg · moderate 2.5–5 mg · severe 5 mg</p>                                       | Every 20 min × 3, then titrate to response; spacer is preferred when the child can use it <sup>8</sup>                                                                                                               |
| Ipratropium bromide           | <p>≤ 5 yr: 80 mcg by spacer (4 puffs of 20 mcg) or 250 mcg nebulized</p> <p>≥ 6 yr: 4–8 puffs (80–160 mcg) or 250–500 mcg nebulized</p> | Moderate–severe: give with each salbutamol dose, every 20 min × 3 in the first hour only <sup>8</sup>                                                                                                                |
| Magnesium sulfate             | 40–50 mg/kg (max 2 g)                                                                                                                   | IV over 20 min, age ≥ 2 yr                                                                                                                                                                                           |

## BEYOND GINA – CRITICAL-CARE THERAPIES (PICU-DIRECTED)

| Therapy                              | Dose                                                     | Notes                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Salbutamol — continuous nebulization | 0.5 mg/kg/h, maximum 20 mg/h (many units use 10–15 mg/h) | NAEPP dose. <sup>338</sup> GINA accepts continuous nebulized salbutamol for severe exacerbations in the emergency department; the mg/kg/h regimen and the high-output nebulizer are critical-care practice. Cardiac + K <sup>+</sup> monitoring. <sup>8</sup>                                        |
| Salbutamol IV                        | bolus then infusion                                      | Not routine (no GINA recommendation); PICU only, cardiac monitoring; watch for hypokalemia and lactic acidosis. <sup>8</sup>                                                                                                                                                                         |
| Aminophylline                        | 5 mg/kg load, then infusion                              | <b>Not recommended (GINA 2026)</b> — poor efficacy, narrow safety margin, potentially fatal toxicity (especially with prior oral theophylline); no reduction in symptoms or length of stay in children. <b>PICU rescue only</b> , after IV salbutamol and magnesium; monitor levels <sup>8,337</sup> |

| Therapy                        | Dose                    | Notes                                                                                                                                                                                                                 |
|--------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heliox (helium–oxygen ≥ 70:30) | driving / breathing gas | No proven benefit in routine exacerbations (Cochrane); may be considered in severe exacerbations in an ICU setting. Needs ≥ 70% helium, so unsuitable when a high FiO <sub>2</sub> is required <sup>175,176,177</sup> |

## CORTICOSTEROIDS – CHOOSE ONE; ORAL BEFORE IV

| Drug                      | Dose                                                                                                                                                                       | Route & notes                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prednisolone</b>       | <p>&lt; 2 yr: 1–2 mg/kg (max 20 mg)</p> <p>2–5 yr: 1–2 mg/kg (max 30 mg)</p> <p>6–11 yr: 1–2 mg/kg (max 40 mg)</p> <p>≥ 12 yr: 1–2 mg/kg (max 50 mg)</p> <p>once daily</p> | PO, 3–5 days; age-specific maxima per GINA <sup>8</sup>                                                                                  |
| <b>Dexamethasone</b>      | 0.3–0.6 mg/kg once daily (max 12–16 mg)                                                                                                                                    | PO/IV, 1–2 days; <b>equal to prednisolone</b> — no difference in relapse, better adherence, substantially less vomiting <sup>8,336</sup> |
| <b>Methylprednisolone</b> | 1–2 mg/kg/day IV in divided doses (max 60 mg/day)                                                                                                                          | IV agent when oral is not tolerated (NAEPP dose; confirm with clinical pharmacy) <sup>338</sup>                                          |

| Drug            | Dose                                | Route & notes                                                                |
|-----------------|-------------------------------------|------------------------------------------------------------------------------|
| Hydro-cortisone | 4 mg/kg every 6 h (max 100 mg/dose) | IV alternative when oral is not tolerated (equivalent to methylprednisolone) |

### ◆ STEROID CHOICE & AVAILABILITY

- Oral prednisolone and oral dexamethasone are equally effective and as effective as IV; reserve IV steroid for the child who cannot take oral.<sup>336</sup>
- When IV is needed, methylprednisolone and hydrocortisone are both acceptable — evidence does not clearly favor one.
- Note that hydrocortisone's mineralocorticoid effect can add to the hypokalemia caused by high-dose salbutamol, and all systemic steroids can raise blood pressure and glucose.
- All four are widely stocked.<sup>8</sup>

### ⚠ CLINICAL PEARLS & RED FLAGS

A **silent chest**, a falling respiratory rate with rising drowsiness, or a normal/rising PaCO<sub>2</sub> in a tiring child is **pre-arrest** — not improvement. **Call PICU urgently:** decisions on sedation (e.g., ketamine),

non-invasive ventilation, imaging, and intubation are made by the critical-care team, aiming to maximize urgent treatment before any intubation decision.<sup>8</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — ongoing oxygen requirement, incomplete response after first-hour therapy, salbutamol needed more often than every 3–4 h, or prior PICU admission/intubation.
- **Discharge** —  $\text{SpO}_2 \geq 92\%$  on room air, salbutamol stretched to  $\geq$  every 4 h, feeding and talking, inhaler and spacer technique confirmed; discharge prescription = salbutamol via spacer (2–6 puffs every 4 h, weaning over days), the oral steroid course, and **an inhaled corticosteroid controller started or stepped up** (GINA: an exacerbation is a failure of controller treatment — no child  $\geq 6$  years leaves on salbutamol alone; preschool wheeze by phenotype); written action plan; review within 1–2 weeks.<sup>8</sup>
- **Refer** — Pulmonology for any PICU-level escalation,  $\geq 2$  admissions per year, or diagnostic doubt; PICU for impending

respiratory failure;  
Allergy/Immunology for  
difficult-to-control atopic disease.

✦ **BEFORE DISCHARGE – SCREEN FOR WHAT  
KEEPS ASTHMA UNCONTROLLED (GINA 2026)**

An exacerbation is the moment to look for the drivers. Ask and act on: **allergic rhinitis / rhinosinusitis** (intranasal corticosteroid — improves asthma control); **obesity** (harder to control; weight management in the plan — even 5–10% loss helps); **obstructive sleep apnea** (habitual snoring, witnessed apnea → sleep referral; see [OSA](#)); **gastro-esophageal reflux** (treat only if symptomatic — anti-reflux therapy does not improve asthma without reflux symptoms); **food allergy** (confirmed food allergy is a risk factor for asthma death → allergy testing and an epinephrine plan); **anxiety or depression** in adolescents; **inducible laryngeal obstruction** when cough or “wheeze” does not respond to bronchodilator; and **exposures** — tobacco, vaping or shisha smoke at home or in the car (parents must never smoke or vape in rooms or cars the child uses), pets, mould, school triggers. Check **adherence and inhaler technique** last — they explain more

exacerbations than any comorbidity. Refer to pulmonology/allergy when asthma stays uncontrolled despite treating these.<sup>8</sup>

## Croup

HOURS

MINUTES if severe / impending failure

6 MO–6 YR

Barking cough, hoarse voice, inspiratory stridor; worse at night. Grade severity with the **Westley score** (0–17:  $\leq 2$  mild · 3–5 moderate · 6–11 severe ·  $\geq 12$  impending respiratory failure), summing level of consciousness, cyanosis, stridor, air entry, and retractions.<sup>18</sup>

+ Calculator: [Westley croup score](#)

KEEP CALM

Agitation worsens obstruction — examine on a parent's lap, minimal handling.



DECISION

Stridor at rest / respiratory distress?

## MILD

Single steroid dose; discharge with clear return-precaution advice (safety-netting).<sup>18</sup>

## MODERATE–SEVERE

Steroid + nebulized L-epinephrine; observe  $\geq 2$ –4 hours for rebound; oxygen if hypoxic.<sup>163</sup>

| Drug                    | Dose                                                                       | Route & notes                                                                                                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dexamethasone           | Mild: 0.15 mg/kg<br>Moderate–severe: 0.6 mg/kg<br>(max 16 mg)              | PO/IM/IV, single dose. 0.15 mg/kg may be as effective as 0.6 mg/kg in mild–moderate croup (Cochrane 2023; 0.6 scored better at 24 h). Dexamethasone halves return visits compared with prednisolone <sup>334</sup> |
| Budesonide (nebulized)  | 2 mg                                                                       | If oral steroid not tolerated                                                                                                                                                                                      |
| Epinephrine (nebulized) | L-epinephrine (adrenaline) 1:1000 (1 mg/mL) 0.5 mL/kg nebulized (max 5 mL) | Moderate–severe; observe for rebound <sup>163</sup>                                                                                                                                                                |



### ◆ CONCENTRATION CHECK

Do not confuse 1:1000 (1 mg/mL) with 1:10 000 (0.1 mg/mL) epinephrine — different concentrations.<sup>163</sup>

### ◆ HELIOX IN SEVERE CROUP

- In **SEVERE** croup with significant distress **NOT SETTLING** after dexamethasone and nebulized L-epinephrine, **HELIOX** (helium–oxygen) may temporize while the steroid takes effect or transfer/airway support is arranged — in small trials its effect was comparable to oxygen plus epinephrine.
- It is a **BRIDGE, NOT A SUBSTITUTE**, and the evidence is very limited (low certainty).
- Needs  $\geq 70\%$  helium, so it is unsuitable when a high oxygen requirement is present. **INVOLVE ICU/ANESTHESIA EARLY.**<sup>174</sup>

### ⚠ DIFFERENTIALS

- Drooling + toxic + no cough → epiglottitis (do not examine the throat).

- Sudden choking → inhaled foreign body.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — stridor at rest persisting after a steroid ± nebulized L-epinephrine, any oxygen requirement, or a repeat epinephrine dose needed — observe  $\geq 2-4$  h after epinephrine for rebound before any discharge.
- **Discharge** — mild croup (no stridor at rest, no distress) after a single steroid dose, feeding well, with reliable carers and safety-netting.
- **Refer** — **early ICU / anesthesia consult** for impending obstruction or failure to respond; ENT if an alternative airway diagnosis is suspected. If intubation is considered, do it in a controlled setting (PICU or operating room) with anesthesia and ENT present and difficult-airway equipment ready — see [Difficult Airway](#).

◆ SEE ALSO [Pediatric ENT Emergencies](#)

(epiglottitis, bacterial tracheitis) in Section 07.

## Retropharyngeal Abscess

HOURS

1–5 YR

Deep neck-space infection after recent upper-respiratory illness — fever, neck pain, refusal to move the neck, drooling; guard the airway.

### RECOGNIZE

- Peak age 1–5 yr (median ~3 yr), usually after upper-respiratory infection.
- Symptoms: fever, sore throat, odynophagia/dysphagia, neck pain and swelling, muffled (“hot-potato”) voice, trismus, and drooling.
- Signs: neck tenderness, limited neck extension or torticollis, and cervical lymphadenopathy. **Neck pain and refusal to move or extend the neck** are the most useful clues.<sup>164</sup>

### ⚠️ GUARD THE AIRWAY

- Stridor, respiratory distress, tripod posture, or pooling of secretions signal an airway threat (it mimics epiglottitis).
- Keep the child calm and sitting upright, give oxygen, do **not** examine the throat or lie the child flat, and involve ENT + anesthesia for a controlled airway before any manipulation. <sup>164</sup>

### ◆ IMAGING

- Lateral neck radiograph is a screening test when the diagnosis is equivocal and there is no airway obstruction (widened prevertebral soft tissue — > 7 mm at C2 or > 14 mm at C6). **Contrast-enhanced CT of the neck is the preferred, confirmatory test** — it separates a drainable abscess (rim-enhancing hypodensity) from retropharyngeal cellulitis/phlegmon and maps the extent. <sup>164</sup>
- No specific laboratory test is usually required.

#### DECISION

**Airway compromise, or a large / loculated abscess? (large = CT cross-sectional area  $> 2 \text{ cm}^2$  or diameter  $> 2.5 \text{ cm}$ ; rim enhancement is the strongest predictor of pus at surgery)<sup>165,376</sup>**

#### YES

Secure the airway with ENT / anesthesia, then **surgical incision & drainage**; send pus for culture.<sup>165</sup>

#### NO – cellulitis / small abscess

**A 48-hour trial of IV antibiotics first** — many children resolve without surgery. Re-image and drain if not improving.<sup>165</sup>

| Drug                                   | Dose                                                                                             | Route & notes                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ceftriaxone + clindamycin — first line | Ceftriaxone 50 mg/kg once daily (max 2 g); clindamycin 10 mg/kg/dose every 8 h (max 900 mg/dose) | IV; together cover group A streptococcus, oral anaerobes, Gram-negatives and MRSA (methicillin-resistant <i>Staphylococcus aureus</i> ) — Kuwait Pediatric Council antibiotic policy. <sup>326,164</sup> |

|                                       |                                                                     |                                                                                                                                                                                                                                                                       |
|---------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Amoxicillin-clavulanate — second line | 30 mg/kg/dose (amoxicillin component) every 8 h IV (max 1.2 g/dose) | IV alternative when the first line is unsuitable. No oral amoxicillin-clavulanate step-down at standard dose — it does not cover <i>S. pneumoniae</i> adequately; choose the oral step-down agent with infectious diseases/ENT per the Council policy. <sup>326</sup> |
|---------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### ADMIT · DISCHARGE · REFER

- **Admit** — all suspected retropharyngeal abscess for IV antibiotics and observation; a monitored bed if there is any airway concern.
- **Discharge** — not from the acute phase; step to an oral agent chosen with infectious diseases/ENT (per

the Council policy) once afebrile, improving and swallowing, with ENT follow-up.

- **Refer — ENT** for drainage and **anesthesia** for airway control; a screening lateral neck film only if it will not delay a needed CT or airway management.

## Difficult Airway MINUTES

Oxygenate first, then work the three zones — optimize (yellow) → intubate (green) → front-of-neck (red); change one thing with every attempt.

+ Calculator: ETT size & depth

### ▲ ANTICIPATE & PREPARE

- **Predictors:** craniofacial anomaly / syndrome (Pierre Robin, Treacher Collins, Goldenhar, Down), limited mouth opening or neck movement, deep-neck infection or airway swelling (retropharyngeal abscess, epiglottitis, angioedema, burns), stridor, a large tongue, or a previously documented difficult airway.<sup>166</sup>

- **Anticipated** — manage in a **controlled setting (PICU / OR)** with **ENT + anesthesia back-up available. Unanticipated** (ineffective bag-mask ventilation, or a failed first direct / video laryngoscopy (DL / VL) attempt) — **call for help now.** <sup>166,374</sup>
- **Muscle relaxants** at the discretion of anesthesia / PICU. **Keep spontaneous breathing** (minimal sedation or inhalational anesthesia) where muscle tone is holding the airway open — anterior mediastinal mass, epiglottitis, large tongue, deep-neck abscess, or known impossible mask ventilation. <sup>375,373</sup>
- **Prepare:** assign the team (1st + 2nd airway managers, assistant, respiratory therapist, drug nurse, circulating nurse); a **time-out** to agree **Plan A + B + C** (1st / 2nd intubator and technique, and the surgical-airway option, with personnel and equipment ready); check equipment and drugs; full monitoring; assess the airway (history + exam). <sup>373</sup>



### ⚠ **OXYGENATION IS THE GOAL – NOT THE TUBE**

- Pre-oxygenate fully and keep **apneic oxygenation** (nasal cannula / high-flow nasal cannula (HFNC)) running through every attempt.
- Change **one variable** with each attempt — operator, device, blade, position, or adjunct — rather than repeating the same one. **If in doubt, take it out** and oxygenate.<sup>374,375</sup>

### **THE THREE-ZONES APPROACH (KUWAIT ALGORITHM)**<sup>372</sup>

#### ✦ **FOR THE GENERAL PEDIATRICIAN**

Recognize the predictors, keep the child oxygenated (yellow zone) and **call anesthesia/ENT early**. The green and red zones below are the airway team's work — shown so you can anticipate and assist, not to lead.

● **YELLOW ZONE · OPTIMIZE** — make bag-mask ventilation effective: **position** (head tilt / chin lift, jaw thrust; **shoulder roll if < 2 yr**), suction, **100% FiO<sub>2</sub>**, change mask size / adjust the seal, **two-person bag-mask ventilation (BMV)**, insert a nasogastric tube (NGT). Add an **oropharyngeal / nasopharyngeal airway (OPA / NPA)**. Add a **supraglottic airway (laryngeal mask airway, LMA)** if ventilation is still ineffective. <sup>374,375</sup>

↓  
saturated & stable  
↓

## ● GREEN ZONE · INTUBATE —

intubate when **this is the best saturation you can get and the child is as stable as possible.**

**Videolaryngoscopy** with an **age-adapted standard blade** is the first choice (use a stylet with a hyperangulated blade);<sup>375,167</sup> **max 4 attempts total (DL + VL).** Between attempts change **operator** or **blade** (straight / curved / hyperangulated), add a **bougie** or a **flexible intubation scope (FIS)** — through a supraglottic airway, via the nasal route, or by a hybrid technique (FIS guided by the videolaryngoscope view) — or try a **smaller endotracheal tube (ETT).** Keep **apneic O<sub>2</sub>** (NC / HFNC) running and **treat laryngospasm.** If **oxygenation falls or the patient deteriorates → go back to the yellow zone** and re-optimize.<sup>374,373</sup>

can't intubate, can't ventilate



● **RED ZONE · CAN'T INTUBATE, CAN'T VENTILATE (CICV)** — declare it early. **ENT available:** rigid bronchoscopy ventilation or tracheostomy. **ENT not available:** **needle cricothyrotomy / cricotracheal puncture (any age)**; percutaneous or “slash” surgical cricothyrotomy in the older child by trained hands. **Call for help if it has not arrived.** <sup>374,373</sup>

✦ **CONFIRM PLACEMENT**

On successful intubation, confirm the tube **primarily with continuous waveform capnography (EtCO<sub>2</sub>)**; support with equal chest rise, rising SpO<sub>2</sub>, bilateral air entry on auscultation, misting of the tube, and a chest X-ray with the tip in the mid-trachea (about 1–2 cm above the carina in infants, 2–3 cm in older children).<sup>1</sup>

▀ **HAVE READY (DIFFICULT-AIRWAY TROLLEY)**

Videolaryngoscope (standard + hyperangulated blades), a range of tubes, bougie and stylets, supraglottic airways (LMA) in graded sizes, a **flexible intubation scope (FIS)**, and a **front-of-neck-access / cricothyrotomy**

**kit** — checked and immediately available before induction.<sup>373</sup>

## Bronchiolitis

HOURS

< 2 YR

Viral (e.g., RSV): coryza → cough, fast breathing, wheeze/crackles, feeding difficulty.

### ▲ SUPPORTIVE CARE IS THE TREATMENT

- Superficial nasal suction when secretions impair breathing or feeding (avoid routine deep suction); small frequent feeds; NG or IV fluids if intake under 50–75% or dehydrated.<sup>333</sup>
- Give oxygen for persistent  $\text{SpO}_2 < 90\%$  (consider  $< 92\%$  in young infants, prematurity, cardiac/lung disease, or severe work of breathing). **Mild** — no oxygen unless  $\text{SpO}_2 < 90\%$ . **Moderate** — low-flow oxygen by nasal cannula first; **high-flow nasal cannula only if failing low-flow** (ward or PICU per local pathway) — in the PARIS trials early high-flow cut escalation in hypoxemic infants but did not shorten oxygen or hospital stay, and in 1–4-year-olds it lengthened stay

and doubled ICU admission. **Severe / impending failure** — high-flow or CPAP in PICU. <sup>17,333,379,380</sup>

✦ **PEARL – FIRST WHEEZE VS THE MIMICKERS**

A **first** episode of wheeze with coryza in a child under 12 months in the bronchiolitis season is bronchiolitis.

**Recurrent or episodic** wheeze, a strong atopic or family history, or wheeze that responds well to a bronchodilator points instead to viral-induced wheeze or pre-school asthma; **persistent focal signs** raise a retained foreign body or a structural airway lesion; and an infant who is failing to feed with hepatomegaly or a murmur may have **heart failure**, not bronchiolitis. <sup>17</sup>

✗ **NOT ROUTINELY RECOMMENDED**

Bronchodilators (salbutamol, epinephrine), corticosteroids, nebulized hypertonic saline, antibiotics, chest physiotherapy, routine chest X-ray or viral testing. <sup>17,333</sup> **Red flags:** apnea, grunting, severe recession, SpO<sub>2</sub> < 90%, dehydration, age < 6–12 weeks, comorbidity.

## ▲ ADMIT · DISCHARGE · REFER

- **Admit** — SpO<sub>2</sub> persistently < 90% (< 92% in high-risk infants), apnea, intake under 50–75% or dehydration, marked work of breathing, — especially if high-risk (age < 8–12 wk, prematurity, or cardiopulmonary disease).
- **Discharge** (NICE NG9, 2021) — clinically stable; taking adequate oral fluids; SpO<sub>2</sub> in room air maintained **for 4 hours including a period of sleep: > 90% if ≥ 6 weeks old, > 92% if < 6 weeks or any underlying health condition**; work of breathing settling; carers confident, with written safety-netting (worsening work of breathing, intake 50–75% of normal or no wet nappy for 12 h, apnea or cyanosis, exhaustion) and a clear route back.<sup>405,333,381</sup>
- **Refer** — PICU (for escalation to high-flow nasal cannula or continuous positive airway pressure (CPAP)) for rising work of breathing, recurrent apnea, or hypoxia despite oxygen.

◆ **SEE ALSO** [Community-Acquired Pneumonia](#), [Complicated Pneumonia](#) /

Empyema, Pertussis and BRUE in Section 04.

## Choking (Foreign-Body Obstruction)

### MINUTES

Sudden cough/stridor/one-sided wheeze with a choking history.

### DECISION

**Mild or severe obstruction?**

#### MILD – effective cough

Coughing effectively, can speak/cry, adequate air movement → **encourage coughing, stay, monitor closely**; no back blows or thrusts.<sup>1</sup>

#### SEVERE – responsive

Weak or silent cough, cannot speak or cry, cyanosis or color change, altered consciousness, apnea → start maneuvers now. **Infant < 1 yr**: 5 back blows → 5 chest thrusts, repeat (no abdominal thrusts). **Child ≥ 1 yr**: 5 back blows → 5 abdominal thrusts, repeat (ILCOR/ERC sequence; AHA teaches abdominal thrusts).<sup>1</sup>



becomes unconscious

Start CPR. Check the mouth before breaths; remove the object only if visible. No blind finger sweeps.<sup>1</sup>

**⚠ REMOVE NOW**

- Foreign body in the larynx or trachea (central airway), stridor at rest, severe distress or cyanosis, or complete obstruction.
- **Your job as the first doctor:** call anesthesia/ENT/PICU at once, oxygenate, continue the maneuvers above, and bring the front-of-neck-access kit to the bedside.<sup>1</sup>

**⚠ AIRWAY TEAM – COMPLETE OBSTRUCTION NOT RELIEVED BY MANEUVERS**

- **Recognize:** cannot speak, cry or cough; clutching the neck; silent chest; paradoxical chest/abdominal movement; loss of consciousness. Act after two cycles of maneuvers fail or as soon as the child becomes unresponsive.<sup>383,1</sup>

- **1 · Laryngoscopy** — remove a visible object with Magill forceps (experienced clinician).<sup>383,382</sup>
- **2 · Not removable and rigid bronchoscopy available** → operating room now: removal by ENT/thoracic surgery with anesthesia.<sup>382</sup>
- **3 · Unstable, no immediate rigid bronchoscopy — act by level. At or above the cords:** never pass a tube blindly past the object; intubate only under direct or video view around/beyond it; if that fails or obstruction becomes complete → **front-of-neck access below the larynx as the last resort — follow the red zone of the [Difficult Airway](#) card** (ENT: tracheostomy). **Below the cords:** intubate and **advance the object into the right main bronchus**, withdraw the tube above the carina, right side down, ventilate the left lung; if intubation is impossible, positive-pressure ventilation to push the object into one bronchus is the last resort.<sup>382,1</sup>
- **Anesthetic principle:** prefer spontaneous breathing and avoid positive pressure while the obstruction is **partial** (it can complete it); once obstruction is

complete, distal displacement is the rescue. Spontaneous and controlled techniques have similar outcomes in the hands of the airway team.<sup>384,375</sup>

#### ▶ **STABLE CHILD – SUSPECTED INHALED FOREIGN BODY**

- **A convincing choking history mandates bronchoscopy even when the chest X-ray is normal** — only about 11% of inhaled objects are radio-opaque and the film is normal in about 17% of children<sup>384</sup>.
- **Chest X-ray:** standard film ± inspiratory and expiratory views (cooperative child) or lateral decubitus views (young child). Look for a radio-opaque object, unilateral hyperinflation/air trapping, segmental or lobar collapse, mediastinal shift, consolidation or pneumothorax; abscess or bronchiectasis suggest a chronic retained object. **A normal examination or X-ray does not exclude an inhaled foreign body;** CT is not a substitute for bronchoscopy.<sup>382</sup>

- **Removal is by rigid bronchoscopy, by ENT or thoracic surgery** with an experienced anesthesia team under optimal conditions, ideally **within 24 hours** (delay increases complications). **Flexible bronchoscopy is for diagnosis only** — the flexible scopes in our hospitals are not retrieval instruments.<sup>382,384</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — after removal, observe for airway injury/aspiration; any hypoxia, stridor, or incomplete relief → ED/PICU.
- **Discharge** — only when a **retained foreign body has been excluded** (by bronchoscopy, or by ENT/respiratory review with follow-up arranged); a child can be fully recovered with an effective cough, normal examination and normal SpO<sub>2</sub> and still have a retained object<sup>382</sup>.
- **Refer** — ENT/anesthesia for suspected retained foreign body → rigid bronchoscopy.

# Obstructive Sleep Apnea

ASSESS

AAP • ERS

Flagged here for its acute peri-operative airway risk (post-adenotonsillectomy compromise, opioid/sedation sensitivity). Habitual snoring and witnessed apnea; in children it looks like hyperactivity, not sleepiness.

## RECOGNIZE

- **Obstructive sleep apnea (OSA)** — habitual snoring (> 3 nights/week), witnessed apnea or gasping, restless sleep, mouth breathing, enuresis; by day **hyperactivity, inattention and poor concentration rather than sleepiness**. Peak 2–8 years (adenotonsillar hypertrophy).<sup>123</sup>
- **High-risk groups:** obesity, Down syndrome, craniofacial anomalies, neuromuscular disease, Prader-Willi, achondroplasia, mucopolysaccharidoses, sickle cell disease.<sup>123,124</sup>
- **Work-up is elective:** polysomnography (PSG; the apnea-hypopnea index, AHI, grades severity) remains the reference standard, or sleep-clinic referral<sup>408</sup>.

Where PSG is not available, BTS accepts overnight pulse oximetry or a cardiorespiratory sleep study as a first-line test — a normal oximetry does not exclude mild–moderate OSA<sup>,407</sup>; the Canadian guideline limits home testing to otherwise healthy children over 5 years and advises against diagnosing OSA on oximetry or questionnaires alone<sup>,408</sup>. ECG/echocardiography if cor pulmonale or pulmonary hypertension is suspected. Chronic treatment (adenotonsillectomy, CPAP, intranasal steroid, weight management) is decided in clinic, not in the emergency setting; the ERS statement supports treatment when  $AHI > 5/h$ , or  $1-5/h$  with morbidity or a complex condition.<sup>,123,124</sup>

#### ▲ RED FLAGS

- **Severe OSA** —  $AHI \geq 10$  obstructive events/h and/or nadir  $SpO_2 < 80\%$  (the AAO-HNS definition used for peri-operative risk)<sup>406</sup>; nocturnal desaturations or hypoventilation.
- **Cor pulmonale / pulmonary hypertension**, failure to thrive — complex, high-risk: refer to a specialist.

- **Cardiorespiratory failure cannot await elective evaluation** — treat urgently.<sup>123</sup>

### **⚠ PERI-OPERATIVE & ACUTE CARE – THE BEDSIDE PRIORITY**

- **Any post-adenotonsillectomy respiratory compromise** → anesthesia review at once; an ongoing oxygen requirement, obstruction or apnea after recovery → monitored bed or PICU.
- **Overnight inpatient monitoring** after tonsillectomy for children **< 3 years** or with **severe OSA** (AAO-HNS); AAP adds obesity, craniofacial/neuromuscular disease, Down syndrome, cor pulmonale, recent upper-respiratory infection and failure to thrive as reasons for a monitored bed.<sup>406,123</sup> **Opioids: no codeine after tonsillectomy under 12 years** (AAO-HNS; FDA — CYP2D6 ultra-rapid metabolizer deaths) and avoid tramadol; ibuprofen and/or acetaminophen are the first-line analgesics.<sup>406,125</sup>
- In moderate-to-severe OSA, minimize opioids and monitor

closely — children with recurrent nocturnal hypoxemia (nadir SpO<sub>2</sub> < 85%) needed **about half the usual morphine dose** for the same analgesia in a randomized study, so start low and titrate with monitoring.<sup>410,123</sup>

- Position ramped, keep airway adjuncts and CPAP ready.

#### ✦ PEARL – IT LOOKS LIKE ADHD

A snoring child who is hyperactive and inattentive may have OSA, not just a behavior problem — ask about snoring and witnessed apnea at every visit.

#### ▀ ADMIT · DISCHARGE · REFER

- **Emergency department — admit** for cardiorespiratory compromise, cor pulmonale or severe hypoventilation; **otherwise discharge home** with an urgent outpatient sleep study and ENT referral (severe OSA alone is an urgent outpatient, not an admission).<sup>123</sup>
- **After adenotonsillectomy — overnight monitored bed** if < 3 years, severe OSA (AHI ≥ 10 and/or



nadir < 80%) or a high-risk comorbidity; otherwise discharge per anesthesia/ENT.<sup>406,123</sup>

- **Refer** — ENT (adenotonsillectomy) and sleep medicine; cardiology if pulmonary hypertension. OSA persists after adenotonsillectomy in up to 40% — re-refer to sleep medicine/ENT if snoring or symptoms continue (options include CPAP, weight management, sleep endoscopy-guided surgery and rapid maxillary expansion)<sup>409</sup>.

## Cow's Milk Protein Allergy

ASSESS

INFANT • ESPGHAN 2024 • WAO DRACMA 2023–24 • CPS 2

The commonest infant food allergy — IgE-mediated (fast) or non-IgE (delayed, cell-mediated).

Over-diagnosis is common: confirmed CMPA affects < 1% of infants.

✚ See also: Acute FPIES (vomiting 1–4 h after a food — criteria and emergency treatment)

## ▶ RECOGNIZE THE TWO PATTERNS

- **IgE-mediated (minutes–2 h):** urticaria, angioedema, vomiting, wheeze, anaphylaxis.<sup>27</sup> **Testing:** skin-prick test or specific IgE to cow's milk when the history suggests an immediate reaction — a positive result shows sensitisation only; a **supervised oral food challenge** confirms the diagnosis, except after anaphylaxis with a high specific IgE.<sup>365</sup>
- **Non-IgE (hours–days):** blood or mucus in the stool, eczema flare, faltering growth, chronic diarrhoea. **Reflux, colic, loose stools and constipation are common in healthy infants and on their own do not indicate CMPA** — consider it only when they fail usual treatment or occur with other features (another organ system, faltering growth, blood in the stool).<sup>365</sup> Distinct from lactose intolerance (an enzymatic, non-immune condition).<sup>193</sup>
- **Confirming non-IgE disease: a 2–4-week diagnostic elimination** (maternal cow's-milk-free diet if breastfed; extensively hydrolyzed formula if formula-fed) **followed by planned reintroduction** —

improvement alone is not enough, because up to 20% of breastfed infants with blood in the stool settle with no change at all.<sup>365</sup> FPIES is the exception: a typical history with resolution on withdrawal is sufficient; a supervised challenge only if the history is unclear.<sup>365</sup>

### ⚠ **WHEN TO START AN ELIMINATION – OBSERVE FIRST**

- A **well, thriving, exclusively breastfed** infant with small streaks of blood in the stool and no other atopic features: **observe for up to a month** without changing the mother's diet — most settle spontaneously.<sup>365</sup>
- **Start a 2–4-week diagnostic elimination** only for: severe or persistent bleeding; faltering growth; moderate–severe eczema not responding to treatment; or a strong overall picture of CMPA.<sup>365</sup>
- Unneeded maternal elimination shortens breastfeeding, and therapeutic formulas taste different and may shape later food preferences — both are reasons to hold the threshold.<sup>365</sup>

## ▲ NON-IGE SUBTYPES – RECOGNIZE THE PATTERN

- **Food protein-induced allergic proctocolitis (FPIAP)** — a well-appearing infant with blood/mucus streaks in the stool, onset **2–8 weeks**; the most common subtype, up to 50% in exclusively breastfed infants; mostly benign and self-limiting. <sup>193,365</sup>
- **Food protein-induced enteropathy (FPE)** — chronic diarrhoea **with** faltering growth and malabsorption (steatorrhoea, hypoalbuminaemia, anaemia); emesis and distension may accompany. A thriving infant with loose stools alone does not have FPE. Uncommon. If the picture fits and CMPA is likely, start the elimination and expect improvement within 1–4 weeks; refer to pediatric GI — endoscopy with biopsy (villous atrophy, crypt hyperplasia) confirms the diagnosis and is prioritised when response is poor, features are severe or another cause is suspected. Exclude infection with stool studies and cystic fibrosis with a sweat test, not biopsy; coeliac testing applies only once gluten is in the diet. <sup>367,193,369</sup>

- **Food protein-induced enterocolitis syndrome (FPIES)** — profuse **repetitive vomiting 1–4 h after the food**, pallor, lethargy ± diarrhoea; ~15% develop hypovolemic shock. Rare; mimics sepsis. Criteria and acute treatment: [Acute FPIES](#).<sup>363</sup>

**MANAGEMENT – BY FEEDING & SUBTYPE (EVERY ELIMINATION IS A 2-4-WEEK DIAGNOSTIC TRIAL, THEN REINTRODUCE TO CONFIRM)**

| Situation                                | First choice                         | Notes                                                                                                                                                                              |
|------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IgE-mediated — anaphylaxis               | Amino acid formula (AAF)             | AAF from the start after an anaphylactic reaction to cow's milk protein. <sup>289</sup>                                                                                            |
| IgE-mediated — milder immediate reaction | Extensively hydrolyzed formula (EHF) | Step up to AAF only per the rule in the Avoid box below. <sup>365</sup>                                                                                                            |
| IgE-mediated, breastfed                  | Continue breastfeeding               | If the infant is symptom-free on breast milk, <b>no maternal elimination</b> ; eliminate maternal cow's milk only if the infant reacts while exclusively breastfed. <sup>366</sup> |

| Situation                        | First choice                                             | Notes                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mixed-fed (breast + formula)     | Change the formula; keep breastfeeding                   | If symptoms occur only with formula and the infant is well on breast milk, use EHF (or AAF per the rows above) and <b>do not restrict the mother's diet</b> — IgE-CMPA and FPIES. <sup>363,366</sup> For FPIAP/FPE the same approach is reasonable; add maternal elimination only if symptoms persist on breast milk alone ( <i>expert consensus — pending Allergy/GI confirmation</i> ). |
| FPIAP, breastfed — threshold met | Maternal cow's-milk-free diet, 2–4 wks, then reintroduce | Cow's milk (and other mammalian milks) first; <b>add soy only if no response</b> after 2–4 weeks (CPS eliminates both from the start; ESPGHAN cow's milk only). <sup>365,193</sup> Continue breastfeeding; dietitian for calcium and vitamin D.                                                                                                                                           |

| Situation                          | First choice                                                    | Notes                                                                                                                                                                                                                   |
|------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FPIAP, formula-fed — threshold met | Extensively hydrolyzed formula (EHF), 2–4 wks, then reintroduce | Whether to eliminate at all in formula-fed FPIAP is debated (a cohort linked it to later IgE-CMPA); if started, reintroduce afterwards. <sup>365</sup> Soy formula is an option > 6 months if tolerated. <sup>193</sup> |
| FPE, breastfed                     | Maternal cow's-milk-free diet, 2–4 wks                          | Add soy if no response; monitor growth and nutrition — resolution takes weeks. No improvement → AAF and pediatric GI. <sup>193,367</sup>                                                                                |
| FPE, formula-fed                   | Amino acid formula (AAF)                                        | First-line (not EHF) given the malabsorption; pediatric GI. <sup>193</sup>                                                                                                                                              |
| FPIES, breastfed                   | Continue breastfeeding                                          | Rare in this group. If reacting to breast milk or faltering: maternal elimination of the trigger with allergy input; if unresolved, <b>consider</b> stopping breastfeeding → EHF/AAF. <sup>363</sup>                    |

| Situation          | First choice                              | Notes                                                                                                                                           |
|--------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| FPIES, formula-fed | EHF first; AAF if no improvement in 2 wks | 10–20% need AAF from the start; AAF preferred if faltering. Strictly avoid <b>all forms</b> (incl. heated/baked) of the trigger. <sup>363</sup> |

### ⚠️ AVOID & WHO DECIDES

- **Goat's / sheep's milk** (high cross-reactivity) and partially hydrolyzed / "comfort" or "HA" formulas.<sup>365</sup>
- **Step up EHF → AAF** only after 2–4 weeks without response **and** after checking the diet for hidden milk and reconsidering the diagnosis; AAF is first-line only for anaphylaxis, FPIES with faltering growth, or FPE.<sup>365,193</sup>
- **Who decides:** the treating paediatrician may start EHF or a maternal trial; the step to AAF and any prolonged maternal elimination are confirmed with allergy/GI, with a dietitian protecting calcium and growth. *Referral thresholds below are provisional pending Allergy/GI wording.*



### ✦ RE-CHALLENGE

After a confirmed diagnosis the first therapeutic elimination lasts **6 months or until 12 months of age, whichever comes first**, then an oral food challenge — supervised for IgE-mediated disease, at home via a milk ladder for most non-IgE disease; the setting is decided by the specialist.<sup>365,27</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Emergency department — admit:** IgE-mediated reaction with anaphylaxis (manage, observe and admit per [Anaphylaxis](#)); acute FPIES with lethargy, dehydration or shock (see [Acute FPIES](#)).
- **Emergency department — home after assessment:** the well infant with blood-streaked stool, mild immediate symptoms that have resolved, or chronic symptoms — with the observe-first advice above, written warning signs, and clinic follow-up within 2–4 weeks; start a diagnostic elimination only if the threshold is met.
- **Clinic / ward:** discharge home once feeding and growing, with the dietary plan, dietitian review and the reintroduction date written down.

- **Refer** (*provisional*) — allergy: anaphylaxis (urgent), IgE-mediated disease for testing and challenge, FPIES, multiple food allergies; pediatric GI: FPE, faltering growth, no response to EHF, diagnostic doubt; dietitian for every infant on an elimination diet.

◆ **BASIS** ESPGHAN 2024 position paper<sup>365</sup>; WAO DRACMA VII (2023, elimination and reintroduction)<sup>27</sup>, X (2023, breastfeeding)<sup>366</sup> and XVI (2024, nutritional management)<sup>289</sup>; Canadian Paediatric Society 2024 (non-IgE subtypes)<sup>193</sup>; international FPIES consensus 2017<sup>363</sup>. Where CPS and ESPGHAN differ (maternal soy; observation before elimination) the card follows ESPGHAN and says so.

# Acute FPIES (Food Protein-Induced Enterocolitis Syndrome)

**HOURS**

INFANT • FPIES CONSENSUS 2017

Repetitive vomiting 1–4 h after a food, with pallor and lethargy — a non-IgE reaction that mimics sepsis; fluids and ondansetron, not epinephrine.

**+ Calculator: Maintenance fluids & bolus**

## ▶ RECOGNIZE – 2017 INTERNATIONAL CRITERIA

- **Major criterion (required):** vomiting 1–4 h after ingestion of the suspect food **and** no classic IgE-mediated skin or respiratory symptoms.<sup>363</sup>
- **Plus ≥ 3 minor criteria:** a second episode after the same food · repetitive vomiting 1–4 h after a different food · extreme lethargy · marked pallor · emergency-department visit · need for IV fluids · diarrhoea within 24 h (usually 5–10 h) · hypotension · hypothermia.<sup>363</sup>
- **Triggers:** cow's milk and soy in infants; rice and oat are the commonest solids; also egg, fish,

poultry, vegetables. Onset is usually with the first exposures to formula or solids, typically at **2–7 months**.<sup>363,368</sup>

- **Looks like:** sepsis, surgical abdomen, gastroenteritis, metabolic crisis. Clues: the timing after a new food, a well child an hour earlier, rapid recovery with fluids. Blood tests may show neutrophilia, thrombocytosis, acidosis and methaemoglobinaemia.<sup>363,368</sup>

## ACUTE MANAGEMENT BY SEVERITY (FPIES CONSENSUS 2017)

| Severity | Features                       | Treatment                                                                                                                                                                 |
|----------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mild     | 1–2 vomits, no lethargy        | Oral rehydration — breastfeeding or clear fluids; if $\geq 6$ months, ondansetron IM 0.15 mg/kg (max 16 mg) may be considered. Can resolve at home. <sup>363</sup>        |
| Moderate | $\geq 3$ vomits, mild lethargy | <b>Ondansetron</b> IM/IV 0.15 mg/kg (max 16 mg); consider IV access and <b>0.9% saline 20 mL/kg</b> , repeated as needed; observe until tolerating fluids. <sup>363</sup> |

| Severity | Features                                                                              | Treatment                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Severe   | ≥ 3 vomits with severe lethargy, hypotonia, ashen or cyanotic appearance, hypotension | IV access; <b>0.9% saline 20 mL/kg rapidly, repeat as needed; ondansetron IV 0.15 mg/kg</b> (max 16 mg); consider <b>methylprednisolone IV 1 mg/kg</b> (max 60–80 mg); blood gas, electrolytes, glucose, CBC; correct acid–base and electrolyte abnormalities and methaemoglobinaemia; manage as shock (see <a href="#">Shock</a> ). <sup>363</sup> |

### ⚠ PITFALLS

- **Epinephrine does not treat FPIES** (not IgE-mediated); give it only if anaphylaxis cannot be excluded — see [Anaphylaxis](#).
- Do not label a sepsis-like infant as FPIES until the timing fits and infection has been considered; when in doubt treat for both.
- Ondansetron ≥ 6 months only; monitor the QT interval if repeated ([QTc](#)).

## ▲ ADMIT · DISCHARGE · REFER

- **Emergency department — home:** mild reaction, tolerating oral fluids, well after observation; written plan to avoid the trigger in **all forms** and to return for repetitive vomiting after any new food.
- **Emergency department — observe:** moderate reaction until drinking and lethargy has resolved, then home as above.
- **Admit:** severe reaction, hypotension, ongoing vomiting or lethargy, or dehydration.
- **Refer** — allergy for every new diagnosis (trigger confirmation, written emergency plan, supervised re-challenge usually 12–18 months after the last reaction); dietitian when a staple food is removed. Chronic feeding choices: see [Cow's Milk Protein Allergy](#).<sup>363,368</sup>

◆ **BASIS** International consensus guidelines for FPIES (Nowak-Węgrzyn 2017)<sup>363</sup>; ESPGHAN 2024 for the diagnostic approach.<sup>365</sup>

# Antibiotic Allergy Assessment & Pretesting

## ASSESS

Most allergy labels are false. Take a history, don't routinely skin-test, and de-label when you can.

### THE LABEL IS USUALLY WRONG

- About **10% of people** carry a penicillin/beta-lactam allergy label, but **~90% are not truly allergic**: only **5–10%** of children with a “penicillin allergy label” have a true penicillin allergy.<sup>107</sup>
- Most labels accumulate from **non-allergic events** — GI upset, headache, family history, or a viral rash during treatment.
- In children, a **maculopapular rash is more often viral than allergic**.<sup>107</sup>
- Routine pre-dose intradermal **"test dosing" is not standard of care, not validated, and should be abandoned** — it delays therapy and misleads (positive predictive value of skin testing only ~50%; it misses dose-independent and enteral reactions).<sup>106</sup>

### ◆ WHY MISLABELING HARMS

A beta-lactam label drives **second-line agents** (vancomycin, clindamycin, fluoroquinolones) that are less effective, broader, and costlier — linked to more *C. difficile* (~+23%), **MRSA** (~+13%), **vancomycin-resistant enterococci (VRE)** (~+30%), longer stays, more surgical-site infections, ICU admission, aminoglycoside-related acute kidney injury, and a ~50-min delay to the first ED antibiotic dose. **No reaction history — or a beta-lactam tolerated since — give the drug. No skin test required.**<sup>106</sup>

## ▲ RISK-STRATIFY THE HISTORY – SEVERITY, TIMING, DRUG CLASS, AND ANY BETA-LACTAM TOLERATED SINCE

### ◆ READING THE TIERS

- Assign the **highest** tier any feature reaches — one high-risk feature makes the patient high-risk.
- Risk rises with reaction **severity** (intolerance → hives → anaphylaxis / severe delayed), **immediacy** (delayed benign → immediate IgE), and **recency** (remote → recent). **None / Low** → give the drug or a direct oral



challenge; **Medium** → skin test ± challenge, pick a non-cross-reacting beta-lactam meanwhile; **High** → avoid the culprit, refer, do not challenge.

- The pediatric category method is preferred; adult scores (e.g., the PEN-FAST penicillin-allergy score) are not robustly validated in children. **The table below applies to a child with a reported penicillin allergy (cephalosporin allergy excluded).** <sup>107</sup>

| Risk                                | History (any of)                                                                                                                                                                 | Action                                                                                                                                 |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| History not consistent with allergy | No previous allergic reaction; side effects / intolerance only (GI upset, headache); a reaction history but tolerated the antibiotic since the index event; family history only. | <b>De-label</b> the record. Give penicillin if indicated.                                                                              |
| Low                                 | Non-IgE-mediated reaction; delayed maculopapular rash starting $\geq$ 3 days into the course.                                                                                    | <b>De-label</b> the record. Give penicillin if indicated.                                                                              |
| Medium                              | Urticaria (hives) or other immediate, IgE-type reaction <b>without anaphylaxis</b> ; a "rash" where hives cannot be excluded.                                                    | Refer to an allergist. Avoid penicillin. See <a href="#">the side-chain chart below</a> for prescribing cephalosporins. <sup>107</sup> |

| Risk | History (any of)                                                                                                                                                                                                                                                                                                                                | Action                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| High | <b>Anaphylaxis;</b><br>recurrent reactions or reactions to <b>multiple</b> beta-lactams; prior positive skin test; any <b>severe delayed</b> reaction — Stevens-Johnson syndrome / toxic epidermal necrolysis (SJS/TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), serum sickness, hemolytic anemia, nephritis, hepatitis. | <b>Do not test or challenge. Refer to an allergist.</b> Avoid penicillins and cephalosporins. <sup>107</sup> |

### **△ WHEN TO INVOLVE AN ALLERGIST**

Any **type-1 (immediate / IgE-mediated)** reaction — urticaria, angioedema, wheeze, or anaphylaxis — prompts **allergist referral**; more nuanced than the

table alone, which is a triage aid, not a substitute for specialist assessment.<sup>106</sup>

## BETA-LACTAM CROSS-ALLERGY – BY SIDE-CHAIN STRUCTURE

### ◆ HOW TO USE THIS CHART

- **Prescribing a cephalosporin for a penicillin-allergic child:** choose one that does **not** share a similar side-chain structure with penicillins (a "✓" against the penicillin columns below).
- Cross-reactivity follows **side-chain similarity** — not the shared beta-lactam ring.  
**Cefazolin has a unique side chain and is safe even after penicillin anaphylaxis.**<sup>106</sup> Use only for **immediate / IgE** histories — **never** for SJS/TEN/DRESS, where all beta-lactams are avoided.<sup>106</sup>

### ▶ CHILDREN WITH A CEPHALOSPORIN ALLERGY

- **Immediate hypersensitivity** (urticaria, angioedema, anaphylaxis) — **avoid all cephalosporins and penicillins**; use an alternative and refer to allergy.

- **Inpatient / ICU** — avoid all cephalosporins and penicillins and use alternatives; consult **Allergy / ID** if a penicillin or cephalosporin is the recommended first-line treatment.
- **Severe cutaneous reaction (SJS/TEN/DRESS)** — **avoid all penicillins, cephalosporins, and carbapenems**; use an alternative and consult allergy or ID.<sup>107</sup>

◆ **BASIS** Kuwait Pediatric Council practice point on skin testing before antibiotic administration; risk table adapted from the Canadian Paediatric Society<sup>107</sup> and Devchand & Trubiano<sup>370</sup>; pathway aligned with the CHOP Penicillin Allergy Clinical Pathway<sup>371</sup> and the AAAAI/ACAAI 2022 drug-allergy practice parameter.<sup>106</sup> Apply to the local antibiogram; verify doses before use.

⚠ **NEVER TEST THESE**

Anaphylaxis, SJS/TEN, DRESS, acute generalized exanthematous pustulosis (AGEP), serum sickness, cytopenias, interstitial nephritis, or hepatitis — no skin test, no challenge; refer to an allergist.<sup>106</sup>

# Hereditary Angioedema – Acute Attack

## MINUTES

Bradykinin-mediated swelling that does **not** respond to epinephrine, antihistamine, or steroid.

### RECOGNIZE

- Recurrent angioedema **without urticaria or itch**; laryngeal swelling, abdominal attacks; family history; known C1-esterase inhibitor deficiency.<sup>269</sup>
- If anaphylaxis cannot be excluded, give intramuscular epinephrine first (see [Anaphylaxis](#)) — failure to respond supports the diagnosis.<sup>269</sup>

Airway is the danger — early anesthetics/ENT for any laryngeal involvement<sup>269</sup>



## SPECIFIC THERAPY

Give a hereditary angioedema (HAE)-specific agent early; earlier treatment means a milder, shorter attack. Every child with HAE should have on-demand therapy available at all times.<sup>269,335</sup>

| Drug                              | Dose                                                                                                                        | Route & notes                                                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1-esterase inhibitor concentrate | 20 units/kg                                                                                                                 | IV; best safety evidence in children — first-line <sup>269</sup>                                                                                                                                                                                                |
| Icatibant                         | Weight-based (max 30 mg):<br>10 mg (12–25 kg)<br>15 mg (26–40 kg)<br>20 mg (41–50 kg)<br>25 mg (51–65 kg)<br>30 mg (>65 kg) | SC (subcutaneous); bradykinin B2-receptor antagonist; ready-to-use prefilled syringe. <b>Not recommended &lt; 2 years or &lt; 12 kg</b> — no dose established (EMA label, licensed from 2 years; the US FDA label covers adults ≥ 18 years only) <sup>269</sup> |

| Drug                                                                       | Dose                                                                                                         | Route & notes                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sebetralstatat (oral; $\geq 12$ years — availability in Kuwait to confirm) | 600 mg (two 300 mg tablets) at the first sign; repeat once after $\geq 3$ h if inadequate; max 1,200 mg/24 h | Oral plasma kallikrein inhibitor; first-line on-demand option from 12 years in the 2026 international pediatric guideline; not approved $< 12$ years. Severe laryngeal attacks were not studied — use a parenteral agent <sup>335</sup> |
| Fresh frozen plasma                                                        | 10 mL/kg                                                                                                     | IV; only if specific agents unavailable <sup>269</sup>                                                                                                                                                                                  |

#### ADMIT · DISCHARGE · REFER

- **Admit** — laryngeal or severe attacks → a monitored bed.
- **Discharge** — after resolution with on-demand therapy and a written plan.
- **Refer** — immunology/allergy (diagnosis, prophylaxis).



# When to Suspect Primary Immunodeficiency (IEI)

## ASSESS

Inborn errors of immunity — think of it before the next live vaccine or transfusion.

### ▀ WHEN TO SUSPECT – THE SPUR PATTERN

- **Severe, Persistent, Unusual** (opportunistic organism or unusually severe course), or **Recurrent** infection.<sup>114</sup>
- **Family history** of immunodeficiency or unexplained early infant death, **parental consanguinity, failure to thrive**, infections needing **intravenous antibiotics** or admission.<sup>114</sup>
- Classic thresholds:  $\geq 2$  pneumonias or  $\geq 4$  **new ear infections** in a year,  $\geq 2$  deep-seated infections (abscess, osteomyelitis), persistent oral/skin thrush after age 1, or little response to  $\geq 2$  months of antibiotics.<sup>114,364</sup>
- **Non-infectious red flags** (increasingly recognized): autoimmune cytopenias — especially chronic, refractory, or multi-lineage — early or treatment-resistant

inflammatory bowel disease (< 2 years), granulomas, lymphoproliferation, severe eczema.<sup>113</sup>

## OVERVIEW – INFECTION PATTERN → LIKELY DEFECT

| Infection pattern                                                                      | Think               | Typical clues                                                                                                       |
|----------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Recurrent sinusopulmonary (encapsulated bacteria), onset mostly after 6 months         | Antibody deficiency | Agammaglobulinemia, common variable immunodeficiency (CVID), IgA deficiency; low immunoglobulins                    |
| Early severe viral/fungal/opportunistic infection, chronic diarrhea, failure to thrive | Combined / T-cell   | Severe combined immunodeficiency (SCID): lymphopenia, may have absent thymic shadow — a pediatric emergency (below) |

| Infection pattern                                                                             | Think                 | Typical clues                                                                                                                      |
|-----------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Deep abscesses, unusual organisms (Serratia, Burkholderia, Aspergillus), poor wound healing   | Phagocyte defect      | Chronic granulomatous disease; leukocyte adhesion deficiency (delayed cord separation)                                             |
| Recurrent Neisseria (meningococcal) or encapsulated-organism infection, or lupus-like disease | Complement deficiency | Early-onset, recurrent or familial meningococcal disease; lupus-like autoimmunity (early classical-pathway defects) <sup>113</sup> |

### ⚠ SUSPECTED SCID IS A PEDIATRIC EMERGENCY

- Infant in the first months of life with severe or persistent infection, chronic diarrhea, failure to thrive, oral thrush, or an **absolute lymphocyte count  $< 2.5\text{--}3 \times 10^9/\text{L}$**

(or a positive newborn T-cell receptor excision circle (TREC) screen) may have severe combined immunodeficiency.<sup>115</sup>

- Diagnose **before** live vaccines, non-irradiated blood, or overwhelming infection — allows a life-saving transplant.<sup>115</sup>
- Phone pediatric immunology the same day.<sup>115</sup>

#### ✦ FIRST-LINE SCREEN

- Full blood count **with differential** (absolute lymphocyte and neutrophil counts), immunoglobulins (IgG, IgA, IgM, IgE), lymphocyte subsets (T/B/NK by flow cytometry), complement (CH50/AH50 — total classical / alternative-pathway complement activity), and specific antibody responses (vaccine titers, isohemagglutinins); **HIV test** — exclude secondary immunodeficiency before labelling an inborn error.<sup>113</sup>
- Add a neutrophil oxidative-burst test (dihydrorhodamine) if a phagocyte defect is suspected.  
**Save a pre-treatment serum**

**sample before giving intravenous immunoglobulin (IVIG).**<sup>113</sup>

- **To diagnose infections do not rely on serology** — antibody production is impaired and IVIG transfers donor antibody; use culture, PCR or antigen detection.

#### **⚠ ACUTE DO / DON'T WHEN IEI IS SUSPECTED**

- **Do not** give live vaccines (BCG, oral polio, rotavirus, MMR, varicella, yellow fever) — they can cause fatal vaccine-strain disease.<sup>115</sup>
- **If blood products are needed, use only irradiated, CMV-negative, leukoreduced products** (prevents transfusion-associated graft-versus-host disease).<sup>331</sup>
- Do not delay empiric antibiotics for fever — treat any infection promptly and aggressively; avoid non-essential steroids/immunosuppression before a diagnosis.<sup>113</sup>

#### **⚠ SUSPECTED SCID – EMERGENCY MANAGEMENT, UNDER PEDIATRIC IMMUNOLOGY**

## DIRECTION

- **A medical emergency:**  
immunology the same day;  
confirmatory testing (lymphocyte subsets  $\pm$  TREC) at a center able to interpret it. <sup>332,331</sup>
- **Protective isolation** of infant and family from the moment of suspicion (single room, strict hand hygiene; laminar-flow unit where available). <sup>332,331</sup>
- **Treat any infection promptly and aggressively.** *Pneumocystis jirovecii* pneumonia (PJP) is frequent — chest X-ray  $\pm$  bronchoalveolar lavage; if suspected or proven, co-trimoxazole (trimethoprim-sulfamethoxazole) 20 mg/kg/day of trimethoprim IV in divided doses. <sup>332</sup>
- **Prophylaxis:** co-trimoxazole 5 mg/kg/day of trimethoprim (from 4–6 weeks of age — in a younger neonate immunology chooses the agent, because sulfonamides displace bilirubin); fluconazole 5 mg/kg/day; palivizumab in the RSV season. <sup>332,331</sup>
- **IVIG replacement** 400 mg/kg every 3 weeks. <sup>332,331</sup>

- Only **irradiated, CMV-negative, leukoreduced** blood products; **no live vaccines**.<sup>332,331</sup>
- Growth failure → parenteral nutrition.<sup>332</sup>
- Plan **hematopoietic cell transplantation** (or gene therapy) immediately once the diagnosis is established.<sup>332,331</sup>

#### REFER

- Pediatric immunology for all suspected inborn errors of immunity — the same day for suspected SCID, any opportunistic infection, or profound lymphopenia.
- IVIG replacement and HLA-typing proceed under specialist direction.<sup>113</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — as dictated by the presenting infection.
- **Discharge** — per the acute problem, with the workup arranged.

- **Refer** — immunology; avoid live vaccines and use irradiated blood products until cleared.

## Tension Pneumothorax & Needle Decompression

MINUTES

APLS/ATLS • NAEMSP 2024

Sudden distress, absent breath sounds and shock — treat before the chest X-ray.<sup>361</sup>

### ▲ RECOGNIZE (CLINICAL)

- Rapidly worsening respiratory distress and **hypoxia** with **hypotension, unilateral absent/reduced breath sounds**, hyper-resonance, distended neck veins, and (late) tracheal deviation — often after trauma, asthma, positive-pressure ventilation, or in a ventilated neonate.<sup>1</sup>

### ⚠️ DECOMPRESS NOW

- Tension pneumothorax is a **clinical diagnosis** — if the child is peri-arrest, **do not wait for a chest X-ray**.



- Perform immediate **needle (or finger) decompression**, then place a chest drain.<sup>361</sup>
- In a ventilated patient who suddenly desaturates and drops their pressure, think of it first.<sup>1</sup>

#### ▲ WORKUP

- **Clinical diagnosis — do not wait for imaging.** Decompress first.
- Confirm **after** decompression with chest X-ray or bedside **lung ultrasound (eFAST — extended Focused Assessment with Sonography for Trauma, adding lung views)**; definitive chest drain follows.<sup>1</sup>

| Step                 | Landmark                                                                                                | Notes                                                                                                                                                                                                                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Needle decompression | 2nd intercostal space, mid-clavicular line – <i>or</i> 4th/5th intercostal space anterior axillary line | Over the top of the rib; use a cannula long enough for chest-wall thickness <sup>361</sup>                                                                                                                                                                                    |
| Cannula size         | Infant: 18–22 G × 25–32 mm<br>Child: 16–20 G × 32–38 mm<br>Adolescent: 14–16 G × 45–50 mm               | Largest bore available (fine cannulas kink); length by chest-wall thickness — 11–19 mm in most children, so a 32–38 mm cannula reaches the pleura, while adult 45–50 mm needles are about twice the depth needed under 13 years; ultrasound if time allows <sup>377,378</sup> |
| Definitive           | Chest tube (tube thoracostomy)                                                                          | Follows decompression; size for age                                                                                                                                                                                                                                           |

#### ✦ NEONATAL CLUE

In a ventilated or term neonate, sudden deterioration with a **shift of the apex beat** and unilateral hyper-resonance suggests tension pneumothorax; **transillumination** can help at the cot-side while you prepare to decompress.<sup>1</sup>

### ✦ PRIMARY SPONTANEOUS PNEUMOTHORAX (ADOLESCENT ≥ 14 Y)

Typically a tall, thin adolescent boy with sudden pleuritic chest pain ± breathlessness and **no tension**; confirm on chest X-ray. **Minimal symptoms and stable** → conservative management with early review (BTS 2023, evidence from patients aged 14–50); **symptomatic or large** → needle aspiration or an ambulatory drainage device; **unstable** → chest drain. No flying or diving until resolved; counsel on recurrence (~30%) and smoking/vaping. Decisions are made with the respiratory/thoracic team — this is adult-derived guidance applied to older adolescents.<sup>362</sup>

### ▀ ADMIT · REFER

- **Admit** — all pneumothoraces needing drainage; PICU/NICU.
- **Call** critical care ± surgery.

# Rheumatologic

Cytokine storm (HLH/MAS) & Kawasaki vasculitis

## Cytokine Storm – HLH / MAS (Systemic Hyperinflammation)

HOURS

EULAR/ACR 2022 • EULAR/PReS 2024

One process, many names: hematology calls it HLH, rheumatology calls it MAS — the joint EULAR/ACR consensus (rheumatology, hematology, immunology, infectious diseases and intensive care) is the common ground this card follows.

+ Calculator: HLH-2004 • MAS-2016 • MS score

### ► RECOGNIZE – A PATTERN THAT EVOLVES, NOT A SINGLE VALUE

- **Persistent fever that looks like sepsis but does not respond, hepatosplenomegaly, falling counts ( $\geq 2$  lines), very high or rising ferritin, rising triglycerides, falling fibrinogen, rising transaminases and LDH, coagulopathy/DIC, and CNS signs (irritability, seizures, altered**

consciousness). Individual findings are non-specific — read them together and over time<sup>396</sup>.

- **Examine perfusion first** — tachycardia with cool extremities, prolonged capillary refill or reduced urine output at a **normal blood pressure** is compensated shock; myocardial dysfunction is common in hyperinflammation, so get an **early echocardiogram**<sup>1,214</sup>.
- **In sJIA / Still's disease:** a **falling ESR** with rising CRP and ferritin, a **falling platelet or white count from a high baseline**, and hypofibrinogenemia are early clues (consumption). On canakinumab or tocilizumab the fever and the ferritin rise may be blunted or delayed; the risk also rises when steroids or biologics are tapered<sup>397,212</sup>.
- **Ferritin:** send it in every child with new or heightened suspicion, even if a previous value was normal, and repeat daily. > 500 ng/mL is 84% sensitive in children (the Histiocyte Society cut-off) but not specific; pediatric studies use 500–2,000; a rapidly rising or very high (> 10,000) ferritin is highly suggestive<sup>396</sup>.

- The same process complicates infection (EBV, CMV, dengue, leishmaniasis, tuberculosis, typhoid, brucellosis), malignancy (leukemia, lymphoma), rheumatic disease (sJIA, lupus, Kawasaki), MIS-C, immunotherapy, and inborn errors (familial HLH, XLP, Chédiak–Higashi, Griscelli)<sup>396</sup>.

## CRITERIA – USEFUL IN THE SETTING THEY WERE BUILT FOR; NONE IS SUFFICIENT EVERYWHERE<sup>396</sup>

| Tool            | Built for                                        | Rule                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HLH-2004</b> | Primary / genetic HLH trials (infants, children) | <b>5 of 8:</b> fever · splenomegaly · cytopenias in $\geq 2$ lines (Hb $< 90$ g/L; platelets $< 100 \times 10^9$ /L; neutrophils $< 1.0 \times 10^9$ /L) · triglycerides $\geq 3.0$ mmol/L (265 mg/dL) and/or fibrinogen $\leq 1.5$ g/L · hemophagocytosis · low/absent NK activity · ferritin $\geq 500$ ng/mL · sCD25 $\geq 2,400$ U/mL <sup>211</sup> |
| <b>MAS-2016</b> | Febrile child with known or suspected sJIA       | Ferritin $> 684$ ng/mL <b>and any 2 of:</b> platelets $\leq 181 \times 10^9$ /L · AST $> 48$ U/L · triglycerides $> 156$ mg/dL · fibrinogen $\leq 360$ mg/dL <sup>212</sup>                                                                                                                                                                              |

| Tool                   | Built for                                       | Rule                                                                                                                                                                               |
|------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>MS score</u></b> | Separating MAS from an sJIA flare               | 7 items (CNS dysfunction, bleeding, active arthritis, platelets, fibrinogen, LDH, ferritin); score $\geq -2.1 \rightarrow$ MAS (sensitivity 0.85, specificity 0.95) <sup>398</sup> |
| <b>HScore</b>          | Adults; infection- or malignancy-associated HLH | Probability from < 1% ( $\leq 90$ ) to > 99% ( $\geq 250$ ); online calculator <sup>399</sup>                                                                                      |

**⚠ DO NOT WAIT TO SATISFY A SET OF CRITERIA BEFORE TREATING A DETERIORATING CHILD**

- In a child with high-risk features or progression, start empiric immunomodulation during the initial evaluation — diagnostic work-up, treatment of the trigger and prevention of complications run in parallel<sup>396,397</sup>.

**▴ WORK-UP – STARTS ON SUSPICION, IN PARALLEL WITH TREATMENT**

- **First tier (any hospital, same day):**  
CBC + film, ferritin, CRP, ESR, fibrinogen, D-dimer, PT/APTT, triglycerides, AST/ALT/GGT/bilirubin/albumin, LDH, creatinine, sodium; blood cultures;

EBV and CMV PCR, HIV; other infections by exposure (dengue or malaria after travel, tuberculosis, leishmaniasis, typhoid, brucellosis) — hyperinflammation can be the presentation of an infection. Send serology **before** IVIG<sup>396</sup>.

- **Second tier (with the specialist teams): sCD25 (soluble IL-2 receptor- $\alpha$ )** — the most useful confirmatory test where available; **bone marrow** (also for malignancy — obtain **before** steroids when malignancy is plausible and it will not delay treatment; hemophagocytosis is neither required nor specific); IL-18 and CXCL9 where available; NK-cell function, CD107a degranulation, perforin/SAP/XIAP flow cytometry<sup>396</sup>.
- **Genetics early in every child** (multigene panel or exome), and in anyone with consanguinity, family history, recurrence, albinism or prominent CNS disease — consanguinity makes this especially relevant in Kuwait; a genetic diagnosis changes treatment, prognosis and family counselling<sup>396</sup>.
- **CNS:** assess every child < 1 year, any suspected familial disease and any neurological sign — CSF (cells,



protein, glucose, cytology) and contrast MRI — **without delaying treatment**<sup>396</sup>.

**TREAT – EMPIRIC IMMUNOMODULATION EARLY;  
TREAT THE TRIGGER AT THE SAME TIME  
(ANTIMICROBIALS BY EXPOSURE; AN ACTIVE  
INFECTION DOES NOT DELAY MAS TREATMENT)**<sup>396,397</sup>

| Step                                 | Drug & dose                                                                                                | Notes                                                                                                                                                          |
|--------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.<br>Glucocorticoid —<br>all forms  | Methylprednisolone pulse<br>15–30<br>mg/kg/day IV<br>(max 1 g) × 3<br>days, then 2<br>mg/kg/day<br>divided | Strongest evidence<br>across all forms of<br>HLH/MAS;<br>shorter-acting agents<br>are preferred while the<br>diagnosis is still<br>evolving <sup>396,397</sup> |
| 1.<br>Alternative —<br>dexamethasone | 10 mg/m <sup>2</sup> /day<br>IV (≈ 2–4<br>mg/kg/day of<br>methyl-<br>prednisolone)                         | Preferred with CNS<br>involvement and in the<br>HLH-94/2004 protocol<br>(better CNS<br>penetration) <sup>396,211</sup>                                         |

| Step                                                                                        | Drug & dose                                                                                                                                                                       | Notes                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 · Add early — MAS / sJIA, Kawasaki or MIS-C overlap, undifferentiated hyperinflammation   | Anakinra (IL-1 receptor antagonist): above the standard 1–2 mg/kg/day – 2–4 mg/kg/day, escalating to 10 mg/kg/day in divided (q6–12 h) or repeated IV doses in the critically ill | Immunomodulation without immunosuppression; short half-life; does not obscure a malignancy work-up <sup>396,397,112</sup>                                                                             |
| 2 · IVIG                                                                                    | 1–2 g/kg                                                                                                                                                                          | Endorsed for empiric use, especially with Kawasaki/MIS-C overlap; a large colloid load — caution with cardiac dysfunction <sup>396</sup>                                                              |
| 3 · Inadequate response at 24–72 h, or severe / rapidly worsening — with the reference team | Ciclosporin (oral or IV, level-guided) ; tacrolimus 0.1 mg/kg/day divided q12h is the EULAR/ACR alternative                                                                       | For MAS; <b>not</b> for primary HLH (proven ineffective). Combination therapy on a background of high-dose glucocorticoid is often needed and may be initial therapy in severe MAS <sup>397,396</sup> |

| Step                                        | Drug & dose                                                                                                                                                                                                                      | Notes                                                                                                                                                                                        |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 ·<br>Emapalu<br>mab<br>(anti-IFN<br>-γ)   | Primary HLH:<br>1 mg/kg IV<br>twice weekly<br>(with<br>dexamethasone<br>) . Still's /<br>sJIA-MAS: 6<br>mg/kg once,<br>then 3 mg/kg<br>every 3 days<br>to day 16,<br>then twice<br>weekly;<br>titrate to<br>max 10<br>mg/kg/dose | FDA-approved for<br>primary HLH and for<br>HLH/MAS in Still's<br>disease after<br>glucocorticoid failure;<br>screen and monitor for<br>TB, CMV and other<br>infection <sup>151,397</sup>     |
| 3 ·<br>Ruxolitinib                          | Specialist,<br>off-label                                                                                                                                                                                                         | JAK inhibition;<br>promising in refractory<br>HLH/MAS <sup>396,397</sup>                                                                                                                     |
| 3 ·<br>Etoposide-based<br>(HLH-94/<br>2004) | Hematology-di<br>rected                                                                                                                                                                                                          | Life-saving in primary<br>HLH and severe<br>EBV-HLH; <b>not<br/>first-line for MAS</b><br>(low-dose only in<br>refractory MAS); not<br>for most non-EBV<br>infections <sup>396,211,397</sup> |
| 3 ·<br>Rituximab                            | Hematology-di<br>rected                                                                                                                                                                                                          | EBV-driven HLH<br>(B-cell depletion) <sup>396</sup>                                                                                                                                          |

### ✦ SUPPORTIVE CARE AND MONITORING

- Sepsis-style resuscitation and broad antimicrobials pending cultures; consider *Pneumocystis*

and antifungal prophylaxis early (part of HLH-94); no live vaccines; irradiated blood products if transplant is possible<sup>396</sup>.

- **Daily** ferritin, CRP, CBC, fibrinogen and ALT ( $\pm$  triglycerides); more often if critical. A ferritin that fails to fall predicts a poor outcome — escalate and re-examine the diagnosis (lymphoma, unrecognized infection, genetic HLH)<sup>396</sup>.

◆ **KUWAIT JOINT PATHWAY – RHEUMATOLOGY · HEMATOLOGY/ONCOLOGY · PICU** (proposed;

pending joint endorsement by the Council's rheumatology and hematology members)

- **SAME FIRST HOUR, WHICHEVER TEAM IS CALLED FIRST:** ferritin + first-tier panel, cultures, methylprednisolone pulse  $\pm$  anakinra. **NO TEAM DELAYS GLUCOCORTICOIDS WAITING FOR THE OTHER.**
- **RHEUMATOLOGY LEADS** when there is a known or strongly suspected rheumatic context (sJIA/Still's, lupus, Kawasaki, MIS-C): MAS pathway — glucocorticoid + anakinra  $\pm$  ciclosporin  $\rightarrow$  emapalumab<sup>397</sup>.

- **HEMATOLOGY/ONCOLOGY LEADS** when: age < 2 years, consanguinity or family history, EBV-driven disease, cytopenias or organomegaly dominant, possible malignancy, or no rheumatic context: HLH pathway — genetics and marrow first where safe; etoposide-based therapy when indicated; emapalumab for primary HLH<sup>396,211</sup>.
- **BOTH ARE CONSULTED WITHIN HOURS IN EVERY CASE**; PICU for organ dysfunction; infectious diseases for the trigger and prophylaxis; immunology for inborn-error evaluation. A multidisciplinary HLH/MAS team improves recognition and reduces over- and under-treatment<sup>396</sup>. Transfer of care is by agreement, not by admission ward.

#### ▲ **ADMIT · DISCHARGE · REFER**

- **Admit** — all → high-dependency/PICU.
- **Discharge** — not acutely; under the leading team with a written taper and monitoring plan.

- **Refer** — hematology and rheumatology within hours; PICU; genetic counselling for every family tested<sup>396</sup>.

## Kawasaki Disease HOURS

Medium-vessel vasculitis; leading cause of acquired heart disease in children.

+ Calculator: [HLH-2004 · MAS-2016 · MS score](#)

### RECOGNIZE

- Fever  $\geq 5$  days +  $\geq 4$  of: bilateral non-exudative conjunctivitis, oral mucosal changes (cracked/red lips, strawberry tongue, diffuse oropharyngeal erythema), polymorphous rash, extremity changes (swelling/peeling), cervical lymphadenopathy  $> 1.5$  cm<sup>35</sup>.
- **Incomplete Kawasaki** is common in infants  $< 6$  months and older children — fewer features but high coronary risk; use inflammatory markers and echocardiography<sup>35</sup>.

#### ✦ PEARL – THINK INCOMPLETE KAWASAKI

- Suspect **incomplete Kawasaki** in any infant **< 6 months** with **≥ 5 days of unexplained fever** even without the full clinical criteria — this is the group most often missed and at the highest risk of coronary aneurysms<sup>35</sup>.
- Check inflammatory markers and arrange echocardiography; treat on a compatible picture rather than waiting for every feature.  
**AHA 2017 pathway:** fever  $\geq 5$  days + 2–3 clinical criteria (or an infant with  $\geq 7$  days of unexplained fever) → check CRP/ESR; if CRP  $\geq 3$  mg/dL and/or ESR  $\geq 40$  mm/h, look for  $\geq 3$  supplementary lab findings (anemia, platelets  $\geq 450 \times 10^9/L$  after day 7, albumin  $\leq 3$  g/dL, raised ALT, WBC  $\geq 15 \times 10^9/L$ , sterile pyuria) or a positive echocardiogram<sup>35</sup> → treat.

**Treat within 10 days of fever onset to protect the coronaries<sup>35</sup>**



#### FIRST-LINE

- IVIG 2 g/kg as a single infusion + **low-dose (antiplatelet) aspirin 3–5 mg/kg/day**; baseline echocardiogram<sup>35</sup>.
- Discuss adjunct corticosteroids with cardiology/rheumatology for high-risk patients (infants < 1 year, coronary changes, high inflammatory markers, or Kawasaki disease shock syndrome)<sup>35</sup> — AHA 2024 defines high risk as **age < 6 months or baseline LAD/RCA z-score  $\geq 2.5$**  (CRP  $\geq 13$  mg/dL and Asian ancestry also predict aneurysms) and supports intensified initial therapy for this group.<sup>291</sup>



| Drug                                          | Dose                                                                                                 | Route & notes                                                                                                                                                                                              |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IVIG                                          | 2 g/kg single infusion over 10–12 h                                                                  | IV. IVIG-resistant fever > 36 h → second IVIG 2 g/kg, or corticosteroids, or a biologic (infliximab, now dosed <b>10 mg/kg</b> IV over 2 h) <sup>35,291</sup>                                              |
| Aspirin                                       | 3–5 mg/kg/day PO once daily from diagnosis; continue 6–8 weeks or until the echocardiogram is normal | <b>Antiplatelet dose only, from the outset.</b> Anti-inflammatory doses (30–100 mg/kg/day) are not recommended — no coronary benefit, more adverse effects <sup>400,401</sup> . See the aspirin note below |
| Methylprednisolone → prednisolone (high-risk) | 2 mg/kg/day, IV then oral, tapered over 2–3 weeks                                                    | Adjunct to IVIG in high-risk patients or Kawasaki disease shock syndrome <sup>35</sup>                                                                                                                     |

#### ⚠ NOTE

- Single-dose methylprednisolone with IVIG is **not** recommended as routine primary therapy<sup>35</sup>.
- Avoid ibuprofen with aspirin (antagonizes antiplatelet effect<sup>35</sup>).
- **Aspirin and varicella/influenza:** withhold aspirin during active

varicella or influenza (Reye syndrome) and use an alternative antiplatelet agent (clopidogrel) meanwhile; give the annual influenza vaccine to every child on aspirin<sup>35</sup>.

#### ◆ WHY THE ANTIPLATELET DOSE

- Anti-inflammatory dosing has never been shown to reduce coronary aneurysms<sup>35</sup>; a randomized trial found IVIG alone non-inferior to IVIG + high-dose aspirin<sup>400</sup>; a network meta-analysis of 1,944 patients found no difference in coronary lesions, IVIG resistance or hospital stay between no aspirin, 3–5, 30–50 and 80–100 mg/kg/day<sup>401</sup>; meta-analyses and cohorts show equal coronary outcomes with low-dose from the outset and fewer adverse effects (GI bleeding, transaminase rise, Reye syndrome)<sup>111,388,403,404</sup>.
- One Japanese database study found more coronary abnormalities at 5 than at 30 mg/kg/day (OR 1.40) — observational and contradicted by the randomized data; a

1,346-patient trial of 30–50 vs 3–5 mg/kg/day is under way<sup>402</sup>.

- AHA 2017 still lists moderate- to high-dose aspirin until afebrile as an option<sup>35</sup>; this guide recommends the antiplatelet dose from diagnosis on the strength of the newer randomized and pooled evidence above.

#### **⚠ WATCH FOR**

##### **Macrophage activation syndrome**

(cytokine storm — see **HLH / MAS**) and

##### **Kawasaki disease shock syndrome**

(hypotension / poor perfusion) — both signal severe, high-risk disease; escalate, add adjunctive steroids, and involve rheumatology/cardiology + PICU<sup>35,212</sup>.

#### **▴ ADMIT · DISCHARGE · REFER**

- **Admit** — all → for IVIG and monitoring.
- **Discharge** — after defervescence and echo, with a plan for repeat echo (at ~1 week and 4–6 weeks)<sup>291</sup> and aspirin.

- **Refer** — cardiology (echo) and rheumatology; IVIG-resistant cases urgently.

## IgA Vasculitis (Henoch-Schönlein Purpura)

ASSESS

SHARE 2019

Palpable purpura on the legs and buttocks — the commonest childhood vasculitis; the kidney is the long game.

### ▶ RECOGNIZE (PURPURA + $\geq 1$ )

- **Palpable purpura** (mandatory), lower-limb/buttock predominant, with normal platelets and coagulation, **plus** at least one of: diffuse abdominal pain · arthritis/arthralgia · biopsy showing IgA deposition · renal involvement (hematuria/proteinuria)<sup>121</sup>.

### ⚠ TWO THINGS THAT TURN IT INTO AN EMERGENCY

- **Intussusception** — usually **ileo-ileal** (easily missed on routine imaging — use ultrasound<sup>213</sup>) in a

child with severe/colicky abdominal pain or GI bleeding.

- And **testicular pain/swelling** can mimic torsion<sup>213</sup>.
- Severe abdominal pain, GI bleeding, scrotal pain or a rapidly rising creatinine/BP all need admission<sup>213</sup>.

#### ✦ THE KIDNEY CAN DECLARE ITSELF LATE

- Nephritis may appear **weeks to months** after the rash.<sup>213</sup>
- Check **blood pressure and urinalysis at every visit** and continue surveillance for **6 months** (schedule below) even after the rash resolves<sup>213,393,394</sup>.
- Persistent proteinuria<sup>213</sup> → nephrology.

#### ▀ WORKUP

- **Urinalysis + urine protein:creatinine ratio** and **blood pressure** at every visit — nephritis screen (monitor for 6–12 months even if initially normal<sup>213</sup>).
- Serum **creatinine / eGFR**<sup>213</sup>.

- CBC + coagulation — platelets are **normal/high** in IgA vasculitis (excludes ITP/HUS<sup>121</sup>).
- **Renal biopsy** only for significant or persistent proteinuria or impaired function<sup>213</sup>.

| Problem                          | Approach                                                        | Notes                                                        |
|----------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------|
| Joint / mild pain                | Acetaminophen ; NSAIDs if no renal involvement or GI bleeding   | Most cases are supportive only <sup>213</sup>                |
| Severe abdominal pain / orchitis | Prednisolone 1–2 mg/kg/day (max 60 mg), short course with taper | Eases pain; does <b>not</b> prevent nephritis <sup>213</sup> |
| Nephritis (significant)          | Nephrology-directed (steroids ± immunosuppression)              | Do not manage severe nephritis alone <sup>213</sup>          |

#### ◆ WHERE TO MANAGE – MOST CHILDREN GO HOME

- **UNCOMPLICATED IGA VASCULITIS** (rash ± joint pain, mild abdominal pain, normal blood pressure, normal or trace urinalysis, eating and walking) is **OUTPATIENT CARE**: analgesia, counselling, and teaching the family to check the urine for protein at home over the next 3–6 months,

with a written follow-up plan. No guideline recommends routine admission — most children need only supportive treatment<sup>213,393,394</sup>.

- **RETURN AT ONCE IF:** severe or worsening abdominal pain, vomiting, blood in the stool, swollen or painful testicle, blood in the urine or foamy urine, swelling of the face/legs, headache or reduced urine output.
- **ADMIT ONLY FOR:** severe abdominal pain (needs parenteral analgesia), GI bleeding or suspected intussusception, unable to walk or drink, scrotal/testicular involvement until torsion is excluded, hypertension, acute kidney injury or a nephrotic/nephritic presentation, CNS or pulmonary involvement, or an unsafe discharge<sup>213,393</sup>.

#### ✦ FOLLOW-UP PLAN FOR THE NEPHRITIS SCREEN – GIVE IT IN WRITING AT THE FIRST VISIT

- **What:** blood pressure + early-morning urine dipstick at every visit; protein  $\geq 1+$  → urine protein:creatinine ratio; creatinine/eGFR at diagnosis and whenever the urine is abnormal<sup>213,394</sup>.

- **When:** diagnosis · **day 7** (a normal day-7 urinalysis predicts a normal renal outcome with 97% negative predictive value<sup>395</sup>) · day 14 · day 28 · months 2, 3 and 6 — at least monthly for 6 months even if normal,<sup>394,393</sup>. Nephritis appears on average ~4 weeks after the rash and can appear late, so early normal results do not shorten the schedule.<sup>213</sup>.
- **Discharge from surveillance** at 6 months if blood pressure and urine have stayed normal; extend to 12 months if any abnormality occurred<sup>213,393</sup>; biopsy-proven nephritis is followed for years by nephrology.<sup>393,394</sup>.
- **Nephrology now:** urine protein:creatinine > 250 mg/mmol at any time (biopsy if it persists 4 weeks); 100–250 mg/mmol persisting (biopsy if 3 months); eGFR < 90 mL/min/1.73 m<sup>2</sup>; hypertension; nephrotic or nephritic syndrome; acute kidney injury<sup>213,393,394</sup>. ACE inhibitor/ARB for proteinuria lasting > 3 months.<sup>213</sup>.
- **Steroids do not prevent nephritis** — no prednisolone for the rash; reserve it for severe abdominal



pain/GI bleeding once intussusception is excluded, orchitis, or other severe organ involvement<sup>213,393</sup>.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — only for the indications above.
- **Discharge** — outpatient follow-up per the schedule, with the return precautions explained.
- **Refer** — nephrology per the thresholds above; surgery if intussusception; urology if the scrotum is involved.

## MIS-C (Multisystem Inflammatory Syndrome)

MINUTES

ACR 2022 · CDC

Post-COVID hyperinflammation that overlaps Kawasaki and toxic shock — and can crash the heart.

✚ Calculator: [HLH-2004 · MAS-2016 · MS score](#)

### ▶ RECOGNIZE (CRITERIA)

- **Fever  $\geq 3$  days** in a child/adolescent, **plus  $\geq 2$  systems** involved (rash/conjunctivitis, hypotension/shock, cardiac dysfunction, coagulopathy, prominent GI symptoms), **raised inflammatory markers** (CRP, ferritin, D-dimer), **evidence of SARS-CoV-2** or recent exposure, and **no alternative diagnosis**<sup>215,214</sup>.

### ⚠ IT LOOKS LIKE SEVERAL KILLERS AT ONCE

- Overlaps **Kawasaki disease, toxic shock syndrome and sepsis** — treat the shock, take cultures and cover for sepsis until excluded<sup>214</sup>.
- Prominent abdominal pain can mimic appendicitis. **Cardiac:** myocardial dysfunction, arrhythmia and **coronary artery aneurysms** — get an early echocardiogram and troponin/BNP<sup>214</sup>.
- **Shock is often compensated at first.** Tachycardia, cool or mottled extremities, prolonged capillary refill, weak pulses, reduced urine output or drowsiness **with a normal blood pressure** is shock

until proven otherwise — hypotension is a late sign<sup>1</sup>. Myocardial dysfunction can be insidious and children can decompensate rapidly: examine perfusion at every review, get the **echocardiogram early** (left-ventricular function, coronary z-scores), and repeat troponin/BNP<sup>214</sup>. Fluid cautiously (5–10 mL/kg, reassess) and start epinephrine early if the heart is the problem.

#### ▲ WORKUP

- **Inflammatory:** CRP, ESR, **ferritin**, procalcitonin, CBC (lymphopenia), fibrinogen<sup>214</sup>, **D-dimer**.
- **Cardiac: troponin, BNP/NT-proBNP, ECG and echocardiogram** (coronary arteries + function)<sup>214</sup>.
- **SARS-CoV-2** PCR and serology; **blood cultures** to exclude sepsis<sup>214</sup>.
- CMP, LDH, triglycerides, coagulation.
- **Tiered approach when the child is stable:** tier 1 (any facility) CBC with differential, metabolic panel, ESR, CRP, SARS-CoV-2 testing → proceed to tier 2 (cardiac markers, ECG,

echocardiogram, ferritin, D-dimer, procalcitonin, cytokine panel if available) when ESR/CRP is raised plus one of lymphopenia, neutrophilia, thrombocytopenia, hyponatremia or hypoalbuminemia. A child with shock is treated while the work-up is completed<sup>214</sup>.

- **Admit for observation** while under investigation if any of: abnormal vital signs, respiratory distress, altered mental status, renal or hepatic injury, marked inflammatory markers, abnormal ECG, or raised troponin/BNP<sup>214</sup>. ECG at least every 48 h in hospital; telemetry if conduction is abnormal<sup>214</sup>.

## TREATMENT PATHWAY – ACR 2022 (STEPWISE; DUAL THERAPY FIRST-LINE FOR MOST ADMITTED CHILDREN)

| Step                                                  | Drug & dose                           | Notes                                                                                                                                                                                                                                          |
|-------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.<br><b>First-line</b><br>— IVIG                     | 2 g/kg (ideal body weight, max 100 g) | Assess cardiac function and fluid status first; with cardiac dysfunction give 1 g/kg/day over 2 days, slow the rate and/or add a diuretic to avoid volume overload <sup>214</sup>                                                              |
| 1.<br><b>First-line</b><br>— plus methyl-prednisolone | 1–2 mg/kg/day IV                      | Dual therapy with IVIG for most hospitalized patients; IVIG alone only in mild disease or when steroids are contraindicated, with intensification at the first sign of worsening. Glucocorticoid monotherapy is not recommended <sup>214</sup> |

| Step                                                                                                                                                    | Drug & dose                                                                                                     | Notes                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 ·<br>Refractor<br>y (no<br>response<br>within 24<br>h:<br>persisten<br>t fever or<br>ongoing<br>organ<br>involvem<br>ent) or<br>shock on<br>inotropes | Methylprednis<br>olone pulse<br>10–30<br>mg/kg/day                                                              | Do <b>not</b> repeat IVIG<br>(hemolysis, volume<br>overload). Children in<br>shock on multiple<br>inotropes/vasopressors<br>respond best to<br>high-dose<br>glucocorticoid <sup>214</sup>   |
| 2 ·<br>Alternati<br>ves                                                                                                                                 | Anakinra > 4<br>mg/kg/day<br>(often 5–10<br>mg/kg/day)<br>IV/subcut ·<br>or infliximab<br>5–10 mg/kg IV<br>once | No infliximab if there<br>are features of<br>macrophage activation<br>syndrome; anakinra<br>also as a<br>steroid-sparing<br>agent <sup>214</sup>                                            |
| Aspirin                                                                                                                                                 | 3–5 mg/kg/day<br>PO (max 81<br>mg)                                                                              | All patients without<br>bleeding or bleeding<br>risk, until platelets are<br>normal and coronaries<br>are normal at ≥ 4<br>weeks; add gastric<br>protection with<br>steroids <sup>214</sup> |

| Step             | Drug & dose                  | Notes                                                                                                                                                                         |
|------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anti-coagulation | Enoxaparin (anti-Xa 0.5–1.0) | Coronary z-score $\geq 10$ ( $\geq 2$ weeks, then warfarin/DOAC) or documented thrombosis; consider with ejection fraction $< 35\%$ ; otherwise individualized <sup>214</sup> |
| Taper            | 2–3 weeks or longer          | Guided by serial CRP/ferritin and cardiac reassessment to avoid rebound <sup>214</sup>                                                                                        |

#### ◆ CARDIAC FOLLOW-UP

- Repeat echocardiogram at 7–14 days and at 4–6 weeks in every child (more often with LV dysfunction or aneurysms); consider one at 1 year if the heart was involved; cardiac MRI at 2–6 months after moderate–severe LV dysfunction<sup>214</sup>.

#### ▴ ADMIT · REFER

- **Admit** all confirmed cases; PICU for shock, cardiac dysfunction or conduction abnormality.
- **Call** pediatric rheumatology/infectious diseases and cardiology from the start

(multidisciplinary team); follow up coronary imaging.

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## Surgical Emergencies

Recognize early · resuscitate · refer fast

### Malrotation / Midgut Volvulus

MINUTES

OFTEN < 1 MO

Bilious (green) vomiting in a well infant — the "drop-everything" abdomen.

#### RECOGNIZE

Bilious vomiting ± distension/pain;  
bloody stool and shock are late  
(ischemia)<sup>217</sup>.



#### RESUSCITATE & CONFIRM

- Nil by mouth, NG decompression, IV fluids, bloods.
- **Urgent upper-GI contrast**  
(abnormal duodeno-jejunal junction, "corkscrew") ± Doppler ultrasound<sup>217</sup>.





#### CONSULT FIRST

- Consult **pediatric surgery immediately** — for definitive care and **safe transport**.
- Do not image at a center that will not operate; if surgery is unavailable on-site, **transfer without delaying for contrast studies**.

**Immediate surgery — Ladd procedure. Peritonitis or shock → straight to the operating room, do not delay for imaging<sup>217</sup>.**

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all suspected → emergency, nil by mouth, fluids.
- **Discharge** — not acutely; post-operative.
- **Refer** — pediatric surgery IMMEDIATELY (bilious vomiting = volvulus until excluded<sup>217</sup>); urgent upper-GI contrast.

# Intussusception

HOURS

6 MO–3 YR

Telescoping bowel; the classic triad is usually incomplete.

## RECOGNIZE

- Intermittent colicky pain (drawing up the legs), vomiting, lethargy between episodes.
- Sausage-shaped mass; "redcurrant jelly" stool is **late**. Unexplained lethargy may be the only clue<sup>218</sup>.

## CONFIRM

Ultrasound — "target/doughnut" sign (highly sensitive and specific in experienced hands)<sup>218</sup>. Resuscitate with IV fluids first.



## DECISION

**Stable, no peritonitis/perforation?**

## YES

Non-operative air or contrast enema reduction — perform only at a center with **pediatric surgical backup** where the child will be definitively treated (perforation needs immediate surgery)<sup>218</sup>.

## NO / failed

Operative reduction; resect non-viable bowel<sup>218</sup>.

### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → for reduction and observation.
- **Discharge** — after successful reduction and an observation period without recurrence.
- **Refer** — pediatric surgery and interventional radiology (air/contrast enema); surgery if reduction fails or peritonitis.

# Hypertrophic Pyloric Stenosis

HOURS

3-6 WK

Progressive non-bilious projectile vomiting in a hungry infant.

## RECOGNIZE & CONFIRM

- "Olive" mass, visible gastric peristalsis, dehydration. Labs: **hypochloremic hypokalemic metabolic alkalosis**.<sup>219</sup> Ultrasound: muscle  $\geq 3-4$  mm, channel  $\geq 14-16$  mm<sup>219</sup>.

## ① CORRECT FIRST

Rehydrate and normalize chloride, potassium and bicarbonate **before** surgery. NG if vomiting<sup>219</sup>.



② Pyloromyotomy once biochemically stable. The electrolytes are the emergency, not the surgery<sup>219</sup>.

## ADMIT · DISCHARGE · REFER

- **Admit** — all → correct fluids/electrolytes before surgery.
- **Discharge** — post-operative, feeding established.

- **Refer** — pediatric surgery (pyloromyotomy) after biochemical correction — it is not a surgical emergency until corrected.

## Acute Appendicitis HOURS

Most common surgical abdomen; young children atypical, high perforation risk.

### + Calculator: Pediatric Appendicitis Score

#### ▶ **RECOGNIZE**

- Periumbilical pain migrating to the right lower quadrant, anorexia, fever, vomiting; guarding/rebound, pain on movement<sup>148</sup>.
- Clinical scores (Alvarado / Pediatric Appendicitis Score) + ultrasound first-line; CT or MRI selectively<sup>148</sup>.

#### **MANAGE**

Nil by mouth · IV fluids · analgesia for pain · **urgent surgical evaluation** → appendectomy.

### ⚠ MIMICS

Examine the testes (torsion), consider DKA (see [Diabetic Ketoacidosis](#)), lower-lobe pneumonia, UTI, ovarian torsion<sup>148</sup>.

### ⚠ DO NOT BYPASS SURGICAL EVALUATION

- **Antibiotics** — not for *suspected* appendicitis before surgical assessment; started before diagnosis is confirmed, they partially treat and mask it (blunted WBC/neutrophils, softer signs, more perforation<sup>148</sup>).
- Start once diagnosis is confirmed and surgical plan agreed — or immediately if the child is septic or perforated. **Analgesia** — never leave a child in pain, but stay opioid-sparing while awaiting surgical review: give non-opioid analgesia first (IV or oral acetaminophen); reserve opioids for severe pain as a single titrated dose, with no repeat opioid until the surgeon has assessed<sup>148</sup>.
- Never give parenteral opioids then discharge an undifferentiated abdomen.

- Analgesia does not reduce diagnostic accuracy — never a reason to delay evaluation<sup>148</sup>.

## ANTIBIOTICS – ONCE DIAGNOSIS CONFIRMED AND SURGICAL PLAN AGREED (OR NOW, IF SEPTIC/PERFORATED)

| Drug                        | Dose                                                                                                              | Route & notes                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Piperacillin-tazobactam     | 100 mg/kg/dose (piperacillin component) IV every 6–8 h (max 4 g/dose)                                             | Single-agent cover including anaerobes; common first choice <sup>148</sup> .          |
| Ceftriaxone + metronidazole | Ceftriaxone 50–75 mg/kg/dose IV once daily (max 2 g) + metronidazole 10 mg/kg/dose IV every 8 h (max 500 mg/dose) | Alternative combination; give the first dose promptly once diagnosed <sup>148</sup> . |

### ◆ DURATION

- Uncomplicated appendicitis: perioperative prophylaxis only — a single pre-operative dose, up to 24 h<sup>148</sup>.
- Complicated or perforated: continue ~4–5 days after adequate source control, or until

afebrile with a normalizing white count and returning bowel function (STOP-IT trial)<sup>160</sup>.

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — all → for surgery and IV antibiotics.
- **Discharge** — post-operative per pathway.
- **Refer** — pediatric surgery; interventional radiology for an abscess (drainage) when non-operative.

## Incarcerated Inguinal Hernia HOURS

Irreducible tender groin or scrotal lump — reduce it or operate.

#### ▲ RECOGNIZE

- Firm, tender, **irreducible swelling** in the groin extending to the scrotum or labia; irritability and vomiting (may mimic bowel obstruction<sup>220</sup>).
- Common in infancy, especially preterm. **Overlying skin erythema or a toxic child suggests**



## strangulation<sup>220</sup>.

### REDUCE

- If no signs of strangulation: analgesia or sedation, head-down (Trendelenburg) position, gentle sustained manual reduction (taxis) by experienced hands<sup>220</sup>.
- If it does not reduce on the first attempt, stop and consult pediatric surgery — repeated attempts worsen edema and make later reduction harder<sup>220</sup>.
- Once reduced, admit for repair on the same admission — re-incarceration risk is high<sup>220</sup>.

### ⚠ DO NOT REDUCE – STRAIGHT TO SURGERY

Skin erythema, systemic toxicity, peritonitis, bloody stools, or prolonged incarceration — signal ischemic bowel or gonad → emergency exploration<sup>220</sup>.

### ▴ ADMIT · DISCHARGE · REFER

- **Admit** — all → for reduction/observation or surgery.
- **Discharge** — after successful reduction, for early elective repair.

- **Refer** — pediatric surgery urgently; emergency if irreducible or strangulated.

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## Toxicology & Environmental

Poisoning · burns · heat & drowning

### Poisoning & Toxic Ingestion HOURS

Resuscitate, identify the agent, call for advice, give the antidote.

+ Calculators: [Paracetamol & iron](#) · [QTc](#) · [Corrected Na](#) · [anion gap](#) · [Ca](#) · [osmolality](#)

☎ Kuwait Poison Control Center (KPCC):  
**1804774** — 24/7. Call early for any exposure:  
agent-specific advice, decontamination,  
antidotes, and disposition. <sup>34</sup>

#### ▴ GENERAL APPROACH (ABCDE FIRST)

- Stabilize airway, breathing, circulation; give oxygen; check glucose and a 12-lead ECG (QRS width, QT interval).

- Identify the agent, amount, time, and intent; in any deliberate ingestion send **acetaminophen and salicylate levels**.<sup>31</sup>
- Decontamination: activated charcoal 1 g/kg (max 50 g) **only** if airway protected and within ~1 h — not for hydrocarbons, caustics, alcohols, iron, or lithium.<sup>34</sup>
- Call KPCC **1804774**; give the specific antidote below; observe for delayed toxicity.<sup>34</sup>

| Poison                        | Antidote / treatment                                         | Notes                                                                                                                                                                                                                                               |
|-------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetaminophen (acetaminophen) | N-acetylcysteine – per Rumack-Matthew nomogram <sup>31</sup> | IV: 150 mg/kg over 1 h → 50 mg/kg over 4 h → 100 mg/kg over 16 h. The nomogram applies to an <b>acute ingestion with a known time</b> , using a level drawn <b>≥ 4 h</b> after ingestion; not for repeated supratherapeutic ingestion <sup>31</sup> |
| Opioids                       | Naloxone 0.1 mg/kg (max 2 mg)                                | IV/IM/IN, repeat; infusion if needed. <b>Titrate to adequate ventilation, not full arousal</b> ; in a known opioid-dependent patient start with a lower titrated dose to avoid precipitated withdrawal                                              |
| Tricyclic antidepressant      | Sodium bicarbonate 1–2 mEq/kg                                | IV for QRS > 100 ms or arrhythmia                                                                                                                                                                                                                   |
| Beta-blocker                  | Glucagon + high-dose insulin/euglycemia                      | + vasopressors                                                                                                                                                                                                                                      |
| Calcium-channel blocker       | Calcium + high-dose insulin/euglycemia                       | + vasopressors                                                                                                                                                                                                                                      |
| Iron                          | Deferoxamine                                                 | IV for severe toxicity                                                                                                                                                                                                                              |

| Poison                                    | Antidote / treatment                                                                                                                                                                                     | Notes                                                                                                                                                                                                                                             |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organo-phosphate / carbamate              | Atropine + pralidoxime                                                                                                                                                                                   | Titrate atropine to drying of secretions                                                                                                                                                                                                          |
| Sulfonylurea                              | Octreotide + dextrose                                                                                                                                                                                    | For recurrent hypoglycemia                                                                                                                                                                                                                        |
| Methanol / ethylene glycol                | Fomepizole                                                                                                                                                                                               | + urgent dialysis                                                                                                                                                                                                                                 |
| Carbon monoxide                           | 100% oxygen ± hyperbaric                                                                                                                                                                                 | High suspicion with heaters/fires                                                                                                                                                                                                                 |
| Cyanide (smoke inhalation, nitroprusside) | Hydroxocobalamin 70 mg/kg IV/IO (max 5 g), may repeat once; ± sodium thiosulfate 0.25 g/kg IV (max 12.5 g); do not delay oxygenation, airway control, or burn/PICU transfer while arranging the antidote | First-line; give early in fire victims — <b>lactate ≥ 8–10 mmol/L plus compatible clinical findings</b> raises strong suspicion; treatment should not depend on lactate alone. Do not co-infuse hydroxocobalamin and thiosulfate in the same line |

| Poison                           | Antidote / treatment                                                                                                                       | Notes                                                                                                                                                                                                                                                                         |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methemoglobinemia                | Methylene blue 1–2 mg/kg                                                                                                                   | IV. <b>Caution in G6PD deficiency</b> (may be ineffective; can precipitate hemolysis — discuss alternatives with toxicology); <b>serotonergic-drug interaction</b> (serotonin syndrome)                                                                                       |
| Local anesthetic toxicity (LAST) | Lipid emulsion 20%<br>– <70 kg: 1.5 mL/kg bolus over 2–3 min, then 0.25 mL/kg/min;<br>≥70 kg: 100 mL bolus, then 200–250 mL over 15–20 min | Call resuscitation/PICU early and keep 20% lipid emulsion immediately available wherever regional anesthesia is used. Repeat bolus and/or double infusion if unstable; max ~12 mL/kg. Avoid vasopressin and calcium-channel/beta-blockers. epinephrine boluses ≤1 mcg/kg only |

### ⚠️ AVOID

- Routine gastric lavage and ipecac; flumazenil (seizure risk) is generally avoided.<sup>34</sup>
- Several agents have **delayed or biphasic** toxicity (acetaminophen, iron, sustained-release products,

sulfonylureas) — observe and recheck.<sup>31</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — symptomatic, a significant/unknown ingestion, or a delayed-toxicity agent (observe).
- **Discharge** — asymptomatic after the appropriate observation window with a non-toxic exposure.
- **Refer** — the poison center/toxicology; psychiatry for intentional ingestion; safeguarding as needed.

## Caustic (Corrosive) Ingestion HOURS

Acid or alkali burns — protect the airway; do no harm.

#### ▲ RECOGNIZE

- Drooling, dysphagia, painful swallowing, oral or lip burns, chest or abdominal pain. Stridor or respiratory distress means airway injury.

- The **absence of oral burns does not exclude esophageal injury**. Alkali (drain cleaner) causes deep liquefactive necrosis; acid causes coagulative necrosis.<sup>270</sup>

**⚠ DO NOT**

No induced emesis, no gastric lavage, **no activated charcoal**, no neutralizing or forced dilution — these re-expose the esophagus and worsen injury.<sup>270</sup>

**DO**

- Airway first — anticipate swelling and intubate early if any compromise (by anesthesia/ENT/PICU)
- Avoid blind nasogastric-tube insertion unless directed after endoscopic/surgical assessment.
- Keep nothing by mouth; give IV fluids, analgesia, antiemetics.
- Chest and abdominal X-ray for free air. **If esophageal perforation is suspected (free air, severe chest or abdominal pain, or sepsis), start IV fluids and consult pediatric surgery immediately.** Contact poison control.<sup>270</sup>



### ◆ ENDOSCOPY

- Upper endoscopy within **12-24 H**, once airway secure, grades the injury; later than 48 h raises perforation risk.<sup>270</sup>
- Watch for delayed strictures over the following weeks.
- A swallowed button battery is a separate emergency<sup>25</sup> — see **Button (Lithium) Battery Ingestion.**

### ▴ ADMIT · DISCHARGE · REFER

- Disposition depends on agent, amount, intent, symptoms and poison-center/GI assessment, and on the ability to tolerate oral intake.
- **Admit** — all symptomatic or significant exposures → nil by mouth, plan imaging/endoscopy.
- **Discharge** — asymptomatic minor exposure after observation.
- **Refer** — ENT/GI/surgery (endoscopy) and PICU; do not blind-pass a nasogastric tube.

# Burns

HOURS

Cool it, size it, replace fluid to urine output — and protect the airway.

## + Calculator: Burns TBSA & fluid

### ▲ ASSESS

- Estimate % total body surface area (TBSA) with a Lund-Browder chart (the child's palm including fingers  $\approx$  1%); assess depth and mechanism.<sup>271</sup>
- Suspect **inhalation injury** (facial burns, soot, hoarseness/stridor, enclosed-space fire) → early airway/anesthetic involvement<sup>143</sup> (see Smoke Inhalation & Carbon Monoxide Poisoning).

### FIRST AID

Stop the burning; cool with running water for 20 min (**not** ice); remove clothing/jewelry; cover; analgesia.<sup>271</sup>



#### FLUIDS (TBSA > 10%)

- Modified Parkland — Lactated Ringer's **3–4 mL/kg × %TBSA** over 24 h,  $\frac{1}{2}$  in the first 8 h from the time of burn, **plus** maintenance (4-2-1, with dextrose in young children)
- 20 mL/kg bolus if shocked.
- Titrate to urine output  $\geq 1$  mL/kg/h in children.<sup>271</sup>



#### REFER TO A BURNS CENTER

**> 10% TBSA, full-thickness, face/hands/feet/perineum/joints, electrical or chemical, inhalation injury, or suspected inflicted injury.**<sup>271</sup>

#### ⚠ AVOID OVER-RESUSCITATION ("FLUID CREEP")

- The formula is only a starting estimate — titrate to urine output, not to the number.<sup>271</sup>
- Don't apply ice or home remedies.

#### ⚠ RAISE SUSPICION FOR INFLECTED INJURY

- A significant share of pediatric burns are inflicted (up to ~1 in 5 in burns-unit series).<sup>168</sup>

- Red flags: a **well-demarcated** burn with **uniform depth** and **no splash marks**; a symmetric immersion ("stocking/glove") pattern with a clear upper margin and skin-fold sparing; burns in normally **protected sites** (groin, perineum, buttocks, back, thighs); delayed presentation or a history that does not fit the burn pattern or the child's developmental stage.
- Look for other injuries and old fractures; safeguard.<sup>168</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — large or deep burns, airway/inhalation injury, electrical/chemical burns, or a safeguarding concern → burns center.
- **Discharge** — small superficial burns with dressing and review.
- **Refer** — a burns unit per criteria; ophthalmology/surgery as needed.

# Heat Stroke

MINUTES

A clinical diagnosis — core  $\geq 40^{\circ}\text{C}$  + CNS dysfunction after heat exposure. Cool fast; antipyretics do not work.

## DIAGNOSE CLINICALLY

- **Core temperature  $\geq 40^{\circ}\text{C}$  with CNS dysfunction** after environmental heat exposure.<sup>272</sup>
- **CNS:** confusion, delirium, irritability, hallucinations, ataxia, slurred speech, seizures, coma. **CVS:** tachycardia, hypotension, dehydration/shock. **GI:** nausea, vomiting ( $\pm$  diarrhea). **Other:** hot skin, tachypnea.

## INVESTIGATIONS – DO NOT DELAY COOLING

RBS (hypoglycemia) · CBC · electrolytes (Na/K/Ca) · urea & creatinine · LFTs · coagulation (DIC) · blood gas (metabolic acidosis  $\pm$  early respiratory alkalosis) · creatine kinase (rhabdomyolysis) · urinalysis (myoglobinuria).<sup>272</sup>

## ▀ MANAGEMENT

- **1 · ABCDE** — secure airway, support breathing and circulation.
- **2 · Immediate rapid cooling (do not delay). Cold-water immersion** preferred if feasible (best for exertional heat stroke). Otherwise **evaporative cooling** — remove clothing, spray/mist tepid water, high-flow fans, mesh stretcher/hammock — plus **ice packs to neck, axillae, and groin**. Room-temperature 0.9% saline for dehydration/shock (avoid overload). **Continuous rectal core-temperature monitoring.** <sup>272</sup>
- **3 · Stop active cooling at core 38.5–39°C** to avoid overshoot hypothermia. <sup>272</sup>
- **4 · No antipyretics** (acetaminophen/ibuprofen) — ineffective in heat stroke and may worsen liver, renal, or GI injury. <sup>272</sup> Treat shivering cautiously with IV midazolam or lorazepam; treat seizures per the pediatric status epilepticus protocol (see **Convulsive Status Epilepticus**).

### ⚠ **DISPOSITION**

- Children with true heat stroke (CNS dysfunction) go to a PICU or high-dependency/PICU-capable setting depending on severity and organ injury — thermal instability and organ injury (rhabdomyolysis, AKI, DIC) can evolve over days.<sup>272</sup>
- High-yield in Kuwait's climate.

### ▀ **ADMIT · DISCHARGE · REFER**

- **Admit** — all with CNS dysfunction or organ injury → high-dependency/PICU.
- **Discharge** — mild heat exhaustion, resolved and hydrated with normal mentation.
- **Refer** — PICU; nephrology for rhabdomyolysis/AKI.

## **Drowning** **MINUTES**

Respiratory impairment from submersion/immersion — ventilate first; rescue breaths are the key intervention.

**+ Calculator: PARDS oxygenation index**

## ▀ INITIAL MANAGEMENT (ABCDE)

- **Airway & breathing:** start **rescue breathing immediately** (the single most important intervention); give 100% oxygen; support ventilation as needed. **A Heimlich maneuver and postural drainage have no role in drowning** — do not perform them; they delay ventilation and risk aspiration.<sup>170</sup>
- **Circulation:** check pulse; if no pulse, start CPR per the **PALS cardiac-arrest algorithm**. In the ED, **check the rhythm and give electrical therapy with a manual defibrillator** if the rhythm is shockable (an AED is a prehospital substitute).<sup>170</sup>
- **Temperature:** remove wet clothing; if core < 35°C, prevent further heat loss and begin active rewarming.<sup>170</sup>
- **Post-ROSC:** manage per PALS post-cardiac-arrest care.



**OBSERVE ALL – 6–8 H, OR PER THE  
LOCAL PEDIATRIC DROWNING PATHWAY /  
RISK STRATIFICATION**

- Every submersion patient, even if asymptomatic: continuous pulse oximetry with serial respiratory and neurological exams<sup>170</sup> (delayed pulmonary edema can appear within 4–8 h)
- No routine chest X-ray or laboratory tests — reserve for symptomatic or hypoxic children.<sup>170</sup>

**STAYS WELL**

Asymptomatic with a normal exam, normal vitals, and normal SpO<sub>2</sub> on room air at 6–8 h → **discharge** with return precautions (cough, breathing difficulty, vomiting, fever, or lethargy).<sup>170</sup>

**SYMPTOMS / HYPOXIA**

Any respiratory symptoms, an oxygen requirement, or altered mental status → **admit**; give oxygen and support and obtain a chest X-ray / blood gas as indicated.

### ▲ ADMIT

- Respiratory distress or increased work of breathing
- persistent or worsening symptoms
- altered mental status
- need for supplemental oxygen or ventilatory support.

### ◆ NOT RECOMMENDED

- Heimlich maneuver
- postural drainage
- routine chest X-ray for discharge decisions
- glucocorticoids
- routine antibiotics
- routine cervical-spine immobilization unless trauma (e.g., a diving injury) is suspected.<sup>170</sup>

### ▀ ADMIT · DISCHARGE · REFER

- **Admit** — any symptomatic patient, or asymptomatic with risk features → observe.
- **Discharge** — asymptomatic with a normal exam and SpO<sub>2</sub> on room air after 6–8 h observation, with return precautions.

- **Refer** — PICU for respiratory compromise; safeguarding as needed.

## Smoke Inhalation & Carbon Monoxide Poisoning

**MINUTES**

FIRE / SMOKE

Fire injures three ways — airway, carbon monoxide, cyanide. Look for and treat all three.

**+ Calculators:** [Burns TBSA & fluid](#) · [PARDS oxygenation index](#)

### ▴ DEFINE

- Fire/smoke injures by three mechanisms: **(1) thermal / chemical airway injury, (2) carbon monoxide (CO), (3) hydrogen cyanide (HCN).** Suspect all three after any **enclosed-space fire**, especially with altered mental status.<sup>143</sup>

### ▴ APPROACH – THE AIRWAY FIRST

- Signs of inhalation injury: **facial/neck burns, singed nasal hairs, soot in the mouth or sputum,**

**carbonaceous sputum, hoarseness, stridor, oropharyngeal edema or blistering.**

- Airway injury is **progressive** — edema worsens over hours. Reassess continuously.<sup>143</sup>

#### **⚠ SECURE THE AIRWAY EARLY**

- Deep facial/neck burns, oropharyngeal blistering or edema, hoarseness/stridor, or large cutaneous burns → **intubate early** (call PICU/anesthesia urgently — expert airway control) before edema makes the airway impassable.<sup>143</sup>
- Avoid over-aggressive IV fluids — they worsen airway edema.<sup>143</sup>

## **CARBON MONOXIDE**

### **▀ RECOGNIZE & MEASURE**

- CO binds hemoglobin ~**200–250× more avidly than oxygen** → tissue hypoxia plus direct cellular toxicity; **symptoms and COHb level correlate poorly.**<sup>141</sup>
- **Pulse oximetry reads falsely normal** (it cannot distinguish carboxyhemoglobin) — measure

**COHb by co-oximetry** (blood gas).  
COHb > 2% (non-smoker) or > 9% (smoker) supports the diagnosis.<sup>141</sup>

- Symptoms: headache, nausea, dizziness, confusion, syncope, seizure, cardiac ischemia, coma.

#### ◆ **HYPERBARIC OXYGEN (HBO) – AVAILABLE LOCALLY**

- High-flow **100% OXYGEN VIA A NON-REBREATHAR** (or ventilator) is first-line and cuts COHb half-life from ~4–5 h to ~60–90 min.
- With a chamber on site, **CONSIDER HBO** for severe poisoning: **COHB > 25%** (> 20% in pregnancy), **LOSS OF CONSCIOUSNESS/SYNCOPE**, neurologic signs, **CARDIAC ISCHEMIA**, severe metabolic acidosis, or **PREGNANCY** — ideally within 6 h.<sup>139,142</sup>
- HBO speeds elimination and may reduce delayed neurocognitive sequelae; trial evidence is mixed, so weigh case by case.<sup>142</sup>
- In children benefit is extrapolated from adult trials (under-16s excluded from the pivotal RCT); let CO exposure duration, COHb level, neurologic signs, and lactate guide the decision.<sup>141</sup>

### ⚠️ CYANIDE – DON'T MISS IT

- Suspect HCN co-poisoning in an enclosed-space fire with soot, **altered mental status, hemodynamic instability, or severe lactic acidosis (lactate  $\geq$  8–10 mmol/L).**
- Treat **empirically — do not wait for levels.**<sup>118</sup>
- Use **hydroxocobalamin**; avoid nitrite-based kits in smoke victims (induced methemoglobinemia worsens hypoxia).<sup>118</sup>

## MANAGEMENT

| Intervention                         | Dose                                  | Notes                                                                                                                                                                         |
|--------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oxygen                               | 100% via non-rebreather (or intubate) | Continue until asymptomatic and COHb normal. <sup>139</sup>                                                                                                                   |
| Hydroxocobalamin (suspected cyanide) | 70 mg/kg/dose IV (max 5 g)            | Empiric in smoke inhalation with shock, or lactate $\geq$ 8–10 mmol/L plus compatible clinical findings (do not rely on lactate alone); repeat once if needed. <sup>118</sup> |

| Intervention       | Dose                         | Notes                                                               |
|--------------------|------------------------------|---------------------------------------------------------------------|
| HBO<br>(severe CO) | 2.5–3.0 ATA,<br>per protocol | Refer to the hyperbaric unit for the criteria above. <sup>139</sup> |

#### ▲ ALSO

- Cardiac monitoring + ECG, serial lactate and COHb, cautious fluid resuscitation; ICU/burns referral.
- See also **Burns** and **Poisoning & Toxic Ingestion** for concurrent burn and antidote management.

#### ✦ PEARL – THE OXIMETER LIES

- A normal SpO<sub>2</sub> does not exclude CO poisoning — measure carboxyhemoglobin.
- In a fire victim with shock and a high lactate, give hydroxocobalamin without waiting for confirmation.<sup>118</sup>

#### ▲ ADMIT · DISCHARGE · REFER

- **Admit** — respiratory symptoms, altered mentation, or a significant carboxyhemoglobin → oxygen and monitoring.
- **Discharge** — asymptomatic with normal levels after observation.

- **Refer** — hyperbaric/PICU for severe CO; burns/ENT for airway injury.

## Envenomation – Scorpion Sting & Snakebite

**MINUTES**

WHO 2016 • regional

Two regional bites: scorpion = an autonomic storm; snake = local tissue injury and coagulopathy.

### ▲ SCORPION STING

- Severe local pain, then an **autonomic storm**: sweating, salivation, vomiting, hypertension and tachycardia, agitation, and in severe cases **pulmonary edema, myocarditis and shock** (young children at highest risk).<sup>273</sup>
- Supportive care and analgesia; **prazosin** for autonomic/cardiovascular toxicity; benzodiazepines for agitation. Give **antivenom** for systemic envenomation per local protocol.<sup>273</sup>



## ▲ SNAKEBITE

- Look for **local** swelling/necrosis and **systemic** effects — coagulopathy/bleeding, neurotoxicity (ptosis, bulbar/respiratory weakness) or rhabdomyolysis, depending on species.<sup>274</sup>
- First aid: reassure; **immobilize/splint the bitten limb and minimize movement**; remove constrictions; move to hospital. Give species-appropriate **antivenom** for systemic envenomation and monitor the airway and clotting.<sup>274</sup>

## ⚠ DON'T MAKE IT WORSE

- **No** tourniquets, incision, suction or ice on snakebites — they increase harm.<sup>274</sup>
- Watch for delayed coagulopathy and compartment syndrome; recheck clotting serially.
- Anticipate **anaphylaxis to antivenom** and have epinephrine ready.<sup>274</sup>

◆ **LOCAL** Use the antivenom stocked for the species in your region and follow the

national/Poison-Control protocol; contact the Poison Control Center early for identification and dosing.

#### ▲ WORKUP

- **Snakebite: 20-minute whole-blood clotting test**, CBC/platelets, PT/INR + fibrinogen, D-dimer, renal function + potassium, CK, urinalysis (myoglobin), ECG, **serial limb marking**, group & crossmatch.<sup>274</sup>
- **Scorpion:** continuous cardiac/BP monitoring, ECG, **troponin + echocardiogram** if cardiac signs, glucose, electrolytes, blood gas (pulmonary edema).<sup>273</sup>

#### ▲ ADMIT · REFER

- **Admit/observe** all systemic envenomation and any child with progressing local signs.
- **Call** critical care and Poison Control; surgery for compartment syndrome/necrosis.



# Quick Reference

Weight · airway · vital signs · master doses

## Time-critical emergencies – quick reference

Solid bar = knowable clock / deadline · dashed bar = unknown onset / prognosis. See each condition's card for detail.

### The Golden Window

What must be done — and by when — for the guide's time-critical emergencies. The intervention above each bar is what the window governs.

• Red "now" = a step that starts immediately, before any window applies (e.g. 100% O<sub>2</sub> in CO poisoning).

■ Solid = clock starts at a knowable moment (recognition / arrival) — a real deadline to beat.

■ ■ ■ Dashed = onset usually unknown — the window is prognostic; act on suspicion, do not treat if it's a countdown.



Solid bar windows are consensus targets for starting the named interventions; dashed bar figures are prognostic. Always act as early as possible. Immediate ABC resuscitation is assumed for every row. Built from the guide's own time-critical cards.

## When to Admit

General triggers first, then condition-specific thresholds.

+ Calculator: Westley croup score

## ▲ ANY ACUTELY UNWELL CHILD

Toxic or septic appearance; shock or unstable vital signs; hypoxia ( $\text{SpO}_2 < 92\%$ ); unable to tolerate oral fluids or medicines; moderate to severe dehydration; fever under 3 months; diagnostic uncertainty in an unwell child; or unreliable follow-up.

| Condition                            | Admit if                                                                                                                                                                                                               |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Febrile young infant</b>          | Any under 28 days; 29–90 days unless low-risk on screening; ill-appearing at any age.                                                                                                                                  |
| <b>UTI</b>                           | Under 3 months; toxic; vomiting or unable to take oral; no response by 48 h or a resistant organism; known uropathy, obstruction, chronic kidney disease, or transplant; suspected renal abscess; acute kidney injury. |
| <b>Gastroenteritis / dehydration</b> | Shock or severe dehydration; failed oral or nasogastric rehydration; persistent vomiting; significant electrolyte disturbance; under 6 months with heavy losses; possible surgical abdomen.                            |
| <b>Bronchiolitis</b>                 | Apnea; $\text{SpO}_2$ persistently under 90–92%; feeding under 50–75% or dehydration;                                                                                                                                  |

severe respiratory distress;  
under 3 months or  
comorbidity (prematurity,  
congenital heart disease,  
chronic lung disease).

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**Croup**

Stridor at rest persisting after  
a corticosteroid; repeated  
nebulized L-epinephrine;  
hypoxia, exhaustion, or toxic  
appearance.

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**Asthma**

Incomplete response to  
bronchodilators; ongoing  
oxygen need; needing  
bronchodilator more often  
than 1–2 hourly; silent chest,  
exhaustion, or altered  
consciousness (→ critical  
care).

---

**Pneumonia**

SpO<sub>2</sub> under 92%; moderate to  
severe distress; under 3–6  
months; not tolerating oral;  
dehydration; effusion or  
empyema; or failed oral  
antibiotics.

---

**Nephrotic  
syndrome**

First presentation;  
hypovolemia (abdominal pain,  
cool peripheries, hypotension,  
oliguria); peritonitis or sepsis;  
thromboembolism; severe  
edema with respiratory  
compromise; acute kidney  
injury; relapse not responding  
to oral steroids.

---

## BRUE

Under 2 months; prematurity; more than one event; event needed CPR or vigorous stimulation; or a concerning history or examination (higher-risk).

## Rabies Post-Exposure Prophylaxis

HOURS

WHO 2018

Dog, cat, or bat bite/scratch — wash, risk-stratify, give vaccine ± immunoglobulin same day.

### ▲ FIRST – WOUND CARE (MOST IMPORTANT)

- Wash/flush the wound **~15 minutes** with soap and running water, then apply a virucidal (povidone-iodine or 70% alcohol).<sup>275</sup>
- Avoid early suturing; if unavoidable, suture loosely **after** immunoglobulin infiltration.<sup>275</sup>
- Give **tetanus prophylaxis**; add antibiotics for a deep or contaminated bite (e.g., amoxicillin-clavulanate).

| WHO category | Exposure                                                                                                               | Action                                       |
|--------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| I            | Touching/feedingNo PEP — wash only licks on intact skin                                                                |                                              |
| II           | Nibbling of uncovered skin; minor scratch/abrasion without bleeding                                                    | Wound care + <b>vaccine</b>                  |
| III          | Transdermal bite/scratch, lick on broken skin, mucous-membrane contamination, or <b>any bat contact</b> <sup>275</sup> | Wound care + <b>vaccine + immunoglobulin</b> |

#### **VACCINE & IMMUNOGLOBULIN**

- **Never previously vaccinated** — cell-culture vaccine IM on days **0, 3, 7, 14** (add day 28 if immunocompromised); the WHO 1-week 2-site intradermal schedule (days 0, 3, 7) is an accepted dose-sparing option.<sup>275</sup>
- **Previously fully vaccinated** — 2 doses only (days 0 and 3); **no immunoglobulin.**<sup>275</sup>

- **Rabies immunoglobulin** (category III, or category II if immunocompromised) — human RIG **20 IU/kg** (equine 40 IU/kg) as a single day-0 dose; infiltrate as much as possible into and around the wound, give any remainder IM at a site distant from the vaccine.<sup>275</sup>
- Repeat exposure within 3 months of a complete course → wound care only (no vaccine or RIG).<sup>275</sup>

**⚠ DO NOT DELAY**

- Start PEP the same day — rabies is almost always fatal once symptomatic.<sup>275</sup>
- Immunoglobulin can blunt live vaccines (separate MMR/varicella by ~4 months).<sup>275</sup>
- Follow the Kuwait MOH / local rabies protocol and notify public health.



## IV → Oral Switch & the DIVA Patient

### ASSESS

In difficult access, every lost cannula is a switch decision — not an automatic re-stick.

#### WHY IT MATTERS IN DIVA

- In **difficult intravenous access (DIVA)**, each re-cannulation means pain, delay, distress, and line risk. Reassess switch eligibility at **every line failure** rather than reflexively re-sticking — for many infections early switch is as effective as completing the IV course.<sup>95</sup>

#### DIVA SCORE – PREDICT DIFFICULT ACCESS

| Variable (add the points)            | Points    |
|--------------------------------------|-----------|
| History of prematurity               | 3         |
| Age: < 1 yr / 1–2 yr / ≥ 3 yr        | 3 / 1 / 0 |
| Vein not palpable (after tourniquet) | 2         |
| Vein not visible (after tourniquet)  | 2         |

#### ◆ INTERPRETATION

- Score  $\geq 4$  (max 10)  $\rightarrow$  > 50% chance of first-attempt cannulation failure.<sup>95</sup>
- Use it early: escalate to an ultrasound-guided / senior inserter or vein-finder — and in a child improving and able to take oral, favor an **IV  $\rightarrow$  ORAL SWITCH** over repeated sticks.
- The simpler **DIVA-3** (age + vein visibility + palpability, dropping prematurity) performs as well.<sup>95</sup>

#### ▲ SWITCH CRITERIA – ALL SHOULD BE MET (COMS)

- **Clinically** improving and responding to treatment.
- **Oral** route works — functioning gut, tolerating intake, not vomiting, no malabsorption.
- **Markers** settling — defervescing, hemodynamically stable, no ongoing sepsis ( $\pm$  falling CRP/white count).
- **Suitable** oral agent — adequate spectrum and bioavailability.

#### DECISION

**Deep-seated / high-risk infection?**

## SWITCH EARLY

Uncomplicated CAP, UTI/pyelonephritis, uncomplicated skin/soft-tissue infection, and acute osteomyelitis/septic arthritis (early oral non-inferior, with fewer line complications).

## KEEP IV

Bacterial meningitis/brain abscess, endocarditis, *S. aureus* bacteremia, febrile neutropenia (until low-risk/recovery criteria), undrained deep abscess, septic shock until stabilized.

### ◆ LEAN ON HIGH ORAL-BIOAVAILABILITY AGENTS

- Near-IV exposure, ideal in DIVA: ciprofloxacin, levofloxacin, linezolid, metronidazole, fluconazole, trimethoprim-sulfamethoxazole, clindamycin, doxycycline, rifampicin.
- Beta-lactams (e.g., high-dose amoxicillin) are usable but dose adequately for their lower oral absorption.

### ⚠️ **DIVA RULE OF THUMB**

- Do not place a PICC or midline solely to finish an IV course that an oral agent could complete.
- Weigh the procedural burden of central access against the switch criteria above.

## Weight & Airway Sizing

Quick estimates — a length-based tape is more reliable.

**+ Calculators:** [ETT size & depth](#) · [Resuscitation dose calculator](#)

| Estimate                     | Formula                                |
|------------------------------|----------------------------------------|
| Weight (1–5 yr)              | $(\text{age} + 4) \times 2 \text{ kg}$ |
| Weight (6–12 yr)             | $(\text{age} \times 3) + 7 \text{ kg}$ |
| Endotracheal tube (uncuffed) | $(\text{age} \div 4) + 4 \text{ mm}$   |
| Endotracheal tube (cuffed)   | $(\text{age} \div 4) + 3.5 \text{ mm}$ |
| Tube depth at lip            | $\text{tube size} \times 3 \text{ cm}$ |
| Hypotension (< 1 yr)         | $\text{systolic BP} < 70 \text{ mmHg}$ |

Hypotension (1–10 yr)

systolic BP < 70 +  
(2 × age) mmHg

Hypotension (> 10 yr)

systolic BP < 90  
mmHg

#### ◆ GROWTH CHARTS

- Plot weight, length/height, and head circumference — [WHO standards](#)

[\(birth–5 yr\)](#)



- [CDC charts \(2–19 yr\)](#)



## Oxygen Delivery Systems & Cylinder Duration

Device → flow → approximate  $\text{FiO}_2$  → when to escalate; and how long a cylinder will last.

+ Calculator: [Oxygen cylinder duration](#)

## OXYGEN DELIVERY SYSTEMS

| Device                                                       | Flow · FiO <sub>2</sub>                                                                          | Notes & escalation                                                                                                                                                                                                           |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LOW</b><br>Nasal cannula (low-flow)                       | Infant 1–2 L/min<br>Child up to 4 L/min<br>Older child up to 6 L/min<br>FiO <sub>2</sub> ~24–45% | Variable — FiO <sub>2</sub> rises with flow, falls with high inspiratory demand / mouth-breathing. Above ~2 L/min the dry gas irritates and dries the nasal mucosa — <b>add humidification</b> (and warming) at these flows. |
| <b>LOW</b><br>Simple face mask                               | 5–10 L/min<br>FiO <sub>2</sub> ~40–60%                                                           | <b>Minimum 5 L/min</b> to flush CO <sub>2</sub> (avoid rebreathing). Variable performance.                                                                                                                                   |
| <b>HIGH O<sub>2</sub></b><br>Non-rebreather (reservoir) mask | 10–15 L/min keep the bag inflated<br>FiO <sub>2</sub> ~60–90%                                    | Highest FiO <sub>2</sub> without support — <b>resuscitation / emergency</b> .                                                                                                                                                |
| <b>FIXED</b><br>Venturi (air-entrainment) mask               | 2–15 L/min per valve<br><b>fixed FiO<sub>2</sub></b> 24 / 28 / 31 / 35 / 40 / 60%                | Delivers a <b>precise FiO<sub>2</sub></b> regardless of breathing pattern — when a controlled FiO <sub>2</sub> matters.                                                                                                      |

| Device                                             | Flow · FiO <sub>2</sub>                                                                   | Notes & escalation                                                                                                                                                                                                                             |
|----------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SUPPORT</b><br>High-flow w nasal cannula (HFNC) | ≈ 2 L/kg/min<br>(to device max, up to ~60 L/min)<br>FiO <sub>2</sub> 21–100% (titratable) | Warmed & humidified; adds <b>1–3 cmH<sub>2</sub>O PEEP</b> + deadspace washout.<br>Can be run on the <b>ward where a clear written escalation protocol exists</b> (defined limits, monitoring, and criteria to step up); otherwise HDU / PICU. |
| <b>SUPPORT</b><br>CPAP / BiPAP (NIV)               | Device-set flow<br>set FiO <sub>2</sub>                                                   | Non-invasive support for rising work of breathing, recurrent apnea, or hypoxaemia despite O <sub>2</sub> .<br><b>PICU-directed.</b>                                                                                                            |

### 🕒 READING THE TABLE

- LOW** low-flow, **VARIABLE** FiO<sub>2</sub> · **HIGH O<sub>2</sub>** / **FIXED** high or **CONTROLLED** FiO<sub>2</sub> · **SUPPORT** also provides **PRESSURE / WASHOUT** → escalation.
- TARGET SPO<sub>2</sub>**: give O<sub>2</sub> for SpO<sub>2</sub> < 90% (consider < 92% in young infants, prematurity, or cardiac / lung disease); titrate to target and wean.
- INFANTS**: if prongs/mask are not tolerated use **BLOW-BY**; a **HEAD BOX / HOOD** (5–15 L/min) gives up to ~80–90% FiO<sub>2</sub>. Neonatal resuscitation:

T-piece or self-inflating bag **WITH RESERVOIR**.

- Low-flow  $\text{FiO}_2$  figures are estimates — the faster the child breathes, the lower the true delivered  $\text{FiO}_2$ .

## READING THE PULSE OXIMETER – PRACTICAL NOTES

### ▶ GET A RELIABLE NUMBER

- **Trust it only with a good trace.**  
Confirm a clean plethysmograph **waveform** and that the oximeter's **pulse rate matches** the palpated / ECG rate — a damped or absent trace means the number is unreliable.
- **Weak signal = poor perfusion.** Cold, shock, vasoconstriction or hypotension drop the signal (under-reads or fails) — warm the hand, try another site (toe, earlobe), check the perfusion index if shown.
- **Motion and bright / ambient light** cause erratic or false readings — settle the limb, shield the probe, use motion-tolerant sensors.
- **Nail polish, henna, dirt** — remove them or turn the probe sideways. In **darker skin** oximeters can **overestimate** true saturation (occult



hypoxemia) — correlate with the clinical picture and a blood gas if in doubt.

**⚠ WHEN THE NUMBER LIES – FALSELY REASSURING**

- **Carbon monoxide poisoning:** a standard oximeter reads **carboxyhemoglobin as oxyhemoglobin**, so SpO<sub>2</sub> looks **normal / high despite real hypoxia**. Suspect it by history; confirm with **co-oximetry / ABG** and give **100% O<sub>2</sub> regardless of the SpO<sub>2</sub>**.
- **Methemoglobinemia** pins SpO<sub>2</sub> around **~85%** whatever the true saturation (a "saturation gap" vs the ABG) — suspect it with cyanosis **unresponsive to O<sub>2</sub>** (topical anesthetics, dapsone).
- **Severe anemia / blood loss:** SpO<sub>2</sub> measures **saturation, not oxygen content** — it can read normal while oxygen delivery is inadequate.

### ✦ **SPO<sub>2</sub> IS NOT PAO<sub>2</sub> – AND IT LAGS**

- On the dissociation curve, **SpO<sub>2</sub> 90% ≈ PaO<sub>2</sub> 60 mmHg**; **below 90%** a small SpO<sub>2</sub> fall means a **large PaO<sub>2</sub> drop** — act quickly. The curve is **flat near the top**, so “100%” hides the reserve and **cannot detect hyperoxia** (important in preterm / neonates).
- SpO<sub>2</sub> **lags** real change by seconds (longer if poorly perfused) — a falling number can already be a **late** sign.
- **Neonates — pre- vs post-ductal:** right hand = pre-ductal, foot = post-ductal; a difference **> 3%** (or lower post-ductal) suggests a right-to-left shunt / **critical CHD** (CCHD screening).

### **OXYGEN CYLINDER – CONTENT & HOW LONG IT LASTS**

| Step                          | Equation                                              |
|-------------------------------|-------------------------------------------------------|
| Usable duration (quick · psi) | (gauge psi – 200) × cylinder factor ÷ flow (L/min)    |
| Usable content (metric)       | capacity (L) × (gauge bar – 14) ÷ full pressure (bar) |

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| Duration<br>(metric) | $\frac{\text{usable content (L)}}{\text{flow (L/min)}}$ |
|----------------------|---------------------------------------------------------|

### ⊙ WHAT THE TERMS MEAN

- **PSI** = pounds per square inch, a unit of gas pressure (what the cylinder gauge reads); **1 BAR ≈ 14.5 PSI**.
- **CYLINDER FACTOR** = litres of oxygen released per 1 psi of pressure (= the cylinder's full capacity ÷ its full pressure). Multiply the **USABLE** pressure (gauge – 200 psi safe residual) by this factor to get the litres available.

### ⊙ SAFE RESIDUAL & CYLINDER DATA

- **SAFE RESIDUAL ≈ 200 PSI (~14 BAR)** — a cylinder is treated as **EMPTY** at this pressure; **SUBTRACT IT FROM THE GAUGE** before calculating, and do not run lower.
- **CONTENT WHEN FULL (BOC):** CD **460 L** (230 bar) · D **340 L** (137 bar) · E **680 L** (137 bar) · F **1360 L** (137 bar). US E ≈ 625–680 L at ~2000 psi. 1 bar ≈ 14.5 psi. **CHECK THE LABEL – CAPACITY/PRESSURE VARY BY SUPPLIER.**
- **CYLINDER FACTORS (L/PSI):** D 0.16 · E 0.28 · M 1.56 · G 2.41 · H/K 3.14.

### ✦ WORKED EXAMPLES

- **Imperial:** E cylinder, gauge **1800 psi**, flow **6 L/min** →  $(1800 - 200) \times 0.28 \div 6 \approx$  **75 min**.
- **Metric:** E cylinder, gauge **100 bar**, flow **4 L/min** →  $680 \times ((100 - 14) \div 137) \approx$  **427 L** →  $\div 4 \approx$  **107 min**.
- **High flow drains fast:** at 30–60 L/min a full E cylinder lasts only **~10–20 min** — use wall O<sub>2</sub> or a large cylinder (and an air blender) for transport.
- An **interactive calculator** is in the **Bedside Calculators** section (app / web).

### ⚠ SAFETY MARGIN

Do not use a cylinder below the **safe residual (~200 psi / ~14 bar)** — treat it as empty. Change or refill at about **~500 psi / ~35 bar** to keep a working reserve, and for transport carry  $\geq 2\times$  the calculated requirement (account for device / blender flush).

### 🕒 BEST CLINICAL RESOURCES (OPEN-ACCESS)

- **OPEN CRITICAL CARE – OXYGEN DELIVERY DEVICES** (evidence-based,

WHO-aligned): [opencriticalcare.org](https://opencriticalcare.org)



- **WHO – OXYGEN THERAPY FOR CHILDREN**

(2016 manual): [who.int](https://who.int)



## Normal Vital Signs by Age

Reference ranges (awake, at rest).

| Age           | Heart rate /min | Respiratory rate /min | Systolic BP mmHg |
|---------------|-----------------|-----------------------|------------------|
| Newborn       | 100–180         | 40–60                 | 60–90            |
| Infant < 1 yr | 100–160         | 30–53                 | 72–104           |
| 1–2 yr        | 90–150          | 22–37                 | 86–106           |
| 3–5 yr        | 80–140          | 20–28                 | 89–112           |
| 6–11 yr       | 70–120          | 18–25                 | 97–120           |
| 12+ yr        | 60–100          | 12–20                 | 110–131          |

◆ **BLOOD-PRESSURE PERCENTILES** Age-, sex-, and height-based BP centiles (elevated BP  $\geq$  90th; hypertension  $\geq$  95th) — [AAP 2017](#)

[BP tables](#)



## Scales, Equations & Synonyms

The scores, formulas, and drug names you reach for at the bedside.

+ Calculators: [Corrected Na](#) · [anion gap](#) · [Ca](#) · [osmolality](#) · [QTc](#) · [eGFR \(Schwartz\)](#) · [Glucose infusion rate](#) · [Glasgow Coma Scale](#) · [BSA & hyperhydration](#)

Glasgow Coma Scale — child values in brackets (best 15, worst 3)

| Pts | Eye         | Verbal<br>(<2 y)                | Motor (infant)                    |
|-----|-------------|---------------------------------|-----------------------------------|
| 6   | —           | —                               | Obeys<br>(spontaneous movement)   |
| 5   | —           | Oriented<br>(coos,<br>babbling) | Localizes<br>(withdraws to touch) |
| 4   | Spontaneous | Confused<br>(irritable cry)     | Withdraws to pain                 |

|          |          |                             |                  |
|----------|----------|-----------------------------|------------------|
| <b>3</b> | To voice | Wrong words (cries to pain) | Abnormal flexion |
| <b>2</b> | To pain  | Sounds only (moans)         | Extension        |
| <b>1</b> | None     | None                        | None             |

APGAR — score at 1 and 5 minutes ( $\leq 3$  low, 4–6 moderate, 7–10 reassuring)

| <b>Sign</b>            | <b>0</b>    | <b>1</b>              | <b>2</b>    |
|------------------------|-------------|-----------------------|-------------|
| <b>Appearance</b>      | blue / pale | body pink, limbs blue | all pink    |
| <b>Pulse</b>           | absent      | < 100                 | $\geq 100$  |
| <b>Grimace</b>         | none        | grimace               | cry / cough |
| <b>Activity (tone)</b> | limp        | some flexion          | active      |
| <b>Respiration</b>     | absent      | slow, irregular       | strong cry  |

Dehydration — severe means shock; give 20 mL/kg

| <b>Sign</b>    | <b>Mild ~3–5%</b> | <b>Moderate ~6–9%</b> | <b>Severe <math>\geq 10\%</math></b> |
|----------------|-------------------|-----------------------|--------------------------------------|
| <b>General</b> | alert, thirsty    | restless, irritable   | lethargic, floppy                    |

|                    |               |             |                        |
|--------------------|---------------|-------------|------------------------|
| Eyes / mucosa      | normal, moist | sunken, dry | deeply sunken, parched |
| Skin pinch         | instant       | < 2 s       | > 2 s                  |
| Cap refill / pulse | normal        | prolonged   | > 3 s, weak, cool      |
| Urine              | normal        | reduced     | minimal or none        |

### Quick equations

| For                         | Formula                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Maintenance fluid           | 4·2·1: 4 mL/kg/h first 10 kg + 2 next 10 + 1 each kg over 20 (Holliday–Segar)                                                     |
| Fluid deficit (mL)          | % dehydration × weight (kg) × 10                                                                                                  |
| Corrected Na (high glucose) | add 2.4 mmol/L per 5.5 mmol/L (100 mg/dL) glucose above 5.5 mmol/L (Hillier; the older 1.6 factor underestimates at high glucose) |
| Corrected Ca                | measured + 0.02 × (40 – albumin g/L) mmol/L                                                                                       |
| Anion gap                   | Na – (Cl + HCO <sub>3</sub> ); normal 8–12 mmol/L                                                                                 |
| Serum osmolality            | 2 × Na + glucose + urea (mmol/L). <b>Osmolar gap = measured – calculated</b>                                                      |



osmolality; **> 10** suggests an unmeasured osmole (methanol, ethylene glycol, ethanol, mannitol).

Expected  
pCO<sub>2</sub> (met.  
acidosis)

Winter:  $1.5 \times \text{HCO}_3 + 8 \pm 2$   
(mmHg) — the **expected  
respiratory compensation** for a  
metabolic acidosis; measured  
PaCO<sub>2</sub> **above** the range = added  
respiratory acidosis, **below** =  
added respiratory alkalosis.

QTc (Bazett)

$\text{QT} \div \sqrt{\text{RR}} \text{ (s)}$ ; prolonged  $> 0.46$   
s

eGFR  
(Schwartz)

$36.5 \times \text{height (cm)} \div \text{creatinine}$   
( $\mu\text{mol/L}$ ) — i.e.  $0.413 \times \text{height} \div$   
creatinine (mg/dL)<sup>349</sup>

Glucose  
infusion rate

$\text{mg/kg/min} = \text{dextrose\%} \times \text{rate}$   
(mL/h)  $\div (6 \times \text{weight kg})$

Names & synonyms

| Generic (US)   | Also known as  |
|----------------|----------------|
| Epinephrine    | epinephrine    |
| Norepinephrine | norepinephrine |
| Albuterol      | salbutamol     |
| Acetaminophen  | acetaminophen  |
| Furosemide     | frusemide      |
| Lidocaine      | lignocaine     |

|                              |                                        |
|------------------------------|----------------------------------------|
| Acyclovir                    | acyclovir                              |
| Amoxicillin–<br>clavulanate  | co-amoxiclav                           |
| Valproic acid                | sodium valproate                       |
| Regular insulin              | soluble insulin                        |
| Normal saline (0.9%<br>NaCl) | isotonic saline                        |
| Lactated Ringer's            | Hartmann's, compound<br>sodium lactate |
| Succinylcholine              | suxamethonium                          |
| Chlorpheniramine             | chlorphenamine                         |

## Procedural Sedation, RSI & Analgesia

Quick weight-based doses — prepare airway, suction, and monitoring first.

**+ Calculators: ETT size & depth · Infusion dose ↔ rate**

### ◆ NON-EMERGENT NEONATAL INTUBATION – PREMEDICATE

- Analgesic/vagolytic premedication is standard of care (not optional) for **all non-emergent** neonatal

intubations; the only exception is delivery-room resuscitation or a life-threatening airway without IV access.<sup>155</sup>

- Use a vagolytic + fast-acting opioid + short-acting relaxant: **atropine 0.01–0.02 mg/kg + fentanyl 2 mcg/kg** slow over 2–5 min + **succinylcholine 2 mg/kg** (avoid succinylcholine in hyperkalemia, malignant-hyperthermia risk, or neuromuscular disease) — CPS.<sup>155</sup>

## ANALGESIA

| Drug          | Dose                                 | Route & notes                                                                |
|---------------|--------------------------------------|------------------------------------------------------------------------------|
| Acetaminophen | 15 mg/kg/dose (max 1 g/dose)         | PO/PR/IV; mild–moderate pain                                                 |
| Ibuprofen     | 10 mg/kg/dose (max 400 mg/dose)      | PO; mild–moderate pain (avoid in dehydration, renal impairment, GI bleeding) |
| Morphine      | 0.05–0.1 mg/kg                       | IV, titrate; for moderate–severe pain                                        |
| Fentanyl      | 1–2 mcg/kg IV; 1.5 mcg/kg intranasal | Rapid onset; useful when no IV access                                        |

## PROCEDURAL SEDATION

Procedural sedation and RSI by credentialed clinicians only — airway-rescue capability, full monitoring, and an institutional sedation policy.

| Drug      | Dose                                                  | Route & notes                                                                                                                       |
|-----------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Ketamine  | 1–2 mg/kg IV;<br>4–5 mg/kg IM                         | Dissociative; usually preserves airway reflexes/respiratory drive — but airway obstruction, laryngospasm, and apnea can still occur |
| Midazolam | 0.05–0.1 mg/kg IV;<br>0.3–0.5 mg/kg buccal/intranasal | Anxiolysis; respiratory depression with opioids                                                                                     |

## RAPID SEQUENCE INTUBATION

| Step      | Dose                                                                                                                                         | Route & notes          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Induction | Ketamine 1–2 mg/kg,<br>propofol 2–3 mg/kg,<br>etomidate 0.3 mg/kg, or<br>midazolam 0.1–0.3 mg/kg<br>(± fentanyl 1–2 mcg/kg as pre-treatment) | Choose by hemodynamics |
| Paralytic | Rocuronium 1–1.2 mg/kg, or<br>succinylcholine 1–2 mg/kg                                                                                      | IV                     |

| Step                     | Dose       | Route & notes                           |
|--------------------------|------------|-----------------------------------------|
| Atropine (pre-treatment) | 0.02 mg/kg | IV in young infants if bradycardia risk |

### ⚠ BEFORE ANY SEDATION/RSI

- Have oxygen, suction, bag-mask, the right tube size  $[(\text{age} \div 4) + 4]$  uncuffed], monitoring, and a reversal plan ready.
- See Weight & Airway Sizing.

## Emergency Drug Doses at a Glance

### MINUTES

Weight-based resuscitation doses — use measured weight or a length-based tape when possible (formulas are estimates); cross-check weight, concentration, and local resuscitation chart before every dose.

**+ Calculator:** [Resuscitation dose calculator](#)

| Drug · indication            | Dose            | Notes                                |
|------------------------------|-----------------|--------------------------------------|
| Epinephrine — cardiac arrest | 0.01 mg/kg (0.1 | Repeat every 3–5 min; max 1 mg/dose. |

|                                      |                                                                                        |                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|                                      | mL/kg of<br>1:10 000)<br>IV/IO                                                         | Endotracheal: 0.1<br>mg/kg (0.1 mL/kg<br>of 1:1000)                                                |
| Epinephrine —<br>anaphylaxis         | 0.01<br>mg/kg<br>(0.01<br>mL/kg of<br>1:1000)<br>IM                                    | Max 0.5 mg;<br>anterolateral thigh;<br>repeat every 5–15<br>min                                    |
| Amiodarone —<br>VF / pulseless<br>VT | 5 mg/kg<br>IV/IO<br>bolus                                                              | May repeat up to<br>twice; max 300<br>mg/dose.<br>Perfusing<br>tachycardia: give<br>over 20–60 min |
| Adenosine —<br>SVT                   | 0.1<br>mg/kg<br>rapid<br>push<br>(max 6<br>mg),<br>then 0.2<br>mg/kg<br>(max 12<br>mg) | Immediate saline<br>flush; use line<br>closest to the<br>heart                                     |
| Atropine —<br>bradycardia            | 0.02<br>mg/kg<br>IV/IO                                                                 | Min 0.1 mg; max<br>0.5 mg/dose; may<br>repeat once                                                 |

|                                                                        |                               |                                                                                                                                                              |
|------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calcium chloride 10% — hyperkalemia, Ca-blocker toxicity, hypocalcemia | 20 mg/kg (0.2 mL/kg) IV/IO    | central line preferred (calcium chloride is sclerosing — high extravasation risk); calcium gluconate 10% 60 mg/kg (0.6 mL/kg) is preferred for peripheral IV |
| Dextrose — hypoglycemia                                                | 0.25–0.5 g/kg                 | D10W 2.5–5 mL/kg (neonate/infant); recheck glucose                                                                                                           |
| Magnesium sulfate — torsades, severe asthma                            | 25–50 mg/kg IV/IO (max 2 g)   | Rapid in torsades; over 15–20 min in asthma                                                                                                                  |
| Naloxone — opioid toxicity                                             | 0.1 mg/kg IV/IM/IN (max 2 mg) | Titrate to breathing; re-dose (short half-life)                                                                                                              |
| Sodium bicarbonate                                                     | 1 mEq/kg IV/IO                | TCA toxicity (QRS > 100 ms), severe acidosis, hyperkalemia                                                                                                   |
| Lidocaine — VF/pVT (alternative to amiodarone)                         | 1 mg/kg IV/IO                 | Then infusion 20–50 mcg/kg/min                                                                                                                               |

|                                              |                                  |                                                                                         |
|----------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------|
| Isotonic fluid bolus — shock                 | 20 mL/kg                         | 10 mL/kg in neonates, cardiac disease, DKA, or severe malnutrition; reassess after each |
| Defibrillation — VF/pVT                      | 2 J/kg →<br>4 J/kg →<br>≥ 4 J/kg | Max 10 J/kg or adult dose (≈ 200 J biphasic / 360 J monophasic)                         |
| Synchronized cardioversion — unstable SVT/VT | 0.5–1 J/kg → 2 J/kg              | Sedate if conscious and time permits                                                    |

### Concentrations & pitfalls

- Epinephrine 1:10 000 = 0.1 mg/mL (IV/IO) versus 1:1000 = 1 mg/mL (IM/endotracheal) — confirm the ampoule before every dose.
- Amiodarone is 150 mg/3 mL.
- Use a length-based (Broselow) tape when weight is unknown.
- Source: American Heart Association / AAP Pediatric Advanced Life Support (PALS), 2025.



# Resuscitation Dose Calculator

**MINUTES**

PALS 2025

Enter the weight — doses compute live and cap at the maximum per dose (works in app and web version).

Weight  kg

Length/height  cm

| Drug · indication              | Rule (per kg · max)     | Dose for this child |
|--------------------------------|-------------------------|---------------------|
| Epinephrine — arrest (IV/IO)   | 0.01 mg/kg · max 1 mg   | —                   |
| Epinephrine — anaphylaxis (IM) | 0.01 mg/kg · max 0.5 mg | —                   |
| Amiodarone — VF/pVT            | 5 mg/kg · max 300 mg    | —                   |
| Adenosine — 1st dose           | 0.1 mg/kg · max 6 mg    | —                   |
| Adenosine — 2nd dose           | 0.2 mg/kg · max 12 mg   | —                   |

| Drug · indication                          | Rule (per kg · max)                    | Dose for this child |
|--------------------------------------------|----------------------------------------|---------------------|
| <b>Atropine</b><br>—<br><b>bradycardia</b> | 0.02 mg/kg ·<br>min 0.1, max<br>0.5 mg | —                   |
| <b>Calcium gluconate 10%</b>               | 60 mg/kg (0.6 mL/kg) · max 2 g         | —                   |
| <b>Dextrose 10% — hypoglycemia</b>         | 2.5–5 mL/kg<br>(0.25–0.5 g/kg)         | —                   |
| <b>Magnesium sulfate</b>                   | 25–50 mg/kg ·<br>max 2 g               | —                   |
| <b>Naloxone — opioid</b>                   | 0.1 mg/kg ·<br>max 2 mg                | —                   |
| <b>Sodium bicarbonate 8.4%</b>             | 1 mEq/kg (1 mL/kg)                     | —                   |
| <b>Lidocaine — VF/pVT alt</b>              | 1 mg/kg · max<br>100 mg                | —                   |
| <b>Isotonic fluid bolus — shock</b>        | 10–20 mL/kg                            | —                   |
| <b>Defibrillation — VF/pVT</b>             | 2 → 4 J/kg ·<br>max ≈ 200 J            | —                   |

| Drug · indication                     | Rule (per kg · max) | Dose for this child |
|---------------------------------------|---------------------|---------------------|
| Synchro<br>nized<br>cardiover<br>sion | 0.5–1 → 2 J/kg      | –                   |

#### ◆ RSI – RAPID SEQUENCE INTUBATION

| Drug · role                                 | Rule (per kg)            | Dose for this child |
|---------------------------------------------|--------------------------|---------------------|
| Fentanyl<br>I —<br>premed<br>/analge<br>sia | 1–2 mcg/kg<br>(slow IV)  | –                   |
| Ketamin<br>e —<br>inductio<br>n             | 1–2 mg/kg IV<br>(4–5 IM) | –                   |
| Etomidate —<br>inductio<br>n                | 0.3 mg/kg IV             | –                   |
| Propofol —<br>inductio<br>n                 | 2–3 mg/kg IV             | –                   |
| Midazolam —<br>inductio<br>n alt            | 0.1–0.3<br>mg/kg IV      | –                   |

| Drug ·<br>role                                       | Rule (per kg)     | Dose for this child |
|------------------------------------------------------|-------------------|---------------------|
| <b>Rocuro<br/>nium —<br/>paralyti<br/>c</b>          | 1–1.2 mg/kg<br>IV | –                   |
| <b>Succiny<br/>lcholine<br/>—<br/>paralyti<br/>c</b> | 1–2 mg/kg IV      | –                   |

Choose one induction agent (by hemodynamics) + one paralytic. Atropine premed (0.02 mg/kg, max 0.5 mg) is in the table above — no minimum dose when used as intubation premedication (the paradoxical-bradycardia rationale for a 0.1 mg floor has been debunked). Credentialed clinicians only — airway rescue ready.

## **VASOACTIVE INFUSIONS – WEIGHT-BASED**

## PREP (VERIFY LOCALLY)

| Infusion                            | Make up to 50 mL | Rate → dose                                                                                                          |
|-------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------|
| Epinephrine—<br>/<br>Norepinephrine |                  | 1 mL/h = 0.1<br>mcg/kg/min · usual<br>0.05–1 mcg/kg/min<br>(0.5–10 mL/h)                                             |
| Milrinone —                         |                  | 1 mL/h = 0.1<br>mcg/kg/min · usual<br>0.25–0.75<br>mcg/kg/min (2.5–7.5<br>mL/h); load 50<br>mcg/kg over 30–60<br>min |
| Dopamine —<br>/<br>Dobutamine       |                  | 1 mL/h = 1<br>mcg/kg/min · usual<br>5–20 mcg/kg/min<br>(5–20 mL/h)                                                   |

Central line for these concentrations;  
for larger children use your unit's  
standard fixed-concentration chart.

### ⚠ ALWAYS CROSS-CHECK

- Confirm weight, ampoule concentration, and the local resuscitation chart before every dose.
- If weight is unknown, entering length gives a **Broselow length-based estimate** — confirm with a scale.

- Static PDF / Word / EPUB copies show the rules and maxima but do not calculate; use the app or web (HTML) file for live values.

## Resuscitation Doses by Weight

MINUTES

PALS 2025

Pre-computed arrest doses — pick the column nearest the weight. Works in every format, including print.

**+ Calculator: Resuscitation dose calculator**

| Drug ·<br>rule<br>(per<br>dose) ·<br>kg▶                                             | 5             | 10         | 20         | 30         | 50         |
|--------------------------------------------------------------------------------------|---------------|------------|------------|------------|------------|
| Epinephrine<br>(adrenaline)<br>0.01<br>mg/kg<br>IV/IO<br>— mg<br>(mL<br>1:10<br>000) | 0.05<br>(0.5) | 0.1<br>(1) | 0.2<br>(2) | 0.3<br>(3) | 0.5<br>(5) |
| Amiodarone<br>5<br>mg/kg<br>(arrest)<br>— mg                                         | 25            | 50         | 100        | 150        | 250        |
| Adenosine<br>0.1<br>mg/kg<br>(1st,<br>max 6)<br>— mg                                 | 0.5           | 1          | 2          | 3          | 5          |
| Atropine<br>0.02<br>mg/kg<br>(0.1–<br>0.5) —<br>mg                                   | 0.1           | 0.2        | 0.4        | 0.5        | 0.5        |

| Drug ·<br>rule<br>(per<br>dose) ·<br>kg▶              | 5   | 10  | 20  | 30  | 50   |
|-------------------------------------------------------|-----|-----|-----|-----|------|
| Calcium gluconate 10% 0.6 mL/kg (max 20) — mL         | 3   | 6   | 12  | 18  | 20   |
| Dextrose 10% 2 mL/kg — mL                             | 10  | 20  | 40  | 60  | 100  |
| Fluid bolus 20 mL/kg (10 if neonate/cardiac/DKA) — mL | 100 | 200 | 400 | 600 | 1000 |
| Defibrillation 4 J/kg (max 200) — J                   | 20  | 40  | 80  | 120 | 200  |



Numbers are per single dose, already capped. Pick the nearest column and adjust to the child’s actual weight; for an exact figure use the interactive Resuscitation Dose Calculator above (app / web).

## Anticoagulation – VTE Prophylaxis & Treatment

ASSESS

Enoxaparin dosing and anti-Xa targets in children.

| Regimen                                     | Dose                                  | Monitoring / notes                                                                                                                                                                                                                 |
|---------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enoxaparin<br>—<br>treatment,<br>< 2 months | 1.5<br>mg/kg/dose<br>SC every 12<br>h | Anti-Xa target 0.5–1.0 IU/mL, drawn 4–6 h post-dose (after the 3rd–4th dose). Treatment anticoagulation is hematology-guided with anti-Xa monitoring; adjust for renal function, age, weight, and bleeding risk <sup>117,387</sup> |
| Enoxaparin<br>—<br>treatment,<br>≥ 2 months | 1.0<br>mg/kg/dose<br>SC every 12<br>h | Infants often need more: term up to 1.7, preterm up to 2 mg/kg/dose                                                                                                                                                                |

|              |             |                    |
|--------------|-------------|--------------------|
| Enoxaparin   | 0.75        |                    |
| —            | mg/kg/dose  | Anti-Xa 0.1–0.3    |
| prophylaxis, | SC every 12 | IU/mL if monitored |
| < 2 months   | h           |                    |

|              |             |                     |
|--------------|-------------|---------------------|
| Enoxaparin   | 0.5         |                     |
| —            | mg/kg/dose  | Routine anti-Xa not |
| prophylaxis, | SC every 12 | required            |
| ≥ 2 months   | h           |                     |

### Before starting

- Baseline complete blood count, renal function, and coagulation screen.
- Avoid with active bleeding, platelets  $< 50 \times 10^9/L$ , or within 24 h of major surgery or a neuraxial procedure.
- Reduce dose if creatinine clearance  $< 30 \text{ mL/min}$ .<sup>117,387</sup>

### Reversal & alternatives

- Protamine partially reverses enoxaparin (about 1 mg per 1 mg given in the prior 8 h; max 50 mg) — anti-Xa activity is only partly neutralized.
- Consider unfractionated heparin when rapid reversibility or severe renal impairment is needed.<sup>117,387</sup>

- Source: ASH 2018 and CHEST pediatric antithrombotic guidelines.

## Maximum Local Anesthetic Doses

### ASSESS

Single-dose ceilings — count total mg/kg across all sites. Use lower limits in neonates/infants, liver disease, or low cardiac output; count the total across all injection sites and when agents are combined.

| Agent                     | Max single dose                                 | Notes                                                                            |
|---------------------------|-------------------------------------------------|----------------------------------------------------------------------------------|
| Lidocaine (plain)         | 4.5 mg/kg<br>(max 300 mg)                       | Onset fast;<br>~30–60 min duration                                               |
| Lidocaine + epinephrine   | 7 mg/kg<br>(max 500 mg)                         | Longer duration;<br>caution on end-artries (digits, pinna, penis) <sup>116</sup> |
| Bupivacaine ± epinephrine | 2.5 mg/kg<br>(single-shot; epinephrine does not | Most cardiotoxic; never IV or Bier block;                                        |

|                 |                              |                                            |
|-----------------|------------------------------|--------------------------------------------|
|                 | raise the pediatric ceiling) | avoid in young infants <sup>116</sup>      |
| Levobupivacaine | 2.5 mg/kg                    | Less cardiotoxic than bupivacaine          |
| Ropivacaine     | 2 mg/kg                      | Lower cardiotoxicity; useful for infusions |
| Mepivacaine     | 4.4 mg/kg (max ~300 mg)      | Duration between lidocaine and bupivacaine |

## CONTINUOUS INFUSION CAPS

Bupivacaine, levobupivacaine, ropivacaine: 0.2 mg/kg/h (< 3 mo), 0.3 mg/kg/h (3 mo–1 yr), up to 0.4 mg/kg/h (> 1 yr).<sup>116</sup>

### Toxicity (LAST)

- Perioral numbness, tinnitus, and agitation progress to seizures, then arrhythmia or arrest.
- Stop injection, secure airway and oxygen, treat seizures with a benzodiazepine, and give 20% lipid emulsion (see [Poisoning & Toxic Ingestion](#)).
- Dilute to the least concentrated effective solution, aspirate before

injecting, and inject slowly in divided doses; reduce maximums in neonates.<sup>116</sup>

- Source: ESRA/ASRA 2018 pediatric recommendations; ASRA 2020 LAST advisory.

## Vital signs — normal vs red zone by age

The escalation threshold, not just the normal range. Ranges: PALS/APLS.

### Vital Signs — Normal vs Red Zone by Age

The number that should trigger escalation. Red-zone values are concern thresholds, not just "abnormal."

Compensated pediatric shock is often normotensive — tachycardia + poor perfusion precede hypotension, which is a late, pre-arrest sign.

| Age         | Normal HR | HR red zone  | Normal RR | RR red zone | Hypotension SBP |
|-------------|-----------|--------------|-----------|-------------|-----------------|
| < 1 month   | 100–205   | <100 or >205 | 30–60     | >60         | <60             |
| 1–12 months | 100–190   | <100 or >190 | 30–53     | >60         | <70             |
| 1–2 years   | 98–140    | <90 or >140  | 22–37     | >40         | <70+(2×age)     |
| 2–5 years   | 80–120    | <80 or >120  | 20–28     | >34         | <70+(2×age)     |
| 5–12 years  | 75–118    | <60 or >120  | 18–30     | >30         | <70+(2×age)     |
| > 12 years  | 60–100    | <60 or >100  | 12–20     | >20         | <90             |

Hypotension (5th-centile SBP):  $70 + (2 \times \text{age in years})$  for 1–10 y; < 90 mmHg for ≥ 10 y. Example — 4-year-old: < 78 mmHg.

Sepsis/shock red flags (any): capillary refill > 2–3 s or flush, mottled/cold or bounding pulses, altered mental status, lactate > 2 mmol/L, reduced urine output.

Ranges: PALS/APLS. Bradycardia and bradypnea in a sick child are ominous late signs — treat as pre-arrest.

# Kuwait Childhood Immunization Schedule

REFERENCE

MOH Kuwait 2025

The routine schedule at a glance — and the ages a missed dose should prompt catch-up.

| Age                  | Vaccine                                                           | Route          |
|----------------------|-------------------------------------------------------------------|----------------|
| Pregnant mother      | Tdap – one dose in this pregnancy, and every subsequent pregnancy | IM             |
| Within 24 h of birth | Hepatitis B (HBV) – 1st dose                                      | IM             |
| End of 2 months      | Hexa (1st) • Pneumococcal (1st) • Rotavirus (1st)                 | IM • IM • Oral |
| End of 3 months      | BCG                                                               | ID             |
| End of 4 months      | Hexa (2nd) • Pneumococcal (2nd) • Rotavirus (2nd)                 | IM • IM • Oral |
| End of 6 months      | Hexa (3rd) • Pneumococcal (3rd) • Rotavirus (3rd)*                | IM • IM • Oral |

| Age                   | Vaccine                                                                           | Route               |
|-----------------------|-----------------------------------------------------------------------------------|---------------------|
| End of 12 months      | OPV (1st) •<br>MMR (1st) •<br>Varicella (1st) •<br>Meningococcal conjugate (ACWY) | Oral • SC • SC • IM |
| At 18 months          | OPV (2nd) •<br>Hexa booster •<br>Pneumococcal booster                             | Oral • IM • IM      |
| End of 2 years        | MMRV (1st)                                                                        | SC                  |
| End of 3.5 years      | DTaP booster                                                                      | IM                  |
| 4–6 years (preschool) | Check immunization status before school admission                                 | —                   |
| 10–12 years (school)  | Tdap booster                                                                      | IM                  |
| 16–18 years (school)  | Tdap booster                                                                      | IM                  |

#### ◆ VACCINE KEY

- **HEXA** = diphtheria, tetanus, acellular pertussis, hepatitis B, inactivated poliovirus (IPV) & Hib

- **MMRV** = measles, mumps, rubella, varicella
- **DTAP/Tdap** = diphtheria, tetanus, acellular pertussis
- **OPV** = oral polio
- \* rotavirus is 2 or 3 doses according to the vaccine type used.

#### ✦ WHAT CHANGED IN THE 2025 SCHEDULE

- **Hexa** replaces the pentavalent (adds IPV)
- **MMRV** at 2 years replaces the old separate 2nd MMR + varicella
- **DTaP** replaces DPT at 3.5 years
- **Tdap** replaces Td at 10–12 and 16–18 years and is now given in **every pregnancy**
- the old OPV booster at 3.5 years and the girls-only MMR booster at 10–12 years were dropped.

#### ✦ USE EVERY EMERGENCY VISIT AS A CATCH-UP TRIGGER

- An unwell or injured child is a chance to check the card.
- A missed **MMR/MMRV** in an outbreak, an incomplete primary series in a young infant with a



fever, or no **tetanus-containing** cover before a dirty wound all change management on the spot.

#### ◆ SOURCE

- Ministry of Health, State of Kuwait — **KUWAIT CHILDHOOD IMMUNIZATION SCHEDULE 2025** (Public Health Sector).
- Transcribed from the official MOH schedule.

## Think Here – Consanguinity & Gulf-Prevalent Conditions

#### REFERENCE

Gulf pre-test probability

The same presentation carries a different differential in Kuwait — recessive disease is not rare here.

#### ▲ WHY THE BASELINE DIFFERS

- Consanguineous marriage is common in Kuwait (population studies report roughly **half** of unions, first-cousin predominant).

Homozygosity sharply raises the pre-test probability of **autosomal-recessive** disease.

✦ **RAISE THESE ON THE DIFFERENTIAL EARLIER THAN THE TEXTBOOK WOULD**

- **Inborn errors of metabolism** in the encephalopathic / acidotic / hypoglycemic neonate or the child who decompensates with fasting or intercurrent illness — check gas, glucose, ammonia, lactate and start **IV dextrose** early. **Hemoglobinopathies** (sickle cell,  $\beta$ -thalassemia) and **G6PD deficiency** in the pale, jaundiced, or acutely hemolyzing child.
- A **recessive syndrome** when a sick infant has an affected sibling or known parental consanguinity.

◆ **LOCAL PROGRAMMES & SOURCES**

- Kuwait runs premarital screening and expanded newborn screening; a family history of consanguinity or an affected sibling is itself a red flag.
- Sources: autosomal-recessive disorders in Kuwait (Teebi/Al-Awadi series); IEM & consanguinity

reviews. [VERIFY LOCAL SCREENING PANEL].

## High-Alert Drugs & Fluid Safety

EVERY SHIFT

ISMP 2024 · AAP 2018

The commonest pediatric harm is not the missed diagnosis — it's the decimal point.

✚ Calculators: Infusion dose ↔ rate · Maintenance fluids & bolus

### ▲ HIGH-ALERT DRUGS – WHERE A DECIMAL KILLS

- **Insulin** (units, U-100 vs U-500) · **opioids** (mcg vs mg) · **concentrated KCl / K-phosphate** (never IV push) · **heparin / LMWH** (1,000 vs 10,000 units/mL) · **chemotherapy** (intrathecal vincristine = fatal) · **digoxin / milrinone** · **magnesium sulfate** (mg vs mmol vs mEq) · **hypertonic saline / dextrose** · **neuromuscular blockers** · **oral methotrexate** (weekly, not daily).
- **Epinephrine concentration: 1 mg/mL** (old 1:1,000, IM) vs **0.1 mg/mL** (old 1:10,000, IV) — the

classic 10-fold pediatric error. State the concentration, never the ratio.

**⚠ THE THREE ERROR MODES TO DOUBLE-CHECK**

- **10-fold / decimal** errors
- **mg vs mL vs microgram vs unit** confusion
- **concentration mix-ups.**
- For any high-alert drug: independent double-check of dose, concentration and pump rate.

**▀ SODIUM – CORRECT WITHIN THE SAFE RAILS**

- **Hyponatremia:** raise  $\text{Na} \leq 8 \text{ mmol/L}$  per 24 h ( $\approx 0.5 \text{ mmol/L/h}$ ) to avoid **osmotic demyelination**. **Acute symptomatic (seizing):** 3% saline **2–5 mL/kg** (max ~100–150 mL) to raise  $\text{Na} \sim 4\text{--}6 \text{ mmol/L}$  and stop the seizure.
- **Hypernatremia:** lower  $\text{Na} \leq 10\text{--}12 \text{ mmol/L}$  per 24 h ( $\approx 0.5 \text{ mmol/L/h}$ ) to avoid **cerebral edema** (pediatric correction-rate evidence is evolving).

## ▀ FLUIDS & POTASSIUM

- **Maintenance fluids (AAP 2018):** for children **28 days–18 years**, use **isotonic** fluid (0.9% saline or a balanced solution) with KCl and dextrose — hypotonic maintenance caused iatrogenic hyponatremia. Volume still by Holliday-Segar.
- **IV potassium:** peripheral  $\leq$  **40–60 mmol/L**; ward rate 0.25–0.5 mmol/kg/h (max 10 mmol/h); PICU up to **1 mmol/kg/h** with **continuous ECG**. Never IV push.

## ✦ EXTRAVASATION – STOP, ASPIRATE, AND KNOW THE TWO ANTIDOTES

- Vesicants: calcium, concentrated potassium, **vasopressors**, hypertonic dextrose/saline, TPN, chemotherapy.
- Immediately **stop the infusion, aspirate, elevate**.
- Antidotes: **hyaluronidase** (most hyperosmolar/calcium/TPN infiltrations) and **phentolamine** (vasopressor extravasation, e.g. norepinephrine/dopamine).

#### ◆ SOURCE

- ISMP High-Alert Medications (Acute Care 2024); AAP maintenance IV fluids (Feld et al., Pediatrics 2018); CJASN hyponatremia 2024; pediatric potassium administration guidance; J Infusion Nursing 2020.
- Confirm concentration/rate ceilings against local protocol.

## Decision-Rules Engine – Cutoff → Action

### REFERENCE

Validated pediatric rules

The rules you already trust, each with its exact operating point and what to do next — in one place.

**+ Calculators:** [Westley croup score](#) · [Pediatric Appendicitis Score](#) · [Glasgow Coma Scale](#)

| Rule                               | Operating point → action                                                                                                                                                                                                                                                 |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PECARN head — < 2 y                | GCS ≤14 / AMS / palpable skull fracture → CT. Occipital-parietal-temporal hematoma, LOC ≥5 s, severe mechanism, or not acting normally → observe vs CT. None → no CT.                                                                                                    |
| PECARN head — ≥ 2 y                | GCS ≤14 / AMS / basilar skull-fracture signs → CT. LOC, vomiting, severe mechanism, or severe headache → observe vs CT. None → no CT.                                                                                                                                    |
| PECARN c-spine (2024)              | GCS 3–8 / abnormal ABC / focal deficit → CT. AMS, substantial head/torso injury, midline neck pain or tenderness → radiographs. None → clinical clearance.                                                                                                               |
| Kocher (septic hip)                | Non-weight-bearing · fever >38.5 · ESR >40 · WBC >12k. Probability by count: 1≈3% · 2≈40% · 3≈93% · 4≈99%. ≥3 → aspirate / ortho; 2 → strongly consider.                                                                                                                 |
| Pediatric Appendicitis Score (/10) | ≤3 low (LR 0.13) → observe/discharge with safety-net; 4–7 equivocal (LR 0.70) → ultrasound ± surgical review; ≥8 high (LR 8.1) → surgical consult. No PAS cut-off rules appendicitis in or out on its own (systematic review of 26 prospective studies) <sup>353</sup> . |

| Rule                              | Operating point → action                                                                                                                                                                                                                                                                       |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Westley croup</u> (/17)</b> | ≤2 mild → dexamethasone, home; 3–5 moderate → dexamethasone + nebulized L-epinephrine, observe; 6–11 severe → epinephrine + steroid, admit; ≥12 → airway / ICU. Bands are the trial convention (Cochrane), not validated cut-offs – scores do not replace clinical assessment <sup>354</sup> . |
| <b>PRAM asthma (/12)</b>          | 0–4 mild · 5–8 moderate · 9–12 severe → escalates bronchodilator frequency, systemic steroid, and admission/ICU.                                                                                                                                                                               |
| <b>PASS asthma</b>                | Pediatric Asthma Severity Score – grades work of breathing, wheezing / air entry, and SpO <sub>2</sub> / ability to speak or feed; higher = more severe. An alternative to PRAM to guide bronchodilator frequency, systemic steroid, and admission; reassess after each step.                  |
| <b>Bacterial Meningitis Score</b> | 1 pt each: CSF Gram+ · CSF ANC ≥1000 · CSF protein ≥80 · blood ANC ≥10k · seizure. Score 0 = very low risk (NPV ~100%). Excludes <2 mo, prior antibiotics, purpura, immunosuppression, shunt.                                                                                                  |



| Rule                           | Operating point → action                                                                                                                                                                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Febrile infant ≤ 60–90 d       | Step-by-Step low risk = well · >21 d · normal UA · PCT <0.5 · CRP ≤20 · ANC ≤10k → discharge with follow-up. PECARN ≤60 d: negative UA + ANC ≤4090 + PCT ≤1.71 → may forgo LP/antibiotics. |
| Gorelick dehydration (/10)     | ≥3 signs → ≥5% dehydration; ≥7 → ≥10%. (4-sign subset: cap refill, tears, mucous membranes, ill appearance.)                                                                               |
| Glasgow-Blatchford (UGI bleed) | Score 0 → low risk, outpatient. Caveat: adult-derived, not validated in children – adjunct only.                                                                                           |

#### ◆ SOURCES

- PECARN head (Kuppermann, Lancet 2009) & c-spine (2024); Kocher 1999 (Caird CRP modification); Samuel PAS 2002; Westley 1978; PRAM; Nigrovic BMS 2007; Step-by-Step (Gomez/Mintegi 2016) & PECARN febrile (Kuppermann 2019); Gorelick 1997; Glasgow-Blatchford 2000.
- Cross-link each row to its card on insertion.

# How This Guide Is Maintained

## GOVERNANCE

Issued June 2026 • review June 2028

A reference is only trusted if it shows its currency, its reviewers, and a way to correct it.

### ▲ CURRENCY & REVIEW

- **Issued** June 2026 • **scheduled review** June 2028, or sooner if a major guideline changes. Each clinical section carries a **"reviewed / next review"** stamp so the reader can see the section's currency at a glance.
- Content is written to the **most recent high-impact guideline** for each topic; the guideline body and year appear on every card.

### ▲ REVIEWERS & GOVERNANCE

- Prepared for the **Kuwait Pediatric Council** and reviewed by its named faculty panel (see the Reviewers page). Section leads sign off their section before release.

### ⚠ **REPORT AN ERROR**

- Found a wrong dose, a broken link, or an out-of-date threshold?  
**Report it** — corrections are logged and issued between scheduled reviews.
- Use the feedback channel on the Reviewers page (a QR / email link is placed here in the published edition).
- Every erratum is dated and credited in the change log below.

### ▀ **CHANGE LOG – THIS EDITION**

- **New clinical cards** — immune thrombocytopenia, IgA vasculitis, pertussis, Guillain–Barré / acute flaccid paralysis, new-diagnosis oncologic emergencies, hyperglycemic hyperosmolar state, vitamin K deficiency bleeding, envenomation, MIS-C, tension pneumothorax, child protection, neonatal abstinence, thyroid storm, rhabdomyolysis, toxic megacolon, neuromuscular disease, massive transfusion.

- **New reference tools** — Kuwait 2025 immunization schedule, high-alert drug & fluid safety, decision-rules engine, symptom front-door navigator; bilirubin nomogram + calculator and age-banded vital-sign red zones.
- **Corrections** — US spelling & drug nomenclature throughout; bilirubin escalation defined as within 2 mg/dL of the exchange line (AAP 2022); immunization schedule updated to the MOH 2025 decree.
- **Editorial** — clickable Contents restored in every PDF; workup block standardized; version label removed from published copies.

#### WEBSITE STATISTICS

- kpc-guide.com counts anonymous visits and downloads (GoatCounter: no cookies, no personal data, country-level location only) to see how the guide is used and to improve it. Offline use is not counted.

#### HOW TO CITE

- Kuwait Pediatric Council. *Pediatric & Neonatal Common & Emergency*

*Care Guide*. Issued June 2026.

- Decision support only — confirm against Ministry of Health protocols and specialist advice.

## Shortest Antibiotic Durations

### STEWARDSHIP

### Short-course RCTs

An improving, immunocompetent child — how few days is actually enough.

### ⚠️ THESE DURATIONS ASSUME

- An **immunocompetent** child who is **improving** (afebrile / clearly better by 48–72 h), with **source control** and **no bacteremia, abscess, or foreign material**.
- Confirm organism and susceptibility; extend if not improving.

| Infection                                                                      | Shortest course                         | Condition                                                                                                          |
|--------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Community-acquired pneumonia (outpatient, non-severe) ( <a href="#">card</a> ) | 5 days                                  | Improving by day 3–4 (SCOUT-CAP). 3 days is emerging for mild disease, not yet standard.                           |
| Acute otitis media ( <a href="#">card</a> )                                    | < 2 y: 10 days · ≥ 2 y: 5–7 days        | A 5-day course was inferior in under-2s; full 10 days for < 2 y and for severe/perforated.                         |
| GAS pharyngitis ( <a href="#">card</a> )                                       | 10 days (penicillin/amoxicillin)        | Unchanged — needed for eradication and rheumatic-fever prevention.                                                 |
| Cystitis (afebrile, lower UTI) ( <a href="#">card</a> )                        | 3–5 days                                | Afebrile lower-tract infection.                                                                                    |
| Febrile UTI / pyelonephritis                                                   | 7–10 days (5 only if clearly improving) | <b>Evidence conflicts</b> — STOP trial: 5 d non-inferior; SCOUT-UTI: more failures at 5 d. Shared decision-making. |

| Infection                                                         | Shortest course                                         | Condition                                                              |
|-------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------|
| Cellulitis / soft-tissue (uncomplicated) ( <a href="#">card</a> ) | 5 days                                                  | Extend only if not improving at day 5.                                 |
| Acute osteomyelitis ( <a href="#">card</a> )                      | ~3 weeks total; IV→oral at 2–4 days                     | Non-MRSA, good response, falling CRP before the oral switch (Peltola). |
| Septic arthritis                                                  | 10 days total; early oral switch                        | Uncomplicated, non-MRSA/non-Salmonella, CRP normalizing (Peltola).     |
| Bacterial meningitis ( <a href="#">card</a> )                     | Meningococcus 5–7 d • Hib 7–10 d • pneumococcus 10–14 d | By organism (GBS 14–21 d; gram-negative/Listeria 21 d).                |
| Perforated appendicitis (post-op) ( <a href="#">card</a> )        | Often 3–5 days                                          | Adequate source control; stop when afebrile ~24 h and tolerating diet. |
| Neonatal culture-negative "rule-out" ( <a href="#">card</a> )     | Stop at 36–48 h                                         | Blood culture negative and infant well — most pathogens grow by 36 h.  |

### ✦ THE STEWARDSHIP HABIT

- Set a **stop or review date** when you start.
- For most improving outpatient infections, **shorter is now standard** — the exceptions are the under-2 ear, the strep throat, and febrile UTI, where the longer course still holds.

### ◆ SOURCES

- SCOUT-CAP (JAMA Peds 2022) & CAP-IT (2021); AAP AOM 2013; IDSA GAS 2012 / CDC; STOP (Pediatrics 2024) & SCOUT-UTI (JAMA Peds 2023); IDSA SSTI 2014; Peltola osteomyelitis (PIDJ 2010) & septic arthritis (CID 2009); AAP Red Book (meningitis durations); pediatric appendicitis pathways.
- Confirm against your current Red Book / local antibiogram.



# Weight ruler – Broselow–Luten color zones

Find the weight → color zone → read across. Doses are per-kg (AHA PALS); ETT sizes are the standardized zone values. Confirm with a length-based tape and your formulary.

## Weight Ruler — Broselow–Luten color zones

Measure or estimate the weight → find the color zone → read across, across a bar the color's weight range are the child's actual weight in brown (kg) and pounds (lb) (AHA PALS)

0-5 kg (11-11 lb) 5-10 kg (11-22 lb) 10-15 kg (22-33 lb) 15-20 kg (33-44 lb) 20-25 kg (44-55 lb) 25-30 kg (55-66 lb) 30-35 kg (66-77 lb) 35-40 kg (77-88 lb) 40-45 kg (88-99 lb)

| Color zone          | ETT (mm)  | ETT (in)  | ETT (mm)  | ETT (in)  | ETT (mm)  | ETT (in)  | ETT (mm)  | ETT (in)  | ETT (mm)  | ETT (in)  |
|---------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 0-5 kg (11-11 lb)   | 4.5-5.5   | 0.18-0.22 | 5.5-6.5   | 0.22-0.26 | 6.5-7.5   | 0.26-0.30 | 7.5-8.5   | 0.30-0.34 | 8.5-9.5   | 0.34-0.38 |
| 5-10 kg (11-22 lb)  | 5.5-6.5   | 0.22-0.26 | 6.5-7.5   | 0.26-0.30 | 7.5-8.5   | 0.30-0.34 | 8.5-9.5   | 0.34-0.38 | 9.5-10.5  | 0.38-0.42 |
| 10-15 kg (22-33 lb) | 6.5-7.5   | 0.26-0.30 | 7.5-8.5   | 0.30-0.34 | 8.5-9.5   | 0.34-0.38 | 9.5-10.5  | 0.38-0.42 | 10.5-11.5 | 0.42-0.46 |
| 15-20 kg (33-44 lb) | 7.5-8.5   | 0.30-0.34 | 8.5-9.5   | 0.34-0.38 | 9.5-10.5  | 0.38-0.42 | 10.5-11.5 | 0.42-0.46 | 11.5-12.5 | 0.46-0.50 |
| 20-25 kg (44-55 lb) | 8.5-9.5   | 0.34-0.38 | 9.5-10.5  | 0.38-0.42 | 10.5-11.5 | 0.42-0.46 | 11.5-12.5 | 0.46-0.50 | 12.5-13.5 | 0.50-0.54 |
| 25-30 kg (55-66 lb) | 9.5-10.5  | 0.38-0.42 | 10.5-11.5 | 0.42-0.46 | 11.5-12.5 | 0.46-0.50 | 12.5-13.5 | 0.50-0.54 | 13.5-14.5 | 0.54-0.58 |
| 30-35 kg (66-77 lb) | 10.5-11.5 | 0.42-0.46 | 11.5-12.5 | 0.46-0.50 | 12.5-13.5 | 0.50-0.54 | 13.5-14.5 | 0.54-0.58 | 14.5-15.5 | 0.58-0.62 |
| 35-40 kg (77-88 lb) | 11.5-12.5 | 0.46-0.50 | 12.5-13.5 | 0.50-0.54 | 13.5-14.5 | 0.54-0.58 | 14.5-15.5 | 0.58-0.62 | 15.5-16.5 | 0.62-0.66 |
| 40-45 kg (88-99 lb) | 12.5-13.5 | 0.50-0.54 | 13.5-14.5 | 0.54-0.58 | 14.5-15.5 | 0.58-0.62 | 15.5-16.5 | 0.62-0.66 | 16.5-17.5 | 0.66-0.70 |

ETT (mm) = 4.5-5.5 (0.18-0.22) 5.5-6.5 (0.22-0.26) 6.5-7.5 (0.26-0.30) 7.5-8.5 (0.30-0.34) 8.5-9.5 (0.34-0.38) 9.5-10.5 (0.38-0.42) 10.5-11.5 (0.42-0.46) 11.5-12.5 (0.46-0.50) 12.5-13.5 (0.50-0.54) 13.5-14.5 (0.54-0.58) 14.5-15.5 (0.58-0.62) 15.5-16.5 (0.62-0.66) 16.5-17.5 (0.66-0.70)

ETT (in) = 0.18-0.22 (0.22-0.26) 0.26-0.30 (0.30-0.34) 0.34-0.38 (0.38-0.42) 0.42-0.46 (0.46-0.50) 0.50-0.54 (0.54-0.58) 0.58-0.62 (0.62-0.66) 0.66-0.70 (0.70-0.74) 0.74-0.78 (0.78-0.82) 0.82-0.86 (0.86-0.90) 0.90-0.94 (0.94-0.98) 0.98-1.02 (1.02-1.06) 1.06-1.10 (1.10-1.14) 1.14-1.18 (1.18-1.22) 1.22-1.26 (1.26-1.30) 1.30-1.34 (1.34-1.38) 1.38-1.42 (1.42-1.46) 1.46-1.50 (1.50-1.54) 1.54-1.58 (1.58-1.62) 1.62-1.66 (1.66-1.70) 1.70-1.74 (1.74-1.78) 1.78-1.82 (1.82-1.86) 1.86-1.90 (1.90-1.94) 1.94-1.98 (1.98-2.02) 2.02-2.06 (2.06-2.10) 2.10-2.14 (2.14-2.18) 2.18-2.22 (2.22-2.26) 2.26-2.30 (2.30-2.34) 2.34-2.38 (2.38-2.42) 2.42-2.46 (2.46-2.50) 2.50-2.54 (2.54-2.58) 2.58-2.62 (2.62-2.66) 2.66-2.70 (2.70-2.74) 2.74-2.78 (2.78-2.82) 2.82-2.86 (2.86-2.90) 2.90-2.94 (2.94-2.98) 2.98-3.02 (3.02-3.06) 3.06-3.10 (3.10-3.14) 3.14-3.18 (3.18-3.22) 3.22-3.26 (3.26-3.30) 3.30-3.34 (3.34-3.38) 3.38-3.42 (3.42-3.46) 3.46-3.50 (3.50-3.54) 3.54-3.58 (3.58-3.62) 3.62-3.66 (3.66-3.70) 3.70-3.74 (3.74-3.78) 3.78-3.82 (3.82-3.86) 3.86-3.90 (3.90-3.94) 3.94-3.98 (3.98-4.02) 4.02-4.06 (4.06-4.10) 4.10-4.14 (4.14-4.18) 4.18-4.22 (4.22-4.26) 4.26-4.30 (4.30-4.34) 4.34-4.38 (4.38-4.42) 4.42-4.46 (4.46-4.50) 4.50-4.54 (4.54-4.58) 4.58-4.62 (4.62-4.66) 4.66-4.70 (4.70-4.74) 4.74-4.78 (4.78-4.82) 4.82-4.86 (4.86-4.90) 4.90-4.94 (4.94-4.98) 4.98-5.02 (5.02-5.06) 5.06-5.10 (5.10-5.14) 5.14-5.18 (5.18-5.22) 5.22-5.26 (5.26-5.30) 5.30-5.34 (5.34-5.38) 5.38-5.42 (5.42-5.46) 5.46-5.50 (5.50-5.54) 5.54-5.58 (5.58-5.62) 5.62-5.66 (5.66-5.70) 5.70-5.74 (5.74-5.78) 5.78-5.82 (5.82-5.86) 5.86-5.90 (5.90-5.94) 5.94-5.98 (5.98-6.02) 6.02-6.06 (6.06-6.10) 6.10-6.14 (6.14-6.18) 6.18-6.22 (6.22-6.26) 6.26-6.30 (6.30-6.34) 6.34-6.38 (6.38-6.42) 6.42-6.46 (6.46-6.50) 6.50-6.54 (6.54-6.58) 6.58-6.62 (6.62-6.66) 6.66-6.70 (6.70-6.74) 6.74-6.78 (6.78-6.82) 6.82-6.86 (6.86-6.90) 6.90-6.94 (6.94-6.98) 6.98-7.02 (7.02-7.06) 7.06-7.10 (7.10-7.14) 7.14-7.18 (7.18-7.22) 7.22-7.26 (7.26-7.30) 7.30-7.34 (7.34-7.38) 7.38-7.42 (7.42-7.46) 7.46-7.50 (7.50-7.54) 7.54-7.58 (7.58-7.62) 7.62-7.66 (7.66-7.70) 7.70-7.74 (7.74-7.78) 7.78-7.82 (7.82-7.86) 7.86-7.90 (7.90-7.94) 7.94-7.98 (7.98-8.02) 8.02-8.06 (8.06-8.10) 8.10-8.14 (8.14-8.18) 8.18-8.22 (8.22-8.26) 8.26-8.30 (8.30-8.34) 8.34-8.38 (8.38-8.42) 8.42-8.46 (8.46-8.50) 8.50-8.54 (8.54-8.58) 8.58-8.62 (8.62-8.66) 8.66-8.70 (8.70-8.74) 8.74-8.78 (8.78-8.82) 8.82-8.86 (8.86-8.90) 8.90-8.94 (8.94-8.98) 8.98-9.02 (9.02-9.06) 9.06-9.10 (9.10-9.14) 9.14-9.18 (9.18-9.22) 9.22-9.26 (9.26-9.30) 9.30-9.34 (9.34-9.38) 9.38-9.42 (9.42-9.46) 9.46-9.50 (9.50-9.54) 9.54-9.58 (9.58-9.62) 9.62-9.66 (9.66-9.70) 9.70-9.74 (9.74-9.78) 9.78-9.82 (9.82-9.86) 9.86-9.90 (9.90-9.94) 9.94-9.98 (9.98-10.02) 10.02-10.06 (10.06-10.10) 10.10-10.14 (10.14-10.18) 10.18-10.22 (10.22-10.26) 10.26-10.30 (10.30-10.34) 10.34-10.38 (10.38-10.42) 10.42-10.46 (10.46-10.50) 10.50-10.54 (10.54-10.58) 10.58-10.62 (10.62-10.66) 10.66-10.70 (10.70-10.74) 10.74-10.78 (10.78-10.82) 10.82-10.86 (10.86-10.90) 10.90-10.94 (10.94-10.98) 10.98-11.02 (11.02-11.06) 11.06-11.10 (11.10-11.14) 11.14-11.18 (11.18-11.22) 11.22-11.26 (11.26-11.30) 11.30-11.34 (11.34-11.38) 11.38-11.42 (11.42-11.46) 11.46-11.50 (11.50-11.54) 11.54-11.58 (11.58-11.62) 11.62-11.66 (11.66-11.70) 11.70-11.74 (11.74-11.78) 11.78-11.82 (11.82-11.86) 11.86-11.90 (11.90-11.94) 11.94-11.98 (11.98-12.02) 12.02-12.06 (12.06-12.10) 12.10-12.14 (12.14-12.18) 12.18-12.22 (12.22-12.26) 12.26-12.30 (12.30-12.34) 12.34-12.38 (12.38-12.42) 12.42-12.46 (12.46-12.50) 12.50-12.54 (12.54-12.58) 12.58-12.62 (12.62-12.66) 12.66-12.70 (12.70-12.74) 12.74-12.78 (12.78-12.82) 12.82-12.86 (12.86-12.90) 12.90-12.94 (12.94-12.98) 12.98-13.02 (13.02-13.06) 13.06-13.10 (13.10-13.14) 13.14-13.18 (13.18-13.22) 13.22-13.26 (13.26-13.30) 13.30-13.34 (13.34-13.38) 13.38-13.42 (13.42-13.46) 13.46-13.50 (13.50-13.54) 13.54-13.58 (13.58-13.62) 13.62-13.66 (13.66-13.70) 13.70-13.74 (13.74-13.78) 13.78-13.82 (13.82-13.86) 13.86-13.90 (13.90-13.94) 13.94-13.98 (13.98-14.02) 14.02-14.06 (14.06-14.10) 14.10-14.14 (14.14-14.18) 14.18-14.22 (14.22-14.26) 14.26-14.30 (14.30-14.34) 14.34-14.38 (14.38-14.42) 14.42-14.46 (14.46-14.50) 14.50-14.54 (14.54-14.58) 14.58-14.62 (14.62-14.66) 14.66-14.70 (14.70-14.74) 14.74-14.78 (14.78-14.82) 14.82-14.86 (14.86-14.90) 14.90-14.94 (14.94-14.98) 14.98-15.02 (15.02-15.06) 15.06-15.10 (15.10-15.14) 15.14-15.18 (15.18-15.22) 15.22-15.26 (15.26-15.30) 15.30-15.34 (15.34-15.38) 15.38-15.42 (15.42-15.46) 15.46-15.50 (15.50-15.54) 15.54-15.58 (15.58-15.62) 15.62-15.66 (15.66-15.70) 15.70-15.74 (15.74-15.78) 15.78-15.82 (15.82-15.86) 15.86-15.90 (15.90-15.94) 15.94-15.98 (15.98-16.02) 16.02-16.06 (16.06-16.10) 16.10-16.14 (16.14-16.18) 16.18-16.22 (16.22-16.26) 16.26-16.30 (16.30-16.34) 16.34-16.38 (16.38-16.42) 16.42-16.46 (16.46-16.50) 16.50-16.54 (16.54-16.58) 16.58-16.62 (16.62-16.66) 16.66-16.70 (16.70-16.74) 16.74-16.78 (16.78-16.82) 16.82-16.86 (16.86-16.90) 16.90-16.94 (16.94-16.98) 16.98-17.02 (17.02-17.06) 17.06-17.10 (17.10-17.14) 17.14-17.18 (17.18-17.22) 17.22-17.26 (17.26-17.30) 17.30-17.34 (17.34-17.38) 17.38-17.42 (17.42-17.46) 17.46-17.50 (17.50-17.54) 17.54-17.58 (17.58-17.62) 17.62-17.66 (17.66-17.70) 17.70-17.74 (17.74-17.78) 17.78-17.82 (17.82-17.86) 17.86-17.90 (17.90-17.94) 17.94-17.98 (17.98-18.02) 18.02-18.06 (18.06-18.10) 18.10-18.14 (18.14-18.18) 18.18-18.22 (18.22-18.26) 18.26-18.30 (18.30-18.34) 18.34-18.38 (18.38-18.42) 18.42-18.46 (18.46-18.50) 18.50-18.54 (18.54-18.58) 18.58-18.62 (18.62-18.66) 18.66-18.70 (18.70-18.74) 18.74-18.78 (18.78-18.82) 18.82-18.86 (18.86-18.90) 18.90-18.94 (18.94-18.98) 18.98-19.02 (19.02-19.06) 19.06-19.10 (19.10-19.14) 19.14-19.18 (19.18-19.22) 19.22-19.26 (19.26-19.30) 19.30-19.34 (19.34-19.38) 19.38-19.42 (19.42-19.46) 19.46-19.50 (19.50-19.54) 19.54-19.58 (19.58-19.62) 19.62-19.66 (19.66-19.70) 19.70-19.74 (19.74-19.78) 19.78-19.82 (19.82-19.86) 19.86-19.90 (19.90-19.94) 19.94-19.98 (19.98-20.02) 20.02-20.06 (20.06-20.10) 20.10-20.14 (20.14-20.18) 20.18-20.22 (20.22-20.26) 20.26-20.30 (20.30-20.34) 20.34-20.38 (20.38-20.42) 20.42-20.46 (20.46-20.50) 20.50-20.54 (20.54-20.58) 20.58-20.62 (20.62-20.66) 20.66-20.70 (20.70-20.74) 20.74-20.78 (20.78-20.82) 20.82-20.86 (20.86-20.90) 20.90-20.94 (20.94-20.98) 20.98-21.02 (21.02-21.06) 21.06-21.10 (21.10-21.14) 21.14-21.18 (21.18-21.22) 21.22-21.26 (21.26-21.30) 21.30-21.34 (21.34-21.38) 21.38-21.42 (21.42-21.46) 21.46-21.50 (21.50-21.54) 21.54-21.58 (21.58-21.62) 21.62-21.66 (21.66-21.70) 21.70-21.74 (21.74-21.78) 21.78-21.82 (21.82-21.86) 21.86-21.90 (21.90-21.94) 21.94-21.98 (21.98-22.02) 22.02-22.06 (22.06-22.10) 22.10-22.14 (22.14-22.18) 22.18-22.22 (22.22-22.26) 22.26-22.30 (22.30-22.34) 22.34-22.38 (22.38-22.42) 22.42-22.46 (22.46-22.50) 22.50-22.54 (22.54-22.58) 22.58-22.62 (22.62-22.66) 22.66-22.70 (22.70-22.74) 22.74-22.78 (22.78-22.82) 22.82-22.86 (22.86-22.90) 22.90-22.94 (22.94-22.98) 22.98-23.02 (23.02-23.06) 23.06-23.10 (23.10-23.14) 23.14-23.18 (23.18-23.22) 23.22-23.26 (23.26-23.30) 23.30-23.34 (23.34-23.38) 23.38-23.42 (23.42-23.46) 23.46-23.50 (23.50-23.54) 23.54-23.58 (23.58-23.62) 23.62-23.66 (23.66-23.70) 23.70-23.74 (23.74-23.78) 23.78-23.82 (23.82-23.86) 23.86-23.90 (23.90-23.94) 23.94-23.98 (23.98-24.02) 24.02-24.06 (24.06-24.10) 24.10-24.14 (24.14-24.18) 24.18-24.22 (24.22-24.26) 24.26-24.30 (24.30-24.34) 24.34-24.38 (24.38-24.42) 24.42-24.46 (24.46-24.50) 24.50-24.54 (24.54-24.58) 24.58-24.62 (24.62-24.66) 24.66-24.70 (24.70-24.74) 24.74-24.78 (24.78-24.82) 24.82-24.86 (24.86-24.90) 24.90-24.94 (24.94-24.98) 24.98-25.02 (25.02-25.06) 25.06-25.10 (25.10-25.14) 25.14-25.18 (25.18-25.22) 25.22-25.26 (25.26-25.30) 25.30-25.34 (25.34-25.38) 25.38-25.42 (25.42-25.46) 25.46-25.50 (25.50-25.54) 25.54-25.58 (25.58-25.62) 25.62-25.66 (25.66-25.70) 25.70-25.74 (25.74-25.78) 25.78-25.82 (25.82-25.86) 25.86-25.90 (25.90-25.94) 25.94-25.98 (25.98-26.02) 26.02-26.06 (26.06-26.10) 26.10-26.14 (26.14-26.18) 26.18-26.22 (26.22-26.26) 26.26-26.30 (26.30-26.34) 26.34-26.38 (26.38-26.42) 26.42-26.46 (26.46-26.50) 26.50-26.54 (26.54-26.58) 26.58-26.62 (26.62-26.66) 26.66-26.70 (26.70-26.74) 26.74-26.78 (26.78-26.82) 26.82-26.86 (26.86-26.90) 26.90-26.94 (26.94-26.98) 26.98-27.02 (27.02-27.06) 27.06-27.10 (27.10-27.14) 27.14-27.18 (27.18-27.22) 27.22-27.26 (27.26-27.30) 27.30-27.34 (27.34-27.38) 27.38-27.42 (27.42-27.46) 27.46-27.50 (27.50-27.54) 27.54-27.58 (27.58-27.62) 27.62-27.66 (27.66-27.70) 27.70-27.74 (27.74-27.78) 27.78-27.82 (27.82-27.86) 27.86-27.90 (27.90-27.94) 27.94-27.98 (27.98-28.02) 28.02-28.06 (28.06-28.10) 28.10-28.14 (28.14-28.18) 28.18-28.22 (28.22-28.26) 28.26-28.30 (28.30-28.34) 28.34-28.38 (28.38-28.42) 28.42-28.46 (28.46-28.50) 28.50-28.54 (28.54-28.58) 28.58-28.62 (28.62-28.66) 28.66-28.70 (28.70-28.74) 28.74-28.78 (28.78-28.82) 28.82-28.86 (28.86-28.90) 28.90-28.94 (28.94-28.98) 28.98-29.02 (29.02-29.06) 29.06-29.10 (29.10-29.14) 29.14-29.18 (29.18-29.22) 29.22-29.26 (29.26-29.30) 29.30-29.34 (29.34-29.38) 29.38-29.42 (29.42-29.46) 29.46-29.50 (29.50-29.54) 29.54-29.58 (29.58-29.62) 29.62-29.66 (29.66-29.70) 29.70-29.74 (29.74-29.78) 29.78-29.82 (29.82-29.86) 29.86-29.90 (29.90-29.94) 29.94-29.98 (29.98-30.02) 30.02-30.06 (30.06-30.10) 30.10-30.14 (30.14-30.18) 30.18-30.22 (30.22-30.26) 30.26-30.30 (30.30-30.34) 30.34-30.38 (30.38-30.42) 30.42-30.46 (30.46-30.50) 30.50-30.54 (30.54-30.58) 30.58-30.62 (30.62-30.66) 30.66-30.70 (30.70-30.74) 30.74-30.78 (30.78-30.82) 30.82-30.86 (30.86-30.90) 30.90-30.94 (30.94-30.98) 30.98-31.02 (31.02-31.06) 31.06-31.10 (31.10-31.14) 31.14-31.18 (31.18-31.22) 31.22-31.26 (31.26-31.30) 31.30-31.34 (31.34-31.38) 31.38-31.42 (31.42-31.46) 31.46-31.50 (31.50-31.54) 31.54-31.58 (31.58-31.62) 31.62-31.66 (31.66-31.70) 31.70-31.74 (31.74-31.78) 31.78-31.82 (31.82-31.86) 31.86-31.90 (31.90-31.94) 31.94-31.98 (31.98-32.02) 32.02-32.06 (32.06-32.10) 32.10-32.14 (32.14-32.18) 32.18-32.22 (32.22-32.26) 32.26-32.30 (32.30-32.34) 32.34-32.38 (32.38-32.42) 32.42-32.46 (32.46-32.50) 32.50-32.54 (32.54-32.58) 32.58-32.62 (32.62-32.66) 32.66-32.70 (32.70-32.74) 32.74-32.78 (32.78-32.82) 32.82-32.86 (32.86-32.90) 32.90-32.94 (32.94-32.98) 32.98-33.02 (33.02-33.06) 33.06-33.10 (33.10-33.14) 33.14-33.18 (33.18-33.22) 33.22-33.26 (33.26-33.30) 33.30-33.34 (33.34-33.38) 33.38-33.42 (33.42-33.46) 33.46-33.50 (33.50-33.54) 33.54-33.58 (33.58-33.62) 33.62-33.66 (33.66-33.70) 33.70-33.74 (33.74-33.78) 33.78-33.82 (33.82-33.86) 33.86-33.90 (33.90-33.94) 33.94-33.98 (33.98-34.02) 34.02-34.06 (34.06-34.

# Bedside Calculators



Offline calculators — every result shows its formula and source. Decision support only; confirm against the drug chart and local protocol before administration.

## Infusion calculator – dose ↔ rate

Weight

kg

Drug

Epinephrine (adrenaline)



Typical: 0.05–1 mcg/kg/min

**Your preparation – enter the actual concentration  
(nothing is assumed)**

4 mg / 50 mL

1 mg / 50 mL

Drug amount

mg

in volume

mL

= concentration

mcg/mL

Target dose

dose

unit

mcg/kg/min



or

Known rate

mL/hr

Enter weight, your preparation, and a dose *or* a rate.

Rate (mL/hr) = dose  $\times$  weight (kg)  $\div$  concentration (mcg/mL), with dose in mcg/kg/hr ( $\times 60$  if per-minute;  $\times 1000$  if mg/kg/hr). Concentrations are **never assumed** — presets are common preparations you can edit, or type your own. Dose ranges are titrated to response. **Source:** AHA/AAP PALS 2025; pediatric Surviving Sepsis Campaign (Weiss, 2020).

## Maintenance fluids & bolus

Weight

kg

Enter weight.

Maintenance — Holliday–Segar: 4-2-1 mL/kg/hr (100-50-20 mL/kg/day) for the first 10 / next 10 / each further kg. Use an **isotonic** fluid with glucose (AAP 2018). Bolus 10–20 mL/kg balanced crystalloid; reassess after each. **Source:** Holliday & Segar 1957; AAP Clinical Practice Guideline 2018; PALS 2025.

## Blood products – volumes & rates

Weight

kg

Current Hb

now

Target Hb

aim

Hb unit

g/L



### Red cells – non-bleeding child (top-up)

Enter weight, current and target Hb.

### Platelets · FFP · cryoprecipitate

Enter weight.

### Major haemorrhage – per-kg aliquots (see [Massive Transfusion](#))

Enter weight.

### Neonatal exchange transfusion

Enter weight.

Red cells (mL) = [target Hb – current Hb (g/L)] × weight (kg) × 0.4 (factor 4 ÷ 10 for g/L; ≈ 4 mL/kg raises Hb by 10 g/L) — aim no more than 20 g/L above the transfusion threshold, double-check the volume is ≤ 20 mL/kg, usually ≤ 1 adult unit (≈ 280 mL), run at ≈ 5 mL/kg/h and finish each pack within 4 h of leaving the fridge; non-bleeding neonate: 15 mL/kg over 4 h. Platelets 10–20 mL/kg (max one adult therapeutic dose, ≈ 200 mL apheresis) and FFP 15–20 mL/kg, cryoprecipitate 5–10 mL/kg (higher volumes when bleeding; max 2 pools), each at 10–20 mL/kg/h. Blood volume ≈ 80 mL/kg (90 in term newborns, 70–80 in older children); massive loss = 80 mL/kg in 24 h, 40 mL/kg in 3 h or 2–3 mL/kg/min; aliquots RBC 20 mL/kg (max 4 units) · FFP 20 mL/kg (max 4 units) · platelets 15–20 mL/kg (max 1 adult dose) after every 40 mL/kg RBC · cryo 10 mL/kg (max 2 pools); targets Hb 80 g/L, fibrinogen > 1.5 g/L, PT ratio < 1.5, platelets > 75 ×10<sup>9</sup>/L. Double-volume exchange 160–200 mL/kg depending on gestational age (removes 85–90 % of red cells, ≈ 50 % of bilirubin). The formula predicts the rise only approximately (donor haematocrit varies) — recheck Hb. **Source:** BSH guideline on transfusion for fetuses, neonates and older children (New 2016)<sup>340</sup>; BSH administration of blood components (Robinson 2018)<sup>341</sup>; TAXI 2018<sup>339</sup>.

## DKA – fluids, insulin & cerebral-injury doses (ISPAD 2022)

Weight

kg

---

Severity (assumed deficit)

**Moderate — pH 7.1–7.19 (7%)**

---



Bolus already given

mL (blank = 10 mL/kg)

---

Replace deficit over

**36 h**

---



Insulin

**0.1 unit/kg/h**

---



**Bolus, deficit & hourly rate**

Enter weight.

**Insulin, potassium, dextrose**

Enter weight.

**Suspected cerebral injury**

Enter weight.

ISPAD 2022 method: bolus 0.9% saline 10–20 mL/kg over 20–30 min (20 mL/kg if perfusion is poor; repeat in shock); assume dehydration 5% mild, 7% moderate, 10% severe; then give the usual maintenance (Holliday–Segar) plus the deficit minus the bolus, evenly over 24–48 h (the guide's algorithm uses 36 h). No fixed cap in ISPAD 2022; in obese children adult protocols may be used (1 L maximum bolus; 500 mL/h infusion). Insulin 0.05–0.1 unit/kg/h (0.03 unit/kg/h may be considered under 5 y with mild DKA) started  $\geq 1$  h after fluids, never as a bolus; 50 units in 50 mL = 1 unit/mL. Potassium 40 mmol/L in the infusate, started after volume expansion and with insulin once  $K^+ < 5.5$  mmol/L and urine has been passed; if  $K^+ < 3.5$  mmol/L give potassium first ( $\leq 0.5$  mmol/kg/h) and defer insulin. Add dextrose 5% at glucose 14–17 mmol/L or a fall  $> 5$  mmol/L/h (10–12.5% may be needed). Bicarbonate only for life-threatening hyperkalemia or pH  $< 6.9$  with compromised contractility. Cerebral injury: mannitol 0.5–1 g/kg over 10–15 min (repeat after 30 min if needed) or 3% NaCl 2.5–5 mL/kg over 10–15 min; the algorithm caps 3% NaCl at 250 mL and mannitol at 100 g. In PICU the Kuwait PICU Task Force protocol governs. **Source:** ISPAD 2022<sup>7</sup>; Kuwait PICU Task Force<sup>172</sup>; algorithm layout TREKK 2023<sup>342</sup> / CPS 2023<sup>343</sup>.

# Burns – Lund-Browder TBSA & resuscitation fluid

Age

< 1 y



Weight

kg

Hours since burn

h

For each area, enter the percentage of **that area** with partial- or full-thickness burn (ignore simple erythema). The area's share of the body for the chosen age is shown in brackets.

Head (19%)

% burned

Neck (2%)

% burned

Anterior trunk (13%)

% burned

Posterior trunk (13%)

% burned

Buttocks (both) (5%)

% burned

Genitalia (1%)

% burned

Upper arms (both) (8%)

% burned

Forearms (both) (6%)

% burned

Hands (both) (5%)

% burned

Thighs (both) (11%)

% burned

Lower legs (both) (10%)

Feet (both) (7%)



% burned

% burned

TBSA **0%**.

Lund–Browder (both sides combined, % of body by age < 1 / 1 / 5 / 10 / 15 / adult): head 19 / 17 / 13 / 11 / 9 / 7 · thighs 11 / 13 / 16 / 17 / 18 / 19 · lower legs 10 / 10 / 11 / 12 / 13 / 14; fixed — neck 2, anterior trunk 13, posterior trunk 13, buttocks 5, genitalia 1, upper arms 8, forearms 6, hands 5, feet 7. Children with > 10% TBSA are formally resuscitated: 2–4 mL/kg/%TBSA of Ringer's lactate over 24 h (the guide uses modified Parkland 3–4 mL/kg/%), half in the first 8 h **from the time of the burn**, the rest over 16 h; add maintenance fluid with glucose; titrate to urine output  $\geq 1$  mL/kg/h — the formula is only a starting estimate. **Source:** Lund & Browder 1944<sup>345</sup>; ISBI 2016<sup>271</sup>; RCH Melbourne burns guideline<sup>346</sup>.

## Hyponatremia – 3% saline & correction ceiling

Weight

kg

---

Enter weight.

### Correction tracker

Starting Na

mmol/L

---

Current Na

mmol/L

---

Hours elapsed

h

---

Enter the starting and current sodium and the hours between them.

Symptomatic hyponatremic encephalopathy: 3% NaCl 2 mL/kg (max 100 mL) over 10 min; repeat 1–2 times until symptoms improve; goal a 5–6 mmol/L rise in the first 1–2 h (each bolus raises Na by at most ~2 mmol/L). Stop boluses once symptoms resolve or Na has risen 10 mmol/L in the first 5 h; total rise in the first 48 h  $\leq$  15–20 mmol/L. For chronic or asymptomatic hyponatremia the guide's ceiling is  $\leq$  8 mmol/L in 24 h (see Acute Electrolyte Emergencies). Sodium-deficit

and free-water formulas are deliberately not included — no validated pediatric source. **Source:** Moritz & Ayus 2010 (Pediatr Nephrol)<sup>347</sup>; ESE/ERA guideline<sup>99</sup>.

## Glucose infusion rate (GIR)

Weight

kg

---

Dextrose

%

---

Rate

mL/h

---

or target GIR

mg/kg/min

---

Enter weight, dextrose % and a rate  
or a target GIR.

GIR (mg/kg/min) = dextrose % × rate (mL/h) ÷ (6 × weight kg) — because 1% dextrose = 10 mg/mL and 1 h = 60 min. Neonatal IV dextrose: GIR 5–8 mg/kg/min, escalate in 2 mg/kg/min steps; needing > 8–10 mg/kg/min to stay euglycemic suggests hyperinsulinism; > 12.5% dextrose needs central access. **Source:** AAP (Adamkin 2011)<sup>82</sup>; Pediatric Endocrine Society (Thornton 2015)<sup>81</sup>.

## Body surface area & hyperhydration

Height

cm

---

Weight

kg

---

Enter height and weight.

$BSA (m^2) = \sqrt{(\text{height cm} \times \text{weight kg} \div 3600)}$ .

Tumor-lysis hyperhydration 2–3 L/m<sup>2</sup>/day (200 mL/kg/day if  $\leq 10$  kg), adjusted to age, cardiac function and urine output; potassium, phosphate and calcium withheld from the fluid initially; target urine output 80–100 mL/m<sup>2</sup>/h (4–6 mL/kg/h if  $\leq 10$  kg), specific gravity  $\leq 1.010$ ; diuretics only if output falls short without hypovolemia or obstruction. **Source:** Mosteller 1987<sup>348</sup>; Coiffier 2008<sup>37</sup>; BCSH TLS guideline (Jones 2015)<sup>66</sup>.

## eGFR – bedside Schwartz (2009)

Height

cm

Creatinine

value

Unit

μmol/L



Enter height and creatinine.

$\text{eGFR (mL/min/1.73 m}^2\text{)} = 0.413 \times \text{height (cm)} \div \text{creatinine (mg/dL)}$ ; with creatinine in  $\mu\text{mol/L}$  the constant is  $0.413 \times 88.4 = 36.5$ . Derived in children 1–16 y with CKD (GFR 30–90) using enzymatic, IDMS-traceable creatinine; less validated at normal GFR, and not valid in AKI when creatinine is not at steady state. **Source:** Schwartz 2009 (J Am Soc Nephrol)<sup>349</sup>.

## Pediatric Appendicitis Score (PAS)

|                                                 |   |                                    |   |
|-------------------------------------------------|---|------------------------------------|---|
| Cough / percussion /<br>hopping tenderness RLQ  |   | Right lower quadrant<br>tenderness |   |
| <b>No (0)</b>                                   | ✓ | <b>No (0)</b>                      | ✓ |
| Anorexia                                        |   | Fever                              |   |
| <b>No (0)</b>                                   | ✓ | <b>No (0)</b>                      | ✓ |
| Nausea / vomiting                               |   | WBC > 10 ×10 <sup>9</sup> /L       |   |
| <b>No (0)</b>                                   | ✓ | <b>No (0)</b>                      | ✓ |
| Neutrophilia (ANC > 7.5<br>×10 <sup>9</sup> /L) |   | Migration of pain to RLQ           |   |
| <b>No (0)</b>                                   | ✓ | <b>No (0)</b>                      | ✓ |

**PAS 0 / 10** — low ( $\leq 3$ ; LR 0.13) —  
observe/discharge with safety-net;  
still not excluded.

Eight items, total 0–10, derived in children 4–15 y. Bands from the systematic review of 26 prospective cohorts (Ebell 2014):  $\leq 3$  low (likelihood ratio 0.13), 4–7 equivocal (0.70),  $\geq 8$  high (8.1) — at usual pre-test probabilities no PAS cut-off rules appendicitis in or out on its own, so pair the score with ultrasound and surgical review. Single-centre validations used  $\leq 2$  / 3–6 /  $\geq 7$  (Goldman 2008) and  $\leq 4$  / 5–7 /  $\geq 8$  (Bhatt 2009). **Source:** Samuel 2002<sup>350</sup>; Ebell 2014 systematic review (Ann Emerg Med)<sup>353</sup>; Goldman 2008 (J Pediatr)<sup>351</sup>; Bhatt 2009<sup>352</sup>.

## HLH-2004 criteria · MAS-2016 · MS score

### HLH-2004 (primary/genetic HLH)

Fever

**No**



Splenomegaly

**No**



Cytopenias  $\geq 2$  lines (Hb  $< 90$  g/L [ $< 100$  if  $< 4$  wk], plt  $< 100$ , ANC  $< 1.0$ )

**No**



Triglycerides  $\geq 3.0$  mmol/L and/or fibrinogen  $\leq 1.5$  g/L

**No**



Hemophagocytosis (marrow, spleen, node)

**No**



Low/absent NK-cell activity

**No**



Ferritin  $\geq 500$   $\mu$ g/L

**No**



Soluble CD25  $\geq 2,400$  U/mL

**No**



**0 of 8** HLH-2004 criteria — not met (needs 5, or a molecular diagnosis).

### Systemic JIA – MAS-2016 classification & MS score

Fever (mandatory)

**Yes**



CNS dysfunction

**No**



Hemorrhagic manifestations

**No**



Active arthritis

**No**



Platelets ( $\times 10^9$ /L)

$\times 10^9$ /L

LDH (U/L)

U/L

Fibrinogen

value

Fibrinogen unit

g/L



Ferritin (μg/L)

μg/L = ng/mL

AST (U/L)

U/L

Triglycerides (mmol/L)

mmol/L

Enter the laboratory values.

HLH-2004: molecular diagnosis, or 5 of 8 criteria.  
MAS-2016 (febrile child with known or suspected sJIA): ferritin > 684 ng/mL plus any 2 of platelets  $\leq 181 \times 10^9/L$ , AST > 48 U/L, triglycerides > 156 mg/dL (1.76 mmol/L), fibrinogen  $\leq 360$  mg/dL (3.6 g/L); abnormalities must not be explained otherwise. MS score = CNS  $\times 2.44$  + hemorrhage  $\times 1.54$  + arthritis  $\times (-1.30)$  + platelets ( $\times 10^9/L$ )  $\times (-0.003)$  + LDH (U/L)  $\times 0.001$  + fibrinogen (mg/dL)  $\times (-0.004)$  + ferritin (ng/mL)  $\times 0.0001$ ;  $\geq -2.1$  indicates MAS (validation sensitivity 89%, specificity 99%); fever is mandatory. Criteria support, never delay, treatment (see Cytokine Storm card). **Source:** Henter 2007<sup>211</sup>; Ravelli 2016<sup>212</sup>; Minoia 2019<sup>398</sup>.



## Shock index, pediatric age-adjusted (SIPA)

Age

years

---

Heart rate

bpm

---

Systolic BP

mmHg

---

Enter age, heart rate and systolic BP.

Shock index = heart rate ÷ systolic BP. Elevated SIPA: > 1.22 at 4–6 y, > 1.0 at 7–12 y, > 0.9 at 13–16 y.

Derived in blunt trauma, ages 4–16 (injury severity score > 15); an elevated SIPA identified more children with ISS > 30, transfusion in 24 h, high-grade liver/spleen injury needing transfusion, and in-hospital death than the unadjusted index. Not validated outside 4–16 y or for non-trauma shock. **Source:** Acker 2015 (J Pediatr Surg)<sup>355</sup>.

# Paracetamol & iron ingestion

## Paracetamol (acetaminophen) – acute single ingestion

Weight

kg

Age

years

Paracetamol taken

mg

Enter weight, age and the amount taken.

Hours since ingestion

h

Level

value

Level unit

mg/L =  $\mu\text{g/mL}$



Enter the hours since ingestion and the level.

Enter weight for the acetylcysteine (21-h) doses.

## Iron

Elemental iron taken

mg

Enter weight and the elemental iron taken (from the label).

Paracetamol: refer to hospital if < 6 y and  $\geq 200$  mg/kg or amount unknown;  $\geq 6$  y and  $\geq 10$  g or 200 mg/kg, whichever is lower. Treatment line 150 mg/L at 4 h, halving every 4 h (37.5 mg/L at 12 h) to 24 h; a level drawn before 4 h cannot be plotted; not for repeated supratherapeutic or unknown-time ingestions (treat if level > 20 mg/L or transaminases abnormal).  $\mu\text{mol/L} = \text{mg/L} \times 6.614$ . Acetylcysteine 21-h: 150 mg/kg over 1 h  $\rightarrow$  50 mg/kg over 4 h  $\rightarrow$  100 mg/kg over 16 h (dose for 100 kg if heavier); diluent  $\leq 20$  kg 3 / 7 / 14 mL/kg; 21–40 kg 100 / 250 / 500 mL; > 40 kg 200 / 500 / 1,000 mL — unadjusted volumes cause hyponatremia and seizures. Iron:  $\geq 40$  mg/kg elemental iron, or severe/persistent symptoms  $\rightarrow$  hospital; < 40 mg/kg with mild symptoms may be observed (call KPCC 1804774). **Source:** Dart 2006 (out-of-hospital)<sup>356</sup>; Dart 2023 consensus<sup>31</sup>; acetylcysteine prescribing information<sup>357</sup>; Manoguerra 2005<sup>358</sup>.

## PARDS – oxygenation index (PALICC-2)

Support

**Invasive ventilation**



FiO<sub>2</sub>

%

Mean airway pressure

cmH<sub>2</sub>O

PaO<sub>2</sub>

mmHg

SpO<sub>2</sub>

%

Choose the support and enter FiO<sub>2</sub>  
with PaO<sub>2</sub> and/or SpO<sub>2</sub>.

OI = mean airway pressure × FiO<sub>2</sub> (%) ÷ PaO<sub>2</sub>; OSI = mean airway pressure × FiO<sub>2</sub> (%) ÷ SpO<sub>2</sub> (use only when SpO<sub>2</sub> ≤ 97%). Invasive ventilation: PARDS if OI ≥ 4 or OSI ≥ 5; mild-moderate OI < 16 / OSI < 12; severe OI ≥ 16 / OSI ≥ 12 (stratify after ≥ 4 h). Non-invasive (full face mask CPAP ≥ 5 cmH<sub>2</sub>O): PaO<sub>2</sub>/FiO<sub>2</sub> ≤ 300 or SpO<sub>2</sub>/FiO<sub>2</sub> ≤ 250. Applies < 18 y without active perinatal lung disease, within 7 days of a known insult, with new infiltrates not fully explained by cardiac failure or overload. **Source:** PALICC-2 (Emeriaud 2023)<sup>359</sup>.

## Corrected age for preterm infants

Date of birth

mm/dd/yyyy



Assessment date

mm/dd/yyyy



Gestation

weeks

+ days

0–6

Enter date of birth and gestation at birth.

Chronological age = time since birth. Postmenstrual age = gestational age + chronological age. Corrected age = chronological age minus the weeks born before 40 weeks; use it for children born preterm up to 3 years of age (growth and development). **Source:** AAP Committee on Fetus and Newborn (Engle 2004)<sup>360</sup>.

## PUCAI – Pediatric Ulcerative Colitis Activity Index

Abdominal pain

**None (0)** 

---

Rectal bleeding

**None (0)**

---

Stool consistency (most stools)

**Formed (0)** 

---

Stools per 24 h

**0–2 (0)** 

---

Nocturnal stools (waking the child)

**No (0)** 

---

Activity level

**No limitation (0)** 

---

**PUCAI 0** — remission (< 10).

Six clinical items, no laboratory values; total 0–85. **< 10** remission · **10–34** mild · **35–64** moderate · **≥ 65** severe = acute severe colitis (admit, IV corticosteroids). On IV steroids: **day 3 > 45** → plan second-line therapy; **day 5 ≥ 65** → start second-line; **35–65** on day 5 → continue steroids 2–5 more days with daily scoring; **< 35** → responding. **Source:** Turner 2007 (Gastroenterology); ESPGHAN/ECCO acute severe colitis guideline 2025 (Assa et al., JPGN).

# Oxygen cylinder duration

Cylinder

— size —



Gauge pressure

now

Units

psi



Flow

L/min

Choose a cylinder, then enter the gauge pressure and flow.

Usable duration = (gauge pressure – safe residual) × cylinder factor ÷ flow. **Safe residual 200 psi (~14 bar)** — treat as empty; change / refill at ~500 psi (~35 bar).

Factors (L/psi): D 0.16 · E 0.28 · M 1.56 · G 2.41 · H/K 3.14 (US/CGA; metric cylinders vary — check the label). **Source:** Egan's Fundamentals of Respiratory Care 12e; AARC; BOC cylinder data.

# Electrolyte & lab corrections

## Corrected sodium (for hyperglycemia)

Measured Na

mmol/L

---

Glucose

mmol/L

---

—

## Anion gap

Na

mmol/L

---

Cl

mmol/L

---

HCO<sub>3</sub>

mmol/L

---

Albumin

g/L (opt)

---

—

## Corrected calcium

Total Ca

mmol/L

---

Albumin



g/L

---

—

### Serum osmolality (calculated)

Na

mmol/L

---

Glucose

mmol/L

---

Urea

mmol/L

---

—

Corrected Na = measured +  $2.4 \times (\text{glucose} - 5.5) \div 5.5$  (Hillier 1999). Anion gap =  $\text{Na} - (\text{Cl} + \text{HCO}_3^-)$ , ref 8–16; albumin-corrected +  $2.5 \times (40 - \text{albumin g/L}) \div 10$ . Corrected Ca (mmol/L) = measured +  $0.02 \times (40 - \text{albumin g/L})$ . Osmolality =  $2 \times \text{Na} + \text{glucose} + \text{urea}$  (mmol/L). **Source:** Hillier, Am J Med 1999; Payne 1973; standard SI formulae.

## QTc (corrected QT)

QT interval

ms

---

Heart rate

bpm

---

Enter QT (ms) and heart rate.

Bazett:  $QTc = QT \div \sqrt{RR}$ . Fridericia:  $QTc = QT \div RR^{1/3}$ , where  $RR (s) = 60 \div HR$ . Prolonged > 450 ms (> 460 ms in females/infants); > 500 ms = high risk of torsades. Bazett over-corrects at extremes of rate.

**Source:** Bazett 1920; Rautaharju/AHA 2009.

## Glasgow Coma Scale

Eye (1–4)

**4 Spontaneous**



Verbal (1–5)

**5 Oriented / coos-babbles**



Motor (1–6)

**6 Obeys / spontaneous**



GCS **15** / 15 (E4 V5 M6) — minor

Pediatric verbal responses shown for pre-verbal children. GCS  $\leq$  8 signals a threatened airway → involve PICU/anesthesia. **Source:** Teasdale & Jennett 1974; pediatric modification (James 1986).

## Westley croup score

Level of consciousness

**Normal**



Cyanosis

**None**



Stridor

**None**



Air entry

**Normal**



Retractions

**None**



**Westley score 0 — mild**

≤ 2 mild · 3–5 moderate · 6–11 severe · ≥ 12 impending respiratory failure. Give dexamethasone 0.15–0.6 mg/kg for all; nebulized L-epinephrine 1:1000 (1 mg/mL) 0.5 mL/kg (max 5 mL) for moderate–severe.

**Source:** Westley 1978.

## APGAR

Appearance

**2 All pink**



Pulse

**2  $\geq 100$**



Grimace

**2 Cry/cough**



Activity

**2 Active**



Respiration

**2 Strong cry**



APGAR **10** / 10

Scored at 1 and 5 min (continue every 5 min to 20 if < 7). APGAR does not guide resuscitation — do not delay NRP steps to assign it. **Source:** Apgar 1953; NRP 8th ed / AHA-AAP 2025.

## Endotracheal tube – size & depth

Age (years)

yr

---

Enter age ( $\geq 1$  year). For neonates use weight-based sizing.

---

Cuffed ID (mm) =  $\text{age} \div 4 + 3.5$ ; uncuffed =  $\text{age} \div 4 + 4$  ( $\geq 1$  yr). Depth at lip (cm) =  $\text{age} \div 2 + 12$ , or  $3 \times \text{tube ID}$ . Confirm with capnography and CXR. Neonate: ID  $\approx 2.5\text{--}3.5$  by weight; depth  $\approx \text{weight (kg)} + 6$  cm.

**Source:** APLS; AHA/AAP PALS 2025.



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Primary guidelines & landmark trials  
(Vancouver)

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Tap to view the full reference list.

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